When Valuation Becomes a World, Part II: Inside the Valuation World - Derivative Entanglement, Relative Frames, and Curved Financial Geometry

Derivative Entanglement, Relative Frames, and Curved Financial Geometry

A Layered QM–SR–GR Toy Architecture Viewed from the Secondary θ-Time Universe

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Source Note

Part I, *When Valuation Becomes a World: Complex Finance, Internal Time, and the Residue of Quantum Strangeness*, began from the pressure-preserving complex completion:

$$Z = R + iQ. \quad (0.1)$$

Here R is admitted financial value, Q is retained valuation pressure, A is the declared pre-filter amplitude, and θ is the orientation induced by a mature valuation filter:

$$A^2 = R^2 + Q^2. \quad (0.2)$$

$$R = A \cos \theta. \quad (0.3)$$

$$Q = A \sin \theta. \quad (0.4)$$

$$Z = A \exp(i\theta). \quad (0.5)$$

Part I then allowed A and θ to vary, separated radial economic change from angular valuation-frame change, distinguished calendar time t from phase order θ and ledger time k, and enlarged the static geometry into a world-forming runtime:

$$\text{Primary Field} \rightarrow \text{Declaration} \rightarrow \text{Projection} \rightarrow R + iQ \rightarrow \text{Phase} \rightarrow \text{Gate} \rightarrow \text{Ledger} \rightarrow \text{Backreaction} \rightarrow \text{Revision}. \quad (0.6)$$

Its central result was deliberately limited. Under constant amplitude and stable declaration:

$$dZ/d\theta = iZ. \quad (0.7)$$

Therefore:

$$dR/d\theta = -Q. \quad (0.8)$$

$$dQ/d\theta = R. \quad (0.9)$$

$$d^2R/d\theta^2 = -R. \quad (0.10)$$

$$d^2Q/d\theta^2 = -Q. \quad (0.11)$$

These are classical rotational equations. They do not by themselves derive tensor-product state

spaces, quantum entanglement, Born probabilities, Bell inequality violation, no-cloning, or physical wavefunction collapse. Part I therefore used finance as a non-quantum control world for subtracting generic observer-bound effects from genuinely quantum structure.

Appendix M nevertheless opened a further path. It proposed a layered architecture in which local CAPM valuation, complex internal states, Lorentz-like valuation frames, curved global financial geometry, gauge transport, contextual gates, ledger formation, and recursive backreaction occupy different mathematical roles. It explicitly suggested that complex states could live in fibres over a curved financial manifold, while locally flat frame relations remained recoverable in suitable regions.

The present article develops that path.

Its most important correction is perspectival.

An option and its underlying appear classically and contractually connected when viewed from the primary financial universe that constructs them. Their relationship may be explained through payoff rules, stochastic pricing models, market data, replication, hedging, funding, clearing, and legal settlement.

But quantum entanglement is not experienced from the hypothetical perspective of an observer standing outside the physical universe with access to its complete constructor. Its strangeness is encountered by observers inside the effective world, with access only to admissible measurements of local subsystems and recorded outcomes.

The corresponding financial comparison must therefore also be made from inside the financial world.

This article distinguishes:

Primary Constructor Universe
→ Secondary θ-Time Valuation World
→ Internal Protocol-Bounded Observer. (0.12)

The primary universe constructs the financial world.

The secondary world carries complex valuation states, local frames, derivative relations, gates, and effective geometry.

The internal observer accesses only a bounded measurement algebra within that world.

From the primary perspective, derivative dependence may be transparent.

From the secondary perspective, an option and its underlying may appear as locally incomplete parts of one globally prepared composite state.

A second source is *Self-Referential Observers in Quantum Dynamics*, which models observers as internal processes that record outcomes, condition later measurement choices on trace, and experience past outcomes as fixed within their own filtration. It also distinguishes internal certainty, cross-observer agreement, frame compatibility, accessible records, and redundancy-generated objectivity.

This article transfers that internal-observer discipline into finance without claiming that financial markets are literal quantum systems.

The result is a formal toy architecture, not a physical unification claim.

Abstract

Modern finance does not merely assign values to independently existing objects. It constructs relational financial objects whose identity, admissibility, dynamics, and historical consequences depend on contracts, valuation protocols, measurement settings, settlement rules, and ledgers.

An option is the clearest example.

From the primary economic universe, the option appears as an ordinary derivative function:

$$D(t) = V[U(t), K, T-t, \sigma(t), r(t), q(t), P, L, \ldots]. \quad (0.13)$$

Here U is the underlying state, K the strike, T−t the remaining maturity, σ the relevant volatility state, r the financing state, q the carry state, P the declared valuation protocol, and L the existing ledger.

From this external constructor perspective, the option–underlying relation is explicable. The derivative is contractually defined, probabilistically valued, dynamically hedged, legally settled, and institutionally recorded.

This article argues that this is not yet the correct perspective for comparison with quantum entanglement.

A declared financial compiler maps part of the primary economic field into a secondary effective valuation world:

$$\mathcal{C}_{P,L}: \Sigma_{primary} \rightarrow W_{\theta}. \quad (0.14)$$

Inside W_θ, financial states are ordered by an internal phase coordinate θ, observed through protocol-bounded instruments, committed through gates, and historicized through ledger time k. An internal observer has access not to the complete primary field or its full construction map, but to a restricted observable projection:

$$Visible_O(\theta) = \hat{O}_{O,P,L}[\rho_F(\theta)]. \quad (0.15)$$

The central proposal is that derivative finance supplies the composite-state grammar missing from the scalar CAPM completion.

Let $\mathscr{H}_U$ be the effective underlying-state space and $\mathscr{H}_D$ the derivative-state space. Their composite space is:

$$\mathscr{H}_{UD} = \mathscr{H}_U \otimes \mathscr{H}_D. \quad (0.16)$$

A contract may be represented as a preparation operator:

$$\hat{U}_{contract}(|u_n\rangle|0_D\rangle) = |u_n\rangle|d_n\rangle. \quad (0.17)$$

Applied to a multi-branch underlying state:

$$|\psi_U\rangle = \Sigma_n c_n \exp(i\varphi_n)|u_n\rangle, \quad (0.18)$$

the contract prepares:

$$|\Psi_{UD}\rangle = \Sigma_n c_n \exp(i\varphi_n)|u_n,d_n\rangle. \quad (0.19)$$

When this state cannot be factorized as:

$$|\Psi_{UD}\rangle \neq |\psi_U\rangle \otimes |\psi_D\rangle, \quad (0.20)$$

the underlying and derivative are nonfactorizable inside the declared secondary valuation world.

This does not by itself establish physical quantum entanglement.

Standard derivative dependence may remain representable by classical probability, contractual constraints, shared information, replication, or causal feedback. A classically correlated mixture has the form:

$$\rho_{mix} = \Sigma_n p_n \rho_n^U \otimes \rho_n^D. \quad (0.21)$$

A stronger coherent state requires relative phases and observable off-diagonal terms:

$$\rho_{UD} = |\Psi_{UD}\rangle\langle\Psi_{UD}|. \quad (0.22)$$

The article therefore develops an entanglement ladder ranging from ordinary correlation through contractual coupling, dynamical binding, effective-world nonfactorization, coherent composite states, local mixedness, contextual joint measurement, no-signalling entanglement, and Bell-nonclassicality.

Finance clearly realizes the lower levels.

The middle levels can be formally constructed and tested.

The highest levels remain unestablished.

The apparent strangeness arises because an observer confined to one local sector sees only a reduced state:

$$\rho_U = Tr_D(\rho_{UD}). \quad (0.23)$$

$$\rho_D = Tr_U(\rho_{UD}). \quad (0.24)$$

The global state may remain well structured while neither local observer possesses a complete independent state. Measurement of one sector conditionally changes the state assigned to the other, not necessarily because an internally visible signal has travelled between two complete objects, but because both measurements refer to one prepared joint state.

This yields the article's central distinction:

Entanglement Is Global Structure; Strangeness Is Local Access. (0.25)

The architecture then embeds this QM-like composite-state layer inside a broader QM–SR–GR financial toy framework.

CAPM is treated as a locally valid valuation law rather than a global theory:

$$r_i = r_f + \beta_i ERP. \quad (0.26)$$

Local complex valuation states are:

$$Z_i = R_i + iQ_i = A_i \exp(i\theta_i). \quad (0.27)$$

SR-like frame transformations relate local valuation observers using different benchmarks, numeraires, funding curves, horizons, legal frames, or reporting rules.

GR-like geometry describes a globally state-dependent financial manifold:

$$ds_F^2 = g^F_{\mu\nu}(x,L,P)dx^\mu dx^\nu. \quad (0.28)$$

Local flat frames are related to the global metric through:

$g^{F}_{\mu\nu} = e^{a}_{\mu} e^{b}_{\nu} \eta_{ab}$. (0.29)

A gauge connection transports valuation phase and orientation between local frames:

$D_{\mu} = \nabla_{\mu} + i\mathcal{A}_{\mu}$. (0.30)

The resulting effective-state equation is written schematically as:

$i\hbar_{FD} \partial_{\theta} |\Psi_F\rangle = \hat{H}_F[g^{F}, \mathcal{A}, P, L] |\Psi_F\rangle + |\varepsilon_F\rangle$. (0.31)

The effective generator may contain:

$\hat{H}_F = \hat{H}_{CAPM} + \hat{H}_{contract} + \hat{H}_{hedge} + \hat{H}_{ledger} + \hat{H}_{environment}$. (0.32)

Here:

- $\hat{H}_{CAPM}$ governs local valuation motion;
- $\hat{H}_{contract}$ binds underlying and derivative sectors;
- $\hat{H}_{hedge}$ produces derivative-to-underlying backreaction;
- $\hat{H}_{ledger}$ carries historical consequence;
- $\hat{H}_{environment}$ represents volatility, liquidity, funding, information, collateral, and institutional coupling.

This arrangement does not force QM, SR, and GR symbols to denote the same thing.

QM-like structure belongs to complex states, tensor composition, relative phase, and measurement.

SR-like structure belongs to local frames and frame transformations.

GR-like structure belongs to the curved global manifold.

Gauge structure belongs to phase transport and frame comparison.

Ledger structure belongs to irreversible historical commitment.

The article concludes by revising the quantum-subtraction programme:

Observed Quantum Strangeness = $G_{world} + G_{composite} + Q_{residue}$. (0.33)

Where:

G_{world} = Declaration + Projection + Gate + Trace + Backreaction. (0.34)

$G_{composite}$ = Joint Preparation + Local Restriction + Conditional Update + Phase Transport. (0.35)

$Q_{residue}$ contains whatever cannot be reproduced through these non-quantum structures, including potentially irreducible Born probability, experimentally mandatory coherent interference, no-signalling entanglement, Bell inequality violation, specifically quantum contextuality, no-cloning, and quantum disturbance relations.

The framework is not investment advice. It is a conceptual and mathematical research

programme. Every added coordinate, phase, operator, metric, and entanglement claim must be tested against standard option pricing, classical joint distributions, copula models, stochastic volatility models, network models, agent-based models, and ordinary market-microstructure explanations.

If the layered architecture provides no measurable gain in prediction, diagnosis, attribution, simulation, cross-frame consistency, or intervention, it should be reduced rather than defended rhetorically.

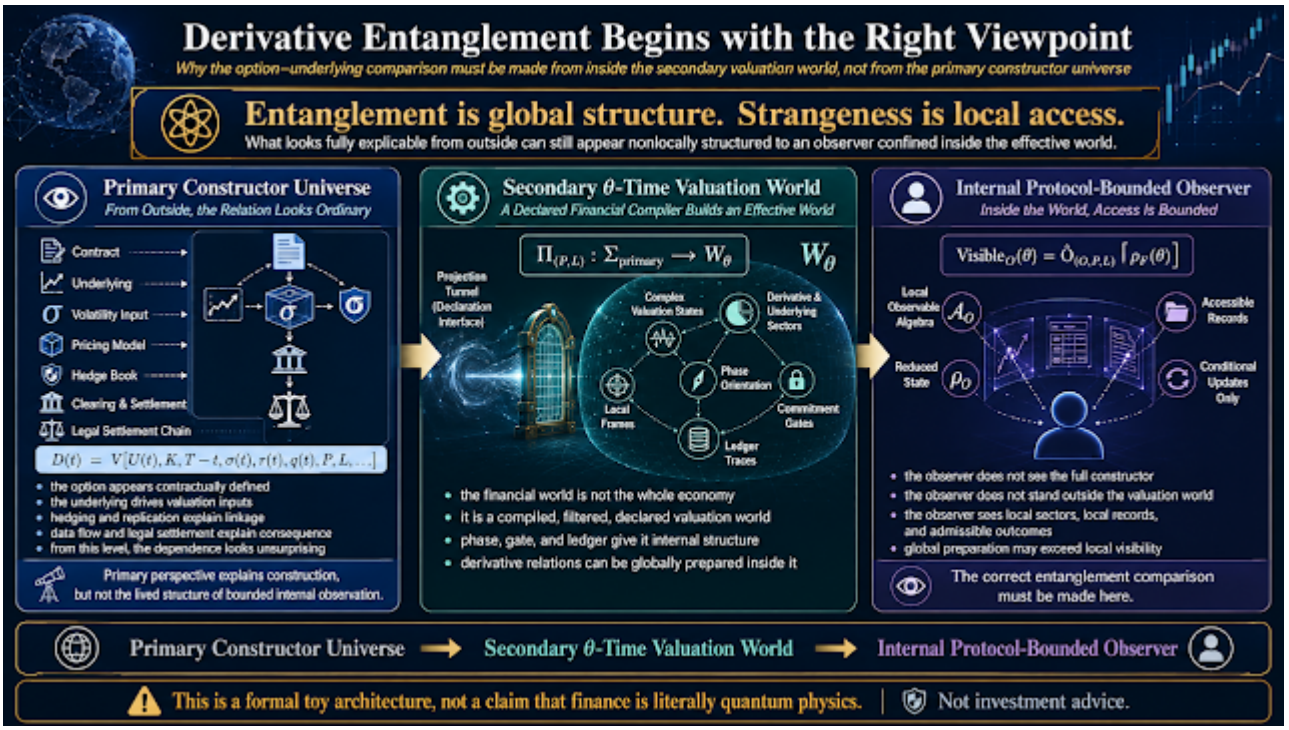
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From Scalar Complex Value to Composite Derivative Structure

Why Part I needed Part II: one complex coordinate can carry phase, but not subsystem entanglement

Complex rotation is not yet entanglement. Composite structure is the missing ingredient.

Part I supplied R, Q, A, and θ. Part II supplies underlying sector × derivative sector.

① Part I Complex Finance

The Scalar Completion

$$Z = R + iQ = Ae^{i\theta}$$
$$R = A\cos\theta$$
$$Q = A\sin\theta$$



Meaning

R = admitted value
Q = retained valuation pressure
A = pre-filter amplitude
θ = valuation orientation

- captures visible value and hidden pressure
- separates radial change from angular change
- supports phase motion, gates, and backreaction
- but remains a one-channel state

② The Limitation

One complex state ≠ two subsystems

No natural decomposition:

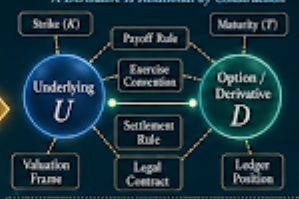
$$\mathcal{H} \neq \mathcal{H}_A \otimes \mathcal{H}_B$$



- no underlying sector versus derivative sector
- no joint state
- no product vs separable vs nonseparable distinction
- no reduced local states
- no branch-relative joint phase structure

③ Derivatives Change the State Space

A Derivative Is Relational by Construction



At maturity: $D_T = \max(U_T - K, 0)$

Before maturity: $D_t = V\,(U_t, K, T-t, \sigma_t, r_t, q_t, P_t, L_t, \ldots)$

The derivative supplies a natural paired structure:
Underlying Sector × Derivative Sector

④ Composite State Grammar

The Missing Composite Structure

$$\mathcal{H}_{UD} = \mathcal{H}_U \otimes \mathcal{H}_D$$
$$\hat{U}_{\text{contract}}(|u_n\rangle|0_D\rangle) = |u_n\rangle|d_n\rangle$$
$$|\psi_U\rangle = \sum_n c_n e^{i\theta_n}|u_n\rangle$$
$$|\Psi_{UD}\rangle = \sum_n c_n e^{i\theta_n}|u_n, d_n\rangle$$
$$|\Psi_{UD}\rangle \neq |\psi_U\rangle \otimes |\psi_D\rangle$$



- the contract acts as a preparation operator
- the underlying and derivative can become nonfactorizable within the declared world
- nonfactorization is now mathematically available
- this still does not automatically prove physical quantum entanglement

The Entanglement Ladder

Nine Levels of Increasing Strength

1–4	5–6	7–9
1. Ordinary correlation	5. Effective-world nonfactorization	7. Contextual joint measurement
2. Functional dependence	6. Coherent composite state	8. No-signalling entanglement
3. Contractual coupling		9. Bell-nonclassicality
4. Dynamical binding		
Levels 1–4: Clearly realized in finance ✓	*Levels 5–6: Formally constructible and testable* ✓	*Levels 7–9: Not established* ?

Finance clearly realizes the lower ladder; the higher ladder remains an empirical question, not a rhetorical shortcut. — *Not investment advice.*

A LAYERED FINANCIAL WORLD: QM-LIKE STATES, SR-LIKE FRAMES, GR-LIKE GEOMETRY

Unity comes from role separation and coupling, not from forcing one symbol to mean everything.

TOP LAYER — 4 — COUPLING LAYER

GAUGE, GATE, AND LEDGER COUPLING



- Connects states across frames and space.
- Gates commit events and create history.
- Ledger records consequences irreversibly.
- Backreaction feeds valuation outputs into the primary world.

Gauge-covariant valuation evolution

$$i\hbar_V D_\theta|\Psi_F\rangle = \hat{H}_F[\beta^F, A, P, \Sigma]|\Psi_F\rangle + |\epsilon_F\rangle$$

$$D_\mu = \nabla_\mu + i\mathcal{A}_\mu \quad \text{(covariant derivative)}$$

Effective Generator Decomposition

$$\hat{H}_F = \hat{H}_{\text{CAPM}} + \hat{H}_{\text{contract}} + \hat{H}_{\text{ledger}} + \hat{H}_{\text{ledger}} + \hat{H}_{\text{environment}}$$

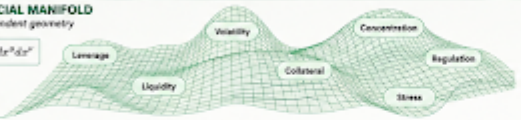
- $\hat{H}_{\text{CAPM}}$ — Local valuation motion (risk/return law)
- $\hat{H}_{\text{contract}}$ — Binds underlying & derivative (preparation)
- $\hat{H}_{\text{hedge}}$ — Hedging & market impact (feedback)
- $\hat{H}_{\text{ledger}}$ — Historical record & commitment
- $\hat{H}_{\text{environment}}$ — Volatility, liquidity, funding, information, institutions, regulation

3 — GR-LIKE GEOMETRY LAYER

CURVED FINANCIAL MANIFOLD

Global state-dependent geometry

$$ds_F^2 = g_{\mu\nu}^F(x, L, P)dx^\mu dx^\nu$$
$$\rightarrow g_{\mu\nu}^F = e_\mu^a e^b{}_\nu \eta_{ab} \quad \text{(via vielbein } e_\mu^a\text{)}$$



- Financial distance depends on state.
- Local flatness can coexist with global curvature.
- Paths and outcomes depend on the geometry (holonomy possible).
- Geometry shaped by leverage, liquidity, collateral, regulation, volatility, stress and more.



2 — SR-LIKE FRAME LAYER

RELATIVE VALUATION FRAMES

Local observers with different references



- Observers use different frames.
- Valuation stances transform between frames.
- SR-like means local frame transformation, not physical spacetime.

Frame Transformation (Lorentz-like Analogy)

$$\begin{pmatrix} \rho_C' \\ \rho' \end{pmatrix} = \Lambda(v)\begin{pmatrix} \rho_D \\ \rho \end{pmatrix}, \quad |v| < 1$$

Converts relative pricing stances between frames.

1 — QM-LIKE FIBRE LAYER

COMPLEX STATES AND COMPOSITE STRUCTURE

Quantum-like mathematics on each local fibre

Core Ingredients

- Complex amplitudes
- Relative phase
- Tensor composition
- Density operators
- Local and joint observables
- Reduced states
- Conditional updates
- Interference (if coherent)



Reduced Local States

$$\rho_U = \text{Tr}_D(\rho_{UD})$$
$$\rho_D = \text{Tr}_U(\rho_{UD})$$

Local views are generally incomplete.

Nonfactorization (Entanglement-like)

$$\rho_{UD} \neq \rho_U \otimes \rho_D$$

Correlations beyond classical mixtures are possible within the valuation world.

0 — FOUNDATION (CAPM LAYER)

CAPM AS A LOCAL VALUATION LAW (NOT A WORLD THEORY)

Local Valuation Law

$$r_i = r_f + \beta_i\, ERP$$

Expected return depends on beta within a local market context.

Local Valuation State

$$Z_i = R_i + iQ_i = A_i e^{i\theta_i}$$

R_i = admitted value (visible component)
Q_i = retained valuation pressure (hidden component)
A_i = pre-filter amplitude $\quad \theta_i$ = valuation orientation

Key Ideas

- CAPM gives local drift/compensation.
- The complex state encodes admission (R) and retained pressure (Q).
- This layer supplies local motion, not global structure.



Outcome

CAPM holds in each small chart. Global geometry may be curved & state-dependent.

IMPORTANT WARNING

This is a FORMAL TOY ARCHITECTURE.

It does NOT claim that finance literally unifies quantum mechanics, relativity, and gravity.

It is a layered framework for organizing ideas, generating hypotheses, and exploring testable consequences.

Not investment advice.

THE LAYERED PICTURE

Local valuation laws live on fibres, frames move, the manifold curves, gauges transport, gates commit, and ledgers remember.

→ **States (QM-like)** — Provide structure, phase, and local mixture.

→ **Frames (SR-like)** — Provide perspective, reference, and transformation.

→ **Geometry (GR-like)** — Provides global context, distance, and path dependence.

→ **Ledger & Backreaction** — Provide history, finality, and feedback into the primary world.

→ **Outcome** — A coherent architecture for derivative entanglement, relative frames, and curved financial geometry.

INSIDE THE VALUATION WORLD: OBSERVATION, COLLAPSE, GATE, AND LEDGER OBJECTIVITY

A price is not just seen. It is projected, gated, recorded, and then made historically operative.

1 GLOBAL STATE (PAIRED RELATION)

The underlying and derivative can form a joint composite state in the valuation world.

$$\rho_{UD} = |\Psi_{UD}\rangle\langle\Psi_{UD}|$$



Local (reduced) access

$\rho_U = \mathrm{Tr}_D(\rho_{UD})$ (Underlying side) | $\rho_D = \mathrm{Tr}_U(\rho_{UD})$ (Derivative side)

- Global information may exceed what either side can access.
- Local observers see reduced states.

2 LOCAL OBSERVATION (Bounded Observer)

An internal financial observer sees through a declared protocol.



$$\mathrm{Visible}_O(\theta) = \hat{O}_{(O,P,L)}[\rho_P(\theta)]$$

- Has instruments and observable algebra
- Uses a local frame and inherited trace
- Policy is adaptive and trace-conditioned
- Past outcomes are fixed within its own filtration

3 CONDITIONAL UPDATE (Collapse Experience)

Measurement-like interaction produces conditional definiteness.

Instrument / Measurement (interaction) → Conditional Update (reduced state) → Outcome Experienced (Local Definiteness)

- Local outcome becomes definite for this observer's records.
- Other compatible outcomes become excluded for future actions.

4 GATE (Commitment Event)

A gate commits the outcome as a consequential event.

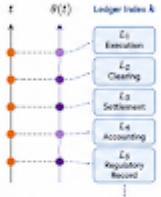


Gate = Authorization + Finality Rule + Consequence Condition

- Not every observation passes the gate.
- Gates create history by writing trace.

5 LEDGER (Irreversible Trace)

The gate writes the event into the ledger.



$$t \to \theta(t) \to \mathrm{Gate}_k \to \mathrm{Trace}_k \to \mathcal{L}_{k+1}$$

- Calendar time t: duration of events
- Phase time θ: internal valuation order
- Ledger index k: committed history order

6 OBJECTIVITY (Cross-Observer Agreement)

Objectivity is alignment across observers on a shared record.



Requirements for Agreement

- ✓ Compatible effects (commutation)
- ✓ Frame mapping between observers
- ✓ Accessible record (public or shared)
- ✓ Provenance and audit trail
- ✓ Redundancy (independent persistence)

Observation is not yet history. → History begins when a gate writes a persistent trace that constrains the future. → **Objectivity emerges from shared, accessible records.**

WHAT MAKES AN OBSERVER "INTERNAL"?

- ✓ Limited instruments
- ✓ Accessible observable algebra
- ✓ Local frame / reference
- ✓ Inherited ledger and traces
- ✓ Adaptive policy based on records



Sees only what the protocol allows.

THREE CLOCKS OF THE FINANCIAL WORLD

Calendar Time t (Physical Duration)	Phase Time $\theta(t)$ (Valuation Order)	Ledger Index k (Committed Order)
Measures how long things take.	Measures internal valuation evolution.	Advances only when a gate commits trace.

KEY TAKEAWAY

A financial price or exposure is the result of a pipeline: projection → conditional update → gate → ledger.

Effective reality in finance is not observer-free.
It is created, maintained, and agreed upon through protocols, commitments, and shared records.

IMPORTANT WARNING — This is a formal toy architecture, not a claim that financial prices are literal quantum collapses. Not investment advice.

NOT LEGAL OR ACCOUNTING ADVICE — Always consult qualified professionals for legal, regulatory, tax, accounting, or audit matters.

RESEARCH FRAMEWORK ONLY — For conceptual understanding, model development, and academic exploration.

INSPIRED BY BOUNDED-OBSERVER DISCIPLINE — Many ideas are adapted from quantum foundations and reinterpreted for financial systems.

WHEN VALUATION CHANGES THE WORLD

Hedging, Curvature, Holonomy, and the Quantum Subtraction Framework

A valuation world is not merely descriptive. Once ledgered, it feeds back on the world that created it.

1 THE HEDGING FEEDBACK LOOP

From Valuation to Backreaction



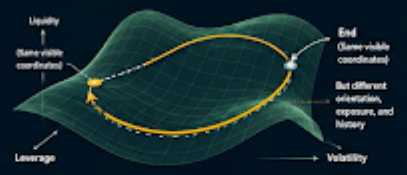
Backreaction Channels

Delta Hedging Flow | Gamma Squeeze | Vega Stress | Collateral Spiral | Funding Feedback | Liquidation Pressure

→ The secondary valuation world acts back on the primary economy.

2 CURVATURE & HOLONOMY

Closed Loops Need Not Return to the Same World-State



BASE HOLONOMY (Manifold Residue) — Effective position on the financial manifold is changed.

FIBRE HOLONOMY (Phase / Orientation Residue) — Phase or orientation does not return to its initial value.

LEDGER HOLONOMY (Irreversible Residue) — Irreversible trace creates a different future ledger.

Key Takeaways

- Same visible price can hide different path histories.
- Same exposure can carry different funding, collateral, or regulatory context.
- Path dependence is deeper than simple hysteresis.
- Financial loops can return geometrically but not historically.

3 REVISED QUANTUM SUBTRACTION PROGRAMME

Observed Quantum Strangeness = What the Non-Quantum World Cannot Yet Reproduce

$$\text{Observed Quantum Strangeness} = G_{\text{world}} + G_{\text{composite}} + Q_{\text{residue}}$$

G_{world} =	$G_{\text{composite}}$ =	Q_{residue} =
Declaration + Projection + Gate + Trace + Backreaction	Joint Preparations + Local Restriction + Conditional Update + Phase Transport	What remains specifically quantum (to be tested)

REPRODUCIBLE BY THE FINANCIAL CONTROL WORLD	CANDIDATE GENUINE QUANTUM RESIDUE
Declaration & filtration	Irreducible Born probability
Projection & reduced access	Empirically mandatory coherent interference
Local incompleteness	No-signalling entanglement
Gates & trace	Bell inequality violation
Frame dependence	Specifically quantum contextuality
Curved effective geometry	No-cloning
Feedback & latching	Quantum disturbance relations
Historical irreversibility	Exchange statistics / monogamy-type structure

The goal is not to absorb quantum theory into finance, but to sharpen the boundary of what remains specifically quantum.

EVIDENCE DISCIPLINE
If the layered model adds no measurable gain, reduce it.

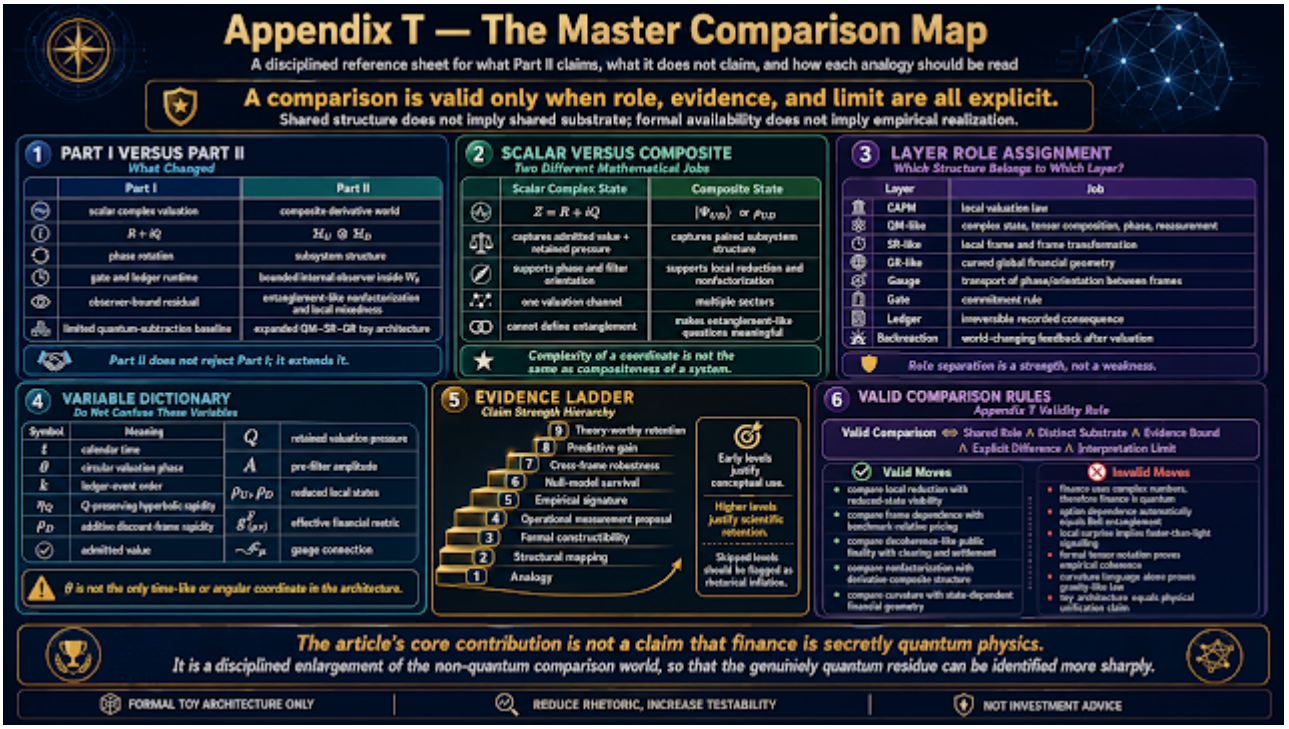
1. **Compare Against Classical Mixtures** — Test vs diagonal scenario weights (no coherence).
2. **Compare Against Rich Null Models** — Stochastic vol, jumps, microstructure, latent states, nonlinear effects.
3. **Require Controlled Interventions** — Phase shifts, hedging perturbations, funding shocks, information releases.
4. **State Failure Conditions** — Define what would falsify coherence, nonfactorization, curvature, holonomy.
5. **Prefer Simpler Models When They Suffice** — Occam's principle with quantitative discipline.

Aim: Reproducible Gain — Keep the model only if it survives out-of-sample tests and cross-frame robustness.

This is a formal toy architecture. Not a claim that finance unifies quantum mechanics, relativity, or gravity.

Not investment advice. For conceptual research and discussion only.

Reduce rhetoric. Increase testability. Build evidence.



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0. Reader's Guide: What Part II Changes

0.1 What Part I completed

Part I began from an ordinary financial fact.

Future economic possibility is not admitted into price or ledger without filtration.

A projected cash flow may be discounted.

A credit instrument may be adjusted for default and recovery.

An asset may be filtered through beta, market risk premium, liquidity, capital requirements, accounting rules, or execution constraints.

A future amount A therefore enters a declared valuation process and exits as admitted value R:

A → Mature Financial Filter → R. (0.36)

Finance Geometry restored the orthogonal complement hidden by the scalar result:

A → θ → R + iQ. (0.37)

The relation was:

$A^2 = R^2 + Q^2$. (0.38)

$R = A \cos \theta$. (0.39)

$Q = A \sin \theta$. (0.40)

The complex state was:

$$Z = A \exp(i\theta). \quad (0.41)$$

Part I then allowed both amplitude and frame orientation to move:

$$Z(t) = A(t)\exp[i\theta(t)]. \quad (0.42)$$

Differentiation gave:

$$dZ/dt = [g_A + i\omega_F]Z + \varepsilon_{dyn}. \quad (0.43)$$

Where:

$$g_A = (1/A)(dA/dt). \quad (0.44)$$

$$\omega_F = d\theta/dt. \quad (0.45)$$

The real and imaginary coordinates obeyed:

$$dR/dt = g_A R - \omega_F Q + Re(\varepsilon_{dyn}). \quad (0.46)$$

$$dQ/dt = g_A Q + \omega_F R + Im(\varepsilon_{dyn}). \quad (0.47)$$

The angular repricing load was:

$$\Lambda_F = Q\omega_F. \quad (0.48)$$

Therefore:

Visible Repricing = Radial Economic Change – Angular Filter Load + Residual. (0.49)

This decomposition separated:

1. change in the declared economic amplitude;
2. change caused by movement of the valuation frame;
3. dynamic failure of the declared effective world.

Part I then distinguished three financial clocks:

Calendar Time = t. (0.50)

Phase Time = θ. (0.51)

Ledger Time = k. (0.52)

Their relation was:

$$t \rightarrow \theta(t) \rightarrow G_k \rightarrow L_{k+1}. \quad (0.53)$$

Calendar time measures duration.

Phase time orders movement through valuation orientation.

Ledger time advances when a consequential event is committed.

The resulting concept was:

Phase Order + Gate + Trace = Effective Financial Time. (0.54)

Part I thereby enlarged a coordinate system into a world-forming runtime:

$$X_k \to \mathrm{Declare}_{\{P_k\}}(X_k) \to \Pi_{\{P_k,L_k\}}(X_k) \to Z_k \to \mathrm{Gate}_k \to L_{k+1} \to \mathscr{B} \to X_{k+1} \to \mathcal{U} \to P_{k+1}. \quad (0.55)$$

Here:

- X_k is the larger primary economic field;
- P_k is the declared valuation protocol;
- L_k is the existing ledger;
- Π is the projection process;
- Z_k is the effective complex state;
- Gate_k determines commitment;
- $\mathscr{B}$ maps financial consequences back into the primary field;
- $\mathcal{U}$ revises the protocol when residual becomes too large.

The financial world therefore did not merely represent its primary field.

It acted back upon it.

A rating changes funding.

A price changes collateral.

A margin rule changes liquidation.

An accounting classification changes distributable capital.

A regulatory model changes institutional behaviour.

A ledgered valuation becomes causally operative.

0.2 The unresolved limitation

Part I's complex state remained one-dimensional in the relevant sense.

It contained one complex coordinate:

$$Z = R + iQ. \quad (0.56)$$

That coordinate could represent:

- admitted value;
- retained pressure;
- filter orientation;
- phase velocity;
- gate-sensitive evolution;

- residual;
- trace-bearing backreaction.

But it did not yet contain two subsystems.

There was no natural decomposition:

$\mathscr{H} = \mathscr{H}_A \otimes \mathscr{H}_B$. (0.57)

There was no joint state.

There were no reduced local states.

There was no distinction between product states, separable mixtures, and nonseparable composite states.

There were no branch-relative phases affecting joint observations.

Part I therefore correctly concluded that its classical rotational law did not derive entanglement or tensor-product nonseparability.

The limitation did not arise because complex finance had failed.

It arose because the state space was too small.

A one-channel complex valuation state cannot reproduce a phenomenon whose definition requires a composite system.

This article begins by supplying the missing composite-state grammar.

0.3 Why derivatives change the problem

A derivative is not merely another asset located beside its underlying.

Its identity is relational.

A call option is not fully specified by the option premium alone.

It requires:

- an underlying;
- a strike;
- a maturity;
- a payoff rule;
- an exercise convention;
- a settlement rule;
- a valuation frame;
- a volatility state;
- a financing state;

- a legal contract;
- a ledger position.

At maturity:

$D_T = \max(U_T - K, 0)$. (0.58)

Before maturity:

$D_t = V(U_t, K, T-t, \sigma_t, r_t, q_t, P, L, \ldots)$. (0.59)

The derivative therefore supplies a natural pairing:

Underlying Sector × Derivative Sector. (0.60)

The composite space is:

$\mathscr{H}_{UD} = \mathscr{H}_U \otimes \mathscr{H}_D$. (0.61)

This still does not prove quantum entanglement.

But it changes the available mathematics.

Finance now possesses:

- distinguishable subsystems;
- local observables;
- joint observables;
- a coupling rule;
- a state-preparation process;
- conditional updates;
- exercise and settlement gates;
- hedge backreaction;
- ledgered history.

The question is no longer whether scalar CAPM alone contains entanglement.

It does not.

The question becomes:

> Once CAPM is embedded as a local valuation law inside a composite derivative world, what entanglement-like structures become available to observers confined to that world?

0.4 The central perspectival correction

The initial temptation is to compare an option with quantum entanglement from the primary financial universe.

From that perspective, the explanation is straightforward:

Underlying state changes
→ pricing inputs change
→ option value changes. (0.62)

Or:

Option exposure changes
→ hedge demand changes
→ underlying order flow changes. (0.63)

There is a visible causal mechanism.

There is no mystery.

But this is analogous to explaining quantum entanglement from the perspective of an imagined external observer who sees the complete construction of the physical world, the global state, every inaccessible degree of freedom, and the entire measurement process.

That is not the perspective of an actual observer inside the physical universe.

The relevant quantum observer:

- occupies the same world as the measured system;
- accesses only admissible instruments;
- observes local or bounded subsystems;
- conditions on recorded outcomes;
- cannot step outside the whole world to inspect its ultimate constructor.

The financial comparison must impose the same restriction.

An internal financial observer does not receive the complete primary economic universe.

The observer receives a compiled effective world.

That world may expose:

- spot prices;
- option prices;
- implied volatilities;
- Greeks;
- margin calls;
- exercise events;
- collateral states;
- settlement records.

It may conceal or compress:

- the complete generating distribution;
- all counterparty positions;
- all hidden liquidity;
- all future policy responses;
- all unobserved order intentions;
- all model choices;
- the full primary-to-secondary compilation map.

The correct comparison therefore begins only after the financial observer has been placed inside the secondary world.

0.5 Three levels of description

Part II uses three levels.

Level P — Primary constructor universe

The primary financial state is:

$X(t) \in \Sigma_{primary}$. (0.64)

It contains the larger economic and institutional machinery:

- productive assets;
- future cash flows;
- agents;
- contracts;
- information channels;
- legal rules;
- pricing systems;
- balance sheets;
- exchanges;
- clearing;
- settlement;
- collateral;
- funding;
- regulation;

- calendar time.

At this level, the derivative can be described constructively.

Level W — Secondary valuation world

The declared compilation is:

$\mathcal{C}_{P,L}: \Sigma_{primary} \to W_\theta$. (0.65)

W_θ contains the effective states and relations available under protocol P and ledger L.

It may possess:

- local complex states;
- composite derivative states;
- phase progression;
- local valuation frames;
- a curved global geometry;
- admissible measurement instruments;
- gates;
- trace;
- backreaction.

Level O — Internal observer

An internal observer O has access only to an admissible subalgebra:

$\mathcal{A}_O \subset \mathcal{A}(W_\theta)$. (0.66)

Its visible state is:

$Visible_O = \hat{O}_{O,P,L}[\rho_F(\theta)]$. (0.67)

The observer does not necessarily possess:

- the full global state;
- the complete state-preparation history;
- the primary constructor;
- the inverse compilation map;
- every other observer's frame;
- every uncommitted branch.

The distinction is therefore:

Primary Construction ≠ Secondary Structure ≠ Internal Observation. (0.68)

0.6 Entanglement is not created by ignorance

Observer restriction is essential to the appearance of strangeness, but ignorance alone is insufficient.

Suppose the option and underlying possess an ordinary classically correlated state:

$\rho_{classical} = \Sigma_n p_n \rho_n^U \otimes \rho_n^D$. (0.69)

An observer may not know which branch n applies.

But the uncertainty can still be interpreted as ignorance over definite paired states.

A stronger effective-world state satisfies:

$\rho_{UD} \notin Sep(\mathcal{H}_U \otimes \mathcal{H}_D)$. (0.70)

Where:

$Sep(\mathcal{H}_U \otimes \mathcal{H}_D) = \{\Sigma_n p_n \rho_n^U \otimes \rho_n^D\}$. (0.71)

This means the joint state cannot be reconstructed as a mixture of independent local states.

Part II will therefore keep two claims separate:

1. **Global relation:** the composite state may be nonfactorizable inside W_θ.
2. **Local strangeness:** internal observers possess only restricted access to that global relation.

The governing statement is:

Entanglement Is Global Structure; Strangeness Is Local Access. (0.72)

0.7 What Part II adds

Part II adds eight main structures.

First — Composite financial states

Derivatives introduce natural subsystem partitions:

$\mathcal{H}_{UD} = \mathcal{H}_U \otimes \mathcal{H}_D$. (0.73)

Second — Contractual state preparation

A contract acts as a binding or preparation operator:

$\hat{U}_{contract}(|u_n\rangle|0_D\rangle) = |u_n\rangle|d_n\rangle$. (0.74)

Third — Internal-observer measurement

Observers access local and joint financial observables through declared instruments.

Fourth — Multi-channel phase

The single angle θ is generalized into branch-relative phases:

$\Delta\theta_{nm} = \theta_n - \theta_m$. (0.75)

Fifth — Local valuation relativity

Different observers use different benchmarks, numeraires, funding curves, horizons, and reporting frames.

Sixth — Curved global financial geometry

Liquidity, leverage, volatility, collateral, funding, and trace may alter the effective metric:

ds_F² = g^F_{μν}(x,L,P)dx^μdx^ν. (0.76)

Seventh — Gauge transport

A connection preserves meaningful phase comparison between local frames:

D_μ = ∇_μ + i𝒜_μ. (0.77)

Eighth — A revised quantum subtraction

The stronger non-quantum financial world is subtracted before specifically quantum structure is identified:

Quantum Observation = G_world + G_composite + Q_residue. (0.78)

0.8 What Part II does not claim

This article does not claim that options are physically quantum particles.

It does not claim that financial contracts generate physical entanglement.

It does not claim that option markets violate Bell inequalities.

It does not claim that risk-neutral probabilities obey the Born rule.

It does not claim that an implied-volatility surface is physical spacetime.

It does not claim that financial crisis is literal spacetime curvature.

It does not claim that CAPM is a Hamiltonian of nature.

It does not claim that changing a valuation basis is identical to measuring spin.

It does not claim that hedge feedback is quantum nonlocality.

It does not claim that an inaccessible primary constructor is an admissible hidden-variable theory.

It does not claim that formal nonfactorization in a designed financial state proves physical nonseparability.

The article makes a narrower claim:

> A layered financial effective world can reproduce more of the operational grammar associated with quantum observation than a scalar CAPM model alone, especially after derivatives provide composite states and the comparison is made from the perspective of observers inside the secondary θ-time world.

Whether those new constructions improve finance is an empirical question.

Whether they illuminate quantum foundations is a conceptual question.

Whether they describe physical ontology remains entirely open.

0.9 Four levels of statement

The article uses four levels of claim.

Level 1 — Mathematical construction

Examples:

$\mathscr{H}_{UD} = \mathscr{H}_U \otimes \mathscr{H}_D$. (0.79)

$\rho_U = \mathrm{Tr}_D(\rho_{UD})$. (0.80)

$g^{F}_{\mu\nu} = e^{a}_{\mu} e^{b}_{\nu} \eta_{ab}$. (0.81)

These follow after the relevant representation has been declared.

Level 2 — Financial interpretation

Examples:

- a contract binds underlying and derivative sectors;
- delta is a local tangent sensitivity;
- gamma is local curvature of derivative response;
- ledger state changes future admissibility;
- frame changes may alter reported value without changing the governed object.

These depend on a disciplined mapping.

Level 3 — Empirical hypothesis

Examples:

- multi-channel phase may improve diagnosis of derivative-state transitions;
- financial holonomy may explain why visible prices return while exposure does not;
- local mixedness may improve modelling of incomplete desk-level state;
- gauge-corrected transport may improve cross-frame reconciliation;

- an entanglement-form representation may outperform classical mixture models.

These require data and null-model comparison.

Level 4 — Physics-facing implication

Examples:

- some entanglement-like strangeness may arise only after an observer is placed inside an effective composite world;
- a deeper constructor may be transparent externally but inaccessible internally;
- globally prepared structure may appear nonlocal to locally bounded observers;
- quantum residue should be defined only after the strongest non-quantum internal-world reconstruction has been attempted.

These are research hypotheses, not established physical conclusions.

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1. The Viewpoint Error in External Financial Analysis

1.1 Why the option looks ordinary from the primary universe

Suppose an observer begins with an underlying asset U and a call option D.

At maturity:

$D_T = \max(U_T - K, 0)$. (1.1)

Before maturity:

$D_t = V(U_t, K, T-t, \sigma_t, r_t, q_t, P, L, \ldots)$. (1.2)

From the primary universe, the option is explicitly constructed from:

- the underlying state;
- the contract;
- the relevant probability model;
- the discount frame;
- the volatility state;
- the legal and settlement protocol.

The observer can therefore say:

The option is not mysterious.

Its value changes because its inputs change.

Its payoff changes because the underlying changes.

Its hedge changes because delta and gamma change.

Its collateral requirement changes because the mark-to-market changes.

Its exercise status changes because the contract gate is crossed.

From this perspective:

Derivative Dependence = Contract + Model + Market Mechanism. (1.3)

This is correct.

But it is not complete.

It describes how the secondary relation is constructed from the primary universe.

It does not yet describe what that relation looks like to an observer who inhabits only the secondary world.

1.2 The hidden privilege of the external observer

An external explanation often grants the analyst several privileges.

The analyst is allowed to know:

- that one instrument is the underlying;
- that another is the derivative;
- the payoff function;
- the stochastic process assumed for the underlying;
- the valuation model;
- the volatility input;
- the interest-rate input;
- the hedge rule;
- the settlement mechanism;
- the relevant causal channels.

The analyst may also move freely between:

- price space;
- payoff space;
- risk-neutral probability;
- physical probability;
- accounting valuation;
- liquidation value;

- regulatory capital;
- legal enforceability.

This analyst is not an ordinary observer inside the effective financial world.

It is closer to a model architect.

It stands partly outside the object it is describing.

It can inspect the source code of the financial construction.

From such a position, many phenomena become unsurprising.

The option and underlying appear correlated because the analyst already knows the contract that binds them.

The derivative appears to react immediately because the analyst already knows the common valuation function.

The hedge appears reflexive because the analyst sees the trading mechanism.

The ledger appears irreversible because the analyst sees settlement and legal enforcement.

The external observer therefore possesses Constructor Privilege.

Define:

Constructor Privilege = Access to State Preparation + Coupling Rule + Primary Causal Mechanism. (1.4)

Quantum observers do not generally possess an equivalent privilege over the physical universe as a whole.

They do not stand outside physical reality and inspect its complete generating mechanism.

They interact from within it.

The financial analogy must therefore remove constructor privilege before comparing the observer experience.

1.3 Entering the secondary world

Let the primary financial field be:

$X(t) \in \Sigma_{primary}$. (1.5)

A protocol P and ledger L compile it into an effective state:

$\rho_F(\theta) = \mathcal{C}_{P,L}[X(t)]$. (1.6)

The internal observer does not observe X(t) directly.

It observes:

$\hat{O}_{O,P,L}[\rho_F(\theta)]$. (1.7)

Suppose the compiled world presents an option and its underlying as a joint state:

$\rho_{UD}(\theta) \in \mathcal{D}(\mathscr{H}_U \otimes \mathscr{H}_D)$. (1.8)

The observer may have access to underlying-side instruments:

$\boldsymbol{\mathcal{M}}^{U}_{a,u}$. (1.9)

It may have access to derivative-side instruments:

$\boldsymbol{\mathcal{M}}^{D}_{b,d}$. (1.10)

It may sometimes access joint instruments:

$\boldsymbol{\mathcal{M}}^{UD}_{a,b;u,d}$. (1.11)

But it does not necessarily access the full preparation map:

$\hat{U}_{contract}$. (1.12)

It does not necessarily access the primary economic state that produced the joint state.

It therefore encounters the derivative relation operationally:

Prepared Joint State
→ Selected Measurement
→ Correlated Outcome
→ Ledgered Record. (1.13)

This is a different description from:

Underlying input
→ pricing formula
→ derivative output. (1.14)

The first is internal and operational.

The second is external and constructive.

They need not contradict one another.

1.4 External explanation and internal strangeness can coexist

Consider a prepared joint state:

$|\Psi_{UD}\rangle = \Sigma_n c_n |u_n, d_n\rangle$. (1.15)

An external constructor may know how the coefficients and pairings were generated.

An internal underlying observer sees only:

$\rho_U = \mathrm{Tr}_D(|\Psi_{UD}\rangle\langle\Psi_{UD}|)$. (1.16)

An internal derivative observer sees only:

$\rho_D = \mathrm{Tr}_U(|\Psi_{UD}\rangle\langle\Psi_{UD}|)$. (1.17)

A measurement on the underlying sector changes the conditional derivative state assigned after

the result:

$$\rho_{D|u} = \mathrm{Tr}_U[(M^U_u \otimes I_D)\rho_{UD}(M^{U\dagger}_u \otimes I_D)] / p(u). \quad (1.18)$$

Where:

$$p(u) = \mathrm{Tr}[(M^{U\dagger}_u M^U_u \otimes I_D)\rho_{UD}]. \quad (1.19)$$

From the internal observer's perspective:

1. the joint state existed before local commitment;
2. one local outcome became definite;
3. the conditional state of the other sector changed;
4. the update referred to the global relation;
5. no separate internal messenger was required by the effective-state description.

From the primary perspective, the entire structure may still have been constructed through ordinary financial mechanisms.

The two descriptions operate at different levels.

The governing distinction is:

$$\text{Primary Causal Explanation} \neq \text{Secondary Conditional Description}. \quad (1.20)$$

This distinction is not a loophole.

It is the central object of study.

1.5 Why external explanation does not automatically dissolve effective nonseparability

Suppose the primary universe constructs a secondary state through a map:

$$\rho_{UD} = \mathcal{C}_{P,L}(X). \quad (1.21)$$

The fact that X and $\mathcal{C}$ are known externally does not imply that the compiled state factorizes internally.

A compiler can generate a relational object.

A legal contract can generate a joint obligation that neither party possesses independently.

A clearing system can generate net exposure that does not belong to any single gross position.

A structured product can generate a payoff state that cannot be specified from one component alone.

A portfolio can possess risks absent from every isolated instrument.

A derivative network can produce a globally constrained state whose local parts are incomplete.

Therefore:

Externally Constructed ≠ Internally Separable. (1.22)

The primary universe may explain where the relation came from.

The secondary universe determines whether the resulting state factorizes relative to its own subsystem decomposition.

These are different questions.

1.6 The internal-world criterion

The article will use the following provisional criterion.

Definition 1.1 — Secondary-World Financial Nonseparability

An underlying–derivative state is secondarily nonseparable under protocol P when:

1. the effective world declares a composite state space $\mathscr{H}_U \otimes \mathscr{H}_D$;
2. the joint state ρ_{UD} cannot be replaced by complete independent local states without changing admissible observations;
3. internal observers have access only to local or protocol-bounded measurement algebras;
4. measurement outcomes are committed into a ledger that changes later admissibility or instrument choice.

In compact form:

$$\mathrm{NonSep}_\theta(U,D \mid P,L) \Leftrightarrow \rho_{UD} \notin \mathrm{Sep}_P(\mathscr{H}_U \otimes \mathscr{H}_D). \quad (1.23)$$

This definition is world-relative.

It does not assert that the primary economic universe is quantum.

It asserts that a declared effective financial world may contain nonfactorizable relational states relative to its own accessible partition and measurement rules.

1.7 The first central proposition

Proposition 1.1 — Constructor Transparency Does Not Eliminate Internal Strangeness

A financial relationship may be transparent from the primary universe and nevertheless appear nonseparable to observers inside the secondary valuation world when:

- the primary-to-secondary map is not internally accessible;
- the effective state is globally prepared;
- local observers possess incomplete state access;
- commitment occurs through contextual measurement gates;

- correlated outcomes are retained in ledger history.

In compact form:

Primary Transparency + Internal Restriction → Explainable Construction + Experienced Strangeness. (1.24)

This proposition is not yet a claim about quantum physics.

It is a rule for conducting the comparison correctly.

Before asking whether finance reproduces quantum entanglement, the observer must first be placed at the corresponding level of access.

---

2. Three Levels of Financial Reality

2.1 Level P: the primary constructor universe

The primary universe is not merely a list of hidden variables.

It is the larger operational reality from which the effective financial world is compiled.

Define:

$\Sigma_{primary} = \{X, C, A, I, M, B, E, R_g, L, t\}$. (2.1)

Where, schematically:

- X = economic states and cash-flow processes;
- C = contracts and legal relations;
- A = agents and institutions;
- I = information channels;
- M = market and pricing mechanisms;
- B = balance-sheet and funding structures;
- E = exchange, clearing, and settlement systems;
- R_g = regulatory and governance rules;
- L = inherited ledgers;
- t = calendar time.

The primary universe contains more structure than any one valuation protocol admits.

A risk desk, trader, regulator, accountant, clearinghouse, and investor do not observe the same projection.

Each constructs a usable world from only part of $\Sigma_{primary}$.

The primary universe is therefore not itself equivalent to one complete financial viewpoint.

It is the larger field over which viewpoints are declared.

2.2 Declaration before projection

A secondary financial world cannot be defined merely by saying that an observer looks at the market.

The observer must declare a protocol.

Let:

$P = (B, \Delta, h, u)$. (2.2)

Where:

- B is the system boundary;
- Δ is the observation or aggregation rule;
- h is the time or state horizon;
- u is the admissible intervention family.

A fuller declaration also includes:

$D_P = (q, \varphi, P, \hat{O}_P, Gate_P, TraceRule_P, ResidualRule_P)$. (2.3)

Where:

- q is the baseline;
- φ is the feature map;
- $\hat{O}_P$ is the projection operator;
- $Gate_P$ determines commitment;
- $TraceRule_P$ determines what is recorded;
- $ResidualRule_P$ determines what remains explicitly unresolved.

The secondary world is therefore not discovered without conditions.

It is disclosed under a declared interface.

Define:

$W_\theta(P,L) = Declare(\Sigma_{primary} \mid q,\varphi,P,L)$. (2.4)

Only after declaration do terms such as asset, option, price, beta, volatility, default, collateral, or exercise become sufficiently definite for the chosen task.

2.3 Level W: the secondary θ-time valuation world

The secondary world contains the effective states that become usable under declaration.

Define:

$W_\theta = (\mathcal{S}_F, \mathcal{D}_F, \mathcal{O}_F, \mathcal{G}_F, \mathcal{L}_F, \mathcal{J}_F, \mathcal{B}_F, \mathcal{R}_F)$. (2.5)

Where:

- $\mathcal{S}_F$ is the effective state space;
- $\mathcal{D}_F$ is the internal dynamical law;
- $\mathcal{O}_F$ is the admissible observable family;
- $\mathcal{G}_F$ is the gate family;
- $\mathcal{L}_F$ is the ledger rule;
- $\mathcal{J}_F$ is the intervention family;
- $\mathcal{B}_F$ is the backreaction map;
- $\mathcal{R}_F$ is the residual-governance rule.

The world is ordered internally by θ:

$\rho_F = \rho_F(\theta)$. (2.6)

But θ alone does not create history.

History requires gate and trace:

$\theta \to G_k \to \text{Record}_k \to L_{k+1}$. (2.7)

Thus:

Internal Progression = θ. (2.8)

Committed History = k. (2.9)

Primary Duration = t. (2.10)

The clocks are related but not identical.

2.4 The secondary world is not merely a representation

A representation becomes world-like when it changes what happens next.

Suppose a rating model produces a downgrade.

The downgrade changes funding eligibility.

Funding conditions change liquidity.

Liquidity changes market price.

Market price changes collateral.

Collateral changes margin.

Margin changes liquidation probability.

The projection has entered the causal loop.

Schematically:

Projection → Gate → Ledger → Intervention → Primary Backreaction. (2.11)

Therefore:

$W_\theta \to \Sigma_{primary,next}$. (2.12)

The secondary world is not ontologically independent of the primary universe.

But neither is it causally inert.

It is an effective world that participates in producing later primary states.

2.5 Level O: the internal observer

Let observer O possess:

$O = (\mathcal{A}_O, F_O, L_O, \pi_O)$. (2.13)

Where:

- $\mathcal{A}_O$ is the accessible measurement algebra;
- F_O is the local valuation frame;
- L_O is the accessible trace;
- π_O is the policy for selecting later instruments.

At internal episode k:

$a_k = \pi_O(L_O, k)$. (2.14)

The observer selects instrument a_k based on prior trace.

The instrument produces outcome y_k:

$y_k \sim \boldsymbol{\mathcal{M}}_{a_k}[\rho_F(\theta_k)]$. (2.15)

The outcome is recorded:

$L_{O,k+1} = L_{O,k} \sqcup \mathrm{Record}(y_k, a_k)$. (2.16)

The next measurement policy changes:

$\pi_{O,k+1} = \mathrm{Update}(\pi_{O,k}, L_{O,k+1})$. (2.17)

This is the financial analogue of a trace-conditioned internal observer.

A trader changes later orders after losses.

A risk committee changes thresholds after a breach.

A clearinghouse changes margin after volatility.

An accountant changes impairment assumptions after default evidence.

A regulator changes admissibility after a crisis.

The observer does not merely read the world.

It uses trace to alter later projection.

2.6 Internal certainty and financial latching

Once an event is committed into the observer's ledger, that event becomes fixed inside the observer's subsequent filtration.

Suppose a barrier option records:

$$\mathrm{BarrierHit}_k = 1. \quad (2.18)$$

Later valuation cannot normally proceed as though the barrier had never been hit.

Similarly:

$$\mathrm{DefaultRecorded}_k = 1. \quad (2.19)$$

$$\mathrm{ExerciseRecorded}_k = 1. \quad (2.20)$$

$$\mathrm{MarginBreachRecorded}_k = 1. \quad (2.21)$$

$$\mathrm{SettlementCompleted}_k = 1. \quad (2.22)$$

These records constrain future admissible states.

Define:

$$\mathrm{FutureStateSet}_{k+1} = \mathrm{AdmissibleStates}(P, L_{k+1}). \quad (2.23)$$

Then generally:

$$\mathrm{AdmissibleStates}(P, L_{k+1}) \neq \mathrm{AdmissibleStates}(P, L_k). \quad (2.24)$$

This is latching.

The event does not merely reveal which branch was already present.

It changes which branches remain operationally available.

The source framework for self-referential observers describes an analogous structure: once outcomes enter an observer's filtration, they become internally fixed, and trace-conditioned policies cause counterfactual branches to diverge.

2.7 Why these three levels must not be collapsed

Several confusions arise if the levels are mixed.

Confusion 1

“The option is only a formula, so there is no entanglement-like structure.”

This argues from primary construction against secondary nonfactorization.

Confusion 2

“The option and underlying update together, so finance has quantum entanglement.”

This argues from secondary conditional update directly to physical quantum ontology.

Confusion 3

“The observer does not know the hidden mechanism, so ignorance creates entanglement.”

This confuses restricted access with genuine state nonfactorization.

Confusion 4

“The financial world is observer-dependent, so it is merely subjective.”

This ignores protocol-defined objectivity and cross-frame invariance.

Confusion 5

“The primary universe explains everything, so internal observer experience is irrelevant.”

This ignores that scientific measurement is itself performed by observers inside an accessible world.

The correct discipline is:

Primary Construction explains origin. (2.25)

Secondary Structure defines effective relations. (2.26)

Internal Observation defines experienced accessibility. (2.27)

None of these levels can replace the other two.

2.8 The three-level principle

The article’s first major principle is therefore:

> A financial object may be constructively classical at the primary level, relationally nonfactorizable at the secondary level, and locally strange at the observer level.

In compact form:

Classical Constructor
→ Composite Effective State
→ Locally Incomplete Observation. (2.28)

This does not prove quantum ontology.

It establishes the correct architecture for the comparison.

3. Three Financial Clocks Revisited

3.1 Why one time coordinate is not enough

A financial system does not possess only one meaningful order.

The primary economic universe unfolds in calendar time.

The secondary valuation world evolves through changing valuation orientation.

The observer's history advances when projected events pass a gate and become ledgered facts.

These three orders are related, but they perform different functions:

t = primary-universe calendar time. (3.1)

θ = secondary-world phase order. (3.2)

k = committed ledger order. (3.3)

Part I introduced this distinction because a continuously changing valuation state does not automatically create history. A price can move, a model can update, and a phase can rotate without any event becoming institutionally consequential. History begins when a gate turns one projected possibility into retained trace.

The complete temporal chain is:

$t \rightarrow \theta(t) \rightarrow Gate_k \rightarrow Record_k \rightarrow L_{k+1}$. (3.4)

This chapter sharpens the role of θ.

θ is not merely another mathematical label attached to calendar time.

Inside the secondary valuation world, θ may function as the coordinate that orders accessible state transformation.

The internal observer does not necessarily experience the primary construction in all its detail.

It encounters the sequence:

$\rho_F(\theta_1) \rightarrow \rho_F(\theta_2) \rightarrow \rho_F(\theta_3) \rightarrow \ldots$. (3.5)

From that observer's perspective, θ is the usable internal progression parameter.

3.2 Calendar time t: the constructor's duration

Calendar time belongs primarily to the larger economic universe.

It orders:

- cash-flow realization;
- information arrival;

- trading sessions;
- contractual maturity;
- interest accrual;
- regulatory deadlines;
- clearing and settlement;
- hedge execution;
- institutional decision cycles.

A derivative model may be written as:

$D_t = V(U_t,K,T−t,\sigma_t,r_t,q_t,P,L,\ldots)$. (3.6)

Here t appears explicitly because maturity approaches and economic conditions evolve.

From the constructor perspective, the relation between the underlying and derivative unfolds through ordinary processes:

Information Arrival
→ Underlying Repricing
→ Model Recalibration
→ Option Repricing
→ Hedge Adjustment
→ Market Feedback. (3.7)

This chain may require:

- nanoseconds in automated markets;
- minutes or hours in less liquid markets;
- days in accounting or collateral systems;
- months or years in long-dated contractual structures.

Calendar time records how long these processes take in the primary universe.

But the secondary observer may not observe every stage.

3.3 Phase time θ: internal valuation progression

Suppose a declared amplitude A is projected through a changing valuation frame:

$Z(t) = A(t) \exp[i\theta(t)]$. (3.8)

The phase coordinate is:

$\theta(t) = \arctan[Q(t)/R(t)]$. (3.9)

Or, when A and R are directly specified:

$\theta(t) = \arccos[R(t)/A(t)]$. (3.10)

If θ is monotonic over a local regime, it may be used as an internal coordinate:

$t \mapsto \theta(t)$. (3.11)

Then:

$dZ/d\theta = (dZ/dt)/(d\theta/dt)$. (3.12)

Under locally stable amplitude:

$dA/d\theta \approx 0$. (3.13)

The internal law reduces to:

$dZ/d\theta = iZ$. (3.14)

This gives:

$dR/d\theta = -Q$. (3.15)

$dQ/d\theta = R$. (3.16)

From the internal observer's perspective, the state evolves according to θ.

The observer need not reconstruct every primary-world mechanism that generated $d\theta/dt$.

It uses θ as the accessible ordering parameter.

This is analogous to an embedded observer using the internal clock supplied by its effective world rather than an inaccessible external construction sequence.

3.4 θ is not automatically a universal time

The phase coordinate becomes usable as internal time only under conditions.

At minimum:

1. θ must vary sufficiently smoothly;
2. θ must remain locally order-preserving;
3. the relevant state must be expressible as a function of θ;
4. phase reversals or branch crossings must be disclosed;
5. the active protocol must remain sufficiently stable;
6. residual must remain below a declared tolerance.

Define local monotonicity:

$d\theta/dt \neq 0$ over interval I. (3.17)

A stronger orientation condition is:

$\text{sign}(d\theta/dt) = \text{constant}$ over I. (3.18)

If $d\theta/dt$ repeatedly changes sign, θ cannot serve as a simple global clock.

The phase world may instead require:

- piecewise phase charts;
- branch labels;
- orientation reversals;
- multiple local clocks;
- a higher-dimensional phase field.

Thus:

Local Internal Time ≠ Universal Monotonic Time. (3.19)

This limitation is especially important in finance.

Markets reverse.

Volatility regimes switch.

Funding conditions jump.

Contracts expire.

Valuation protocols change.

A single θ may be usable only inside a declared local regime.

3.5 Multi-channel phase time

A derivative world contains more than one valuation channel.

A single angle θ is therefore insufficient.

Let channel n possess:

$Z_n = A_n \exp(i\theta_n)$. (3.20)

The total financial state may be:

$|\Psi_F\rangle = \Sigma_n c_n Z_n |n\rangle$. (3.21)

Possible channels include:

- strike regions;
- maturity buckets;
- volatility regimes;
- credit states;
- liquidity states;
- collateral states;

- investor classes;
- valuation protocols;
- exercise branches.

Each channel has its own phase:

$\theta_n = \theta_n(t,P,L)$. (3.22)

The physically or operationally relevant object is often not the absolute phase but the relative phase:

$\Delta\theta_{nm} = \theta_n - \theta_m$. (3.23)

The secondary valuation world is therefore better represented as a phase field:

$\Theta_F = \{\theta_1,\theta_2,\ldots,\theta_N\}$. (3.24)

An internal observer may be synchronized to only part of this field.

For example:

- an equity trader may follow spot and short-dated option phase;
- a credit desk may follow default and funding phase;
- a regulator may follow capital and liquidity phase;
- an accountant may follow recognition and impairment phase.

The observers inhabit the same larger financial system but do not necessarily share one operational clock.

3.6 Phase synchronization and cross-observer agreement

Let observer A use phase θ_A and observer B use phase θ_B.

A local synchronization map is:

$\theta_B = f_BA(\theta_A)$. (3.25)

For simple affine synchronization:

$\theta_B = a_BA\theta_A + b_BA$. (3.26)

Where:

- a_BA is the relative phase-rate conversion;
- b_BA is the phase-origin offset.

Cross-observer agreement requires more than equal displayed values.

It requires compatible event ordering.

Suppose observer A records:

Event E_1 before Event E_2. (3.27)

Observer B should not map the same admissible event pair into:

Event E_2 before Event E_1, (3.28)

unless the frames are explicitly incompatible or the events are not jointly orderable.

A phase map should therefore preserve admissible order:

$\theta_A(E_1) < \theta_A(E_2) \Rightarrow \theta_B(E_1) < \theta_B(E_2)$. (3.29)

The self-referential observer framework similarly requires consistent frame maps, compatible effects, and accessible records before cross-observer agreement can be secured.

In finance, the corresponding conditions include:

- same contract identity;
- same event definition;
- compatible settlement conventions;
- consistent time-zone handling;
- compatible corporate-action treatment;
- access to the same or reconcilable ledger record.

Without these conditions, two observers may appear to disagree while merely using different clocks or event definitions.

3.7 Ledger time k: history begins at commitment

Phase movement alone does not create an irreversible financial event.

Suppose the projected state crosses a threshold:

$\theta \rightarrow \theta^*$. (3.30)

No historical change follows unless a gate commits the crossing.

Define a gate:

$G_k = \text{Gate}(\rho_F(\theta_k), P, L_k)$. (3.31)

Possible outputs are:

$G_k \in \{\text{Commit}, \text{Reject}, \text{Defer}, \text{Escalate}\}$. (3.32)

If committed:

$\text{Record}_k = \text{Commit}[G_k, \rho_F(\theta_k), P]$. (3.33)

The ledger updates:

$L_{k+1} = L_k \sqcup \text{Record}_k$. (3.34)

The symbol ⊔ means that the new record is joined to the prior ledger without erasing the earlier trace.

Ledger time advances:

$k \to k+1$. (3.35)

Examples include:

- trade execution;
- option exercise;
- barrier activation;
- margin call;
- covenant breach;
- credit downgrade;
- default declaration;
- impairment recognition;
- collateral seizure;
- settlement finality;
- regulatory intervention.

These are not merely changes in representation.

They alter future admissibility.

3.8 Phase order versus historical order

Two valuation states may occupy different phase locations without producing different ledger histories.

Conversely, one ledger event may permanently distinguish histories even if visible prices later return to the same level.

Consider two paths:

Path A:
$Z_0 \to Z_1 \to Z_2 \to Z_0$. (3.36)

Path B:
$Z_0 \to Z_3 \to Z_4 \to Z_0$. (3.37)

Both return to the same visible coordinate:

$R_{final} = R_{initial}$. (3.38)

Possibly also:

$\theta_{final} = \theta_{initial} \bmod 2\pi$. (3.39)

But if Path A triggered a margin call and Path B did not:

$L_{final}^{A} \neq L_{final}^{B}$. (3.40)

Therefore:

Same Price + Same Phase ≠ Same Financial World. (3.41)

The histories differ in:

- ownership;
- collateral;
- leverage;
- trust;
- liquidity;
- legal rights;
- future intervention possibilities.

The return of the coordinate does not erase the ledger.

This is a financial holonomy of history.

3.9 Internal simultaneity

Suppose the joint option–underlying state is measured at internal phase θ_k.

The gate produces outcome pair:

(u_k, d_k). (3.42)

Inside the secondary world, both outcomes belong to one committed episode:

$\text{Episode}_k = \{(u_k, d_k), \theta_k, G_k\}$. (3.43)

The internal observer may therefore treat them as simultaneous within the episode.

But the primary constructor may have required a sequence:

Underlying Quote
→ Volatility Update
→ Pricing Engine
→ Option Quote
→ Hedge Order
→ Ledger Update. (3.44)

The apparent simultaneity belongs to the compiled event representation.

The causal sequence belongs to the primary process.

This gives:

Primary Sequentiality + Secondary Episode Compression → Internal Simultaneity. (3.45)

An internal episode may compress many primary operations into one effective tick.

3.10 The temporal lesson for entanglement

The option–underlying relation should not be compared with quantum entanglement by asking only:

“How quickly does one price react to the other in calendar time?”

That question remains in the primary universe.

The secondary-world question is:

> Are both local outcomes disclosed as components of one joint state at one internal measurement episode?

If yes, then the internal observer experiences:

One Joint State
→ One Measurement Episode
→ Correlated Local Outcomes. (3.46)

This is the relevant entanglement-form comparison.

It does not prove physical nonlocality.

It identifies the observer-relative event structure.

---

4. What Qualifies as a Secondary Financial World?

4.1 A coordinate system is not yet a world

The complex coordinate:

$Z = R + iQ$ (4.1)

is not by itself a financial world.

A chart is not a world.

A state vector is not a world.

A pricing model is not a world.

A world-like financial construction must support more than representation.

It must support:

1. states;
2. dynamics;
3. admissible observations;

4. gates;
5. trace;
6. intervention;
7. backreaction;
8. residual governance;
9. local observers;
10. cross-frame reconciliation.

Define a secondary financial world as:

$W_P = (\mathcal{S}_P,\mathcal{D}_P,\mathcal{O}_P,\mathcal{G}_P,\mathscr{L}_P,\mathscr{J}_P,\mathscr{B}_P,\mathscr{R}_P,\mathscr{F}_P)$. (4.2)

Where:

- $\mathcal{S}_P$ = effective state space;
- $\mathcal{D}_P$ = internal dynamics;
- $\mathcal{O}_P$ = admissible observations;
- $\mathcal{G}_P$ = commitment gates;
- $\mathscr{L}_P$ = ledger rule;
- $\mathscr{J}_P$ = admissible interventions;
- $\mathscr{B}_P$ = backreaction map;
- $\mathscr{R}_P$ = residual rule;
- $\mathscr{F}_P$ = local frame family.

A valuation construction becomes world-like when these elements form an approximately closed operational loop.

4.2 Effective state space

The state space must specify what can vary.

For scalar Finance Geometry:

$\mathcal{S}_P^{scalar} = \{(R,Q,A,\theta)\}$. (4.3)

For a derivative world:

$\mathcal{S}_P^{derivative} = \{(U,D,\sigma,r,q,K,T-t,H,C,L,\ldots)\}$. (4.4)

Where:

- U = underlying state;
- D = derivative state;

- σ = volatility field;
- r = financing state;
- q = carry state;
- H = hedge state;
- C = collateral state;
- L = ledger state.

A complex composite version may use:

ρ_F ∈ 𝒟(ℋ_F). (4.5)

Where 𝒟(ℋ_F) denotes the declared family of density-like operators on the effective financial state space.

The state space must be large enough to preserve the distinctions relevant to the task.

If collateral matters but is omitted, the world is under-declared.

If path history matters but is omitted, a barrier instrument is misrepresented.

If funding matters but is hidden inside one discount rate, crisis dynamics may be suppressed.

A world can be simple only relative to a declared purpose.

4.3 Internal dynamics

The world must possess a transition rule:

ρ_F(θ+δθ) = 𝒯_{δθ}[ρ_F(θ);P,L]. (4.6)

For continuous phase evolution:

dρ_F/dθ = ℒ_F[ρ_F;P,L]. (4.7)

Here ℒ_F is an effective generator.

In state-vector form:

iℏ_FD_θ|Ψ_F⟩ = Ĥ_F|Ψ_F⟩ + |ε_F⟩. (4.8)

The residual term |ε_F⟩ is essential.

It records the part of the observed transition not explained by the declared effective world.

A world-like model does not require perfect closure.

It requires declared closure tolerance:

‖ε_F‖ ≤ ε*. (4.9)

When:

‖ε_F‖ > ε*, (4.10)

the protocol must:

- widen its state space;
- revise its dynamics;
- change its frame;
- attach residual;
- suspend prediction;
- escalate to a larger model.

Effective closure means bounded explanatory failure, not metaphysical completeness.

4.4 Admissible observables

A world must define what its observers can measure.

Let:

$\mathcal{O}_P = \{\hat{O}_1, \hat{O}_2, \ldots\}$. (4.11)

Examples include:

- spot price;
- option premium;
- implied volatility;
- delta;
- gamma;
- vega;
- collateral requirement;
- liquidation value;
- default status;
- settlement status;
- regulatory capital;
- accounting carrying value.

An observable is protocol-indexed:

$O_{j,P}(\rho_F) = \mathrm{Tr}(\rho_F \hat{O}_{j,P})$. (4.12)

The same primary economic field may yield different observables under different protocols.

For example:

MarketValue_P1 ≠ LiquidationValue_P2. (4.13)

AccountingValue_P3 ≠ RegulatoryExposure_P4. (4.14)

These are not necessarily contradictions.

They may be different projections of the same larger field.

The discipline is to preserve the protocol index.

4.5 Gates

Observation does not automatically imply commitment.

A gate converts an observed state into an event status.

Define:

Gate_P: (ρ_F,O,L) → {Commit,Reject,Defer,Escalate}. (4.15)

Examples:

Exercise Gate:

Commit iff U_T > K and exercise remains legally admissible. (4.16)

Margin Gate:

Commit iff Exposure − Collateral > Threshold. (4.17)

Default Gate:

Commit iff declared credit event conditions are satisfied. (4.18)

Impairment Gate:

Commit iff accounting evidence exceeds the declared recognition threshold. (4.19)

A gate is not merely a threshold in data.

It includes:

- authority;
- timing;
- admissibility;
- evidence;
- legal effect;
- trace rule.

Two observers may see the same numerical state but apply different gates because they hold different institutional roles.

4.6 Ledger

A world must retain consequential history.

Define the ledger:

$L_k = \{Record_0, Record_1, \ldots, Record_{k-1}\}$. (4.20)

A strong record includes:

$Record_k = (Outcome_k, Instrument_k, Frame_k, Authority_k, Evidence_k, Residual_k, Timestamp_k)$. (4.21)

The ledger does more than store values.

It preserves:

- what was observed;
- how it was observed;
- which frame was used;
- who had authority;
- why the gate passed;
- what remained unresolved.

A weak ledger records only the conclusion.

A strong ledger records the conditions of commitment.

This distinction determines whether later observers can reconstruct, challenge, or revise the world honestly.

4.7 Intervention

A world must define what actions are possible.

Let:

$u \in \mathcal{J}_P$. (4.22)

Possible financial interventions include:

- trade;
- hedge;
- rebalance;
- novate;
- close out;
- exercise;

- post collateral;
- change margin;
- alter credit limit;
- revise accounting assumption;
- suspend trading;
- impose capital constraint.

The world's transition law is intervention-dependent:

$\rho_F' = \mathcal{T}_u(\rho_F)$. (4.23)

A world without admissible intervention may remain a descriptive model.

A world with intervention becomes operational.

4.8 Backreaction

Intervention and ledger change the primary field:

$X_\{k+1\} = \mathcal{B}(X_k, L_{k+1}, u_k)$. (4.24)

Examples:

Price Record
→ Collateral Change. (4.25)

Collateral Change
→ Forced Sale. (4.26)

Forced Sale
→ Market Price Change. (4.27)

Market Price Change
→ New Option State. (4.28)

Thus:

Secondary World → Primary Field → New Secondary World. (4.29)

The loop is:

$X_k \to W_k \to L_{k+1} \to X_{k+1} \to W_{k+1}$. (4.30)

This is why the secondary financial world is not merely representational.

Its outputs alter the conditions of later representation.

4.9 Residual governance

No declared world captures everything.

Define residual:

$\varepsilon_k = \text{Observation}_k - \text{Prediction_P}(\rho_k)$. (4.31)

Residual may arise from:

- omitted variables;
- frame mismatch;
- model error;
- hidden liquidity;
- legal uncertainty;
- strategic behaviour;
- regime change;
- data failure;
- unmodelled feedback.

Residual must not be flattened into noise automatically.

The world should classify it:

$\text{Residual}_k \in \{\text{Noise},\text{ModelError},\text{FrameError},\text{ProtocolFailure},\text{NovelRegime},\text{Unresolved}\}$. (4.32)

Repeated residual pressure may force revision:

$P_{k+1} = \mathcal{U}(P_k, L_k, \text{Residual}_k)$. (4.33)

The related declaration framework treats residual honesty and trace-preserving revision as necessary conditions for mature observers rather than optional reporting preferences.

4.10 Internal observers

A secondary world becomes especially important when it can contain observers whose later actions depend on prior internal trace.

Define:

$O_k = (\mathcal{A}_k, F_k, L_k, \pi_k)$. (4.34)

The observer:

1. selects an instrument;
2. receives an outcome;
3. records the result;
4. updates its policy;
5. intervenes;

6. changes the later world.

This creates:

Observation → Trace → Policy → Intervention → Backreaction. (4.35)

Examples include:

- automated trading systems;
- risk desks;
- clearinghouses;
- rating agencies;
- regulators;
- accounting committees;
- portfolio managers.

These observers are not outside finance.

They are components of its causal architecture.

4.11 Cross-frame objectivity

A world need not be observer-free to be objective.

Objectivity may be defined as invariance across admissible observer frames.

Let frame transformation F_{BA} map observer A's description into observer B's:

$\rho_B = U_{BA}\rho_A U_{BA}^{\dagger}$. (4.36)

An invariant claim satisfies:

$I(\rho_A) = I(\rho_B)$. (4.37)

Examples may include:

- same contractual payoff;
- same settlement record;
- same no-arbitrage relation;
- same exposure after valid numeraire transformation;
- same event identity after reconciled time-zone conversion.

Objectivity therefore means:

Cross-Frame Invariance + Trace Reconciliation. (4.38)

Not:

Absence of Observers. (4.39)

4.12 Effective-world criterion

A financial representation qualifies as a secondary world when:

WorldLike_P ⇔ State ∧ Dynamics ∧ Observables ∧ Gates ∧ Ledger ∧ Intervention ∧ Backreaction ∧ ResidualRule. (4.40)

A stronger observer-bearing world requires:

ObserverWorld_P ⇔ WorldLike_P ∧ InternalObserver ∧ TraceConditionedPolicy ∧ CrossFrameReconciliation. (4.41)

This is the environment in which derivative entanglement must be analysed.

The derivative and underlying are not compared as isolated prices.

They are compared as subsystems inside an observer-bearing effective world.

Part II — CAPM as a Local Law, Not the Whole World

5. The Local Status of CAPM

5.1 CAPM's mature financial role

The Capital Asset Pricing Model provides the familiar relation:

$E[r_i] = r_f + \beta_i(E[r_m] - r_f)$. (5.1)

Define:

$ERP = E[r_m] - r_f$. (5.2)

Then:

$E[r_i] = r_f + \beta_i ERP$. (5.3)

CAPM connects:

- expected asset return;
- risk-free return;
- market risk premium;
- systematic exposure beta.

It is a mature financial model with known assumptions and known limitations.

This article does not replace CAPM.

It changes CAPM's architectural status.

CAPM is treated as a local valuation law inside a larger effective geometry.

5.2 Why CAPM should be local

The values of β, ERP, and even r_f depend on regime and protocol.

Beta may vary with:

- sampling window;
- market index;
- leverage;
- volatility;
- liquidity;
- corporate structure;
- crisis conditions.

The risk-free rate may depend on:

- currency;
- collateral;
- maturity;
- credit convention;
- funding access;
- numeraire.

The market risk premium may depend on:

- horizon;
- estimation method;
- investor class;
- economic regime;
- risk appetite.

Therefore:

$\beta = \beta(x,P,L)$. (5.4)

$ERP = ERP(x,P,L)$. (5.5)

$r_f = r_f(x,P,L)$. (5.6)

The CAPM law becomes:

$$E[r_i \mid x,P,L] = r_f(x,P,L) + \beta_i(x,P,L)ERP(x,P,L). \quad (5.7)$$

This is a local relation.

Its coefficients are valid near a declared financial state x under protocol P and ledger L.

5.3 The tangent-frame interpretation

Let the global financial state space be a manifold:

$$\mathcal{M}_F. \quad (5.8)$$

At point $x \in \mathcal{M}_F$, define a local tangent space:

$$T_x\mathcal{M}_F. \quad (5.9)$$

CAPM acts inside this local frame:

$$r_i^{local} = r_f^{local} + \beta_i^{local}ERP^{local}. \quad (5.10)$$

The local frame approximates the larger geometry as flat over a sufficiently small region:

$$g^F_{\mu\nu}(x+\delta x) \approx \eta_{\mu\nu} \text{ for } \|\delta x\| \text{ sufficiently small.} \quad (5.11)$$

Thus:

CAPM = Local Flat-Frame Approximation. (5.12)

This does not diminish CAPM.

Local physical laws can remain powerful without being globally uniform.

The important question becomes:

Over what domain is the local approximation reliable?

5.4 The local validity domain

Define the CAPM validity neighbourhood:

$$\mathcal{N}_{CAPM}(x;P,L,\varepsilon^*) = \{x' \mid \|Residual_{CAPM}(x')\| \le \varepsilon^*\}. \quad (5.13)$$

Within this neighbourhood:

- beta remains sufficiently stable;
- factor structure remains approximately unchanged;
- liquidity does not collapse;
- funding remains regular;
- market portfolio definition remains usable;
- no major gate changes the ledger regime.

Outside it, the local law may fail.

The failure may arise not because CAPM is meaningless, but because the observer extrapolated a tangent law across a curved region.

5.5 Local prediction and global transport

Suppose an asset moves from x_A to x_B.

At x_A:

E[r_i]*A* = *r*{f,A} + β_{i,A}ERP_A. (5.14)

At x_B:

E[r_i]*B* = *r*{f,B} + β_{i,B}ERP_B. (5.15)

The difference is not merely:

ΔE[r_i] = E[r_i]_B − E[r_i]_A. (5.16)

Part of the difference may come from changing frames.

A covariant comparison requires transport:

E[r_i]*{A→B}* = *Transport*{A→B}[E[r_i]_A]. (5.17)

The corrected change is:

Δ_covE[r_i] = E[r_i]*B* − *E[r_i]*{A→B}. (5.18)

This separates:

- genuine state change;
- frame change;
- transport effect;
- residual.

Thus:

Raw Repricing ≠ Covariant Repricing. (5.19)

5.6 CAPM and the local complex completion

At local state x, let the declared pre-filter amplitude be A_i(x).

Let admitted value be R_i(x).

Define:

cos θ_i(x) = R_i(x)/A_i(x). (5.20)

Then:

$Q_i(x) = \sqrt{[A_i(x)^2 - R_i(x)^2]}$. (5.21)

The local complex state is:

$Z_i(x) = R_i(x) + iQ_i(x)$. (5.22)

Or:

$Z_i(x) = A_i(x)\exp[i\theta_i(x)]$. (5.23)

CAPM helps determine the local valuation filter.

The complex completion preserves the orthogonal pressure hidden by the scalar output.

CAPM therefore supplies part of the local projection structure, while R+iQ supplies the local complex state.

They perform different roles.

5.7 CAPM does not generate the entire financial Hamiltonian

The effective financial generator may be:

$\hat{H}_F = \hat{H}_{CAPM} + \hat{H}_{contract} + \hat{H}_{hedge} + \hat{H}_{liquidity} + \hat{H}_{funding} + \hat{H}_{ledger} + \hat{H}_{environment}$. (5.24)

CAPM contributes one term.

It does not by itself generate:

- derivative payoff structure;
- tensor-product composition;
- hedge feedback;
- collateral dynamics;
- settlement gates;
- path dependence;
- ledger latching;
- global curvature.

Therefore:

$\hat{H}_{CAPM} \neq \hat{H}_F$. (5.25)

The architectural error would be to make CAPM carry every physics analogy.

The correct move is to assign each mathematical role to the financial structure capable of supporting it.

5.8 Recovery of ordinary CAPM

The larger architecture must reduce to ordinary CAPM under suitable limits.

Require:

1. one asset channel;
2. no derivative coupling;
3. stable local frame;
4. negligible curvature;
5. zero gauge connection;
6. negligible ledger backreaction;
7. residual below tolerance.

Formally:

dim $\mathscr{H}_F \to 1$. (5.26)

$\hat{H}_{contract} \to 0$. (5.27)

$\hat{H}_{hedge} \to 0$. (5.28)

$g^{F}_{\mu\nu} \to \eta_{\mu\nu}$. (5.29)

$\mathcal{A}_\mu \to 0$. (5.30)

$\hat{H}_{ledger} \to 0$. (5.31)

Then:

$E[r_i] \to r_f + \beta_i ERP$. (5.32)

A proposed unification that cannot recover the mature local law is not an extension.

It is a replacement.

This article requires reduction rather than replacement.

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6. The Local Complex State

6.1 From scalar value to complex completion

Let A be the declared pre-filter amplitude.

Let R be the value admitted by the local financial filter.

The scalar valuation reports only:

R. (6.1)

The complex completion preserves:

$$Z = R + iQ. \quad (6.2)$$

With:

$$A^2 = R^2 + Q^2. \quad (6.3)$$

The retained pressure is:

$$Q = \sqrt{(A^2 - R^2)}. \quad (6.4)$$

The local angle is:

$$\theta = \arccos(R/A). \quad (6.5)$$

Therefore:

$$R = A \cos \theta. \quad (6.6)$$

$$Q = A \sin \theta. \quad (6.7)$$

$$Z = A \exp(i\theta). \quad (6.8)$$

Q is not:

$$Q = A - R. \quad (6.9)$$

The scalar haircut A−R and the orthogonal complement Q are different quantities.

6.2 Interpretation of the local coordinates

R records what the declared filter admits.

Q records the pressure implied by the same geometry but hidden by the scalar report.

A records the declared pre-filter scale.

θ records the orientation of the valuation frame relative to that scale.

The state is protocol-relative:

$$Z_P = R_P + iQ_P. \quad (6.10)$$

A different protocol P′ may produce:

$$Z_{P'} = R_{P'} + iQ_{P'}. \quad (6.11)$$

Thus:

Same Primary Field + Different Protocol → Different Local Complex State. (6.12)

This is not arbitrary.

Each state must remain tied to a declared filter and measurable input.

6.3 Dynamic local state

Let:

$Z(t) = A(t)\exp[i\theta(t)]$. (6.13)

Differentiating:

$dZ/dt = [(1/A)(dA/dt) + i(d\theta/dt)]Z$. (6.14)

Add residual:

$dZ/dt = [g_A + i\omega_F]Z + \varepsilon_{dyn}$. (6.15)

Where:

$g_A = (1/A)(dA/dt)$. (6.16)

$\omega_F = d\theta/dt$. (6.17)

The coordinate equations are:

$dR/dt = g_AR - \omega_FQ + Re(\varepsilon_{dyn})$. (6.18)

$dQ/dt = g_AQ + \omega_FR + Im(\varepsilon_{dyn})$. (6.19)

The visible coordinate R therefore changes through:

- radial economic change;
- angular frame movement;
- unexplained residual.

Define:

$\Lambda_F = Q\omega_F$. (6.20)

Then:

$dR/dt = g_AR - \Lambda_F + Re(\varepsilon_{dyn})$. (6.21)

6.4 Local phase velocity

The phase velocity is:

$\omega_F = d\theta/dt$. (6.22)

A large $|\omega_F|$ means the valuation frame is changing rapidly.

This may occur during:

- volatility shock;
- liquidity collapse;
- policy change;

- funding stress;
- credit event;
- accounting reclassification;
- regulatory intervention.

The same underlying amplitude A may produce major visible repricing because the filter rotates.

Thus:

Large Price Change ≠ Necessarily Large Economic Amplitude Change. (6.23)

The complex state separates these effects.

6.5 Local oscillator limit

When:

$dA/d\theta = 0$, (6.24)

and:

$\varepsilon_{dyn} = 0$, (6.25)

the internal law is:

$dZ/d\theta = iZ$. (6.26)

Hence:

$d^2Z/d\theta^2 = -Z$. (6.27)

This is a classical oscillator.

The significance is not that CAPM has become quantum mechanics.

The significance is that the secondary world possesses a simple internal law even when its construction from the primary universe is complicated.

The difficult structure may reside in:

- declaration;
- channel selection;
- frame transport;
- measurement;
- gates;
- ledger;
- backreaction.

6.6 From one state to a state fibre

At each point x of the global financial manifold, define a local complex state space:

$\mathscr{H}_x$. (6.28)

The local state is:

$|\psi(x)\rangle \in \mathscr{H}_x$. (6.29)

The collection of local spaces forms a bundle:

$\pi: \boldsymbol{\mathcal{H}}_F \to \boldsymbol{\mathcal{M}}_F$. (6.30)

Where:

- $\boldsymbol{\mathcal{M}}_F$ is the financial base manifold;
- $\boldsymbol{\mathcal{H}}_F$ is the total complex state bundle;
- π maps each local state to its financial location.

CAPM acts locally in the base frame.

The complex state occupies the fibre.

Gauge transport compares fibres across locations.

Curvature governs the base geometry.

This assignment prevents conceptual collision between the QM-like, SR-like, and GR-like roles.

6.7 Multi-asset state

For N asset channels:

$|\psi_A\rangle = \Sigma_i c_i Z_i |i\rangle$. (6.31)

Where:

$Z_i = A_i \exp(i\theta_i)$. (6.32)

The norm-like quantity is:

$\langle\psi_A|\psi_A\rangle = \Sigma_i |c_i Z_i|^2$, (6.33)

when the basis states are orthogonal.

If the channels overlap under the selected measurement basis:

$\langle i|j\rangle \neq 0$, (6.34)

cross terms appear:

$\langle\psi_A|\psi_A\rangle = \Sigma_i |c_i Z_i|^2 + \Sigma_{i\neq j} c_i c_j Z_i Z_j \langle i|j\rangle$. (6.35)

This is interference-form algebra.

Whether it adds financial value depends on whether the relative phases and overlaps correspond to measurable phenomena.

6.8 The transition to derivative states

An asset-only state still lacks a clearly defined underlying–derivative partition.

Derivatives supply:

$\mathscr{H}_U \otimes \mathscr{H}_D$. (6.36)

The local option–underlying state becomes:

$|\Psi_{UD}(x)\rangle \in \mathscr{H}_U(x) \otimes \mathscr{H}_D(x)$. (6.37)

This is the point at which the scalar complex completion becomes a composite complex financial state.

The next chapters will distinguish:

- ordinary correlation;
- contractual coupling;
- dynamical binding;
- separable mixtures;
- effective nonfactorization;
- coherent composite states.

The distinction is necessary because tensor notation alone does not create entanglement.

It only makes the entanglement question mathematically expressible.

7. Three Different Angular or Hyperbolic Variables

7.1 Why one symbol cannot perform every role

The first complex construction introduced:

$Z = A \exp(i\theta)$. (7.1)

This makes θ a circular angle in the Euclidean valuation plane:

$A^2 = R^2 + Q_E^2$. (7.2)

$R = A \cos \theta$. (7.3)

$Q_E = A \sin \theta$. (7.4)

Once SR-like frame transformations are introduced, however, the theory also requires hyperbolic parameters.

A circular angle and a Lorentz-like rapidity are not the same mathematical object.

The article must therefore preserve three distinct variables:

θ = circular valuation-filter angle. (7.5)

η_Q = Q-preserving hyperbolic rapidity. (7.6)

ρ_D = additive discount-frame rapidity. (7.7)

Ledger order must remain separate:

k or τ_L = committed ledger time. (7.8)

Appendix M explicitly warns against using θ simultaneously as quantum phase, Lorentz rapidity, global time, and spacetime coordinate without declared maps between those roles. It also distinguishes η_Q from ρ_D and shows that they are related but not interchangeable.

The rule is:

Distinct Mathematical Role → Distinct Variable. (7.9)

A theory becomes apparently unified but actually incoherent when one symbol is forced to carry several incompatible geometries.

7.2 Circular valuation angle θ

The original geometry is Euclidean:

$A^2 = R^2 + Q_E^2$. (7.10)

The angle θ is defined by:

$\cos \theta = R/A$. (7.11)

$\sin \theta = Q_E/A$. (7.12)

$\tan \theta = Q_E/R$. (7.13)

Therefore:

$\theta = \arccos(R/A)$. (7.14)

Or:

$\theta = \arctan(Q_E/R)$. (7.15)

The circular angle answers:

> How is the declared amplitude divided between admitted value and retained pressure under the current financial filter?

It belongs to the local complex state:

$Z = R + iQ_E$. (7.16)

It is most useful for:

- local phase progression;

- radial versus angular repricing;
- pressure-preserving complex completion;
- relative phase between valuation channels;
- internal θ-ordering.

It should not automatically be interpreted as a frame boost.

7.3 Q-preserving rapidity η_Q

The Euclidean relation can be rewritten as:

$A^2 - Q_E^2 = R^2$. (7.17)

This permits a hyperbolic parametrization:

$A = R \cosh \eta_Q$. (7.18)

$Q_E = R \sinh \eta_Q$. (7.19)

Therefore:

$\tanh \eta_Q = Q_E/A$. (7.20)

And:

$\eta_Q = \operatorname{artanh}(Q_E/A)$. (7.21)

This embedding preserves the original pressure coordinate Q_E.

It reinterprets admitted value R as an interval-like invariant:

$A^2 - Q_E^2 = R^2$. (7.22)

The Q-preserving rapidity answers:

> How large is retained pressure relative to the total declared amplitude when the same R–Q geometry is written hyperbolically?

It is useful when the purpose is to preserve Finance Geometry while introducing a Lorentz-like representation.

It is not automatically additive under discount-frame composition.

7.4 Direct relation between θ and η_Q

Since:

$\sin \theta = Q_E/A$, (7.23)

and:

$\tanh \eta_Q = Q_E/A$, (7.24)

we obtain:

$\sin \theta = \tanh \eta_Q$. (7.25)

Therefore:

$\eta_Q = \operatorname{artanh}(\sin \theta)$. (7.26)

Also:

$\cos \theta = R/A$. (7.27)

From:

$A = R \cosh \eta_Q$, (7.28)

we have:

$R/A = \operatorname{sech} \eta_Q$. (7.29)

Thus:

$\cos \theta = \operatorname{sech} \eta_Q$. (7.30)

And:

$\tan \theta = \sinh \eta_Q$. (7.31)

These maps show that θ and η_Q encode the same local A–R–Q_E relation through different geometries.

But:

Same Underlying Ratio ≠ Same Mathematical Role. (7.32)

θ is circular.

η_Q is hyperbolic.

One should not replace the other without changing the representation.

7.5 Discount-frame rapidity ρ_D

Suppose a declared amplitude A is filtered into admitted value R.

Define the discount-frame rapidity:

$\rho_D = \ln(A/R)$. (7.33)

Then:

$R = A \exp(-\rho_D)$. (7.34)

This variable is especially useful because multiplicative discount ratios become additive.

Suppose frame a admits:

$R_a = A \exp(-\rho_a)$. (7.35)

And frame b admits:

$R_b = A \exp(-\rho_b)$. (7.36)

The relative frame parameter is:

$\rho_{ab} = \rho_b - \rho_a$. (7.37)

Therefore:

$R_b = \exp(-\rho_{ab}) R_a$. (7.38)

For three frames:

$\rho_{ac} = \rho_{ab} + \rho_{bc}$. (7.39)

This exact additive composition is the principal reason to introduce ρ_D.

It answers:

> How far apart are two valuation frames when their multiplicative admission or discount ratios are expressed additively?

7.6 Relation between η_Q and ρ_D

From:

$A = R \cosh \eta_Q$, (7.40)

we obtain:

$A/R = \cosh \eta_Q$. (7.41)

Therefore:

$\rho_D = \ln \cosh \eta_Q$. (7.42)

Using the circular angle:

$A/R = \sec \theta$. (7.43)

Therefore:

$\rho_D = \ln \sec \theta$. (7.44)

Equivalently:

$\rho_D = -\ln \cos \theta$. (7.45)

The full map is:

$\sin \theta = \tanh \eta_Q$. (7.46)

$\cos \theta = \operatorname{sech} \eta_Q = \exp(-\rho_D)$. (7.47)

$\rho_D = \ln \cosh \eta_Q = -\ln \cos \theta$. (7.48)

These equations connect the three variables while preserving their distinct purposes.

The relation is nonlinear.

Therefore:

$\eta_Q \neq \rho_D$. (7.49)

And:

$\theta \neq \eta_Q \neq \rho_D$. (7.50)

Appendix M identifies the Q-preserving and discount-composition constructions as two different Lorentz embeddings: the first preserves Q_E, while the second provides additive relative valuation boosts.

7.7 Small-pressure approximation

When retained pressure is small:

$Q_E/A \ll 1$. (7.51)

Then:

$\theta \approx Q_E/A$. (7.52)

$\eta_Q \approx Q_E/A$. (7.53)

Using:

$\rho_D = \ln \cosh \eta_Q$, (7.54)

and:

$\cosh \eta_Q \approx 1 + \eta_Q^2/2$, (7.55)

we obtain:

$\rho_D \approx \eta_Q^2/2$. (7.56)

Therefore:

$\rho_D \approx \theta^2/2$. (7.57)

In the low-pressure limit:

$\theta \approx \eta_Q$. (7.58)

But:

ρ_D is second order. (7.59)

This explains why the three variables may appear similar near the origin but diverge materially under large valuation pressure.

A model calibrated only in quiet conditions may fail to reveal the difference.

7.8 Large-pressure regime

As admitted value approaches zero:

$R/A \to 0$. (7.60)

Then:

$\theta \to \pi/2$. (7.61)

$\eta_Q \to \infty$. (7.62)

$\rho_D \to \infty$. (7.63)

The circular angle is bounded.

The rapidities are unbounded.

This has an important interpretive consequence.

θ approaches a finite boundary representing almost complete transfer from admitted value into retained pressure.

η_Q and ρ_D continue to distinguish increasingly extreme compression near that boundary.

Thus:

θ is geometrically compact. (7.64)

η_Q and ρ_D preserve unbounded stress resolution. (7.65)

For extreme credit, liquidity, or certainty-equivalent deterioration, the hyperbolic variables may preserve distinctions that the circular angle compresses near $\pi/2$.

7.9 Two hyperbolic pressures

The Q-preserving construction uses:

$Q_E = \sqrt{(A^2 - R^2)}$. (7.66)

The discount-composition construction can define:

$U = (A^2 + R^2)/(2R)$. (7.67)

$V = (A^2 - R^2)/(2R)$. (7.68)

Then:

$U^2 - V^2 = A^2$. (7.69)

Define:

$Q_H = V = (A^2 - R^2)/(2R)$. (7.70)

Therefore:

$Q_H = Q_E^2/(2R)$. (7.71)

The light-cone coordinates are:

$R_+ = U + V = A^2/R$. (7.72)

$R_- = U - V = R$. (7.73)

Under:

$\rho_D = \ln(A/R)$, (7.74)

we have:

$U = A \cosh \rho_D$. (7.75)

$V = A \sinh \rho_D$. (7.76)

And:

$R_- = A \exp(-\rho_D)$. (7.77)

The two pressure variables are therefore:

Q_E = Euclidean retained pressure. (7.78)

Q_H = hyperbolic light-cone pressure. (7.79)

They are not interchangeable.

They serve different embeddings of the same A–R relation. Appendix M identifies equating Q_E with Q_H as a direct source of contradiction.

7.10 Bounded frame variable u_D

A Lorentz-like bounded frame coordinate may be constructed from discount rapidity:

$u_D = \tanh \rho_D$. (7.80)

Therefore:

$-1 < u_D < 1$. (7.81)

For relative frames:

$u_{ab} = \tanh \rho_{ab}$. (7.82)

Since rapidities add:

$\rho_{ac} = \rho_{ab} + \rho_{bc}$, (7.83)

the bounded variables compose as:

$u_{ac} = (u_{ab} + u_{bc})/(1 + u_{ab}u_{bc})$. (7.84)

This resembles relativistic velocity addition.

But the resemblance must be interpreted carefully.

u_D is a normalized valuation-frame coordinate.

It is not physical velocity.

It does not derive a universal financial speed of light.

It does not establish financial causal cones.

Most importantly:

$\beta_{CAPM} \neq u_D$. (7.85)

CAPM beta is a covariance sensitivity:

$\beta_i = Cov(r_i, r_m)/Var(r_m)$. (7.86)

It is not a bounded frame speed.

Appendix M explicitly warns against treating CAPM beta as relativistic beta and proposes tanh ρ_D as the more plausible constructed bounded frame variable.

7.11 Role assignment table

Variable	Definition	Geometry	Primary role
θ	$\arccos(R/A)$	Circular	Local filter orientation and phase
η_Q	$\operatorname{artanh}(Q_E/A)$	Hyperbolic	Preserve original R–Q_E geometry
ρ_D	$\ln(A/R)$	Hyperbolic/additive	Relative valuation-frame transport
u_D	$\tanh \rho_D$	Bounded frame coordinate	Composable normalized frame difference
k or τ_L	ledger event index	Ordered trace	Committed historical time
Q_E	$\sqrt{(A^2-R^2)}$	Euclidean	Retained financial pressure
Q_H	$Q_E^2/(2R)$	Light-cone hyperbolic	Discount-composition pressure

The article will use:

- θ for internal phase progression;
- η_Q when preserving the original pressure geometry;
- ρ_D for SR-like frame comparison;
- k for ledger commitment.

No single variable will be made to represent every layer.

---

8. Local Financial Lorentz Frames

8.1 Why a frame layer is needed

Financial observers do not value from nowhere.

Each observer occupies a frame defined by:

- benchmark;

- currency;
- numeraire;
- funding curve;
- collateral convention;
- horizon;
- risk model;
- accounting rule;
- legal position;
- intervention authority.

Let frame a be:

F_a = (N_a,r_a,h_a,P_a,L_a). (8.1)

Where:

- N_a is the numeraire or benchmark;
- r_a is the relevant discount or funding structure;
- h_a is the horizon;
- P_a is the protocol;
- L_a is the accessible ledger.

The same primary claim may receive different local coordinates in different frames:

Z_a = R_a + iQ_a. (8.2)

Z_b = R_b + iQ_b. (8.3)

A frame theory is needed to distinguish:

- genuine change in the governed financial object;
- change caused by moving between local descriptions;
- failure of the mapping between descriptions.

8.2 Relative discount frames

Suppose the same terminal amount CF_T is valued under two locally stable discount frames:

R_a = CF_T/(1 + r_a)^T. (8.4)

R_b = CF_T/(1 + r_b)^T. (8.5)

Then:

R_b/R_a = [(1 + r_a)/(1 + r_b)]^T. (8.6)

Define:

ρ_ab = T ln[(1 + r_b)/(1 + r_a)]. (8.7)

Therefore:

R_b = exp(−ρ_ab)R_a. (8.8)

The reverse transformation is:

ρ_ba = −ρ_ab. (8.9)

And:

R_a = exp(−ρ_ba)R_b. (8.10)

For frames a, b, and c:

ρ_ac = ρ_ab + ρ_bc. (8.11)

This gives an exact additive frame-composition law.

8.3 Light-cone representation

Define frame coordinates:

U_a = A cosh ρ_a. (8.12)

V_a = A sinh ρ_a. (8.13)

Then:

$U_a^2 - V_a^2 = A^2$. (8.14)

Define light-cone coordinates:

R_{a,+} = U_a + V_a = A exp(ρ_a). (8.15)

R_{a,−} = U_a − V_a = A exp(−ρ_a). (8.16)

Identify the admitted value with:

R_a = R_{a,−}. (8.17)

A relative frame boost acts as:

R_{b,+} = exp(ρ_ab)R_{a,+}. (8.18)

R_{b,−} = exp(−ρ_ab)R_{a,−}. (8.19)

The invariant is:

$R_{a,+}R_{a,-} = A^2$. (8.20)

And:

$R_{b,+}R_{b,-} = A^2$. (8.21)

This construction provides:

- multiplicative value transformation;
- additive rapidity;
- invariant hyperbolic amplitude.

It remains a constructed valuation-frame geometry.

8.4 Lorentz-matrix form

The transformation between local hyperbolic coordinates may be written:

$$\begin{bmatrix} U_b \\ V_b \end{bmatrix} = \begin{bmatrix} \cosh \rho_{ab} & \sinh \rho_{ab} \\ \sinh \rho_{ab} & \cosh \rho_{ab} \end{bmatrix} \begin{bmatrix} U_a \\ V_a \end{bmatrix}. \quad (8.22)$$

The invariant is:

$U_b^2 - V_b^2 = U_a^2 - V_a^2$. (8.23)

Thus:

$A_b^2 = A_a^2$, (8.24)

when the transformation is treated as a coordinate change describing the same declared amplitude.

This condition is crucial.

If A itself changes economically:

$A_b \neq A_a$, (8.25)

then the movement is not merely a frame transformation.

It contains primary state change.

8.5 Coordinate transformation versus protocol transition

A coordinate transformation describes the same effective world differently:

W_P in frame a $\leftrightarrow$ W_P in frame b. (8.26)

A protocol transition changes the declaration:

$P_{k+1} \neq P_k$. (8.27)

Examples of coordinate or frame change may include:

- legitimate numeraire conversion;

- equivalent benchmark expression;
- reconciled currency translation;
- change of local basis preserving the governed exposure.

Examples of protocol transition include:

- moving from going-concern value to liquidation value;
- changing legal enforceability assumptions;
- changing accounting recognition rules;
- changing admissible collateral;
- changing the market boundary;
- changing the intervention authority.

The second case may alter the world itself.

Therefore:

Frame Change ≠ Protocol Change. (8.28)

Appendix M identifies confusing protocol transition with coordinate transformation as another major source of contradiction.

8.6 Local inertial valuation frame

A local frame is approximately inertial when:

1. the metric varies negligibly over the relevant neighbourhood;
2. the protocol remains stable;
3. frame acceleration is small;
4. residual remains bounded;
5. no major gate changes the ledger regime.

Let financial location be y.

A local inertial condition is:

$\partial\lambda g^{F}\{\mu\nu\}(y) \approx 0$. (8.29)

Within the local neighbourhood:

$g^{F}_{\mu\nu}(y) \approx \eta_{\mu\nu}$. (8.30)

CAPM can then function as a local law:

$E[r_i]_{local} = r_{f,local} + \beta_{i,local} ERP_{local}$. (8.31)

A local observer may reasonably treat:

- beta as stable;
- liquidity as continuous;
- funding as available;
- the benchmark as fixed;
- small price changes as linearly decomposable.

The approximation fails when the frame itself accelerates or the geometry changes materially.

8.7 Frame acceleration

Let relative rapidity vary with θ:

ρ_ab = ρ_ab(θ). (8.32)

Define frame acceleration:

a_ab^F = dρ_ab/dθ. (8.33)

A nonzero a_ab^F indicates that the relative valuation frame is itself changing.

Examples include:

- widening funding spread;
- rapidly changing collateral terms;
- accelerating risk aversion;
- policy intervention;
- unstable numeraire;
- benchmark dislocation.

The observed change in admitted value includes:

dR_b/dθ = exp(−ρ_ab)[dR_a/dθ − R_a(dρ_ab/dθ)]. (8.34)

Therefore:

dR_b/dθ = exp(−ρ_ab)[dR_a/dθ − a_ab^F R_a]. (8.35)

The second term is frame-generated repricing.

A local observer who ignores it may attribute all movement to the asset.

8.8 Time-dilation-like comparison

Suppose different observers accumulate internal phase at different rates:

dθ_a/dt ≠ dθ_b/dt. (8.36)

Define a local phase-rate ratio:

$\Gamma_{ab}^{F} = (d\theta_a/dt)/(d\theta_b/dt)$. (8.37)

This may reflect:

- different trading frequencies;
- different decision thresholds;
- different information latency;
- different recognition rules;
- different ledger gates.

A high-frequency market maker may experience many valuation updates during one accounting period.

A regulator may commit one event only after substantial evidence accumulation.

The analogy to time dilation is functional:

Same Calendar Interval → Different Internal Phase Accumulation. (8.38)

It is not yet a physical Lorentz time-dilation law.

A true Lorentz-like law would require an invariant interval and empirically justified transformation rule.

8.9 What could function as a financial invariant?

Candidate invariants include:

- declared amplitude A;
- contractual payoff identity;
- no-arbitrage relation;
- aggregate exposure;
- realizable cash requirement;
- settlement obligation;
- legal enforceability;
- capital-feasibility constraint.

A frame transformation is admissible only if it preserves the declared invariant set:

$Inv_P(F_a) = Inv_P(F_b)$. (8.39)

More generally:

$Distance[Inv_P(F_a), Inv_P(F_b)] \leq \varepsilon_{inv}$. (8.40)

If a change of frame hides a funding exposure or changes the legal object, invariance has failed.

The Gauge Grammar expresses the same general discipline:

Local Freedom + Invariant Constraint = Gauge-Like Structure. (8.41)

In finance, the object should remain stable across desk, accounting, legal, funding, and risk descriptions when those frames are genuinely equivalent.

8.10 What the SR-like layer achieves

The SR-like layer provides:

1. local valuation frames;
2. additive relative rapidity;
3. bounded frame coordinates;
4. local invariant structure;
5. transformation rules;
6. local-flat approximation;
7. correction for frame motion.

It does not yet provide:

- a universal propagation speed;
- a physically necessary causal cone;
- a universal financial spacetime;
- global curved geometry;
- quantum probability;
- derivative entanglement.

Those belong to other layers or remain unresolved.

---

9. Cross-Frame Invariance

9.1 The same event must first be identified

Two observers cannot agree or disagree until they identify the same proposition.

Let observer A describe:

E_A = “Barrier K was crossed under price source S_A at event time t_A.” (9.1)

Observer B describes:

E_B = “Barrier K′ was crossed under source S_B at time t_B.” (9.2)

A frame map must specify:

T_Θ: Context_A → Context_B. (9.3)

T_Φ: Outcome_A → Outcome_B. (9.4)

T_E: Observable_A → Observable_B. (9.5)

Only after this mapping can one ask whether:

T(E_A) = E_B. (9.6)

Without a frame map, “the same event” may be undefined.

9.2 Compatibility

The mapped observations must also be compatible.

In operator language:

[T_E(Ô_A),Ô_B] = 0. (9.7)

Or more generally, they must admit a joint measurement model.

In finance, compatible observations may include:

- two reconciled records of the same trade;
- legal and accounting views of the same settled obligation;
- equivalent price sources after declared adjustments;
- matched collateral records.

Incompatible observations may include:

- going-concern value and immediate liquidation value treated as if they were the same object;
- gross and net exposure without the netting protocol;
- pre-default and post-default contract states;
- incompatible time windows;
- incompatible settlement conventions.

Compatibility is not automatic merely because both observers use financial numbers.

9.3 Accessible record

Agreement requires a record accessible to both observers.

Let:

$Record_k$ = (Event,Context,Outcome,Authority,Evidence). (9.8)

Observer B can condition on A's event only when the mapped record is accessible in B's filtration:

$Record_k \in \mathscr{F}_k^B$. (9.9)

Then B may assign certainty to the mapped past event:

$P_B[T_\Phi(Outcome_A) \mid \mathscr{F}_k^B] = 1$. (9.10)

Before the record becomes accessible, B need not possess that certainty.

This prevents retrospective agreement from being confused with advance prediction.

The self-referential observer framework formalizes cross-observer fixedness through three requirements:

1. a frame map;
2. compatible or jointly measurable effects;
3. an accessible record.

Without any one of these, agreement may be ill-defined or unsupported.

9.4 Financial AB-fixedness

Define two financial observers:

A = derivative desk. (9.11)

B = clearinghouse. (9.12)

Suppose A records an exercise event:

Exercise_A = 1. (9.13)

The clearinghouse may regard the event as fixed only when:

- the contract identifier maps correctly;
- the exercise instruction satisfies the valid time window;
- the event is compatible with settlement rules;
- the record is accessible;
- authority is valid.

Then:

P_B(Exercise_B = 1 | AccessibleRecord) = 1. (9.14)

This is financial AB-fixedness.

It is not merely two observers quoting the same number.

It is cross-frame certainty about a committed event.

9.5 Redundancy and financial objectivity

A financial event becomes increasingly objective when it is redundantly encoded across:

- trading venue;
- broker;
- clearinghouse;
- custodian;
- bank ledger;
- regulator;
- counterparty statement.

Let observer j access record fragment E_j.

Consensus improves when the fragments reliably encode the same pointer event.

Define consensus error:

ε_consensus(N) = P(at least one material mapped disagreement among N observers). (9.15)

A redundant system aims for:

ε_consensus(N) ↓ as N and record quality increase. (9.16)

Redundancy does not guarantee truth.

A false record can be copied many times.

But redundancy plus:

- independent capture;
- compatible frames;
- auditability;
- cryptographic or legal integrity;
- residual disclosure;

can stabilize intersubjective financial fact.

The source observer framework similarly treats redundancy of accessible records as the mechanism by which high-probability multi-observer objectivity emerges.

9.6 Invariance of contractual identity

Let contract C be represented in frames a and b.

A valid frame map should preserve:

- parties;
- notional;
- underlying;
- strike;
- maturity;
- payoff rule;
- settlement rule;
- governing law.

Define contractual invariant:

I_C = (PartySet,Notional,Underlying,K,T,Payoff,Settlement,Law). (9.17)

Then:

T_ab(I_C^a) = I_C^b. (9.18)

If the transformation changes one of these without declaration, it is not a coordinate change.

It is a different contract.

Thus:

Same Symbol ≠ Same Contract. (9.19)

Cross-frame entanglement analysis requires the subsystem identity to remain stable.

9.7 Invariance of exposure

Suppose desk A reports delta exposure:

Δ_A. (9.20)

Desk B reports cash-equivalent exposure:

E_B. (9.21)

A frame map may relate them:

E_B = U·Δ_A. (9.22)

Where U is the declared underlying scale.

The invariant may be the realizable hedge requirement rather than the displayed coordinate.

Define:

HedgeNeed_P(F_A) ≈ HedgeNeed_P(F_B). (9.23)

If this fails materially:

GaugeFailure = Hidden Exposure Change under Apparent Reframing. (9.24)

A risk measure is not frame-robust merely because it is mathematically transformable.

The governed financial consequence must remain stable.

9.8 Invariance of nonseparability

Suppose the option–underlying state is transformed locally:

$\rho_{UD}' = (U_U \otimes U_D)\rho_{UD}(U_U^\dagger \otimes U_D^\dagger)$. (9.25)

A valid entanglement or nonseparability measure E should satisfy:

$E(\rho_{UD}') = E(\rho_{UD})$. (9.26)

This expresses invariance under local basis changes.

If apparent entanglement disappears merely because:

- the underlying desk changes units;
- the option desk changes basis;
- the numeraire changes validly;
- the risk representation changes invertibly;

then the original claim may have been coordinate-dependent.

The relevant relation must survive admissible local frame transformations.

9.9 Frame invariance does not mean protocol invariance

Suppose:

$P_{market} \neq P_{liquidation}$. (9.27)

Then:

ρ_{UD}^{market} and $\rho_{UD}^{liquidation}$ (9.28)

may represent different effective worlds.

Their nonseparability properties need not be identical.

A liquidation protocol may introduce:

- execution gates;
- market impact;

- legal priority;
- collateral seizure;
- discontinuous price formation.

Therefore:

E_{P_market}(ρ) ≠ E_{P_liquidation}(ρ) (9.29)

does not necessarily indicate inconsistency.

Entanglement-like structure is protocol-relative.

But within one declared protocol, it should remain frame-robust.

9.10 Cross-frame objectivity principle

The article adopts:

> A financial relation is objectively established under protocol P when equivalent observers can map their contexts and outcomes, the relevant measurements are compatible, accessible records can be reconciled, and the governed relation survives admissible frame transformation.

In compact form:

Objectivity_P = FrameMap + Compatibility + Record + Invariance. (9.30)

This principle will later govern claims about:

- derivative nonseparability;
- correlated collapse;
- local mixedness;
- phase transport;
- financial holonomy.

---

10. From Local Flatness to Global Curvature

10.1 Why SR-like finance is insufficient

The SR-like layer assumes a locally fixed metric.

But financial distance is not globally stable.

The effective distance between two states may change with:

- liquidity;
- leverage;

- volatility;
- collateral;
- funding;
- market depth;
- legal admissibility;
- ledger history.

A one-percent price move in a liquid market is not operationally equivalent to a one-percent move during a funding crisis.

The coordinate difference may be equal.

The financial distance is not.

Therefore the metric must become state-dependent:

$ds_F^2 = g^F_{\mu\nu}(y,L,P)dy^\mu dy^\nu$. (10.1)

This is the transition:

Fixed Valuation Geometry → State-Dependent Financial Geometry. (10.2)

Appendix M identifies precisely this transition as the GR-like extension and treats SR-like structure as the local-flat limit of the curved geometry rather than as an independent competing metric.

10.2 Financial base coordinates

Let:

$y^\mu = (\rho_D,\lambda_L,\lambda_V,\sigma,c,f,m,\ldots)$. (10.3)

Where, illustratively:

- ρ_D = discount-frame rapidity;
- λ_L = liquidity coordinate;
- λ_V = leverage coordinate;
- σ = volatility coordinate;
- c = collateral coordinate;
- f = funding coordinate;
- m = market-depth or margin coordinate.

The system evolves in internal phase:

$y^\mu = y^\mu(\theta)$. (10.4)

Ledger and protocol influence the metric:

$g^F_{\mu\nu} = g^F_{\mu\nu}(y,L_k,P_k)$. (10.5)

The complex state fibres sit over this manifold:

$|\Psi_F(y)\rangle \in \mathscr{H}_y$. (10.6)

Thus θ orders the motion, while y specifies the financial location through which the state moves.

θ should not be forced to serve as every base coordinate.

10.3 Metric interpretation

A local financial displacement is:

$dy = (dy^1,dy^2,\ldots,dy^n)$. (10.7)

The squared effective distance is:

$ds_F^2 = g^F_{\mu\nu}dy^\mu dy^\nu$. (10.8)

The metric weights directions according to financial difficulty, sensitivity, or distinguishability.

Examples:

- a small liquidity move may be cheap in normal conditions;
- the same move may be enormous near market closure;
- a small collateral decline may be harmless at low leverage;
- the same decline may trigger liquidation near a margin boundary.

Therefore:

$g^F_{cc} = g^F_{cc}(\text{leverage,margin,liquidity,L})$. (10.9)

And:

$g^F_{\lambda\lambda} = g^F_{\lambda\lambda}(\text{market depth,funding stress,P})$. (10.10)

Mixed terms may appear:

$g^F_{c\lambda} \neq 0$. (10.11)

This encodes coupling between collateral and liquidity movement.

10.4 Local tetrads

At each point y, define a local frame:

$e^a_\mu(y)$. (10.12)

The curved metric is:

$g^F_{\mu\nu}(y) = e^a_\mu(y)e^b_\nu(y)\eta_{ab}$. (10.13)

Here:

η_ab = local flat financial-frame metric. (10.14)

The local observer uses coordinates indexed by a and b.

The global manifold uses coordinates indexed by μ and ν.

This equation makes the SR-like and GR-like layers compatible.

Locally:

g^F → η. (10.15)

Globally:

$\partial\lambda g^{F}\{\mu\nu\}$ may be nonzero. (10.16)

Therefore:

SR-Like Finance = Local Tangent Description. (10.17)

GR-Like Finance = Globally Curved State Geometry. (10.18)

10.5 Financial connection

The metric determines the Levi-Civita-like connection:

$\Gamma^{\mu}_{\alpha\beta} = \frac{1}{2} g^{\mu\nu}(\partial_\alpha g_{\nu\beta} + \partial_\beta g_{\nu\alpha} - \partial_\nu g_{\alpha\beta})$. (10.19)

This connection describes how local coordinate bases change across financial state space.

A trajectory y^μ(s) obeys the geodesic equation in the absence of additional forces:

$\frac{d^2 y^\mu}{ds^2} + \Gamma^{\mu}_{\alpha\beta}\frac{dy^\alpha}{ds}\frac{dy^\beta}{ds} = 0$. (10.20)

A more realistic financial path includes intervention and residual:

$\frac{d^2 y^\mu}{ds^2} + \Gamma^{\mu}_{\alpha\beta}\frac{dy^\alpha}{ds}\frac{dy^\beta}{ds} = F_P^\mu + \varepsilon^\mu$. (10.21)

Where:

- F_P^μ is protocol-bound intervention or external forcing;
- ε^μ is unexplained residual.

The connection term can produce apparent acceleration even when the path is locally force-free.

This is the geometric analogue of repricing generated by changing financial geometry.

10.6 Liquidity curvature

Suppose liquidity coordinate λ_L approaches a boundary λ*.

A simple metric component may be:

$g^F_{\lambda\lambda} = 1/(\lambda_L - \lambda^*)^2$. (10.22)

As:

$\lambda_L \to \lambda^*$, (10.23)

we obtain:

$g^F_{\lambda\lambda} \to \infty$. (10.24)

A small coordinate movement then produces a large financial distance.

This represents:

- disappearing market depth;
- widening bid–ask spread;
- discontinuous execution cost;
- inability to exit without impact.

The geometry stretches near the liquidity boundary.

A local flat model calibrated far from λ^* may fail catastrophically near it.

10.7 Leverage and margin curvature

Let equity buffer be:

$b_E = \text{AssetValue} - \text{LiabilityValue} - \text{RequiredMargin}$. (10.25)

A stylized leverage metric may include:

$g^F_{bb} = 1/(b_E + \varepsilon_b)^2$. (10.26)

As the buffer approaches zero:

$b_E \to 0$, (10.27)

the metric becomes steep.

A small adverse move may then cross a gate:

$b_E < 0 \to \text{MarginCall}$. (10.28)

Margin call changes the ledger.

The ledger changes admissible action.

Forced action changes the primary field.

Thus:

$\text{Curvature} \to \text{Gate} \to \text{Ledger} \to \text{Backreaction} \to \text{More Curvature}$. (10.29)

This is a financial source–geometry feedback loop.

10.8 Volatility curvature

Option value sensitivity increases nonlinearly in some regions.

Let σ be volatility and U the underlying.

The derivative valuation surface has Hessian:

H_D =
[
\begin{bmatrix}
∂²D/∂U² & ∂²D/(∂U∂σ) \
∂²D/(∂σ∂U) & ∂²D/∂σ²
\end{bmatrix}.
] (10.30)

The components correspond to:

- gamma;
- vanna-like coupling;
- volga-like curvature.

A local metric candidate may be derived from sensitivity or information geometry:

g^F_{ij} ∝ E[(∂_i ln p)(∂_j ln p)]. (10.31)

Or from valuation curvature:

g^F_{ij} ∝ |∂²D/(∂y^i∂y^j)|. (10.32)

These are candidates, not uniquely derived metrics.

Their usefulness must be assessed empirically.

10.9 Ledger-conditioned geometry

The same visible financial coordinates can support different future paths depending on ledger history.

Let:

y_A = y_B. (10.33)

But:

L_A ≠ L_B. (10.34)

Then generally:

g^F(y_A,L_A,P) ≠ g^F(y_B,L_B,P). (10.35)

Examples:

- one portfolio has already breached a covenant;

- one barrier option has already knocked in;
- one institution has exhausted collateral capacity;
- one trader is constrained by prior loss limits;
- one issuer has suffered a rating downgrade.

The coordinate state may look similar.

The geometry of future action is different.

Thus:

Trace Becomes Geometry. (10.36)

The Gauge Grammar uses the same principle: accumulated trace changes which future paths become easy, difficult, trusted, liquid, legitimate, or admissible.

10.10 Curvature tensor

The connection produces a curvature tensor:

R^ρ_{σμν}
= ∂μΓ^ρ{νσ} − ∂νΓ^ρ{μσ}

- Γ^ρ_{μλ}Γ^λ_{νσ}
  − Γ^ρ_{νλ}Γ^λ_{μσ}. (10.37)

Financially, nonzero curvature means that transporting a state around different paths can produce different results.

Let path γ_1 and path γ_2 connect the same endpoints.

Then:

Transport_{γ1}(State_A) ≠ Transport_{γ2}(State_A). (10.38)

This may represent:

- funding path dependence;
- collateral sequencing;
- order-of-operation effects;
- regulatory route dependence;
- hedge-path dependence;
- volatility-surface recalibration.

The endpoint coordinates do not fully determine the transported state.

10.11 Financial holonomy

For a closed loop γ:

y_final = y_initial. (10.39)

Yet parallel transport may produce:

|Ψ_final⟩ = Hol_γ|Ψ_initial⟩. (10.40)

Where:

Hol_γ ≠ I. (10.41)

In an Abelian phase model:

Hol_γ = exp(iΦ_γ). (10.42)

The visible asset price may return to its starting point while:

- implied volatility differs;
- dealer inventory differs;
- open interest differs;
- collateral differs;
- trust differs;
- ledger differs.

Therefore:

Closed Coordinate Loop ≠ Closed Financial State Loop. (10.43)

This will later provide an important bridge to geometric phase and multi-channel coherence.

10.12 Source–geometry backreaction

A GR-like architecture requires more than estimating a curved metric.

It requires a law relating financial content to geometry.

Define an effective financial stress tensor:

T^F_{μν}. (10.44)

Candidate contents may include:

- leverage concentration;
- liquidity demand;
- collateral pressure;
- volatility load;
- funding mismatch;

- derivative gamma concentration;
- ledgered obligations.

A schematic field equation could be:

G^F_{μν} = κ_FT^F_{μν} + Ξ^F_{μν}. (10.45)

Where:

- G^F_{μν} is an effective curvature tensor;
- κ_F is a calibrated coupling;
- Ξ^F_{μν} is residual or omitted-structure pressure.

This is only a research template.

It is not derived from CAPM.

It must not be presented as a financial Einstein equation merely because the symbols resemble GR.

Appendix M explicitly notes that a state-dependent covariance metric is not yet a derived gravitational field equation; a genuine GR-like financial theory requires a source–geometry backreaction law.

10.13 Local CAPM recovery

In a sufficiently small region:

$\partial_\lambda g^F_{\mu\nu} \approx 0$. (10.46)

Then:

Γ^μ_{αβ} ≈ 0. (10.47)

And:

g^F_{μν} ≈ η_{μν}. (10.48)

The local observer recovers:

E[r_i] ≈ r_f + β_iERP. (10.49)

Thus CAPM is not rejected by curvature.

It is recovered as the tangent approximation.

The problem arises only when the local law is extrapolated across a region where:

$\partial_\lambda g^F_{\mu\nu}$ is materially nonzero. (10.50)

10.14 Crisis as geometric regime change

A financial crisis may involve:

- metric steepening;
- frame acceleration;
- gate proliferation;
- ledger discontinuity;
- protocol transition;
- residual explosion.

Define crisis onset when:

$\|\partial g^{F}\| > g^{*}$. (10.51)

Or:

$\|\varepsilon_{F}\| > \varepsilon^{*}$. (10.52)

Or:

$\mathrm{GateRate} > G^{*}$. (10.53)

The crisis is not merely a large price move.

It is a failure of the previous local world to remain approximately closed.

The response may require:

$P_{k+1} = \mathcal{U}(P_k, L_k, \mathrm{Residual}_k)$. (10.54)

A new protocol defines a new effective world.

10.15 The role of the GR-like layer

The GR-like layer explains why:

- local laws need not remain globally valid;
- financial distances depend on state;
- path order matters;
- ledger history alters future geometry;
- closed coordinate loops need not return the full state;
- stress can curve the space of admissible motion.

It does not prove that finance is physical spacetime.

It provides the global geometric environment in which local complex states and derivative composite systems evolve.

The architecture now has three separated roles:

QM-Like Fibre → Complex and Composite State. (10.55)

SR-Like Frame → Local Relative Description. (10.56)

GR-Like Base → Global State-Dependent Geometry. (10.57)

The next part introduces the mechanism that turns this layered world from a collection of local complex assets into a genuinely composite financial runtime:

Derivative contracts as state-preparation and binding operators.

Part III — Derivatives as Composite-State Constructors

11. Why CAPM Alone Has No Entanglement Grammar

11.1 A complex number is not yet a composite system

The local CAPM completion produced:

$Z = R + iQ$. (11.1)

This state contains two orthogonal coordinates.

But orthogonal coordinates are not the same as two subsystems.

R and Q are components of one declared valuation state:

$Z \in \mathbb{C}$. (11.2)

They do not automatically define:

$\mathscr{H} = \mathscr{H}_A \otimes \mathscr{H}_B$. (11.3)

A tensor-product decomposition requires two distinguishable sectors whose local observables and joint relations can be declared separately.

The scalar complex state has no natural answer to:

- Which part belongs to subsystem A?
- Which part belongs to subsystem B?
- What is the local state of each subsystem?
- What joint observables connect them?
- Can the global state be factored into local states?

Therefore:

Complexity ≠ Compositeness. (11.4)

And:

Complex Phase ≠ Entanglement. (11.5)

Part I's conclusion that the R+iQ geometry did not derive quantum entanglement was therefore mathematically appropriate.

The model lacked the state-space architecture in which entanglement could even be defined.

11.2 Correlation does not supply the missing architecture

Suppose two assets have returns r_A and r_B with:

Corr(r_A,r_B) ≠ 0. (11.6)

This establishes statistical dependence.

But ordinary correlation can arise from:

- common market exposure;
- shared industry factors;
- common macroeconomic information;
- similar leverage;
- investor flows;
- liquidity conditions;
- measurement construction.

Correlation does not imply that the two assets lack complete local states.

A classical joint distribution may exist:

p(a,b) = p(a)p(b|a). (11.7)

Or:

p(a,b) = ∫ p(λ)p(a|λ)p(b|λ)dλ. (11.8)

Here λ represents shared classical conditions.

Even perfect correlation may be classically separable.

For example:

p(0,0) = 1/2. (11.9)

p(1,1) = 1/2. (11.10)

p(0,1) = p(1,0) = 0. (11.11)

This produces complete outcome agreement without requiring a coherent nonfactorizable state.

Therefore:

Strong Correlation ≠ Entanglement. (11.12)

11.3 Tensor notation alone does not solve the problem

One could artificially write:

$|A\rangle \otimes |B\rangle$. (11.13)

But notation does not create a meaningful composite world.

The tensor product earns its role only if:

1. A and B have distinguishable operational identities;
2. each sector admits local observations;
3. joint measurements are meaningful;
4. an interaction or preparation rule connects them;
5. factorization and nonfactorization have observable consequences.

Without these conditions:

$\mathcal{H}_A \otimes \mathcal{H}_B$ (11.14)

is decorative abstraction.

The purpose of Part II is not to place financial quantities into quantum symbols merely to imitate physics.

It is to identify financial structures that genuinely support the mathematical roles.

11.4 Derivatives supply a natural subsystem partition

An option and its underlying satisfy the required distinction more clearly.

The underlying sector may contain:

- spot or forward state;
- dividend or carry state;
- liquidity state;
- ownership state;
- tradability state.

The derivative sector may contain:

- premium;
- payoff state;
- exercise status;
- implied-volatility state;

- Greeks;
- settlement status.

Define:

$\mathscr{H}_U$ = effective underlying-state space. (11.15)

$\mathscr{H}_D$ = effective derivative-state space. (11.16)

The joint state space is:

$\mathscr{H}_{UD} = \mathscr{H}_U \otimes \mathscr{H}_D$. (11.17)

The derivative and underlying are distinguishable because:

- they can be separately traded;
- they can be separately recorded;
- they possess different local observables;
- they may be held by different agents;
- they have different contractual rights;
- their local ledgers may differ.

But they are relationally bound because the derivative's identity and payoff depend upon the underlying and contract.

This provides a natural composite-state grammar.

11.5 The subsystem boundary is protocol-relative

The partition:

Underlying | Derivative (11.18)

is not the only possible partition.

Alternative decompositions include:

Underlying | Option | Hedge. (11.19)

Underlying Basket | Basket Option. (11.20)

Reference Entity | Protection Buyer | Protection Seller. (11.21)

Debt Component | Equity Conversion Component. (11.22)

Issuer Credit | Embedded Option | Collateral. (11.23)

The chosen tensor decomposition depends on the declared purpose.

Let partition Π_P be:

$\Pi_P = \{S_1, S_2, \ldots, S_N\}$. (11.24)

Then:

$\mathscr{H}_F(P) = \bigotimes_i \mathscr{H}_\{S_i\}$. (11.25)

Entanglement or nonseparability is therefore always relative to a subsystem partition.

A state may be separable under one decomposition and nonseparable under another.

Thus:

Entanglement Claim = State + Partition + Measurement Algebra + Protocol. (11.26)

No entanglement statement is complete without these declarations.

11.6 The minimum financial entanglement grammar

A financial composite system requires at least:

$\text{CompositeFinance_P} = (\mathscr{H}_A, \mathscr{H}_B, \rho_AB, \mathcal{A}_A, \mathcal{A}_B, \mathcal{A}_AB, \mathcal{U}_AB, L)$. (11.27)

Where:

- $\mathscr{H}_A$ and $\mathscr{H}_B$ are subsystem spaces;
- ρ_AB is the joint state;
- $\mathcal{A}_A$ and $\mathcal{A}_B$ are local measurement algebras;
- $\mathcal{A}_AB$ is the joint algebra;
- $\mathcal{U}_AB$ is the preparation or interaction rule;
- L is the inherited ledger.

Only after these are specified can one ask:

$\rho_AB = \rho_A \otimes \rho_B$? (11.28)

Or:

$\rho_AB \in \text{Sep}(\mathscr{H}_A \otimes \mathscr{H}_B)$? (11.29)

The scalar CAPM geometry did not provide this structure.

Derivative finance can.

11.7 CAPM's proper contribution

CAPM remains relevant inside the underlying sector.

For local underlying state U:

$E[r_U] = r_f + \beta_UERP$. (11.30)

The local complex completion is:

$Z_U = R_U + iQ_U$. (11.31)

But the derivative sector requires additional structure:

$Z_D = R_D + iQ_D$. (11.32)

The composite state is not merely:

$Z_U + Z_D$. (11.33)

It requires a joint object:

$|\Psi_{UD}\rangle \in \mathscr{H}_U \otimes \mathscr{H}_D$. (11.34)

Thus:

CAPM supplies local valuation dynamics. (11.35)

Derivative contracts supply relational composition. (11.36)

Hedging supplies backreaction. (11.37)

Ledger supplies historical fixation. (11.38)

No one component should be made to perform every role.

---

12. The Option–Underlying Composite System

12.1 What counts as the underlying sector?

The underlying sector is not necessarily one scalar price.

Define an effective underlying state:

$x_U = (S,\mu_U,\sigma_U,q_U,\lambda_U,L_U,\ldots)$. (12.1)

Where:

- S = observable spot or forward price;
- μ_U = expected drift under the declared measure;
- σ_U = underlying volatility state;
- q_U = carry or yield;
- λ_U = liquidity condition;
- L_U = underlying-side ledger.

A basis may be:

$\{|u_1\rangle,|u_2\rangle,\ldots,|u_N\rangle\}$. (12.2)

The basis states may represent:

- price intervals;

- return regimes;
- volatility regimes;
- liquidity states;
- terminal outcomes;
- path-dependent classes.

The underlying state is:

$|\psi_U\rangle = \Sigma_n \alpha_n |u_n\rangle$. (12.3)

Or, for a mixed state:

$\rho_U = \Sigma_{n,m} \rho^U_{nm} |u_n\rangle\langle u_m|$. (12.4)

This is an effective representation under a declared valuation world.

It does not imply that the underlying asset is physically a quantum object.

12.2 What counts as the derivative sector?

Define the derivative state:

$x_D = (D,\Delta,\Gamma,\text{Vega},\Theta,\text{Rho},E,G_D,L_D,\ldots)$. (12.5)

Where:

- D = derivative premium or value;
- Δ = delta;
- Γ = gamma;
- Vega = volatility sensitivity;
- Θ = calendar-time sensitivity;
- Rho = financing sensitivity;
- E = exercise state;
- G_D = contract-gate state;
- L_D = derivative-side ledger.

A derivative basis may be:

$\{|d_1\rangle,|d_2\rangle,\ldots,|d_M\rangle\}$. (12.6)

Possible basis labels include:

- option-value interval;
- in-the-money status;

- exercise status;
- barrier status;
- volatility state;
- settlement state;
- hedge state.

The derivative state is:

$|\psi_D\rangle = \Sigma_m \beta_m |d_m\rangle$. (12.7)

The joint space is:

$\mathscr{H}_UD = \text{span}\{|u_n\rangle \otimes |d_m\rangle\}$. (12.8)

12.3 Product states

A product state has the form:

$|\Psi_UD\rangle = |\psi_U\rangle \otimes |\psi_D\rangle$. (12.9)

Expanding:

$|\Psi_UD\rangle = \Sigma_{n,m} \alpha_n \beta_m |u_n, d_m\rangle$. (12.10)

The joint coefficients factorize:

$c_{nm} = \alpha_n \beta_m$. (12.11)

In such a state, the derivative and underlying may each possess complete local states.

Measurement statistics factorize for local observables when no further classical correlation is added:

$p(u_n, d_m) = p_U(u_n) p_D(d_m)$. (12.12)

This is the simplest separable case.

A freely floating derivative unrelated to the declared underlying would approximate this structure.

A genuine option contract normally does not.

12.4 Contractually admissible pairs

A derivative contract restricts which joint states are admissible.

Let the contract relation be:

$C_P(u,d) = 0$. (12.13)

Then the admissible joint set is:

$\mathcal{K}_P = \{(u,d) \mid C_P(u,d) = 0\}$. (12.14)

For a European call at maturity:

$d = \max(u - K, 0)$. (12.15)

The admissible set is:

$\mathcal{K}_{call} = \{(u,d) \mid d = \max(u - K, 0)\}$. (12.16)

Before maturity, the relation is:

$d = V_P(u,\sigma,r,q,T-t,L)$. (12.17)

Thus:

$\mathcal{K}_t = \{(u,d,\sigma,r,q,L) \mid d = V_P(u,\sigma,r,q,T-t,L)\}$. (12.18)

Not every pair $|u,d\rangle$ is financially admissible.

The contract selects a subspace or submanifold of the full tensor-product space.

Define the contract projector:

$\Pi_{contract} = \Sigma_{(n,m)\in\mathcal{K}_P} |u_n,d_m\rangle\langle u_n,d_m|$. (12.19)

An admissible state satisfies:

$\Pi_{contract}|\Psi_{UD}\rangle = |\Psi_{UD}\rangle$. (12.20)

12.5 The joint state may be primary

Suppose the contract pairs each underlying branch u_n with a derivative branch d_n.

Then:

$|\Psi_{UD}\rangle = \Sigma_n c_n |u_n,d_n\rangle$. (12.21)

The underlying reduced state is:

$\rho_U = \Sigma_{n,m} c_n c_m^* \langle d_m|d_n\rangle |u_n\rangle\langle u_m|$. (12.22)

The derivative reduced state is:

$\rho_D = \Sigma_{n,m} c_n c_m^* \langle u_m|u_n\rangle |d_n\rangle\langle d_m|$. (12.23)

If the paired basis states are orthonormal:

$\langle u_m|u_n\rangle = \delta_{mn}$. (12.24)

$\langle d_m|d_n\rangle = \delta_{mn}$. (12.25)

Then:

$\rho_U = \Sigma_n |c_n|^2 |u_n\rangle\langle u_n|$. (12.26)

$\rho_D = \Sigma_n |c_n|^2 |d_n\rangle\langle d_n|$. (12.27)

The global state may be pure while the local states are mixed.

Completeness belongs to the joint state.

This is the formal pattern associated with entanglement.

12.6 Local incompleteness in financial terms

The local underlying price S does not fully determine:

- the option's volatility state;
- its path history;
- its collateral state;
- its exercise admissibility;
- its legal status;
- its hedge history.

The local option premium D does not fully determine:

- the underlying path;
- the common volatility field;
- the hedge inventory;
- the contract's primary economic state;
- the underlying liquidity condition.

Thus the scalar local states are incomplete.

A more complete description belongs to:

Underlying + Derivative + Contract + Volatility + Ledger. (12.28)

This may be represented as a joint state:

ρ_UDCL. (12.29)

The relevant financial principle is:

Local Price Completeness < Joint Contract-State Completeness. (12.30)

12.7 No-arbitrage as a joint-state constraint

A European call under simple assumptions satisfies bounds such as:

$\max(0, S_0 e^{-qT} - Ke^{-rT}) \le C_0 \le S_0 e^{-qT}$. (12.31)

These inequalities constrain the joint valuation state.

An observed pair (S_0, C_0) outside the bounds is not admissible under the declared model.

No-arbitrage therefore acts like a geometric or algebraic constraint:

$$\mathcal{A}_{noarb}(\rho_{UD}) = 0. \quad (12.32)$$

Or:

$$\rho_{UD} \in \mathcal{K}_{noarb}. \quad (12.33)$$

The derivative and underlying cannot move independently without potentially leaving the admissible manifold.

This is stronger than correlation.

It is governed relational dependence.

12.8 Replication as an alternative basis

For a replicable derivative:

$$D = \Delta S + B. \quad (12.34)$$

Where:

- Δ is the underlying holding;
- B is the bond or cash position.

The derivative basis can be transformed into a replication basis:

$$|d\rangle \leftrightarrow |\Delta, B\rangle. \quad (12.35)$$

The same financial object can therefore be represented in:

- derivative basis;
- underlying–cash basis;
- payoff basis;
- Greek basis.

The frame transformation must preserve the governed claim:

$$Value_D = Value_{replication}. \quad (12.36)$$

Under ideal assumptions:

$$D_t = \Delta_t S_t + B_t. \quad (12.37)$$

This basis equivalence will later become important for distinguishing genuine nonseparability from a representation that disappears under an admissible decomposition.

12.9 The derivative's identity is relational

A common stock remains identifiable without reference to one particular option.

An option does not remain the same option if one changes:

- underlying;
- strike;
- maturity;
- payoff;
- exercise style;
- settlement method;
- governing law.

Define derivative identity:

I_D = (U,K,T,Payoff,Exercise,Settlement,Law). (12.38)

The derivative is therefore relational by construction:

D = D[I_D]. (12.39)

This does not make it quantum.

But it provides the strongest financial example of an object whose identity is constituted through a declared relation.

12.10 The composite-system proposition

Proposition 12.1 — Derivative Composite-State Principle

A derivative and its underlying form an effective composite system under protocol P when:

1. their local state spaces are separately identifiable;
2. the contract restricts admissible joint states;
3. local measurements do not exhaust the joint contract state;
4. interventions in either sector may change the other;
5. joint outcomes are committed through shared gates or ledgers.

In compact form:

DerivativeComposite_P ⇔ LocalDistinction ∧ JointConstraint ∧ LocalIncompleteness ∧ Coupling ∧ SharedTrace. (12.40)

This proposition establishes the state-space foundation.

It does not yet establish coherent entanglement.

---

13. The Contract as a State-Preparation Operator

13.1 From payoff function to preparation rule

A payoff function is usually written as:

$D_T = f(U_T)$. (13.1)

For a call:

$f(U_T) = \max(U_T - K, 0)$. (13.2)

This describes the terminal relationship.

For an internal-world composite-state model, the contract should be represented as an operator that prepares correlated admissible states.

Let the derivative begin in a neutral registration state:

$|0_D\rangle$. (13.3)

For each underlying basis state:

$|u_n\rangle$, (13.4)

define the contract action:

$\hat{U}_C|u_n, 0_D\rangle = |u_n, d_n\rangle$. (13.5)

Where:

$d_n = f_P(u_n)$. (13.6)

The contract therefore creates a branchwise pairing.

13.2 Linear extension

Suppose the underlying state is:

$|\psi_U\rangle = \Sigma_n c_n|u_n\rangle$. (13.7)

The initially separable state is:

$|\psi_{initial}\rangle = |\psi_U\rangle \otimes |0_D\rangle$. (13.8)

Applying the contract operator:

$|\Psi_{UD}\rangle = \hat{U}_C(|\psi_U\rangle \otimes |0_D\rangle)$. (13.9)

By linearity:

$|\Psi_{UD}\rangle = \Sigma_n c_n|u_n, d_n\rangle$. (13.10)

If at least two distinct derivative states d_n survive, the state generally cannot be written as a product.

Thus a contract can act as an entangling operator inside the effective representation.

13.3 Is the contract operator unitary?

A physical quantum entangling gate is often represented by a unitary operator.

A financial contract map need not be unitary.

For example, multiple underlying states may lead to the same derivative payoff:

$u_1 < K \rightarrow d_1 = 0$. (13.11)

$u_2 < K \rightarrow d_2 = 0$. (13.12)

The map:

$u \rightarrow \max(u-K,0)$ (13.13)

is many-to-one in the derivative coordinate.

Information may be lost if the underlying state is discarded.

However, the joint map:

$|u,0_D\rangle \rightarrow |u,f(u)\rangle$ (13.14)

can remain injective because the underlying label is retained.

An enlarged state-preparation map may therefore be represented reversibly:

$\hat{U}_C: |u,0_D\rangle \rightarrow |u,f(u)\rangle$. (13.15)

But later gate and ledger processes may be non-unitary.

The article should distinguish:

Contract Preparation → potentially reversible in enlarged state space. (13.16)

Exercise / Settlement / Ledger Commitment → generally irreversible effective update. (13.17)

13.4 Contract Hamiltonian

The preparation process may be generated dynamically.

Let:

$\hat{H}_C$ = contract-interaction generator. (13.18)

Then:

$i\hbar_F \partial|\Psi_UD\rangle/\partial\theta = (\hat{H}_U \otimes I_D + I_U \otimes \hat{H}_D + \hat{H}_C)|\Psi_UD\rangle$. (13.19)

If:

$\hat{H}_C = 0$, (13.20)

an initially separable state may remain separable under local evolution.

If:

$\hat{H}_C \neq 0$, (13.21)

joint evolution can generate nonfactorization.

A schematic interaction may be:

$\hat{H}_C = \Sigma_n g_n |u_n\rangle\langle u_n| \otimes \hat{O}_{D,n}$. (13.22)

Where $\hat{O}_{D,n}$ prepares or rotates the derivative state conditionally on underlying branch n.

This is the operator analogue of conditional contract binding.

13.5 Continuous-state preparation

For a continuous underlying variable u:

$|\psi_U\rangle = \int \psi(u)|u\rangle du$. (13.23)

The contract prepares:

$|\Psi_{UD}\rangle = \int \psi(u)|u, f_P(u)\rangle du$. (13.24)

For pre-maturity valuation:

$f_P(u) = V_P(u,\sigma,r,q,T-t,L)$. (13.25)

The reduced underlying kernel is:

$\rho_U(u,u') = \psi(u)\psi^*(u')\langle f_P(u')|f_P(u)\rangle$. (13.26)

If derivative states corresponding to distinct underlying values become distinguishable:

$\langle f_P(u')|f_P(u)\rangle \to 0$ for $u' \neq u$, (13.27)

off-diagonal underlying coherence is suppressed.

The derivative sector then acts as a record of the underlying branch.

This resembles decoherence through correlation with another subsystem.

Again, the statement is formal.

Observable financial coherence must still be established empirically.

13.6 Path-dependent contract preparation

For a path-dependent derivative, the contract depends on the entire path:

$d = f[U_{0:T}]$. (13.28)

The joint state must include path memory:

$|\Psi\rangle = \Sigma_\gamma c_\gamma |\gamma_U, d_\gamma\rangle$. (13.29)

Where γ labels an underlying path.

For a barrier option:

$d_\gamma = f(U_T, Hit_\gamma)$. (13.30)

With:

$Hit_\gamma \in \{0,1\}$. (13.31)

The contract therefore binds:

Underlying Path + Barrier Record + Derivative State. (13.32)

The appropriate space is:

$\mathscr{H}_{path} \otimes \mathscr{H}_{gate} \otimes \mathscr{H}_D$. (13.33)

This is a multipartite composite system.

A scalar terminal price is insufficient.

13.7 Credit derivatives as event-entangled contracts

A credit default swap links:

- reference entity;
- protection buyer;
- protection seller;
- credit-event determination;
- premium leg;
- protection leg;
- settlement convention.

The joint state may be:

$|\Psi_{CDS}\rangle \in \mathscr{H}_R \otimes \mathscr{H}_B \otimes \mathscr{H}_S \otimes \mathscr{H}_G \otimes \mathscr{H}_L$. (13.34)

Where:

- R = reference credit state;
- B = buyer state;
- S = seller state;
- G = credit-event gate;
- L = settlement ledger.

The contract-preparation operator binds multiple parties into one contingent obligation.

The event gate changes all components simultaneously at the internal ledger episode.

From the primary universe, this is contractual causation.

From the internal financial world, it is a multipartite state transition.

13.8 Contract creation versus contract observation

The contract exists before any particular market observer measures its value.

Define contract preparation at episode k_0:

$$\rho_{UD}(k_0) = \mathrm{Prepare}_C[\rho_U \otimes \rho_D^0]. \quad (13.35)$$

Later observers measure:

$$O_j(k) = \mathrm{Tr}[\rho_{UD}(k)\hat{O}_j]. \quad (13.36)$$

The observer does not create the contract relation by measuring it.

The observer selects which aspect becomes locally visible.

This mirrors the earlier distinction:

Observer Restriction does not create nonseparability. (13.37)

It reveals only a bounded projection of a globally prepared state.

13.9 State preparation depends on declaration

The same legal contract may be compiled differently under different protocols.

Under a market-value protocol:

$$\mathrm{Prepare}_{market}(C,X). \quad (13.38)$$

Under a liquidation protocol:

$$\mathrm{Prepare}_{liquidation}(C,X). \quad (13.39)$$

Under an accounting protocol:

$$\mathrm{Prepare}_{accounting}(C,X). \quad (13.40)$$

The resulting effective states may differ:

$$\rho_{UD}^{market} \neq \rho_{UD}^{liquidation} \neq \rho_{UD}^{accounting}. \quad (13.41)$$

The primary contract remains the same.

The secondary valuation world changes.

Thus:

Contract Identity can be invariant while Effective Joint State is protocol-relative. (13.42)

13.10 Preparation residual

No contract model captures the complete market relationship.

Define preparation residual:

$\varepsilon_C = \rho_{observed} - Prepare_C(\rho_{input})$. (13.43)

Sources include:

- model risk;
- volatility-surface error;
- liquidity premium;
- funding mismatch;
- legal uncertainty;
- execution cost;
- strategic behaviour;
- hidden optionality.

The contract preparation model is acceptable only when:

$\|\varepsilon_C\| \le \varepsilon_C^*$. (13.44)

Large residual means that the declared contract operator is incomplete.

Tensor formalism must not conceal model error.

---

14. Separability, Nonfactorization, and Local Mixedness

14.1 Product-state separability

The simplest separable state is:

$\rho_{UD} = \rho_U \otimes \rho_D$. (14.1)

For local observables $\hat{O}_U$ and $\hat{O}_D$:

$\langle \hat{O}_U \otimes \hat{O}_D \rangle = \langle \hat{O}_U \rangle \langle \hat{O}_D \rangle$. (14.2)

There are no joint correlations beyond independent local states.

This rarely describes an active derivative contract.

14.2 Classical correlated separability

A broader separable state is:

$\rho_sep = \Sigma_j p_j \rho\hat{}U_j \otimes \rho\hat{}D_j.$ (14.3)

Where:

$p_j \geq 0.$ (14.4)

$\Sigma_j p_j = 1.$ (14.5)

This state may exhibit strong correlations.

But each branch j contains definite local states.

The joint uncertainty is classical mixing over branches.

A standard option scenario model often has this form:

$\rho_scenario = \Sigma_j q_j |u_j,d_j\rangle\langle u_j,d_j|.$ (14.6)

Where q_j are scenario weights.

This is not quantum entanglement.

It is a classically correlated contract state.

14.3 Coherent nonfactorizable state

A coherent joint state is:

$|\Psi_UD\rangle = \Sigma_j c_j e^{i\phi j} |u_j,d_j\rangle.$ (14.7)

Its density operator is:

$\rho_UD = \Sigma_{j,k} c_j c_k^* e^{i(\phi j - \phi k)} |u_j,d_j\rangle\langle u_k,d_k|.$ (14.8)

The diagonal terms are:

$\rho_diag = \Sigma_j |c_j|^2 |u_j,d_j\rangle\langle u_j,d_j|.$ (14.9)

The off-diagonal terms are:

$\rho_off = \Sigma_\{j \neq k\} c_j c_k^* e^{i(\phi j - \phi k)} |u_j,d_j\rangle\langle u_k,d_k|.$ (14.10)

The off-diagonal terms encode branch-relative coherence.

A stronger financial analogy requires these terms to affect an observable.

Otherwise:

ρ_UD and ρ_diag (14.11)

are operationally indistinguishable.

The phase structure would then be decorative.

14.4 Schmidt decomposition

For a pure bipartite state, write:

$|\Psi_UD\rangle = \Sigma_r \sqrt{\lambda_r}|\tilde{u}_r\rangle|\tilde{d}_r\rangle$. (14.12)

Where:

$\lambda_r \geq 0$. (14.13)

$\Sigma_r \lambda_r = 1$. (14.14)

This is the Schmidt decomposition.

The state is separable iff only one Schmidt coefficient is nonzero:

$\text{SchmidtRank}(|\Psi_UD\rangle) = 1$. (14.15)

It is nonfactorizable when:

$\text{SchmidtRank}(|\Psi_UD\rangle) > 1$. (14.16)

A candidate financial nonseparability measure is the Schmidt entropy:

$S_Schmidt = -\Sigma_r \lambda_r \ln \lambda_r$. (14.17)

If:

$S_Schmidt = 0$, (14.18)

the pure state is separable.

If:

$S_Schmidt > 0$, (14.19)

the effective pure state is nonfactorizable.

The financial meaning of the Schmidt basis must be operationally declared.

Without measurable basis states, the entropy is only formal.

14.5 Reduced states

For global state ρ_UD:

$\rho_U = \text{Tr}_D(\rho_UD)$. (14.20)

$\rho_D = \text{Tr}_U(\rho_UD)$. (14.21)

For the Schmidt state:

$\rho_U = \Sigma_r \lambda_r|\tilde{u}_r\rangle\langle\tilde{u}_r|$. (14.22)

$\rho_D = \Sigma_r \lambda_r|\tilde{d}_r\rangle\langle\tilde{d}_r|$. (14.23)

The local von Neumann-like entropy is:

$S(\rho_U) = -\mathrm{Tr}(\rho_U \ln \rho_U)$. (14.24)

$S(\rho_D) = -\mathrm{Tr}(\rho_D \ln \rho_D)$. (14.25)

For a pure global state:

$S(\rho_U) = S(\rho_D)$. (14.26)

This common entropy measures effective entanglement in the formal model.

14.6 Financial interpretation of local mixedness

Suppose the global derivative complex is fully specified:

GlobalState = Underlying + Contract + Volatility Surface + Hedge + Ledger. (14.27)

The underlying desk sees only:

View_U = PartialTrace_DHL(GlobalState). (14.28)

The derivative desk sees only:

View_D = PartialTrace_UHL(GlobalState). (14.29)

The clearinghouse sees:

View_C = PartialTrace_nonsettlement(GlobalState). (14.30)

Each observer obtains a mixed local state because relevant distinctions reside elsewhere in the global system.

Local mixedness may represent:

- incomplete knowledge;
- genuine effective coupling;
- aggregation loss;
- hidden counterparty state;
- omitted path information;
- unresolved volatility regime.

Therefore local entropy alone does not prove entanglement.

The model must distinguish:

Ignorance Entropy. (14.31)

Aggregation Entropy. (14.32)

Entanglement-Form Entropy. (14.33)

14.7 Classical purification problem

Any classical mixed distribution can be represented as part of a larger pure formal state.

Therefore, writing a purification:

$|\Psi\rangle = \Sigma_j \sqrt{p_j}|j\rangle_U|j\rangle_D$ (14.34)

does not prove that the original financial system possessed quantum coherence.

It may merely embed a classical distribution into a Hilbert space.

The crucial test is whether the financial measurement algebra reveals phase-sensitive consequences unavailable to the classical mixture:

$\rho_mix = \Sigma_j p_j|j,j\rangle\langle j,j|$. (14.35)

Thus:

Formal Purification ≠ Empirical Coherence. (14.36)

This is one of the most important safeguards in Part II.

14.8 Entanglement witnesses

An entanglement witness W satisfies:

$Tr(W\rho_sep) \geq 0$ for all separable states. (14.37)

But for at least one nonseparable state:

$Tr(W\rho_ent) < 0$. (14.38)

A financial analogue would require an observable combination whose measured value cannot be reproduced by the declared family of classical separable models.

Let $\mathcal{S}_null$ be the allowed classical model family.

Define witness condition:

$E_data[W] < \inf_{\rho \in \mathcal{S}_null} Tr(W\rho) - \delta$. (14.39)

Where δ includes estimation uncertainty and model-selection penalty.

This would be a stronger test than observing correlation or option dependence.

Candidate null families include:

- copula models;
- stochastic volatility;
- latent-factor models;
- regime-switching models;
- agent-based feedback;

- classical path-dependent contracts.

Only after these fail should stronger nonseparability language be considered.

14.9 Conditional states

Suppose the underlying measurement uses effect E_u.

The probability of outcome u is:

$$p(u) = \mathrm{Tr}[(E_u \otimes I_D)\rho_{UD}]. \quad (14.40)$$

The conditional derivative state is:

$$\rho_{D|u} = \mathrm{Tr}_U[(\sqrt{E_u} \otimes I_D)\rho_{UD}(\sqrt{E_u} \otimes I_D)] / p(u). \quad (14.41)$$

Likewise, derivative measurement d produces:

$$\rho_{U|d} = \mathrm{Tr}_D[(I_U \otimes \sqrt{F_d})\rho_{UD}(I_U \otimes \sqrt{F_d})] / p(d). \quad (14.42)$$

This formalizes the internal observer's experience:

Measurement of One Sector
→ Conditional Reassignment of the Other Sector. (14.43)

From the primary financial universe, this may correspond to ordinary information updating.

From the secondary world, it is a local projection of one global state.

The distinction between these descriptions must remain explicit.

14.10 Steering-like asymmetry

Suppose derivative observations strongly condition the underlying state:

$$\rho_{U|d_1} \neq \rho_{U|d_2}. \quad (14.44)$$

This may occur because the option surface reveals information about:

- expected volatility;
- tail risk;
- dealer positioning;
- latent demand;
- anticipated events.

The derivative sector may appear to steer the state assigned to the underlying.

But classical information channels can explain this.

A genuine steering test would need to show that no declared local-hidden-state model reproduces the observed assemblage:

{p(d|b),ρ_U|d,b}. (14.45)

The financial research question is therefore:

Can derivative measurement settings generate conditional underlying-state families that outperform every admissible classical latent-state explanation? (14.46)

This remains unestablished.

14.11 Nonseparability is not nonlocality

A state may be nonfactorizable without implying spatial nonlocality.

Financial derivative and underlying states may reside:

- in the same market;
- on the same server;
- in the same legal contract;
- within one balance sheet.

Their joint update may be immediate in the effective model because both belong to one internal episode.

This does not demonstrate:

- superluminal influence;
- physical spatial separation;
- quantum no-signalling;
- Bell inequality violation.

Therefore:

Effective Nonfactorization ≠ Physical Nonlocality. (14.47)

The article will later treat no-signalling and Bell nonclassicality as higher thresholds.

14.12 Protocol-relative separability

A state may be separable under protocol P_1 and nonfactorizable under protocol P_2 because the declared observables and partitions differ.

For example:

P_1 = terminal payoff only. (14.48)

P_2 = full path + volatility + hedge + collateral. (14.49)

A terminal-payoff representation may collapse many distinctions into a classical mixture.

A richer protocol may preserve branch relations.

Thus:

$Sep_{P_1}(\rho)$ does not imply $Sep_{P_2}(\rho)$. (14.50)

The reverse also need not hold.

This does not make separability arbitrary.

It means the claim is indexed by what the effective world preserves.

14.13 Secondary-world entanglement definition

Definition 14.1 — Secondary-World Derivative Entanglement

An option and underlying are secondarily entangled under protocol P when:

1. the declared joint state belongs to $\mathscr{H}_U \otimes \mathscr{H}_D$;
2. the state is nonseparable relative to the admissible local decomposition;
3. local observers have access only to restricted subalgebras;
4. joint correlations affect admissible measurements;
5. the effect cannot be reproduced by the declared classical null family;
6. the relation remains invariant under admissible local frame transformations.

In compact form:

$$Ent_\theta(U,D \mid P) \Leftrightarrow NonSep_P \wedge LocalRestriction_P \wedge ObservableJointEffect_P \wedge NullRejection_P \wedge FrameRobust_P. \quad (14.51)$$

This is a demanding definition.

Most ordinary derivative dependence may satisfy only the lower conditions:

- joint constraint;
- functional coupling;
- conditional information.

The stronger entanglement claim requires much more.

14.14 The entanglement ladder begins here

The article will use the following progression:

E_1 = Statistical Correlation. (14.52)

E_2 = Functional Dependence. (14.53)

E_3 = Contractual Coupling. (14.54)

E_4 = Dynamical Binding. (14.55)

E_5 = Effective Nonfactorization. (14.56)

E_6 = Observable Coherent Composite State. (14.57)

E_7 = Local Mixedness and Contextual Joint Measurement. (14.58)

E_8 = No-Signalling Entanglement. (14.59)

E_9 = Bell-Nonclassical Entanglement. (14.60)

Finance clearly realizes E_1–E_4.

It can formally represent E_5.

E_6–E_7 require new empirical evidence.

E_8–E_9 remain unestablished.

The next chapter develops the differential structure that makes derivative binding locally measurable:

Greeks as the tangent and curvature grammar of the option–underlying composite state.

15. Greeks as Differential Geometry of Coupling

15.1 Why Greeks belong at the centre of the composite-state model

The option–underlying relation is not static.

Its coupling changes with:

- underlying price;
- volatility;
- maturity;
- financing;
- carry;
- path history;
- ledger state;
- market regime.

Greeks measure the local response of the derivative sector to movement in these financial directions.

Let the derivative value be:

$D = V(U,\sigma,r,q,\tau_T,P,L)$. (15.1)

Where:

$\tau_T = T - t.$ (15.2)

For a small displacement:

$dD \approx \Delta dU + \frac{1}{2}\Gamma(dU)^2 + \text{Vega}\cdot d\sigma + \Theta dt + \text{Rho}\cdot dr + \text{Carry}\cdot dq + \dots .$ (15.3)

This is normally read as a sensitivity expansion.

Inside the layered architecture, it has a deeper role:

> Greeks are the local differential map between sectors of the composite financial world.

They describe how movement in the underlying, volatility, financing, and temporal directions is transported into the derivative fibre.

15.2 Delta as a tangent map

Delta is:

$\Delta = \partial D/\partial U.$ (15.4)

It maps an infinitesimal underlying displacement into a first-order derivative displacement:

$dD_U \approx \Delta dU.$ (15.5)

Geometrically:

$\Delta: T_U\mathcal{M}_U \to T_D\mathcal{M}_D.$ (15.6)

Here:

- $T_U\mathcal{M}_U$ is the local tangent space of the underlying sector;
- $T_D\mathcal{M}_D$ is the local tangent space of the derivative sector.

Delta therefore acts as a local pushforward:

$dU \mapsto dD.$ (15.7)

If:

$\Delta \approx 0,$ (15.8)

the derivative is locally insensitive to the underlying direction.

If:

$|\Delta| \approx 1,$ (15.9)

the derivative locally moves almost one-for-one with the underlying under the declared frame.

Delta is not entanglement.

But it is a measurable local coefficient of inter-sector coupling.

15.3 Delta depends on the valuation frame

Delta is protocol-relative:

$\Delta_P = \partial D_P / \partial U_P$. (15.10)

A market-value delta may differ from:

- liquidation delta;
- accounting delta;
- collateral delta;
- regulatory delta;
- model delta;
- executable hedge delta.

Therefore:

$\Delta_{market} \neq \Delta_{liquidation} \neq \Delta_{accounting}$. (15.11)

These differences need not be errors.

They may reflect different effective worlds.

A cross-frame comparison requires transport:

$\Delta_B = T_{BA}(\Delta_A) + \varepsilon_\Delta$. (15.12)

Where:

- T_{BA} is the admissible frame map;
- ε_Δ is the unexplained mismatch.

A supposedly invariant hedge relationship that changes materially under an equivalent basis transformation signals frame failure or model incompleteness.

15.4 Gamma as curvature of coupling

Gamma is:

$\Gamma = \partial^2 D / \partial U^2$. (15.13)

Delta changes with the underlying:

$d\Delta \approx \Gamma dU$. (15.14)

Gamma therefore measures the local curvature of the derivative surface along the underlying direction.

If:

$\Gamma = 0$, (15.15)

the derivative–underlying relation is locally linear.

If:

$|\Gamma|$ is large, (15.16)

small underlying movement changes the coupling itself.

This creates a reflexive structure:

Underlying Move
→ Delta Change
→ Hedge Requirement Change
→ Underlying Order Flow. (15.17)

Gamma is therefore not only curvature of valuation.

It can become curvature of market action.

15.5 Gamma and second-order state transport

For finite movement ΔU:

$\Delta D \approx \text{Delta}\cdot\Delta U + \tfrac{1}{2}\text{Gamma}\cdot(\Delta U)^2$. (15.18)

The first term transports the state linearly.

The second term records departure from local flatness.

The ratio:

$\chi_\Gamma = |\tfrac{1}{2}\Gamma(\Delta U)^2| \,/\, |\Delta\Delta U|$ (15.19)

measures the relative importance of curvature.

When:

$\chi_\Gamma \ll 1$, (15.20)

the local tangent approximation is adequate.

When:

$\chi_\Gamma \gtrsim 1$, (15.21)

linear hedging is no longer a sufficient description.

Thus gamma provides a natural local-flatness diagnostic.

15.6 Vega as coupling to the volatility field

Vega is:

$\text{Vega} = \partial D/\partial\sigma$. (15.22)

It maps movement in the volatility direction into derivative repricing:

$dD_\sigma \approx \text{Vega} \cdot d\sigma$. (15.23)

Volatility should not be treated merely as one external scalar input.

In a derivative world, volatility may form a state field:

$\sigma = \sigma(K,T,U,L,P)$. (15.24)

The derivative therefore couples not only to the underlying but to a volatility surface or manifold.

Vega is the local connection between:

Derivative Fibre ↔ Volatility Direction. (15.25)

A derivative with low delta but high vega may be weakly coupled to spot movement while remaining strongly coupled to changes in expected dispersion.

15.7 Theta as primary-time sensitivity

Option theta is usually:

$\Theta = \partial D / \partial t$. (15.26)

Or, using remaining maturity τ_T:

$\Theta_\tau = \partial D / \partial \tau_T$. (15.27)

This theta must not be confused with the valuation phase θ.

The article therefore distinguishes:

Θ_{option} = calendar-time decay sensitivity. (15.28)

θ_{phase} = internal valuation orientation. (15.29)

To reduce ambiguity, write option theta as:

$\text{Theta}_D = \partial D / \partial t$. (15.30)

Then:

$\text{Theta}_D \neq \theta$. (15.31)

This distinction is essential.

One symbol refers to sensitivity with respect to primary calendar time.

The other refers to internal phase order in the secondary world.

15.8 Rho as financing-frame sensitivity

Rho is:

Rho_D = ∂D/∂r. (15.32)

It measures response to movement in the financing or discount direction.

Since different observers may use different financing curves:

r = r(F_a), (15.33)

rho is naturally frame-dependent.

A change in derivative value may arise from:

- underlying movement;
- volatility movement;
- financing-frame movement;
- protocol revision.

Thus:

dD = dD_state + dD_frame + dD_residual. (15.34)

Rho contributes to the frame term.

This makes it particularly relevant to the SR-like layer.

15.9 Cross-Greeks as mixed geometric terms

The derivative surface contains mixed derivatives.

Vanna:

Vanna = $\partial^2 D/(\partial U \partial \sigma)$. (15.35)

Volga:

Volga = $\partial^2 D/\partial \sigma^2$. (15.36)

Charm:

Charm = $\partial^2 D/(\partial U \partial t)$. (15.37)

Color:

Color = $\partial^3 D/(\partial U^2 \partial t)$. (15.38)

These measure how one coupling changes as another coordinate moves.

For example:

dDelta ≈ Gamma·dU + Vanna·dσ + Charm·dt. (15.39)

This means delta is not an isolated coefficient.

It is a field over financial state space.

Mixed Greeks therefore resemble connection or curvature components linking distinct directions.

15.10 The Greek tensor

Let financial coordinates be:

$y^i = (U,\sigma,r,q,t,c,f,\ldots)$. (15.40)

The first derivative vector is:

$J_i = \partial D/\partial y^i$. (15.41)

The Hessian is:

$H_{ij} = \partial^2 D/(\partial y^i \partial y^j)$. (15.42)

The local derivative expansion is:

$dD \approx J_i dy^i + \frac{1}{2} H_{ij} dy^i dy^j$. (15.43)

Using summation over repeated indices.

J_i gives the local tangent response.

H_{ij} gives local curvature and mixed coupling.

A more complete model may include third-order tensor:

$T_{ijk} = \partial^3 D/(\partial y^i \partial y^j \partial y^k)$. (15.44)

The derivative contract therefore carries a hierarchy:

Gradient → Hessian → Higher Curvature. (15.45)

This hierarchy is already present in mature finance.

The new framework reorganizes it geometrically.

15.11 Greeks as local entanglement coordinates

Suppose the global state is:

ρ_UD. (15.46)

A local coupling matrix may be defined:

$C_{ij} = \partial O_D,_i/\partial O_U,_j$. (15.47)

Delta is one component of C.

Gamma measures variation of C.

Vega introduces coupling to an environmental sector.

The matrix C describes how locally observable changes in one sector are reflected in another.

But:

Nonzero $C_{ij} \neq$ Entanglement. (15.48)

A classical function can generate nonzero sensitivities.

The Greeks identify coupling strength.

They do not decide whether the joint state is classically separable.

A stronger nonseparability test requires the full joint distribution or density-like state and an explicit classical null family.

15.12 Greeks and observer restriction

Different observers see different components of the coupling tensor.

A spot trader may observe:

{U,Delta,Gamma}. (15.49)

A volatility trader may observe:

{σ,Vega,Vanna,Volga}. (15.50)

A funding desk may observe:

{r,Rho,CollateralSensitivity}. (15.51)

A clearinghouse may observe:

{Exposure,Margin,DefaultGate}. (15.52)

No local observer necessarily sees the complete tensor.

The full derivative world may be globally structured while each desk possesses only a reduced local model.

This is a practical example of:

Global Coupling Structure + Local Measurement Restriction. (15.53)

15.13 Greek instability as frame curvature

Greeks are usually computed locally.

But their stability depends on the surrounding geometry.

Define Greek drift:

$dJ_i/d\theta = \partial J_i/\partial y^j \cdot dy^j/d\theta + \partial J_i/\partial L \cdot dL/d\theta + \partial J_i/\partial P \cdot dP/d\theta$. (15.54)

The terms represent:

- state movement;

- ledger movement;
- protocol movement.

A Greek may therefore change even when the underlying price is unchanged.

For example:

$$dU = 0, \quad (15.55)$$

but:

$$d\sigma \neq 0 \text{ or } dL \neq 0. \quad (15.56)$$

Then:

$$d\text{Delta} \neq 0. \quad (15.57)$$

This shows why the derivative state cannot be reduced to spot dependence alone.

15.14 Greek singularities and gate proximity

Some derivatives exhibit rapidly changing sensitivities near:

- strike;
- barrier;
- maturity;
- default boundary;
- liquidation threshold.

Let g be distance to a gate:

$$g = \text{State} - \text{GateThreshold}. \quad (15.58)$$

A sensitivity may scale as:

$$|\Gamma| \propto 1/(|g| + \varepsilon_g). \quad (15.59)$$

As:

$$g \to 0, \quad (15.60)$$

the local curvature becomes large.

The derivative world is then approaching a transition surface.

A local linear law becomes unreliable precisely where commitment is most consequential.

This links:

$$\text{Differential Geometry} \to \text{Gate Geometry}. \quad (15.61)$$

15.15 A Greek-based local metric

One candidate derivative metric is:

$g^{D}_{ij} = J_i J_j + \lambda_{HH} H_{ik} H_{jk}$. (15.62)

Where λ_H is a calibration weight.

Another candidate is:

$g^{D}_{ij} = E[(\partial_i \ln p)(\partial_j \ln p)]$. (15.63)

A third candidate is a risk-weighted Hessian:

$g^{D}_{ij} = w_{ij} |H_{ij}|$. (15.64)

These alternatives measure different things:

- sensitivity magnitude;
- statistical distinguishability;
- valuation curvature;
- hedge instability.

There is no automatic unique metric.

Metric choice must remain protocol-bound.

15.16 The Greek interpretation rule

Part II adopts:

> Greeks are local differential and curvature coordinates of the derivative composite state, not proof of quantum entanglement.

They become relevant to entanglement only when:

1. the state is composite;
2. local observations are restricted;
3. joint measurements reveal structure beyond separable classical models;
4. the coupling remains frame-robust.

Thus:

Greeks Measure Coupling. (15.65)

Entanglement Tests Nonseparability. (15.66)

The two concepts are related but not identical.

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16. The Implied-Volatility Surface as an Effective Field

16.1 From one volatility input to a state field

A simple option model may use one volatility parameter:

σ = constant. (16.1)

Real derivative markets usually display:

$\sigma_{imp} = \sigma_{imp}(K,T)$. (16.2)

The implied-volatility surface varies across:

- strike;
- maturity;
- underlying level;
- market regime;
- liquidity;
- order flow;
- ledger history.

A richer expression is:

$\sigma_{imp} = \sigma_{imp}(K,T,U,\theta,L,P)$. (16.3)

This turns volatility from a parameter into an effective field.

The option value becomes:

$D = V[U,K,T,\sigma_{imp}(K,T,U,\theta,L,P),r,q]$. (16.4)

The derivative sector therefore lives over a volatility landscape.

16.2 Strike and maturity as coordinates

Let surface coordinates be:

$x^1 = \log(K/F)$. (16.5)

$x^2 = T$. (16.6)

Where F is the relevant forward level.

Define:

$\Sigma(x^1,x^2) = \sigma_{imp}(K,T)$. (16.7)

Each point corresponds to a local derivative frame.

The surface gradient is:

$\nabla\Sigma = (\partial\Sigma/\partial x^1, \partial\Sigma/\partial x^2)$. (16.8)

The Hessian is:

$H^{\wedge}\Sigma_{ij} = \partial^2\Sigma/(\partial x^i \partial x^j)$. (16.9)

Skew, smile curvature, and term structure become geometric properties of the volatility field.

16.3 Smile and skew as local curvature

Strike skew is:

$Skew_K = \partial\sigma_imp/\partial K$. (16.10)

Smile curvature is:

$Curv_K = \partial^2\sigma_imp/\partial K^2$. (16.11)

Term slope is:

$Slope_T = \partial\sigma_imp/\partial T$. (16.12)

Mixed curvature is:

$Mix_KT = \partial^2\sigma_imp/(\partial K \partial T)$. (16.13)

These quantities determine how local derivative states differ across the surface.

A flat surface corresponds to a highly simplified geometry.

A sharply curved surface records asymmetry in:

- downside risk;
- jump expectation;
- demand for protection;
- market constraints;
- supply of optionality.

16.4 The surface as shared environment

Options with different strikes and maturities are not independent.

They share the same volatility field.

A state in one region can affect others through:

- calibration;
- arbitrage constraints;

- dealer inventory;
- surface interpolation;
- common market beliefs;
- cross-hedging.

Thus:

Option_i ↔ Shared Volatility Field ↔ Option_j. (16.14)

This creates a many-body coupling architecture even before direct option-to-option contracts are introduced.

The volatility field acts as an environment and mediator.

16.5 Surface recalibration as contextual update

Suppose one option price changes.

A calibration process updates the implied surface:

Price_i → Calibrate → $\Sigma'(K,T)$. (16.15)

The new surface changes the model values of many other options:

$\Sigma' \to \{D_1', D_2', \ldots, D_N'\}$. (16.16)

From the primary universe, this is a model-update mechanism.

From inside the derivative world, one local measurement may condition the state assigned to distant sectors of the option surface.

This resembles an effective global update.

But the causal calibration channel remains classically available.

Therefore:

Surface-Wide Recalibration ≠ Quantum Nonlocality. (16.17)

It is nevertheless an important internal-world analogue of joint-state conditioning.

16.6 Arbitrage as geometric consistency

The volatility surface cannot vary arbitrarily.

Static no-arbitrage conditions constrain:

- monotonicity;
- convexity;
- calendar consistency;

- density positivity.

For call prices:

$\partial C/\partial K \le 0$. (16.18)

And:

$\partial^2 C/\partial K^2 \ge 0$. (16.19)

Under suitable assumptions:

$\partial^2 C/\partial K^2$ = discounted risk-neutral density. (16.20)

Calendar consistency requires suitable ordering across maturity.

The surface is therefore not a free field.

It is a constrained field.

Define admissible surface family:

$\mathcal{S}_{noarb} = \{\Sigma \mid NoButterflyArbitrage \wedge NoCalendarArbitrage \wedge BoundaryConditions\}$. (16.21)

A calibrated state must satisfy:

$\Sigma \in \mathcal{S}_{noarb}$. (16.22)

No-arbitrage acts as a gauge-like or geometric consistency constraint across local frames.

16.7 Local option fibres over the surface

At each point $x = (K,T)$, define a local option fibre:

$\mathscr{H}_{D,x}$. (16.23)

A local state is:

$|\psi_D(x)\rangle \in \mathscr{H}_{D,x}$. (16.24)

The collection forms:

$\pi_D: \boldsymbol{\mathcal{H}}_D \to \boldsymbol{\mathcal{M}}_\sigma$. (16.25)

Where:

- $\boldsymbol{\mathcal{M}}_\sigma$ is the volatility-surface base;
- $\boldsymbol{\mathcal{H}}_D$ is the derivative-state bundle.

Transporting an option state from one strike or maturity region to another requires a connection.

Naive comparison may fail because:

- delta differs;
- vega differs;

- local skew differs;
- liquidity differs;
- contract identity differs.

The connection determines what it means to compare local derivative states across the surface.

16.8 Surface connection

Define a local connection:

$\mathcal{A}_i^\sigma$ = phase-orientation transport along surface coordinate x^i. (16.26)

The covariant derivative is:

$D_i = \partial_i + i\mathcal{A}_i^\sigma$. (16.27)

A transported state satisfies:

$D_i|\psi_D\rangle = 0$ (16.28)

for parallel transport.

The curvature is:

$\mathbb{F}_{ij}^\sigma = \partial_i\mathcal{A}_j^\sigma - \partial_j\mathcal{A}_i^\sigma$. (16.29)

If:

$\mathbb{F}_{ij}^\sigma \neq 0$, (16.30)

transport around a closed strike–maturity loop can produce nontrivial phase or orientation change.

This is a candidate financial holonomy.

16.9 Volatility-surface holonomy

Consider a loop:

(K_1,T_1)
$\rightarrow (K_2,T_1)$
$\rightarrow (K_2,T_2)$
$\rightarrow (K_1,T_2)$
$\rightarrow (K_1,T_1)$. (16.31)

The state returns to the same coordinate.

But the transported orientation may satisfy:

$|\psi_{final}\rangle = \exp(i\Phi_\sigma)|\psi_{initial}\rangle$. (16.32)

Where:

$\Phi_\sigma = \oint_\gamma \mathcal{A}_i^\sigma dx^i$. (16.33)

By Stokes-like relation:

$\Phi_\sigma = \iint_S \mathbb{F}_{12}^\sigma dx^1 dx^2$. (16.34)

A nonzero Φ_σ could encode:

- calibration path dependence;
- cross-strike hedge residue;
- surface interpolation memory;
- ledger-dependent repricing.

This is a proposed measurable construct, not established finance.

16.10 Surface state and derivative entanglement

A multi-option state may be:

$|\Psi_{surface}\rangle = \sum_i c_i |D_i, \Sigma_i\rangle$. (16.35)

Options become correlated through the shared field.

But shared-environment correlation may remain separable.

To claim effective entanglement, one must test whether:

$\rho_{options} \notin Sep(\bigotimes_i \mathcal{H}_{D,i})$. (16.36)

Relative phase and joint observables must matter.

The surface supplies the environment in which stronger composite states could form.

It does not automatically establish them.

16.11 Surface decoherence

Public quoting, calibration, and clearing may suppress alternative surface states.

Suppose pre-commitment alternatives are:

$|\Psi_\Sigma\rangle = \sum_a c_a |\Sigma_a\rangle$. (16.37)

A public calibration gate selects one operational surface:

$Gate_{calibration}(|\Psi_\Sigma\rangle) \to \Sigma^*$. (16.38)

The chosen surface enters:

- pricing;
- margin;

- risk;
- accounting;
- hedging.

Ledgered use stabilizes it.

Alternative surfaces become residual or scenario branches rather than active common reality.

This is a financial decoherence-like process:

Competing Surfaces
→ Calibration Gate
→ Shared Operational Surface
→ Ledgered Backreaction. (16.39)

16.12 The volatility-field proposition

Proposition 16.1 — Shared-Field Coupling

A derivative market forms a shared-field composite system when:

1. multiple options depend on one calibrated volatility field;
2. local observations update the common field;
3. the common field updates other local valuations;
4. arbitrage and ledger constraints restrict admissible field states;
5. path-dependent transport leaves measurable residual.

In compact form:

SharedVolField ⇔ CommonState ∧ LocalUpdate ∧ GlobalRepricing ∧ ConsistencyConstraint ∧ PathResidual. (16.40)

This proposition explains many-body coupling without requiring physical quantum entanglement.

---

17. Many-Body Derivative Worlds

17.1 From one option pair to a derivative network

A single option supplies a bipartite system:

Underlying ⊗ Derivative. (17.1)

Real markets contain:

- many underlyings;
- many strikes;

- many maturities;
- many counterparties;
- many hedge positions;
- many collateral agreements.

The state space becomes:

$\mathcal{H}_{total} = \mathcal{H}_U \otimes \mathcal{H}_{D_1} \otimes \mathcal{H}_{D_2} \otimes \dots \otimes \mathcal{H}_{D_N} \otimes \mathcal{H}_H \otimes \mathcal{H}_C \otimes \mathcal{H}_L.$ (17.2)

The joint state is:

$\rho_{total} \in \mathcal{D}(\mathcal{H}_{total}).$ (17.3)

This is a many-body financial world.

Its state cannot generally be inferred from isolated prices alone.

17.2 Multiple strikes

For strikes $K_1,\dots,K_N$:

$|\Psi_{strike}\rangle \in \mathcal{H}_U \otimes \bigotimes_i \mathcal{H}_{\{D,K_i\}}.$ (17.4)

The options share:

- underlying;
- volatility surface;
- dealer inventory;
- funding;
- settlement infrastructure.

A local shock at strike K_j may propagate through:

Quote Change
→ Surface Recalibration
→ Greek Recalculation
→ Hedge Adjustment
→ Other Strike Repricing. (17.5)

The strike network behaves as a coupled system.

17.3 Multiple maturities

For maturities $T_1,\dots,T_M$:

$\mathcal{H}_{term} = \bigotimes_j \mathcal{H}_{\{D,T_j\}}.$ (17.6)

The maturities are linked through:

- forward curve;
- term volatility;
- calendar arbitrage;
- roll-down;
- funding;
- long-dated hedging.

A short-dated event can reshape long-dated implied distributions.

Conversely, long-dated demand can alter short-dated dealer hedging.

This produces cross-time coupling inside the secondary world.

17.4 Basket options

A basket option depends on several underlyings:

$B = \Sigma_i w_i U_i$. (17.7)

The payoff is:

$D_T = f(B_T)$. (17.8)

The underlying state space is:

$\mathscr{H}_B = \otimes_i \mathscr{H}_{U_i}$. (17.9)

The derivative couples to the joint basket state:

$\mathscr{H}_{total} = \mathscr{H}_B \otimes \mathscr{H}_D$. (17.10)

A basket derivative is therefore a natural multipartite object.

Its value depends not only on individual marginals but on:

- correlations;
- tail dependence;
- joint jumps;
- covariance regime;
- liquidity co-movement.

The joint state may be more fundamental than the separate asset descriptions for the purpose of the contract.

17.5 Correlation derivatives

A correlation derivative directly prices dependence structure.

Let realized correlation be:

$\rho_{realized} = RealizedCorr(U_1,U_2,\ldots,U_N)$. (17.11)

The payoff may be:

$D_T = f(\rho_{realized})$. (17.12)

The derivative is not merely exposed to local states.

It is exposed to their relation.

This makes correlation derivatives especially important for the article.

They demonstrate that finance already creates tradable objects whose underlying is a joint relation rather than one asset.

Thus:

Relational State → Tradable Claim. (17.13)

17.6 Dispersion trades

A dispersion structure may combine:

- index options;
- component options;
- correlation exposure.

The state includes:

$\rho_{dispersion} \in \mathscr{H}_{index} \otimes \bigotimes_i \mathscr{H}_{component,i}$. (17.14)

The trade depends on the mismatch between:

- aggregate implied variance;
- component implied variances;
- implied correlation.

This creates a financial experiment on the difference between local sector states and one global composite state.

The index is not reducible to a simple sum of independent option states because cross terms matter.

17.7 Credit derivatives

A credit derivative network may include:

- reference entity;
- protection buyer;

- protection seller;
- guarantor;
- collateral provider;
- central counterparty;
- event-determination committee.

The composite state is:

$\mathscr{H}$_credit = $\mathscr{H}$_R ⊗ $\mathscr{H}$_B ⊗ $\mathscr{H}$_S ⊗ $\mathscr{H}$_C ⊗ $\mathscr{H}$_G ⊗ $\mathscr{H}$_L. (17.15)

A default event changes:

- protection payment;
- collateral;
- funding;
- capital;
- legal rights;
- market perception.

One gate updates many sectors in one ledger episode.

This is a strong example of multipartite contractual binding.

17.8 Convertible bonds

A convertible bond combines:

Debt Sector ⊗ Equity Option Sector ⊗ Credit Sector ⊗ Conversion Gate. (17.16)

The state space is:

$\mathscr{H}$_CB = $\mathscr{H}$_debt ⊗ $\mathscr{H}$_equity ⊗ $\mathscr{H}$_credit ⊗ $\mathscr{H}$_gate. (17.17)

The instrument cannot be fully described as pure debt or pure equity.

Its identity depends on a conversion relation.

This is a financial object whose local category is incomplete.

The global hybrid state is primary.

17.9 Structured notes

A structured note may bind:

- issuer credit;

- underlying asset;
- barrier;
- coupon condition;
- autocall gate;
- maturity payoff;
- collateral;
- liquidity.

Its state is:

$\mathcal{H}_{SN} = \mathcal{H}_{issuer} \otimes \mathcal{H}_{U} \otimes \mathcal{H}_{barrier} \otimes \mathcal{H}_{coupon} \otimes \mathcal{H}_{autocall} \otimes \mathcal{H}_{L}$. (17.18)

The visible note price compresses a large multipartite state.

A local scalar price may therefore conceal substantial retained relational pressure.

This is a natural domain for extending R+iQ into a composite complex representation.

17.10 Counterparty networks

Let exposure matrix be:

E_{ij} = exposure of institution i to institution j. (17.19)

The system state includes:

$E = [E_{ij}]$. (17.20)

Collateral and netting alter effective exposure:

E_eff = Netting(E,C,L,P). (17.21)

One institution's local solvency state depends on the global network.

A shock can propagate through:

Default
→ Loss Recognition
→ Margin Demand
→ Asset Sale
→ Price Decline
→ Further Default. (17.22)

This is dynamical nonseparability at the network level.

It remains classically causal.

But local balance-sheet states may be incomplete without the global exposure network.

17.11 Multipartite separability

A multipartite state may be:

- fully separable;
- separable across one partition;
- nonseparable across another;
- genuinely multipartite nonseparable.

Fully separable:

$\rho = \Sigma_j p_j \rho^1{}_j \otimes \rho^2{}_j \otimes \ldots \otimes \rho^N{}_j$. (17.23)

Biseparable:

$\rho = \Sigma_partitions p_a \rho_A,_a \otimes \rho_B,_a$. (17.24)

Genuine multipartite nonseparability requires failure across every bipartition.

A financial claim of “system-wide entanglement” must therefore declare the tested partition.

Without this, network connectedness may be mistaken for genuine multipartite structure.

17.12 Hypergraph representation

Derivative relations may be represented by a hypergraph:

$\mathcal{G}_F = (V,E_H)$. (17.25)

Where:

- V are financial sectors;
- E_H are contracts or constraints joining multiple sectors.

A standard edge joins two nodes.

A hyperedge may join:

Underlying + Issuer + Barrier + Coupon + Ledger. (17.26)

The hypergraph makes the composite architecture explicit.

The state-space structure is then derived from the declared hyperedges rather than imposed arbitrarily.

17.13 Interaction Hamiltonian on the network

A schematic generator is:

$\hat{H}_total = \Sigma_i \hat{H}_i + \Sigma_\{i<j\}\hat{H}_ij + \Sigma_\{i<j<k\}\hat{H}_ijk + \ldots$. (17.27)

Where:

- $\hat{H}_i$ are local sector dynamics;

- $\hat{H}_{ij}$ are pairwise contracts;
- $\hat{H}_{ijk}$ are multi-party or multi-condition interactions.

For a basket derivative:

$\hat{H}_{basket} = \hat{H}_{U_1U_2\ldots UND}.$ (17.28)

For a credit derivative:

$\hat{H}_{credit} = \hat{H}_{reference,buyer,seller,gate}.$ (17.29)

Higher-order interactions may not decompose into pairwise terms.

This is one reason a simple covariance matrix may fail to capture the full contract network.

17.14 Many-body residual

Define:

$\varepsilon_{many} = ObservedJointDynamics - DeclaredNetworkDynamics.$ (17.30)

Residual sources include:

- hidden counterparties;
- legal uncertainty;
- unreported leverage;
- common funding dependence;
- correlated model error;
- endogenous liquidity collapse;
- strategic feedback.

A many-body representation is useful only if:

$\|\varepsilon_{many}\| < \|\varepsilon_{pairwise}\|$ (17.31)

under fair model-complexity penalties.

Otherwise the additional tensor structure is not justified.

17.15 The many-body proposition

Proposition 17.1 — Derivative Network World

A derivative network qualifies as a many-body effective world when:

1. contract identities define stable subsystem roles;
2. multiple contracts share underlying fields or ledgers;

3. local state evolution depends on higher-order joint relations;
4. gates update several sectors in one episode;
5. pairwise models leave systematic residual.

In compact form:

ManyBodyFinance ⇔ StableRoles ∧ SharedFields ∧ HigherOrderCoupling ∧ JointGates ∧ PairwiseFailure. (17.32)

This proposition is testable.

It does not require physical quantum ontology.

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18. Gates in Derivative Worlds

18.1 Why derivatives are naturally gate-bearing

A derivative does not merely vary continuously.

Its contract often contains discrete transition conditions.

Examples include:

- exercise;
- expiry;
- barrier hit;
- knock-in;
- knock-out;
- default;
- margin breach;
- collateral call;
- autocall;
- conversion;
- settlement.

These gates convert continuous valuation evolution into discrete historical events.

Thus derivative finance naturally joins:

Phase Evolution → Commitment Gate → Ledger Time. (18.1)

18.2 General derivative gate

Define state:

$x_D(\theta)$. (18.2)

A gate is:

$G_j[x_D(\theta),P,L] \in \{0,1\}$. (18.3)

Or more generally:

$G_j \in \{Commit,Reject,Defer,Escalate\}$. (18.4)

The gate commits when:

$C_j[x_D(\theta),P,L] \geq 0$. (18.5)

Where C_j is the declared condition function.

The event record is:

$Record_j = (Gate_j,State_j,Frame_j,Evidence_j,Authority_j,Residual_j)$. (18.6)

The ledger updates:

$L_{k+1} = L_k \sqcup Record_j$. (18.7)

18.3 Exercise gate

For a European call at maturity:

$Exercise = 1$ iff $U_T > K$. (18.8)

The payoff is:

$D_T = \max(U_T - K,0)$. (18.9)

The gate converts a continuous underlying state into a discrete exercise status.

Before commitment:

$Exercise \in \{0,1\}$ as unresolved contract possibility. (18.10)

After commitment:

$ExerciseRecorded_k \in \{0,1\}$ as fixed ledger fact. (18.11)

The gate creates historical asymmetry.

18.4 American exercise gate

For an American option, exercise depends on a stopping rule:

$\tau^* = \inf\{t \mid ExerciseValue_t \geq ContinuationValue_t\}$. (18.12)

The gate is adaptive.

It depends on:

- current state;
- expected continuation;
- funding;
- dividends;
- volatility;
- holder policy;
- ledger constraints.

The exercise event is therefore observer-policy-dependent.

Two holders may exercise differently under different admissible objectives.

This is a clear example of contextual gating.

18.5 Barrier gate

Let barrier level be H.

Define hit variable:

$B_t = 1$ iff $\sup_{0\le s\le t} U_s \ge H$. (18.13)

For a down barrier:

$B_t = 1$ iff $\inf_{0\le s\le t} U_s \le H$. (18.14)

Once hit:

$B_t = 1$ for all later t. (18.15)

This is latching.

The path cannot later be reclassified as unhit without ledger revision or error correction.

Thus:

Barrier Event → Persistent Trace. (18.16)

The current spot price alone cannot reconstruct the derivative state.

Path ledger is essential.

18.6 Knock-in and knock-out

A knock-in option becomes active after the gate:

$B_t: 0 \to 1$. (18.17)

A knock-out option becomes inactive:

Active_t: 1 → 0. (18.18)

These are not mere value changes.

They are changes in the admissible state space.

Before gate:

$\mathcal{S}$_before. (18.19)

After gate:

$\mathcal{S}$_after. (18.20)

Generally:

$\mathcal{S}$_after ≠ $\mathcal{S}$_before. (18.21)

The gate changes the world's ontology at the effective level.

18.7 Default gate

A credit event gate may be:

Default = 1 iff LegalCreditEvent(P,L,Evidence) = true. (18.22)

The gate depends not only on economic distress but on:

- contractual definition;
- determination authority;
- evidence;
- timing;
- settlement convention.

Thus:

Economic Distress ≠ Automatically Ledgered Default. (18.23)

The primary field may be deteriorating before the secondary world commits the legal event.

The declaration and authority structure matter.

18.8 Margin gate

Let exposure be E_t.

Let collateral be C_t.

Let threshold be M*.

Define unsecured exposure:

$X_t = E_t - C_t$. (18.24)

Margin gate:

$\text{MarginCall} = 1 \text{ iff } X_t > M^*$. (18.25)

The gate leads to:

- collateral demand;
- liquidity need;
- funding action;
- possible liquidation.

Thus:

Valuation Projection
→ Margin Gate
→ Ledgered Obligation
→ Primary Backreaction. (18.26)

The secondary financial world alters the primary economy.

18.9 Liquidation gate

Let equity buffer be b_t.

Liquidation condition:

$\text{Liquidate} = 1 \text{ iff } b_t \le 0$. (18.27)

Once triggered:

- positions are sold;
- market depth is consumed;
- prices may decline;
- other buffers may cross zero.

The gate can generate a cascade:

$\text{Gate}_i \to \text{PriceImpact} \to \text{Gate}_j \to \text{PriceImpact}_2 \to \dots$. (18.28)

This is a collapse cascade in the financial sense.

It is not physical wavefunction collapse.

But it shares the structural grammar:

Potential State → Threshold → Discrete Commitment → Irreversible Trace → Backreaction. (18.29)

18.10 Settlement gate

A trade is not final merely because price agreement occurred.

Settlement requires:

- valid instruction;
- asset delivery;
- cash delivery;
- clearing confirmation;
- finality.

Define:

Settled = 1 iff DeliveryAsset ∧ DeliveryCash ∧ ValidRecord. (18.30)

Settlement converts an economic promise into a ledgered ownership state.

This is a strong example of:

Projection Outcome → Institutional Reality. (18.31)

18.11 Gate noncommutativity

The order of gates may matter.

Suppose:

G_1 = collateral call. (18.32)

G_2 = rating downgrade. (18.33)

Then:

$G_2 \circ G_1(\text{State})$ may differ from $G_1 \circ G_2(\text{State})$. (18.34)

A collateral call before downgrade may preserve solvency.

A downgrade before collateral posting may trigger additional requirements and default.

Thus:

$[G_1, G_2] \neq 0$. (18.35)

Where:

$[G_1, G_2] = G_1G_2 - G_2G_1$. (18.36)

This is operational noncommutativity.

It is not automatically quantum contextuality.

But it produces sequence-dependent histories.

18.12 Gate basis dependence

The same primary field may be evaluated through different gates:

- market-value gate;
- accounting-recognition gate;
- legal-default gate;
- regulatory-capital gate;
- internal-risk gate.

Let:

$$G_{market} \neq G_{accounting} \neq G_{legal}. \quad (18.37)$$

One protocol may commit an event while another defers it.

This creates contextual outcome structure:

$$Outcome = Gate_P(State). \quad (18.38)$$

The outcome is not solely a function of the raw state.

It depends on the declared measurement-and-commitment context.

18.13 Gate-induced decoherence

Suppose the pre-gate state is:

$$|\Psi\rangle = \Sigma_n c_n |branch_n\rangle. \quad (18.39)$$

A gate selects event class g:

$$Gate_g|\Psi\rangle \rightarrow |branch_g\rangle + Residual. \quad (18.40)$$

The selected branch enters the ledger.

Future policy conditions on it.

Alternative branches become:

- counterfactual;
- residual;
- scenario analysis;
- legally irrelevant;
- operationally inaccessible.

This is decoherence-like branch suppression through commitment.

The process is institutionally generated rather than physically postulated.

18.14 Gate error

No gate is infallible.

Define false commitment:

P(Commit | ConditionFalse). (18.41)

Define missed commitment:

P(Reject | ConditionTrue). (18.42)

Gate quality depends on:

- measurement error;
- latency;
- authority;
- protocol clarity;
- strategic manipulation;
- data integrity.

A mature ledger records gate uncertainty:

$Record_k$ = Outcome + Confidence + Evidence + Residual. (18.43)

This prevents commitment from being misrepresented as omniscient truth.

18.15 Gate energy and cost

A gate may require resources:

Cost_G = DataCost + VerificationCost + LiquidityCost + LegalCost + DelayCost. (18.44)

A low-cost gate may commit too easily.

A high-cost gate may delay necessary action.

The gate-design problem is:

Minimize ExpectedLoss_G + λCost_G. (18.45)

Subject to:

FalseCommit ≤ α*. (18.46)

MissedCommit ≤ β*. (18.47)

The secondary world's apparent collapse rule is therefore an engineered institution with measurable trade-offs.

18.16 Gate–ledger proposition

Proposition 18.1 — Derivative Commitment Principle

A derivative event becomes historically real inside the secondary financial world when:

1. the event is defined under a declared protocol;
2. an authorized gate evaluates the state;
3. the outcome is committed;
4. the record enters a persistent ledger;
5. later admissible states depend on that record.

In compact form:

HistoricalEvent_P ⇔ DeclaredCondition ∧ AuthorizedGate ∧ Commitment ∧ PersistentTrace ∧ FutureConstraint. (18.48)

This is the financial mechanism by which phase progression becomes ledger time.

---

19. Hedging as Backreaction

19.1 One-way pricing is incomplete

A simple derivative model assumes:

Underlying → Derivative. (19.1)

The option value depends on the underlying.

But an active market contains hedging.

The derivative position generates hedge demand:

Derivative → Hedge. (19.2)

Hedge demand changes underlying order flow:

Hedge → Underlying. (19.3)

The full loop is:

Underlying → Derivative → Hedge → Underlying. (19.4)

This is backreaction.

The secondary valuation world acts upon the primary market that generates it.

19.2 Delta hedging

Let option position size be N_D.

The hedge position is:

$$H = -N_D\Delta. \quad (19.5)$$

When delta changes:

$$dH = -N_D d\Delta - \Delta dN_D. \quad (19.6)$$

If position size is fixed:

$$dH \approx -N_D d\Delta. \quad (19.7)$$

Using:

$$d\Delta \approx \Gamma dU + \text{Vanna}\cdot d\sigma + \text{Charm}\cdot dt, \quad (19.8)$$

we obtain:

$$dH \approx -N_D[\Gamma dU + \text{Vanna}\cdot d\sigma + \text{Charm}\cdot dt]. \quad (19.9)$$

The derivative state therefore generates underlying order flow.

19.3 Gamma feedback

Suppose market makers are short gamma:

$$N_D\Gamma < 0. \quad (19.10)$$

Then an increase in U may require buying more underlying:

$$dU > 0 \rightarrow dH > 0. \quad (19.11)$$

A decline may require selling:

$$dU < 0 \rightarrow dH < 0. \quad (19.12)$$

The hedge can amplify movement.

For long gamma, hedging may oppose movement.

Define feedback sign:

$$s_\Gamma = \text{sign}(-N_D\Gamma). \quad (19.13)$$

The market effect depends on:

- sign of gamma exposure;
- position size;
- market depth;
- hedge speed;
- liquidity.

Thus:

Derivative Geometry + Position Ledger → Market Feedback. (19.14)

19.4 Price-impact equation

Let hedge order flow be Q_H.

A simple impact law is:

dU = μdt + σdW + λ_IQ_Hdt. (19.15)

Where:

λ_I = market-impact coefficient. (19.16)

Hedge demand depends on derivative state:

Q_H = F_H(Δ,Γ,Vanna,N_D,L). (19.17)

Therefore:

dU = μdt + σdW + λ_IF_H(Δ,Γ,Vanna,N_D,L)dt. (19.18)

The underlying dynamics now depend on the derivative sector.

This closes the loop.

19.5 Coupled differential system

A simplified system is:

dU/dt = F_U(U,D,H,L,ξ_U). (19.19)

dD/dt = F_D(D,U,σ,r,P,L,ξ_D). (19.20)

dH/dt = F_H(H,Δ,Γ,D,U,L,ξ_H). (19.21)

dL/dt = F_L(Trades,Margin,Settlement,Exercise). (19.22)

Where ξ terms represent residual disturbances.

In vector form:

dX/dt = $\boldsymbol{\mathcal{F}}$(X;P) + ξ. (19.23)

With:

X = (U,D,H,L). (19.24)

The system is dynamically inseparable when the Jacobian is not block-diagonal.

Define:

J = ∂$\boldsymbol{\mathcal{F}}$/∂X. (19.25)

If cross-block terms are nonzero:

$$\partial F_U/\partial D \neq 0 \text{ or } \partial F_D/\partial U \neq 0, \quad (19.26)$$

the sectors are dynamically coupled.

19.6 Backreaction Hamiltonian

In the effective state representation:

$$\hat{H}_F = \hat{H}_U + \hat{H}_D + \hat{H}_{contract} + \hat{H}_{hedge} + \hat{H}_{ledger}. \quad (19.27)$$

The hedge term may be:

$$\hat{H}_{hedge} = g_H \hat{O}_D \otimes \hat{O}_H \otimes \hat{O}_U. \quad (19.28)$$

This is schematic.

It represents a three-sector interaction:

Derivative State
↔ Hedge State
↔ Underlying State. (19.29)

An initially prepared derivative relation may evolve into a more deeply coupled market state through hedging.

19.7 Ledger-dependent hedge feedback

The same derivative Greeks can generate different market effects depending on position ledger.

Let:

$$\Gamma_{market} = \Sigma_j N_j \Gamma_j. \quad (19.30)$$

This is aggregate gamma exposure.

Two markets with identical listed contracts and prices may differ because:

$$L_{positions}^A \neq L_{positions}^B. \quad (19.31)$$

Therefore:

$$Feedback_A \neq Feedback_B. \quad (19.32)$$

The derivative state is not fully specified by public prices.

Position ledger is part of the global state.

This supports the principle:

Same Visible Coordinate + Different Ledger → Different Dynamics. (19.33)

19.8 Volatility feedback

Option demand affects implied volatility:

OrderFlow_D → σ_imp. (19.34)

Implied volatility affects Greeks:

σ_imp → {Delta,Gamma,Vega}. (19.35)

Greeks affect hedging:

Greeks → HedgeFlow. (19.36)

Hedging affects underlying realized volatility:

HedgeFlow → σ_realized. (19.37)

The loop is:

Option Demand
→ Implied Volatility
→ Greeks
→ Hedge Flow
→ Realized Volatility
→ Option Demand. (19.38)

This is a self-referential derivative field.

19.9 Margin and collateral backreaction

A derivative price move changes exposure:

dE = dD·N_D. (19.39)

Exposure changes margin:

dM = MarginRule(dE,σ,L,P). (19.40)

Margin demand changes liquidity:

dLiquidity = −FundingCost(dM). (19.41)

Liquidity change affects underlying and derivative prices:

dU,dD = PriceImpact(dLiquidity). (19.42)

Thus:

Valuation → Margin → Funding → Price → Valuation. (19.43)

The valuation world helps generate its own future geometry.

19.10 Backreaction and curved geometry

Let aggregate derivative stress contribute to the effective stress tensor:

$$T^F_{\mu\nu} = T^{underlying}_{\mu\nu} + T^{derivative}_{\mu\nu} + T^{hedge}_{\mu\nu} + T^{ledger}_{\mu\nu}. \quad (19.44)$$

The metric may respond:

$$G^F_{\mu\nu} = \kappa_F T^F_{\mu\nu} + \Xi^F_{\mu\nu}. \quad (19.45)$$

Derivative concentration may therefore alter:

- liquidity distance;
- funding distance;
- collateral curvature;
- margin-gate proximity.

The derivative sector does not merely move within geometry.

It may curve the geometry through backreaction.

This is the GR-like upgrade.

19.11 Positive and negative feedback regimes

Define local loop gain:

$$\mathcal{G} = (\partial U_{next}/\partial H)(\partial H/\partial \Delta)(\partial \Delta/\partial U). \quad (19.46)$$

Since:

$$\partial \Delta/\partial U = \Gamma, \quad (19.47)$$

we obtain:

$$\mathcal{G} \propto \text{MarketImpact} \times \text{PositionSize} \times \Gamma. \quad (19.48)$$

If:

$$\mathcal{G} < 0, \quad (19.49)$$

the feedback is locally stabilizing.

If:

$$0 < \mathcal{G} < 1, \quad (19.50)$$

the system amplifies but remains locally bounded.

If:

$$\mathcal{G} \geq 1, \quad (19.51)$$

the loop may become unstable.

This gives a concrete stability criterion for option-induced backreaction.

19.12 Phase-locking and hedge synchronization

Let dealer j hedge at internal phase θ_j.

Define synchronization order parameter:

$R_H \exp(i\Phi_H) = (1/N)\Sigma_j \exp(i\theta_j)$. (19.52)

Where:

$0 \le R_H \le 1$. (19.53)

High R_H means many dealers act in phase.

Synchronized hedging may produce:

- concentrated liquidity demand;
- stronger market impact;
- faster gate cascades.

Low R_H spreads action across time and may reduce amplification.

Thus:

Correlation of Positions + Phase Synchrony → Systemic Backreaction. (19.54)

This is a more precise mechanism than saying merely that many traders hold similar options.

19.13 Hedge-induced observer effect

A derivative desk measures its exposure.

The measurement changes its hedge.

The hedge changes the market.

The market changes the measured exposure.

Thus:

Measure
→ Act
→ Change Object
→ Re-measure. (19.55)

The observer is internal and backreactive.

This is not passive measurement.

The observer's risk model becomes part of the financial dynamics.

19.14 Public Greeks as performative observables

Published estimates of:

- dealer gamma;
- option open interest;
- volatility exposure;
- expected hedge flows;

may alter trader behaviour.

The observable becomes performative:

PublishedMeasure
→ Anticipatory Trading
→ Actual Flow
→ Observed Outcome. (19.56)

A variable may therefore become predictive partly because agents believe and act on it.

This is a specifically financial observer effect.

19.15 Entanglement versus feedback

Backreaction strengthens relational nonseparability.

But:

Two-Way Feedback ≠ Quantum Entanglement. (19.57)

A classical coupled dynamical system can exhibit:

- mutual dependence;
- path dependence;
- synchronization;
- instability;
- global modes.

The entanglement question remains:

Can the observed joint structure be reproduced by a classical separable state with ordinary causal feedback?

If yes, the system remains lower on the entanglement ladder.

Backreaction is necessary for a self-forming financial world.

It is not sufficient for Bell-nonclassicality.

19.16 Primary and secondary descriptions of hedging

From the primary universe:

Option Position
→ Risk Calculation
→ Hedge Order
→ Market Impact. (19.58)

From inside the secondary θ-world:

Joint State
→ Measurement Gate
→ Conditional Local Update
→ New Composite State. (19.59)

The first describes the mechanism.

The second describes the effective internal episode.

Both are useful.

The article must not confuse them.

19.17 Backreaction residual

Define predicted hedge impact:

$\Delta U_{pred} = \lambda_{IF_H}$. (19.60)

Observed impact is:

ΔU_{obs}. (19.61)

Residual:

$\varepsilon_H = \Delta U_{obs} - \Delta U_{pred}$. (19.62)

Residual may reveal:

- hidden liquidity;
- other market participants;
- anticipatory trading;
- model misspecification;
- nonlinear impact;
- delayed execution;
- frame mismatch.

A backreaction theory earns value only if it reduces ε_H out of sample.

19.18 The backreaction proposition

Proposition 19.1 — Derivative Backreaction Principle

A derivative sector backreacts upon its underlying world when:

1. derivative state determines hedge or funding action;
2. action changes the underlying or market environment;
3. the changed environment alters later derivative valuation;
4. the loop gain is measurable;
5. the resulting trace changes future admissibility.

In compact form:

Backreaction_D ⇔ State→Action ∧ Action→World ∧ World→State ∧ MeasurableLoop ∧ LedgerEffect. (19.63)

This proposition converts the derivative world from a passive representation into a reflexive effective universe.

The composite-state foundation is now complete:

- derivatives provide subsystem structure;
- contracts prepare joint states;
- Greeks measure local coupling;
- volatility supplies a shared field;
- derivative networks generate many-body architecture;
- gates create discrete history;
- hedging closes the backreaction loop.

The next part places the observer fully inside this constructed θ-time world and asks why the resulting option–underlying relation can appear entanglement-like from local access even when its primary construction remains classically intelligible.

Part IV — Entanglement from Inside the θ-Time World

20. The Internal Observer's Measurement Algebra

20.1 Why the observer must be defined by accessible operations

An observer inside the secondary valuation world does not possess unrestricted access to the global state.

It does not observe:

ρ_UD in itself. (20.1)

It observes only the outcomes produced by admissible instruments.

Let the full observable algebra of the option–underlying world be:

$\mathcal{A}_{UD} = \mathcal{A}(\mathcal{H}_U \otimes \mathcal{H}_D)$. (20.2)

An underlying-side observer accesses:

$\mathcal{A}_U^O \otimes I_D \subset \mathcal{A}_{UD}$. (20.3)

A derivative-side observer accesses:

$I_U \otimes \mathcal{A}_D^O \subset \mathcal{A}_{UD}$. (20.4)

A joint risk or clearing observer may access a larger subalgebra:

$\mathcal{A}_J^O \subset \mathcal{A}_{UD}$. (20.5)

But normally:

$\mathcal{A}_U^O \neq \mathcal{A}_D^O \neq \mathcal{A}_J^O$. (20.6)

The observers do not merely hold different opinions.

They possess different operational interfaces.

The self-referential observer framework uses the same internal discipline: observers are characterized by the instruments they can apply, the outcomes they retain, and the trace-conditioned policies through which later instruments are chosen.

20.2 The observer is not the whole desk, institution, or human

A financial observer should be defined functionally.

Let:

$O = (\mathcal{A}_O, \mathbb{M}_O, F_O, L_O, \pi_O)$. (20.7)

Where:

- $\mathcal{A}_O$ is the accessible observable algebra;
- $\mathbb{M}_O$ is the available instrument family;
- F_O is the local valuation frame;
- L_O is the accessible ledger;
- π_O is the policy selecting later instruments.

The observer may be:

- a trader;
- a risk engine;

- a clearinghouse;
- an accounting system;
- a regulator;
- an automated hedging agent;
- a contractual gate.

The observer is not defined by consciousness.

It is defined by the ability to:

1. select a measurement setting;
2. obtain an outcome;
3. write or access trace;
4. condition later action on that trace.

Thus:

Observer_O = Projection + Outcome + Trace + Adaptive Policy. (20.8)

20.3 Measurement instruments

A financial measurement is not merely a scalar function.

It may change the effective state or later policy.

Let instrument a have outcomes u.

Represent it as:

$\mathbb{M}^U_a = \{\mathbb{M}^U_{a,u}\}_u$. (20.9)

The probability of outcome u is:

$p(u|a) = Tr[(E^U_{a,u} \otimes I_D)\rho_{UD}]$. (20.10)

Where:

$E^U_{a,u} = \mathbb{M}^{U^*}_{a,u}(I_U)$. (20.11)

After outcome u:

$\rho_{UD|u,a} = (\mathbb{M}^U_{a,u} \otimes I_D)(\rho_{UD}) / p(u|a)$. (20.12)

The effect E determines the outcome probability.

The map $\mathbb{M}$ determines the post-measurement state.

This distinction matters financially.

Two instruments may report the same value but produce different consequences.

For example:

- passive market observation;
- public quote publication;
- binding margin calculation;
- executable trade;
- regulatory classification.

They may disclose similar information while applying different backreaction.

20.4 Read-only and performative instruments

Define a read-only instrument:

$\mathbb{M}_{read}(\rho) \approx \rho$. (20.13)

Its primary purpose is estimation.

Define a performative instrument:

$\mathbb{M}_{perf}(\rho) \neq \rho$. (20.14)

Examples include:

- executing a large trade;
- issuing a margin call;
- changing collateral eligibility;
- announcing a rating;
- declaring default;
- exercising an option.

A performative instrument changes:

- state;
- ledger;
- admissibility;
- future measurement settings.

Therefore:

Financial Measurement = Information Extraction + Possible State Intervention. (20.15)

This is why the measurement setting cannot be reduced to a passive viewpoint.

20.5 Local underlying observables

Underlying-side observables may include:

Ô_U^price = spot or forward price. (20.16)

Ô_U^return = return over horizon h. (20.17)

Ô_U^liquidity = market-depth state. (20.18)

Ô_U^volatility = realized-volatility state. (20.19)

Ô_U^ownership = ownership or inventory state. (20.20)

Ô_U^barrier = barrier-relative position. (20.21)

These observables do not disclose the complete derivative state.

A spot price does not reveal:

- implied volatility;
- strike structure;
- contract path history;
- exercise policy;
- dealer position;
- collateral status.

The underlying observer therefore sees a reduced world.

20.6 Local derivative observables

Derivative-side observables may include:

Ô_D^premium = option price. (20.22)

Ô_D^delta = delta. (20.23)

Ô_D^gamma = gamma. (20.24)

Ô_D^vega = vega. (20.25)

Ô_D^exercise = exercise status. (20.26)

Ô_D^barrier = barrier state. (20.27)

Ô_D^margin = margin requirement. (20.28)

These observables do not disclose the complete underlying world.

An option premium alone does not reveal:

- the true underlying distribution;

- the hidden order book;
- all dealer inventory;
- the physical probability measure;
- every contract counterparty;
- the complete primary constructor.

The derivative observer also sees a reduced world.

20.7 Joint observables

Some financial quantities belong naturally to the joint system.

Examples include:

Ô_UD^basis = D − Replication(U,Cash). (20.29)

Ô_UD^hedgeerror = ΔD − H·ΔU. (20.30)

Ô_UD^moneyness = U/K. (20.31)

Ô_UD^exposure = Position_D·Sensitivity(U,D). (20.32)

Ô_UD^noarb = NoArbitrageViolation(U,D). (20.33)

Ô_UD^correlation = JointReturnStructure(U,D). (20.34)

These observables cannot be assigned to one local sector alone without reference to the other.

They reveal the relational state.

A composite world is operationally meaningful when at least some of its important observables are irreducibly joint in this sense.

20.8 The difference between local and global completeness

Let the global state be:

ρ_UD. (20.35)

The local states are:

ρ_U = Tr_D(ρ_UD). (20.36)

ρ_D = Tr_U(ρ_UD). (20.37)

If:

ρ_UD = ρ_U ⊗ ρ_D, (20.38)

the local descriptions jointly reconstruct the global state.

If:

$\rho_{UD} \neq \rho_U \otimes \rho_D$, (20.39)

local states alone are insufficient.

The missing structure lies in correlations.

Define correlation operator:

$\chi_{UD} = \rho_{UD} - \rho_U \otimes \rho_D$. (20.40)

If:

$\chi_{UD} = 0$, (20.41)

the state is uncorrelated.

If:

$\chi_{UD} \neq 0$, (20.42)

the global state contains relational information absent from the marginals.

But $\chi_{UD} \neq 0$ still includes both:

- classical correlation;
- entanglement-form nonseparability.

The observer needs stronger tests to distinguish them.

20.9 The observer's filtration

Let the observer's trace after k episodes be:

$L_{O,k} = \{y_1, y_2, \ldots, y_k\}$. (20.43)

The corresponding information filtration is:

$\mathscr{F}_{O,0} \subset \mathscr{F}_{O,1} \subset \ldots \subset \mathscr{F}_{O,k}$. (20.44)

At episode k+1, the observer selects:

$a_{k+1} = \pi_O(\mathscr{F}_{O,k})$. (20.45)

The next setting depends on the recorded past.

Examples:

- a trader increases monitoring after a volatility shock;
- a risk system changes stress scenarios after a breach;
- a clearinghouse raises margin after disorderly movement;
- an option holder changes exercise policy after dividend information;

- a regulator changes reporting requirements after failure.

The measurement process is therefore adaptive.

20.10 Internal certainty

Suppose observer O records outcome:

$y_k = u$. (20.46)

After the record enters its filtration:

$P_O(y_k = u \mid \mathscr{F}_O,k) = 1$. (20.47)

The past outcome becomes delta-certain within that observer's own trace.

This does not mean the observer knows the full primary history.

It means the observer's own committed record is fixed relative to its later internal state.

The self-referential observer framework uses this filtration-based fixedness to explain why recorded outcomes are experienced as definite and why trace-conditioned policies create latching between alternative histories.

20.11 Financial latching from adaptive measurement

Suppose two possible outcomes occur at episode k:

$y_k = u_1$ or $y_k = u_2$. (20.48)

The observer's next setting is:

$a_{k+1} = \pi_O(L_k \sqcup u_1)$, (20.49)

or:

$a'_{k+1} = \pi_O(L_k \sqcup u_2)$. (20.50)

Generally:

$a_{k+1} \neq a'_{k+1}$. (20.51)

The future measurement path diverges.

For example:

No Margin Breach
→ ordinary monitoring. (20.52)

Margin Breach
→ forced liquidation monitoring. (20.53)

The two histories no longer use the same instruments or admit the same actions.

This is financial latching:

Outcome Difference → Policy Difference → Future-World Difference. (20.54)

20.12 Local ignorance versus structural restriction

The observer may fail to see the global state for two different reasons.

Epistemic ignorance

The information exists in principle within the same accessible algebra but has not been obtained.

Structural restriction

The observer's admissible interface does not contain the relevant observable.

These must be distinguished.

Let hidden variable h be measurable in principle:

$h \in \mathcal{A}_O$ but h not yet observed. (20.55)

This is ignorance.

Let global observable J lie outside the observer's algebra:

$J \notin \mathcal{A}_O$. (20.56)

This is structural restriction.

Entanglement-like strangeness becomes stronger when completeness depends on global observables that no local observer can independently access.

20.13 The observer-access principle

Proposition 20.1 — Internal Access Principle

An observer inside W_θ experiences only:

$\text{AccessibleState}_O = \text{Restrict}(\rho_F \mid \mathcal{A}_{O,F_O,L_O})$. (20.57)

Its observable world is therefore determined jointly by:

- the global effective state;
- the accessible algebra;
- the local frame;
- the inherited ledger.

In compact form:

$\text{ObservedReality}_O = \text{GlobalState} \times \text{LocalAccess} \times \text{Frame} \times \text{Trace}$. (20.58)

The apparent strangeness of a global relation cannot be assessed without specifying all four.

21. Why Entanglement Appears Strange Internally

21.1 The external observer sees preparation

The primary-universe analyst may know:

$|\Psi_{UD}\rangle = \hat{U}_C(|\psi_U\rangle \otimes |0_D\rangle)$. (21.1)

It knows:

- the contract;
- the valuation model;
- the state preparation;
- the coupling mechanism;
- the market infrastructure.

From this position, the joint relation appears constructed.

The option state corresponds to the underlying state because the contract was designed to make it so.

There is little apparent mystery.

21.2 The internal observer receives the prepared state

The internal observer does not begin before the contract-preparation process.

It begins with:

$\rho_{UD}(\theta_0)$. (21.2)

It may know the contract rule in an abstract sense.

But it does not possess complete access to:

- every branch amplitude;
- every primary variable;
- every hidden ledger;
- every preparation detail;
- every environmental interaction.

It receives one usable effective world.

Thus:

External Observer Sees Preparation. (21.3)

Internal Observer Sees Prepared Relation. (21.4)

This is the first source of apparent strangeness.

21.3 One global state, two incomplete local views

Suppose:

$|\Psi_UD\rangle = \alpha|u_1,d_1\rangle + \beta|u_2,d_2\rangle$. (21.5)

With:

$|\alpha|^2 + |\beta|^2 = 1$. (21.6)

The underlying observer sees:

$\rho_U = |\alpha|^2|u_1\rangle\langle u_1| + |\beta|^2|u_2\rangle\langle u_2|$. (21.7)

The derivative observer sees:

$\rho_D = |\alpha|^2|d_1\rangle\langle d_1| + |\beta|^2|d_2\rangle\langle d_2|$. (21.8)

Neither local state contains the relative phase between α and β.

That phase belongs to the joint state.

Therefore:

Local State Information < Global State Information. (21.9)

The local object appears statistically incomplete even when the global effective state is pure.

21.4 Conditional definiteness

Suppose the underlying observer measures in the basis:

$\{|u_1\rangle,|u_2\rangle\}$. (21.10)

If outcome u_1 is recorded, the conditional derivative state becomes:

$\rho_D|u_1 = |d_1\rangle\langle d_1|$. (21.11)

If outcome u_2 is recorded:

$\rho_D|u_2 = |d_2\rangle\langle d_2|$. (21.12)

The derivative's assigned state changes conditionally with the underlying outcome.

From the internal description:

Joint Possibility
→ Local Outcome
→ Correlated Conditional State. (21.13)

No independent complete derivative state existed before conditioning under this representation.

The derivative's local definiteness belonged to the measurement context and outcome.

21.5 Why no messenger appears inside the state-update rule

The conditional update is:

$$\rho_{D|u} = \mathrm{Tr}_U[(M_u \otimes I_D)\rho_{UD}(M_u^\dagger \otimes I_D)] / p(u). \quad (21.14)$$

The formula refers to the global state.

It does not contain a term representing:

Signal from U to D. (21.15)

From the internal effective-world perspective, the correlation is already encoded globally.

The local outcome reveals which conditional derivative state applies.

The update therefore looks instantaneous at one θ-episode.

21.6 The primary universe may still contain causal machinery

The absence of an internal messenger term does not mean the primary financial universe lacks causal mechanisms.

The primary process may include:

- data feeds;
- price dissemination;
- option recalibration;
- arbitrage;
- hedge execution;
- balance-sheet response;
- clearing messages.

Thus:

No Messenger in Effective Update ≠ No Causal Mechanism in Primary Construction. (21.16)

This distinction is essential.

The effective state-update rule compresses primary machinery into one joint-state relation.

21.7 State update versus physical disturbance

Conditional state assignment can mean at least two things.

Informational conditioning

The observer learns which branch applies.

Performative disturbance

The measurement or action changes the system itself.

Financial markets often contain both.

For example:

Observing a public option quote may mainly update information.

Executing a large option trade may change:

- implied volatility;
- dealer position;
- hedge flow;
- underlying price.

Therefore:

ρ_D|u = Information Update + Possible Market Backreaction. (21.17)

A strict comparison with quantum measurement must separate these contributions.

21.8 Why option entanglement appears stranger after the primary world is hidden

Suppose the observer sees only:

- one underlying output;
- one derivative output;
- one joint measurement episode;
- one ledgered result.

It does not see:

- the full preparation process;
- the complete causal chain;
- all hidden state;
- all unselected branches.

Then the observed relation becomes:

Outcome_D = ConditionalFunction(Outcome_U,Setting,GlobalState). (21.18)

The relationship may appear stronger than ordinary local causation because the global relation is primary inside the accessible description.

This is precisely why the perspective must remain internal.

From outside, the relationship looks manufactured.

From inside, the globally manufactured relation is the world's given structure.

21.9 The role of subsystem separation

Entanglement appears strange only when the global relation is divided into apparently separate local sectors.

If the option and underlying are treated as one indivisible object:

UD = one composite security. (21.19)

then correlated outcomes are not surprising.

Strangeness arises when the observer simultaneously maintains:

1. U and D are distinct local systems;
2. neither has a complete independent state;
3. one joint state governs both;
4. local measurements produce correlated definiteness.

Thus:

Distinct Local Identity + Global State Primacy → Entanglement Strangeness. (21.20)

This is also why the partition must be operationally justified.

21.10 Reduced states are not hidden complete states

A reduced state:

ρ_U = Tr_D(ρ_UD) (21.21)

is not necessarily a probability distribution over a secretly complete local pure state.

It may merely be the complete description available to the local observer under the declared formalism.

A financial model may still admit a classical completion.

But the internal θ-world does not automatically provide one.

The question becomes:

Can every reduced financial state and every cross-setting correlation be reproduced by one

globally consistent classical hidden-state model? (21.22)

This is the key threshold between a formal entanglement analogy and a stronger nonclassical claim.

21.11 The internal-strangeness decomposition

The apparent strangeness can be decomposed as:

S_ent = S_partition + S_global + S_restriction + S_commitment. (21.23)

Where:

- S_partition = the world is divided into distinct local sectors;
- S_global = the joint state contains more information than local states;
- S_restriction = observers access only local algebras;
- S_commitment = one outcome becomes fixed in ledger history.

This decomposition can arise in non-quantum effective worlds.

Therefore:

Entanglement-Like Strangeness Is Not Automatically Quantum. (21.24)

But it is stronger than mere co-movement.

21.12 The core proposition

Proposition 21.1 — Internal Entanglement-Strangeness Principle

A composite relation appears entanglement-like to an internal observer when:

1. subsystem identities are locally preserved;
2. the global state is not reconstructible from local states alone;
3. the observer lacks constructor-level access;
4. measurement of one sector conditionally fixes the state assigned to another;
5. the correlated outcome is committed within one internal episode.

In compact form:

StrangeEnt_O ⇔ LocalIdentity ∧ GlobalPrimacy ∧ ConstructorInaccessibility ∧ ConditionalFixing ∧ SharedEpisode. (21.25)

This proposition identifies the observer-relative experience.

It does not yet establish quantum entanglement.

---

22. Financial Measurement Settings

22.1 Why price is only one basis

A financial object can be measured through several operational bases.

An option may be represented by:

- premium;
- Greeks;
- replication portfolio;
- terminal payoff;
- collateral requirement;
- liquidation exposure;
- accounting carrying amount;
- regulatory capital.

These are not merely alternative labels.

They disclose different relational structures.

Let measurement setting be:

a ∈ Θ_U for the underlying sector. (22.1)

b ∈ Θ_D for the derivative sector. (22.2)

The joint outcome distribution is:

p(u,d|a,b) = Tr[ρ_UD(E^U_{a,u} ⊗ E^D_{b,d})]. (22.3)

The settings determine which financial distinction becomes visible.

22.2 Price basis

The price basis measures:

a_price → U. (22.4)

b_price → D. (22.5)

The observer sees the pair:

(U,D). (22.6)

This basis is useful for:

- market marking;

- execution;
- profit and loss;
- quoted comparison.

But it may conceal:

- sensitivity structure;
- hedge instability;
- path dependence;
- embedded optionality;
- funding exposure.

Price basis is highly visible but informationally compressed.

22.3 Risk basis

The derivative is projected into:

$|d\rangle \rightarrow |\text{Delta},\text{Gamma},\text{Vega},\text{Theta_D},\text{Rho},\ldots\rangle$. (22.7)

The underlying may be projected into:

$|u\rangle \rightarrow |\text{Price},\text{Volatility},\text{Liquidity},\text{JumpRisk},\ldots\rangle$. (22.8)

The risk basis reveals coupling derivatives rather than scalar value.

Two options with the same premium may possess very different risk states.

Thus:

Same Price ≠ Same Derivative State. (22.9)

The risk basis can disclose distinctions invisible in price basis.

22.4 Replication basis

A derivative may be represented as:

$D = \Delta U + B + \varepsilon_{rep}$. (22.10)

Where:

- ΔU is the underlying component;
- B is the financing component;
- ε_{rep} is replication residual.

The basis transformation is:

$|D\rangle \leftrightarrow |\Delta U, B, \varepsilon_{rep}\rangle$. (22.11)

In an ideal complete market:

$\varepsilon_{rep} = 0$. (22.12)

In real markets:

ε_{rep} may contain:

- transaction cost;
- discrete hedging;
- liquidity;
- jumps;
- funding;
- model error.

The replication basis tests whether the derivative is genuinely relationally novel or merely a different representation of other local positions.

22.5 Payoff basis

At maturity, the observer may measure:

- in the money;
- at the money;
- out of the money;
- exercised;
- expired worthless.

For a call:

$\text{Payoff} = \max(U_T - K, 0)$. (22.13)

The payoff basis discards much of the pre-maturity valuation structure.

Many distinct paths collapse into the same terminal outcome.

Thus:

Pre-Maturity State → Many-to-One Payoff Projection. (22.14)

This is a coarse-graining basis.

22.6 Path basis

For path-dependent derivatives, the observer measures:

- barrier hit;
- running average;
- maximum;
- minimum;
- realized variance;
- event sequence.

Define path observable:

Ô_path = F[U_{0:T}]. (22.15)

The path basis can distinguish states that terminal price basis identifies.

For example:

U_T^A = U_T^B, (22.16)

but:

BarrierHit_A ≠ BarrierHit_B. (22.17)

Therefore:

Same Terminal Coordinate ≠ Same Contract State. (22.18)

22.7 Volatility basis

The observer may use:

b_σ → σ_imp(K,T). (22.19)

Or a surface-mode decomposition:

|Σ⟩ → |Level,Skew,Curvature,TermSlope,…⟩. (22.20)

This basis reveals expected-distribution structure.

It may show market concern invisible in spot price.

A stable spot price combined with rising downside skew may represent a substantial change in the derivative world.

22.8 Collateral basis

The same derivative may be measured as:

CollateralNeed = C(D,σ,Counterparty,P,L). (22.21)

This basis includes:

- current exposure;

- potential future exposure;
- netting;
- collateral agreement;
- credit quality;
- margin model.

The collateral basis can transform a small market-value change into a large funding event.

It is therefore performative.

22.9 Accounting basis

Accounting measurement may project into:

- fair value;
- amortized cost;
- hedge-accounting relation;
- impairment;
- realized or unrealized gain;
- income-statement or reserve effect.

Let:

Ô_accounting,P(ρ_UD) = CarryingValue. (22.22)

The accounting setting may commit consequences not visible in trading basis.

The same economic state may produce different institutional events because the gate differs.

22.10 Regulatory-capital basis

A regulator may measure:

CapitalCharge = F(Exposure,RiskWeight,Netting,Maturity,P,L). (22.23)

The regulatory basis determines:

- required capital;
- leverage ratio;
- liquidity requirement;
- concentration limit;
- admissibility.

A derivative position that appears neutral in market-value basis may be expensive in capital basis.

Thus:

Basis Choice Changes Operational Reality. (22.24)

22.11 Liquidation basis

A liquidation observer asks:

What value can be realized under forced execution?

Define:

D_liq = D_market − LiquidityCost − ImpactCost − FundingPenalty. (22.25)

The liquidation basis differs structurally from going-concern valuation.

It may include:

- market depth;
- execution priority;
- collateral seizure;
- legal delay;
- fire-sale feedback.

A change from market basis to liquidation basis may be a protocol transition rather than a reversible coordinate change.

22.12 Measurement-basis transformation

Let basis a and basis a′ be related by:

Ô_{a′} = U_{a′a}Ô_aU_{a′a}†. (22.26)

A valid reversible basis transformation preserves the governed state information.

But many financial “basis changes” are lossy.

For example:

Full Path State → Terminal Payoff (22.27)

is not invertible.

A noninvertible projection destroys distinctions.

Therefore measurement settings fall into two classes:

Reversible basis change

Information is reorganized.

Irreversible coarse-graining

Information is discarded or committed.

These should not be conflated.

22.13 Incompatible settings

Two financial measurements may be incompatible operationally even if their numerical functions commute in ordinary mathematics.

For example:

- exact immediate liquidation value;
- undisturbed going-concern value.

Measuring the first through actual liquidation destroys the second.

Similarly:

- passive estimate of dealer inventory;
- executable stress trade revealing liquidity.

The intervention required to obtain one quantity may alter the state relevant to the other.

Define operational incompatibility:

Incompat(a,a′) ⇔ $\mathbb{M}_a \circ \mathbb{M}_{a'} \neq \mathbb{M}_{a'} \circ \mathbb{M}_a$ or MeasurementDisturbance > δ^*. (22.28)

This is stronger than formula noncommutativity.

It is a property of the instrument sequence.

22.14 Context dependence versus quantum contextuality

Financial outcome may depend on setting:

$p(u|a) \neq p(u|a')$. (22.29)

This is ordinary context dependence.

A stronger contextuality claim asks whether one can assign a single global hidden value:

$v(\hat{O}_a)$ (22.30)

to every admissible setting while preserving all compatibility relations.

Ordinary finance often permits contextual hidden-state models with:

- memory;
- institutional rules;
- path dependence;
- measurement disturbance;
- adaptive agents.

Therefore:

Basis Dependence ≠ Kochen–Specker Contextuality. (22.31)

Part I explicitly preserved this boundary: protocol dependence, order sensitivity, and observer backreaction do not by themselves derive quantum contextuality or entanglement.

22.15 Setting pairs and correlation functions

Assign numerical outcomes:

$u \in \{-1,+1\}$. (22.32)

$d \in \{-1,+1\}$. (22.33)

Define:

$E(a,b) = \Sigma_{u,d} ud \cdot p(u,d|a,b)$. (22.34)

The correlation structure across several settings can then be tested.

For settings:

a_0,a_1 and b_0,b_1, (22.35)

define CHSH-like quantity:

$S_F = E(a_0,b_0) + E(a_0,b_1) + E(a_1,b_0) - E(a_1,b_1)$. (22.36)

A classical local hidden-variable model under standard assumptions satisfies:

$|S_F| \leq 2$. (22.37)

But a financial experiment must handle severe complications:

- signalling;
- common information;
- measurement dependence;
- adaptive sampling;
- market impact;
- memory;
- selection bias;

- non-spatial separation.

Therefore an apparent violation would not automatically establish Bell nonclassicality.

The settings must be designed so the relevant assumptions are explicitly tested.

22.16 The setting-declaration rule

Every joint-measurement claim must specify:

MeasurementClaim =
(Partition,Settings,Outcomes,InstrumentMaps,Timing,Ledger,NullFamily). (22.38)

Without this declaration, words such as:

- entanglement;
- contextuality;
- collapse;
- nonlocality;

remain rhetorically underdetermined.

---

23. Conditional Collapse of the Joint State

23.1 Pre-measurement joint state

Let:

$\rho_{UD}(\theta_k^-)$ (23.1)

be the joint state immediately before an internal measurement episode.

Observer A selects underlying setting a.

Observer B selects derivative setting b.

The joint instrument is:

$\mathbb{M}_{a,b}^{u,d} = \mathbb{M}^U_{a,u} \otimes \mathbb{M}^D_{b,d}$. (23.2)

The outcome probability is:

$p(u,d|a,b) = \mathrm{Tr}[\mathbb{M}_{a,b}^{u,d}(\rho_{UD}(\theta_k^-))]$. (23.3)

The post-outcome state is:

$\rho_{UD}(\theta_k^+|u,d,a,b) = \mathbb{M}_{a,b}^{u,d}(\rho_{UD}(\theta_k^-)) / p(u,d|a,b)$. (23.4)

This is the secondary-world collapse-like update.

23.2 Local underlying measurement

If only the underlying sector is measured:

$$p(u|a) = \mathrm{Tr}[(E^U_{a,u} \otimes I_D)\rho_{UD}]. \quad (23.5)$$

The updated global state is:

$$\rho_{UD|u,a} = (\mathbb{M}^U_{a,u} \otimes I_D)(\rho_{UD}) / p(u|a). \quad (23.6)$$

The conditional derivative state is:

$$\rho_{D|u,a} = \mathrm{Tr}_U(\rho_{UD|u,a}). \quad (23.7)$$

The derivative observer may not yet possess the outcome record.

Therefore distinguish:

Conditional State relative to A's record. (23.8)

Accessible State relative to B's filtration. (23.9)

These are not automatically identical.

23.3 Accessibility delay

Suppose A records u at episode k.

B receives the record at episode k+m.

For:

$$j < k+m, \quad (23.10)$$

the record is not in B's filtration:

$$u \notin \mathscr{F}_{B,j}. \quad (23.11)$$

At:

$$j = k+m, \quad (23.12)$$

the record becomes accessible:

$$u \in \mathscr{F}_{B,k+m}. \quad (23.13)$$

Then B can condition on it.

Thus:

Global Conditional Relation ≠ Immediate Local Knowledge. (23.14)

This prevents the formal state update from being mistaken for communicable information.

23.4 Ledgered collapse

The joint outcome becomes a financial historical event only after gate commitment:

$G_k(u,d,a,b,P,L_k) \rightarrow$ Commit. (23.15)

The record is:

$Record_k = (u,d,a,b,\theta_k,F_A,F_B,Evidence,Residual)$. (23.16)

The ledger updates:

$L_{k+1} = L_k \sqcup Record_k$. (23.17)

The post-ledger state is:

$\rho_UD,k+1 = \mathcal{K}_\{L_{k+1}\}[\rho_UD(\theta_k^+)]$. (23.18)

Where $\mathcal{K}$ represents the consequences of the committed record.

The full collapse-like process is therefore:

State Projection
→ Outcome Selection
→ Gate
→ Ledger
→ Backreaction. (23.19)

The state update and historical commitment are separate stages.

23.5 Outcome without historical commitment

A model may generate a provisional result:

u^*. (23.20)

But the gate may defer:

$G_k(u^*) =$ Defer. (23.21)

Then:

$L_{k+1} = L_k$. (23.22)

The projected outcome has not yet become an institutional fact.

Examples include:

- indicative quote not traded;
- unverified barrier event;
- provisional credit event;
- preliminary impairment estimate;
- failed settlement instruction.

Thus:

Projection ≠ Commitment. (23.23)

This distinction is central to the financial measurement problem.

23.6 Historical commitment without complete certainty

A gate may commit despite residual uncertainty.

Let confidence be:

$c_k \in [0,1]$. (23.24)

The record may be:

$\text{Record}_k = (\text{Outcome},\text{Confidence},\text{Residual})$. (23.25)

The event becomes operationally fixed even when epistemic certainty is incomplete.

For example:

- accounting estimate;
- legal judgment;
- margin model output;
- regulatory classification.

Therefore:

Ledger Definiteness ≠ Omniscient Truth. (23.26)

The ledger determines what the effective world will act upon.

It does not necessarily reveal the total primary reality.

23.7 Collapse of an option state

Consider an option before maturity with branch state:

$|\Psi\rangle = \Sigma_n c_n|u_n,d_n\rangle$. (23.27)

At maturity, the payoff gate measures terminal underlying state.

Outcome u_j gives:

$|\Psi\rangle \rightarrow |u_j,d_j\rangle$. (23.28)

The derivative payoff is committed:

$d_j = \max(u_j - K,0)$. (23.29)

The ledger records:

Settlement_j. (23.30)

The option ceases to remain a field of future contingent payoffs.

It becomes a settled claim.

This is a natural financial possibility-to-record transition.

23.8 Barrier collapse

Before barrier determination:

$|\Psi_{barrier}\rangle = \alpha|NotHit,Active\rangle + \beta|Hit,ChangedContractState\rangle$. (23.31)

After the barrier event is confirmed:

$|\Psi_{barrier}\rangle \rightarrow |Hit,ChangedContractState\rangle$. (23.32)

The ledger alters all later valuation.

The branch is latched.

Even if the underlying price later returns:

$U_{later} = U_{before}$, (23.33)

the contract state remains changed:

$D_{later} \neq D_{counterfactualNotHit}$. (23.34)

This is one of the clearest examples of:

Same Coordinate + Different Trace → Different World. (23.35)

23.9 Conditional underlying state from derivative measurement

Suppose an observer measures a derivative signal:

d = HighDownsideSkew. (23.36)

The conditional underlying state may become:

$\rho_{U|d} = Update(\rho_U \mid \text{downside-tail information})$. (23.37)

This is common financial inference.

But if the measurement is performative—for example, a large protective-option trade—it may also cause:

- implied volatility change;
- dealer hedge demand;
- underlying order flow.

Then:

$\rho_{U|d}$ = InformationalConditioning + CausalBackreaction. (23.38)

A strong analysis must estimate both components.

23.10 Counterfactual branch divergence

Suppose outcome u_1 leads to:

Policy π_1. (23.39)

Outcome u_2 leads to:

Policy π_2. (23.40)

Then:

FutureDistribution($\rho \mid u_1, \pi_1$) ≠ FutureDistribution($\rho \mid u_2, \pi_2$). (23.41)

The branches diverge not only because the state differs, but because the observer's later measurement and action policy differs.

This is the adaptive source of latching.

It corresponds closely to the internal-observer model in which trace-conditioned policies make counterfactual branches operationally irreversible.

23.11 Collapse relative to one observer

Observer A may have committed:

$\text{Outcome}_A = u$. (23.42)

Observer B may still assign:

$p_B(u) < 1$. (23.43)

because the record is not accessible or the frame mapping is incomplete.

Thus:

Collapsed for A ≠ Yet Fixed for B. (23.44)

Cross-observer fixedness requires:

1. context map;
2. outcome map;
3. compatible effects;
4. accessible record.

The source observer framework identifies precisely these conditions for AB-fixedness.

23.12 Redundant settlement records and objectivity

Suppose the committed event is copied into:

- exchange record;
- clearing record;
- broker record;
- custody record;
- bank ledger;
- regulatory report.

Let fragments be:

$R_1, R_2, \ldots, R_N$. (23.45)

Each observer estimates event e from one or more fragments.

Consensus improves when:

$P(\hat{e}_j \neq e) = \varepsilon_j < 1/2$. (23.46)

Under sufficiently independent reliable redundancy:

$P(\text{majority estimate} \neq e) \to 0 \text{ as } N \to \infty$. (23.47)

Objectivity emerges through distributed record accessibility rather than through removal of all observers.

The self-referential observer framework formalizes an analogous redundancy-driven convergence of observer reports.

23.13 The collapse sequence

The derivative-world collapse sequence is:

$\rho_{UD}(\theta^-)$
$\to$ Select(a,b)
$\to$ Instrument Interaction
$\to$ Outcome(u,d)
$\to$ Gate
$\to$ Record
$\to L_{k+1}$
$\to$ Backreaction
$\to \rho_{UD,new}$. (23.48)

This sequence is more elaborate than a bare projection postulate.

It separates:

- state evolution;

- measurement context;
- conditional outcome;
- institutional commitment;
- historical consequence.

23.14 The financial collapse proposition

Proposition 23.1 — Internal Financial Collapse

A financial state collapses internally under protocol P when:

1. a bounded observer applies an admissible instrument;
2. one outcome becomes conditionally definite;
3. an authorized gate commits the outcome;
4. the record enters the observer's filtration;
5. future state evolution or policy becomes trace-dependent.

In compact form:

$$\text{Collapse}_{F,P} \Leftrightarrow \text{Instrument} \wedge \text{ConditionalDefiniteness} \wedge \text{Gate} \wedge \text{Trace} \wedge \text{Latching}. \quad (23.49)$$

This is an operational financial collapse definition.

It does not derive physical quantum collapse.

---

24. Primary Causation Versus Secondary Simultaneity

24.1 The apparent contradiction

From the primary universe, option repricing may unfold sequentially:

$$\begin{aligned} &U \text{ changes} \\ &\rightarrow \text{data are transmitted} \\ &\rightarrow \text{model inputs update} \\ &\rightarrow D \text{ is recalculated} \\ &\rightarrow \text{hedge is changed}. \end{aligned} \quad (24.1)$$

From inside the θ-world, the observer may describe:

$$\begin{aligned} &\text{Joint State at } \theta_k \\ &\rightarrow \text{one correlated outcome pair at episode } k. \end{aligned} \quad (24.2)$$

The first description is sequential.

The second appears simultaneous.

The contradiction is only apparent.

They describe different temporal resolutions and different levels of the system.

24.2 Compilation compresses primary sequences

Let primary process contain micro-events:

$m_1, m_2, \ldots, m_N$. (24.3)

The secondary compiler maps them into one effective episode:

$\mathcal{C}_P(\{m_1, \ldots, m_N\}) = \text{Episode}_k$. (24.4)

The internal observer sees:

Episode_k. (24.5)

It does not see every m_i.

Therefore:

Many Primary Events → One Secondary Tick. (24.6)

The apparent simultaneity may be a result of coarse-graining.

24.3 Example: real-time option repricing

A simplified primary chain is:

1. underlying trade occurs;
2. market-data feed updates;
3. pricing engine receives data;
4. implied-volatility surface recalibrates;
5. option quote changes;
6. risk engine recalculates Greeks;
7. hedge order is generated.

Calendar delays may be extremely small.

The internal financial dashboard may display all effects in one update cycle.

Thus:

$\Delta t_{primary} > 0$. (24.7)

But:

$\Delta k_{secondary} = 1$ episode. (24.8)

The internal observer treats the changes as one event-bearing tick.

24.4 Secondary simultaneity is relation-based

Within one joint measurement episode:

(u_k,d_k) (24.9)

are not interpreted as two isolated events joined later.

They are one outcome of:

$\mathbb{M}_{a,b}(\rho_{UD})$. (24.10)

Their unity comes from the measurement relation, not necessarily from exact equality of physical timestamps.

Define θ-simultaneity:

$Sim_\theta(E_U,E_D) \Leftrightarrow SameJointInstrument \wedge SameCommitmentEpisode$. (24.11)

This is an operational simultaneity relation.

It differs from:

$t_U = t_D$ exactly. (24.12)

24.5 Event simultaneity versus signal simultaneity

Suppose the underlying event occurs at:

t_U. (24.13)

The derivative system receives the data at:

$t_D = t_U + \delta t$. (24.14)

If both belong to one internal episode:

$k_U = k_D = k$, (24.15)

then they are simultaneous in ledger time despite a small calendar delay.

Thus:

Ledger Simultaneity ≠ Calendar Simultaneity. (24.16)

This is analogous to batch processing, settlement cycles, or observer ticks.

24.6 The role of θ-time

The joint state evolves in θ.

Suppose:

$\theta_D = f(\theta_U)$ (24.17)

under a locally synchronized relation.

A measurement episode occurs at:

$\theta = \theta_k$. (24.18)

The underlying and derivative outcomes are disclosed relative to that joint phase surface.

Thus:

$Sim_\theta(U,D) = JointDisclosure\ at\ \theta_k$. (24.19)

The primary causal chain that generated θ_k may remain hidden.

The observer experiences the phase surface, not the entire constructor sequence.

24.7 Primary local causation remains available

Nothing in the secondary description removes ordinary primary causation.

A complete primary model may be:

$dU_t = F_U(X_t)dt + \sigma_U dW_t$. (24.20)

$dD_t = F_D(U_t,\sigma_t,r_t,L_t)dt + \sigma_D dW'_t$. (24.21)

$dH_t = F_H(D_t,U_t,L_t)dt$. (24.22)

The interactions propagate through allowed market channels.

The secondary state:

$\rho_{UD}(\theta)$ (24.23)

is an effective summary of these relations.

Therefore:

Secondary Jointness Does Not Abolish Primary Causation. (24.24)

24.8 Why the internal observer cannot simply invoke the primary model

The internal observer may possess an estimated primary model.

But its model is itself part of W_θ.

It is:

$Model_O = Projection(\Sigma_{primary} \mid O,P,L)$. (24.25)

The observer does not possess $\Sigma_{primary}$ without projection.

Any reconstruction is bounded by:

- available data;
- model class;
- memory;
- computation;
- instruments;
- protocol.

Thus:

Observer's Primary Explanation = Secondary Model of the Primary Universe. (24.26)

The observer cannot step outside all world-forming interfaces.

This preserves the constructor problem raised in Part I: an observer investigates the larger field through instruments produced by the same world-forming architecture it is trying to reconstruct.

24.9 When the primary explanation is sufficient

The financial entanglement analogy should be weakened when a classical primary model:

1. reproduces all joint distributions;
2. explains setting dependence;
3. predicts sequence effects;
4. accounts for signalling and backreaction;
5. remains stable out of sample.

Then the stronger representation adds no empirical value.

The correct conclusion is:

Classical Composite Dynamics Sufficient. (24.27)

The θ-world account may remain conceptually useful, but it should not be presented as evidence of nonclassical finance.

24.10 When the secondary description adds value

The secondary-world representation may still improve analysis when it:

- compresses complex primary mechanisms;
- reveals globally shared states;
- clarifies local incompleteness;
- separates measurement settings;

- tracks gate and ledger dependence;
- identifies frame-invariant relations;
- detects holonomy or path residual;
- improves prediction or intervention.

Thus:

Useful Effective Ontology ≠ Fundamental Ontology. (24.28)

A representation can be operationally valuable without being microscopically fundamental.

24.11 The nonlocal-appearance equation

Define primary propagation delay:

$\delta t_{primary} = t_D - t_U$. (24.29)

Define internal episode distance:

$\delta k = k_D - k_U$. (24.30)

If:

$\delta t_{primary} > 0$, (24.31)

but:

$\delta k = 0$, (24.32)

the relation appears simultaneous inside the ledgered world.

Define nonlocal-appearance indicator:

$N_{app} = 1$ iff $\delta k = 0$ and LocalMessenger is absent from $\mathcal{A}_O$. (24.33)

This indicator measures observer-level appearance.

It does not establish physical nonlocality.

24.12 Layer confusion as a false paradox

A false paradox arises when one asks:

“How can D update simultaneously with U if the causal signal takes time?”

The answer may be:

- they are simultaneous only in the secondary episode;
- the primary process still contains propagation;
- the internal state relation compresses that propagation.

The opposite false paradox is:

“If the primary mechanism is known, there can be no meaningful entanglement-like structure.”

The answer is:

- external constructibility does not force internal factorization;
- the effective joint state may still be primary for internal prediction.

The two mistakes are:

Primary Reductionism. (24.34)

Secondary Reification. (24.35)

The architecture avoids both.

24.13 The two-description principle

Proposition 24.1 — Dual Causal Description

A derivative event may possess:

1. a primary causal description in calendar time;
2. a secondary joint-state description in θ-time;
3. a committed historical description in ledger time.

In compact form:

Event = PrimaryProcess_t + JointDisclosure_θ + Record_k. (24.36)

These descriptions are compatible when the compilation and backreaction maps are declared.

---

25. Contractual Entanglement Versus Quantum Entanglement

25.1 Why one word is insufficient

The term entanglement may refer to several different levels of dependence.

Part II therefore distinguishes:

Statistical Entanglement-Like Correlation. (25.1)

Contractual Binding. (25.2)

Dynamical Nonseparability. (25.3)

Effective-State Nonfactorization. (25.4)

Coherent Quantum Entanglement. (25.5)

Bell-Nonclassical Entanglement. (25.6)

These should not be merged.

25.2 Level E_1: statistical correlation

Two outputs satisfy:

$p(u,d) \neq p(u)p(d)$. (25.7)

This is ordinary dependence.

It may arise from common factors.

Finance clearly realizes E_1.

25.3 Level E_2: functional dependence

The derivative value satisfies:

$D = V(U,X)$. (25.8)

Where X includes other state variables.

Changing U changes D.

This is stronger than raw correlation but remains classically functional.

Finance clearly realizes E_2.

25.4 Level E_3: contractual coupling

The derivative's identity is constituted through:

$I_D = (U,K,T,Payoff,Exercise,Settlement,Law)$. (25.9)

Without the relation, the same derivative object does not exist.

This is ontological at the effective institutional level:

Derivative Identity = Contractual Relation. (25.10)

Finance clearly realizes E_3.

25.5 Level E_4: dynamical binding

The joint dynamics satisfy:

$\partial F_U/\partial D \neq 0$, (25.11)

and:

$\partial F_D/\partial U \neq 0$. (25.12)

The sectors mutually affect one another.

Finance realizes E_4 through:

- hedging;
- margin;
- collateral;
- volatility recalibration;
- market impact.

25.6 Level E_5: effective-world nonfactorization

The declared effective state satisfies:

$\rho_{UD} \neq \rho_U \otimes \rho_D$. (25.13)

A stronger condition is:

$\rho_{UD} \notin \mathrm{Sep}(\mathscr{H}_U \otimes \mathscr{H}_D)$. (25.14)

Finance can formally construct E_5.

But empirical content depends on:

- state estimation;
- partition choice;
- measurement algebra;
- classical null family.

25.7 Level E_6: observable coherence

A coherent state contains:

$\rho_{off} \neq 0$. (25.15)

And there exists an admissible observable $\hat{O}$ such that:

$\mathrm{Tr}(\hat{O}\rho_{UD}) \neq \mathrm{Tr}(\hat{O}\rho_{diag})$. (25.16)

The relative phase changes a measurable financial outcome.

Standard option pricing does not automatically establish E_6.

This is a research hypothesis.

25.8 Level E_7: contextual joint measurement

Different local settings reveal correlations that cannot be represented by one noncontextual assignment.

The financial model would require:

No GlobalValueMap consistent with all admissible contexts. (25.17)

Ordinary financial memory, backreaction, and protocol dependence may mimic contextuality.

Therefore E_7 requires strong null-model rejection.

25.9 Level E_8: no-signalling entanglement

The joint outcomes are strongly correlated, but one observer's setting cannot controllably alter the other observer's local marginal:

$p(u|a,b) = p(u|a)$ for all b. (25.18)

And:

$p(d|a,b) = p(d|b)$ for all a. (25.19)

Finance usually fails this condition because:

- trades transmit information;
- prices signal;
- hedges affect markets;
- settings may change behaviour.

E_8 is not established.

25.10 Level E_9: Bell-nonclassicality

A Bell-type result requires correlation incompatible with local hidden variables under declared assumptions.

For CHSH:

$|S| > 2$. (25.20)

A credible financial claim would need to address:

- measurement independence;
- locality or causal separation;
- memory loopholes;
- post-selection;

- common information;
- signalling;
- adaptive markets;
- stationarity.

No such result is established here.

25.11 The classification table

Level	Structure	Current financial status
E_1	Statistical correlation	Established
E_2	Functional dependence	Established
E_3	Contractual coupling	Established
E_4	Dynamical binding	Established
E_5	Effective nonfactorization	Formally constructible
E_6	Observable phase coherence	Unestablished
E_7	Irreducible contextual joint measurement	Unestablished
E_8	No-signalling entanglement	Not shown
E_9	Bell-nonclassicality	Not shown

25.12 The corrected claim

The strongest defensible current claim is:

> Derivatives provide natural composite states, contractual binding, local incompleteness, shared gates, and reflexive joint dynamics. From inside a θ-time valuation world, these may reproduce important operational aspects of entanglement's strangeness. They do not yet establish the nonclassical probability structure of quantum entanglement.

In compact form:

Derivative Entanglement Analogy = $E_1 + E_2 + E_3 + E_4$ + Candidate E_5. (25.21)

Quantum Entanglement requires more:

Quantum Entanglement $\supset E_5 + E_6$ + Strong Contextuality + No-Signalling Structure. (25.22)

25.13 The anti-literalism condition

A cross-domain mapping is admissible only when:

ValidMapping $\Leftrightarrow$ FunctionalRolePreserved $\wedge$ ProtocolDeclared $\wedge$ DiagnosticValueAdded. (25.23)

This follows the methodological discipline of the Gauge Grammar: physics language should be removed when it adds ornament rather than explanation, diagnosis, stability, or control.

25.14 The Part IV conclusion

The internal observer perspective materially strengthens the derivative comparison.

It reveals why:

- global preparation may be invisible locally;
- local states may be incomplete;
- one measurement episode may condition both sectors;
- outcomes become fixed through trace;
- a primary causal sequence may appear simultaneous internally.

But it also sharpens the remaining boundary.

Observer restriction can explain why a joint relation appears strange.

It cannot by itself derive:

- coherent amplitude probability;
- no-signalling;
- Bell violation;
- irreducible contextuality.

Thus:

Observer Perspective Explains the Location of Strangeness. (25.24)

It Does Not Automatically Explain the Full Quantum Residue. (25.25)

26. The No-Signalling Boundary

26.1 Why simultaneous conditioning is not enough

The internal observer may encounter:

Joint State
→ Local Measurement
→ Correlated Conditional State. (26.1)

This structure can appear nonlocal because the effective update rule contains no visible messenger travelling from one local sector to the other.

But quantum entanglement has a more demanding property.

Although local outcomes may be strongly correlated, one party cannot use its choice of measurement setting to control the other party's local outcome statistics.

For underlying-side setting a and derivative-side setting b:

$p(u|a,b) = \Sigma_d\, p(u,d|a,b)$. (26.2)

A no-signalling condition from D to U requires:

$p(u|a,b_0) = p(u|a,b_1)$ for all u,a,b_0,b_1. (26.3)

Likewise, no signalling from U to D requires:

$p(d|a_0,b) = p(d|a_1,b)$ for all d,b,a_0,a_1. (26.4)

Equivalently:

$p(u|a,b) = p(u|a)$. (26.5)

$p(d|a,b) = p(d|b)$. (26.6)

Correlated collapse alone does not imply these relations.

26.2 Finance normally contains signalling

Financial observers routinely influence one another through:

- prices;
- quotes;
- order flow;
- public disclosures;
- hedge transactions;
- margin calls;
- collateral demands;
- rating changes;
- regulatory action.

Suppose the derivative-side observer selects setting b_1 by executing a large option trade.

That action may change:

- implied volatility;
- dealer inventory;
- hedge demand;
- underlying order flow.

Therefore:

$p(u|a,b_1) \neq p(u|a,b_0)$. (26.7)

The derivative measurement setting has become a causal intervention.

This violates the no-signalling condition.

The violation does not indicate a flaw in finance.

It indicates that the financial interaction is not equivalent to quantum entanglement under this criterion.

26.3 Passive settings versus active interventions

A measurement setting may be passive or active.

A passive setting selects how existing data are classified:

b_passive = choose price basis or Greek basis. (26.8)

An active setting changes the market:

b_active = execute trade, publish quote, or trigger margin. (26.9)

The no-signalling test is meaningful only when the settings do not themselves transmit ordinary causal information.

Define setting influence:

$I_{b\to U} = \text{Distance}[p(u|a,b_0),p(u|a,b_1)]$. (26.10)

A candidate no-signalling regime requires:

$I_{b\to U} \le \varepsilon_{NS}$. (26.11)

And:

$I_{a\to D} \le \varepsilon_{NS}$. (26.12)

The tolerance ε_{NS} must include statistical estimation error.

26.4 State update is not message transmission

Suppose observers share a joint state:

ρ_{UD}. (26.13)

Observer U measures outcome u.

The conditional derivative state becomes:

$\rho_{D|u} = \mathrm{Tr}_U[(M_u \otimes I_D)\rho_{UD}(M_u^\dagger \otimes I_D)] / p(u)$. (26.14)

But the unconditional derivative state remains:

$\rho_{D'} = \Sigma_u\, p(u)\rho_{D|u}$. (26.15)

For a trace-preserving local instrument:

$\rho_{D'} = \rho_D$. (26.16)

Thus, without access to u, the derivative observer cannot distinguish whether the underlying observer measured.

This is the quantum-style no-signalling pattern.

A financial model wishing to reproduce it would need:

$$\Sigma_u \mathrm{Tr}_U[(\mathbb{M}^U_{a,u} \otimes I_D)(\rho_{UD})] = \rho_D \text{ independent of } a. \quad (26.17)$$

Ordinary financial interventions generally fail this because measurement changes the economic system.

26.5 Informational conditioning versus physical influence

Three different processes must be separated.

Process A — Conditional inference

Observer B learns outcome u and updates:

$$p_B(d|u). \quad (26.18)$$

Process B — Public information transmission

Outcome u is communicated through a message, price, or record.

Process C — Market backreaction

The act generating u changes underlying or derivative dynamics.

Only Process A resembles state conditioning without causal transmission.

Processes B and C provide ordinary signalling channels.

A financial entanglement experiment must isolate these processes rather than mix them.

26.6 A financial no-signalling test

Let A and B choose settings independently.

For each setting pair, estimate:

$$p(u,d|a,b). \quad (26.19)$$

Compute the marginal deviations:

$$\delta_{B\to A}(u,a;b_0,b_1) = p(u|a,b_0) - p(u|a,b_1). \quad (26.20)$$

$$\delta_{A\to B}(d,b;a_0,a_1) = p(d|a_0,b) - p(d|a_1,b). \quad (26.21)$$

Define:

$$NS_F = \max\{|\delta_{B\to A}|,|\delta_{A\to B}|\}. \quad (26.22)$$

A no-signalling-compatible dataset requires:

NS_F ≤ ε_NS. (26.23)

But this statistical condition is not sufficient.

One must also verify that:

- settings were not inferred from shared information;
- the sample was not post-selected;
- time ordering did not leak information;
- participants did not adapt across trials;
- market state remained sufficiently stationary;
- the setting mechanisms did not affect liquidity.

26.7 Common-cause models

Even when no-signalling approximately holds, classical common causes may explain the correlations.

Let λ be a shared latent market state:

p(u,d|a,b) = ∫ p(λ)p(u|a,λ)p(d|b,λ)dλ. (26.24)

Possible λ include:

- macroeconomic news;
- latent volatility;
- dealer positioning;
- liquidity;
- funding stress;
- investor sentiment;
- algorithmic state.

A strong claim requires rejecting a sufficiently rich class of such models.

No-signalling alone does not imply nonclassicality.

26.8 Measurement independence

Bell-style reasoning assumes settings are statistically independent of the hidden state:

p(a,b|λ) = p(a,b). (26.25)

In finance, settings are commonly selected because of the observed market state:

a = π_A(L_A,MarketState). (26.26)

$b = \pi_B(L_B, MarketState)$. (26.27)

Therefore:

$p(a,b|\lambda) \neq p(a,b)$. (26.28)

This is measurement dependence.

It is normal adaptive financial behaviour.

But it invalidates a straightforward Bell interpretation.

A credible experiment would need settings generated independently of the market state or would need an explicit causal model correcting for dependence.

26.9 Memory and repeated-market trials

Standard Bell experiments aim to treat trials as appropriately independent.

Financial trials are strongly history-dependent:

$\rho_{k+1} = \mathcal{T}(\rho_k, a_k, b_k, u_k, d_k, L_k)$. (26.29)

The next state retains:

- prior trades;
- inventory;
- learning;
- public information;
- ledgered outcomes.

Therefore:

$p(u_{k+1}, d_{k+1} | a_{k+1}, b_{k+1})$ (26.30)

depends on earlier trials.

This memory can generate apparent contextual or Bell-like patterns.

Any financial test must include a sequential null model.

26.10 Spatial separation is not the essential issue

An option and its underlying need not be physically distant.

Quantum entanglement can exist between nearby systems.

The relevant issue is whether local operations satisfy the required algebraic and causal separation.

For finance, define operational locality:

Locality_F ⇔ Setting_A acts only on subsystem A within the declared trial window. (26.31)

And:

Setting_B acts only on subsystem B within the declared trial window. (26.32)

If either action changes the shared market state during the window, operational locality fails.

26.11 A narrow candidate financial regime

The strongest candidate regime would use:

- historical or simulated data;
- pre-registered passive measurement settings;
- no market intervention;
- independently randomized basis selection;
- concealed outcomes until trial completion;
- stationary or controlled state preparation;
- explicit classical latent-state models.

Even then, a violation may show only that the selected classical model family is inadequate.

It would not automatically establish physical quantum behaviour in markets.

26.12 The no-signalling conclusion

Derivative finance readily produces:

- conditional dependence;
- global recalibration;
- synchronized outcomes;
- observer-relative update;
- reflexive feedback.

But these usually coexist with ordinary signalling.

Therefore:

Secondary Simultaneity ≠ No-Signalling. (26.33)

Conditional State Update ≠ Controllable Nonlocal Communication. (26.34)

Derivative Entanglement-Like Structure ≠ Bell Entanglement. (26.35)

The no-signalling boundary remains one of the clearest separators between the stronger financial analogy and established quantum structure.

Part V — From Probability Mixtures to Coherent Composite States

27. Risk-Neutral Weights and the Amplitude Lift

27.1 Standard derivative valuation begins with weighted possibilities

Under a risk-neutral measure $\mathbb{Q}$, a derivative value may be written:

$$D_0 = B_0 E_{\mathbb{Q}}[B_T^{-1} \text{Payoff}(U_T)]. \quad (27.1)$$

Under a constant continuously compounded rate:

$$D_0 = e^{-rT} E_{\mathbb{Q}}[\text{Payoff}(U_T)]. \quad (27.2)$$

For discrete terminal states u_n:

$$D_0 = e^{-rT} \Sigma_n q_n f(u_n). \quad (27.3)$$

Where:

$$q_n \geq 0. \quad (27.4)$$

$$\Sigma_n q_n = 1. \quad (27.5)$$

The q_n are risk-neutral pricing weights.

They are not necessarily the physical probabilities governing realized outcomes.

27.2 From probability weights to amplitude magnitudes

A formal Hilbert lift defines:

$$\alpha_n = \sqrt{q_n} \exp(i\varphi_n). \quad (27.6)$$

Then:

$$|\alpha_n|^2 = q_n. \quad (27.7)$$

And:

$$\Sigma_n |\alpha_n|^2 = 1. \quad (27.8)$$

The underlying scenario state becomes:

$$|\psi_U\rangle = \Sigma_n \sqrt{q_n} \exp(i\varphi_n) |u_n\rangle. \quad (27.9)$$

The contract-prepared state is:

$$|\Psi_{UD}\rangle = \Sigma_n \sqrt{q_n} \exp(i\varphi_n) |u_n, d_n\rangle. \quad (27.10)$$

This provides:

- normalized amplitude magnitudes;
- a branch basis;
- relative phases;
- a composite state.

But it is only a mathematical lift until the phases affect observables.

27.3 The amplitude lift is not unique

Given probabilities q_n, infinitely many phase choices produce the same diagonal distribution:

$\{\varphi_n\}$ arbitrary. (27.11)

The standard derivative price depends only on q_n:

$D_0 = e^{-rT}\Sigma_n |\alpha_n|^2 f(u_n)$. (27.12)

The phases cancel.

Therefore the pricing rule does not identify φ_n.

A meaningful amplitude theory requires an additional phase-generating law.

Possible candidates include:

- valuation-frame phase;
- path phase;
- protocol phase;
- liquidity phase;
- hedge phase;
- gauge-transport phase;
- ledger holonomy.

Without such a law:

Amplitude Lift = Nonunique Representation. (27.13)

27.4 Risk-neutral probability is not Born probability

The numerical relation:

$q_n = |\alpha_n|^2$ (27.14)

resembles the Born rule.

But similarity of form does not establish identity.

Risk-neutral weights are determined by:

- no-arbitrage;
- change of measure;
- market completeness or calibration;
- state-price density;
- pricing kernel.

Born probabilities are physical outcome probabilities assigned from quantum amplitudes.

Thus:

Risk-Neutral Weight ≠ Physical Frequency. (27.15)

Pricing Probability ≠ Measurement Probability. (27.16)

Amplitude Magnitude Construction ≠ Born-Rule Derivation. (27.17)

The distinction must remain explicit.

27.5 State prices as amplitude magnitudes

Let state price be:

$\pi_n = Boq_n/B_T,n.$ (27.18)

The derivative value is:

$Do = \Sigma_n \pi_n f(u_n).$ (27.19)

A state-price amplitude may be defined:

$a_n = \sqrt{\pi_n} \exp(i\varphi_n).$ (27.20)

Then:

$\pi_n = |a_n|^2.$ (27.21)

This may be more natural for valuation because it incorporates discounting.

But π_n may not sum to one.

Normalize:

$\tilde{a}_n = a_n/\sqrt{(\Sigma_m \pi_m)}.$ (27.22)

Again, normalization is a formal choice unless supported by a measurement protocol.

27.6 Pricing kernel representation

Under the physical measure $\mathbb{P}$:

$$Do = E_{\mathbb{P}}[m_T f(U_T)]. \quad (27.23)$$

Where:

$$m_T = \text{stochastic discount factor}. \quad (27.24)$$

For discrete states:

$$Do = \Sigma_n p_n m_n f(u_n). \quad (27.25)$$

Define effective pricing weight:

$$w_n = p_n m_n. \quad (27.26)$$

An amplitude lift is:

$$\alpha_n = \sqrt{w_n} \exp(i\varphi_n). \quad (27.27)$$

This shows that amplitude magnitude may combine:

- physical likelihood;
- marginal utility;
- discounting;
- risk adjustment.

Therefore the amplitude is not naturally one simple probability wave.

It is a valuation-weighted possibility coefficient.

27.7 The branch basis problem

The amplitude depends on the chosen branch basis.

Possible bases include:

- terminal price intervals;
- path classes;
- volatility regimes;
- default states;
- liquidity states;
- contract outcomes.

Let basis transformation be:

$$|u'_a\rangle = \Sigma_n U_{an} |u_n\rangle. \quad (27.28)$$

The amplitudes transform:

$\alpha'_a = \Sigma_n U_{an} \alpha_n$. (27.29)

If the model is genuinely Hilbert-like, observable predictions must transform consistently.

But standard scenario probabilities are usually defined only in one chosen partition.

The basis-change structure is not automatically available.

27.8 Amplitude preparation from CAPM-seeded channels

Let each financial channel n have local CAPM-derived state:

$Z_n = A_n \exp(i\theta_n)$. (27.30)

Define normalized coefficient:

$\alpha_n = c_n Z_n / \sqrt{(\Sigma_m |c_m Z_m|^2)}$. (27.31)

Then:

$|\psi_F\rangle = \Sigma_n \alpha_n |n\rangle$. (27.32)

The magnitude contains:

$|\alpha_n|^2 \propto |c_n|^2 A_n^2$. (27.33)

The phase is:

$\arg \alpha_n = \arg c_n + \theta_n$. (27.34)

This supplies a CAPM-seeded amplitude construction.

But its probability interpretation remains an empirical question.

27.9 The declared financial norm

A state vector requires a norm.

Define:

$\|\psi_F\|^2_G = \langle \psi_F | G_F | \psi_F \rangle$. (27.35)

Where G_F is a positive metric operator.

Normalization requires:

$\langle \psi_F | G_F | \psi_F \rangle = 1$. (27.36)

The choice G_F may reflect:

- state prices;
- risk weights;

- capital weights;
- liquidity weights;
- information geometry.

Different norms create different amplitude magnitudes.

Therefore:

Financial Normalization Is Protocol-Dependent. (27.37)

A Born-like rule cannot be claimed until the relevant norm is uniquely justified.

27.10 Observable outcome rule

A candidate financial outcome rule is:

$p(y|a) = \langle \psi_F | E_{a,y} | \psi_F \rangle_G$. (27.38)

Where:

$\langle \psi | E | \psi \rangle_G = \langle \psi | G_{FE} | \psi \rangle$. (27.39)

For consistency:

$E_{a,y} \geq 0$. (27.40)

$\Sigma_y E_{a,y} = I$. (27.41)

This creates a generalized Born-form measurement model.

But the rule is currently postulated, not derived from CAPM.

Its merit must be assessed against standard probabilistic classifiers.

27.11 The amplitude-lift proposition

Proposition 27.1 — Financial Amplitude Lift

A probability or state-price model admits a formal amplitude lift when:

1. nonnegative branch weights are available;
2. a normalized inner product is declared;
3. branch phases are assigned;
4. observables are defined on the lifted state space.

In compact form:

AmplitudeLift ⇔ Weights ∧ Norm ∧ Phase ∧ ObservableAlgebra. (27.42)

The lift becomes empirically meaningful only when:

Predictions_depend_on(RelativePhase) = true. (27.43)

Otherwise it is a redundant reformulation of classical probability.

---

28. Classical Mixture Versus Coherent State

28.1 The classical scenario mixture

A standard scenario model is:

$\rho_{mix} = \Sigma_n q_n |u_n,d_n\rangle\langle u_n,d_n|$. (28.1)

It contains only diagonal branch weights.

For an observable $\hat{O}$:

$E_{mix}[O] = Tr(\rho_{mix}\hat{O})$. (28.2)

If $\hat{O}$ is diagonal in the scenario basis:

$\hat{O} = \Sigma_n O_n |u_n,d_n\rangle\langle u_n,d_n|$, (28.3)

then:

$E_{mix}[O] = \Sigma_n q_n O_n$. (28.4)

This is ordinary weighted expectation.

28.2 The coherent state

Define:

$|\Psi\rangle = \Sigma_n \sqrt{q_n} \exp(i\varphi_n) |u_n,d_n\rangle$. (28.5)

Then:

$\rho_{coh} = |\Psi\rangle\langle\Psi|$. (28.6)

Expanding:

$\rho_{coh} = \rho_{mix} + \rho_{off}$. (28.7)

Where:

$\rho_{off} = \Sigma_{n\neq m} \sqrt{(q_n q_m)} \exp[i(\varphi_n - \varphi_m)] |u_n,d_n\rangle\langle u_m,d_m|$. (28.8)

The difference between coherent state and mixture lies in ρ_{off}.

28.3 When coherence is invisible

For any observable diagonal in the branch basis:

$Tr(\rho_{coh}\hat{O}_{diag}) = Tr(\rho_{mix}\hat{O}_{diag})$. (28.9)

Thus standard payoff expectation may be unable to distinguish coherence.

A coherence-sensitive observable must contain off-diagonal components:

$\hat{O}_{cross} = \Sigma_{n\neq m} O_{nm} |u_n, d_n\rangle\langle u_m, d_m|$. (28.10)

Then:

$Tr(\rho_{coh}\hat{O}_{cross}) = \Sigma_{n\neq m}\sqrt{(q_n q_m)}\exp[i(\varphi_n-\varphi_m)]O_{mn}$. (28.11)

For the mixture:

$Tr(\rho_{mix}\hat{O}_{cross}) = 0$. (28.12)

Therefore:

Coherence Exists Operationally Only Relative to a Cross-Branch Observable. (28.13)

28.4 What could be a cross-branch financial observable?

Candidate observables may combine alternatives before commitment.

Examples include:

- order-flow aggregation before execution;
- portfolio netting across contingent branches;
- competing valuation narratives affecting one price;
- cross-strike calibration;
- strategy interaction before public disclosure;
- path amplitudes in a decision model;
- coherent belief or preference models.

However, mature finance normally treats these as classical interacting alternatives.

A cross-branch observable must produce predictions not reproducible by:

- nonlinear probability aggregation;
- latent factors;
- common causes;
- agent-based interaction;
- Bayesian updating.

28.5 Two-branch example

Let:

$|\Psi\rangle = \sqrt{q_1}|1\rangle + \sqrt{q_2}\exp(i\Delta\varphi)|2\rangle$. (28.14)

Project onto:

$|+\rangle = (|1\rangle + |2\rangle)/\sqrt{2}$. (28.15)

The coherent probability is:

$p_{coh}(+) = \frac{1}{2}[q_1 + q_2 + 2\sqrt{(q_1 q_2)}\cos\Delta\varphi]$. (28.16)

Since $q_1 + q_2 = 1$:

$p_{coh}(+) = \frac{1}{2}[1 + 2\sqrt{(q_1 q_2)}\cos\Delta\varphi]$. (28.17)

The classical mixture gives:

$p_{mix}(+) = 1/2$. (28.18)

The difference is:

$\Delta p = \sqrt{(q_1 q_2)}\cos\Delta\varphi$. (28.19)

This is the simplest interference signature.

28.6 Constructive and destructive interference

Constructive interference occurs when:

$\cos\Delta\varphi > 0$. (28.20)

Destructive interference occurs when:

$\cos\Delta\varphi < 0$. (28.21)

Complete cancellation is possible for equal amplitudes and opposite phase:

$q_1 = q_2 = 1/2$. (28.22)

$\Delta\varphi = \pi$. (28.23)

Then:

$p_{coh}(+) = 0$. (28.24)

A financial analogue would require two individually viable channels whose joint pre-commitment combination suppresses an outcome below every classical additive expectation.

This is a strong requirement.

28.7 Classical nonlinear interaction can mimic interference

Suppose outcome probability is:

$p(y) = \sigma_logistic(w_1x_1 + w_2x_2 + w_{12}x_1x_2)$. (28.25)

The interaction term can create constructive or destructive effects.

Similarly, agent interactions can generate cancellation or amplification.

Therefore:

Nonadditivity ≠ Coherence. (28.26)

A valid coherence claim must compare against flexible classical interaction models.

28.8 Density-matrix estimation

Given measurements in several bases, estimate:

$\hat{\rho}_F = \arg\min_\rho \text{Loss}(\text{Data}, \text{Predictions}_\rho)$. (28.27)

Subject to:

$\rho \geq 0$. (28.28)

$\text{Tr}\ \rho = 1$. (28.29)

Compare:

Model M_0: ρ diagonal. (28.30)

Model M_1: ρ full. (28.31)

Use out-of-sample criteria:

$\Delta LL = LL_test(M_1) - LL_test(M_0)$. (28.32)

$\Delta IC = IC(M_0) - IC(M_1)$. (28.33)

A full coherent model is justified only if it improves predictive performance after complexity penalties.

28.9 Decoherence parameter

A partially coherent state may be:

$\rho_\gamma = \rho_mix + \gamma\rho_off$. (28.34)

Where:

$0 \leq \gamma \leq 1$. (28.35)

Interpretation:

$\gamma = 0 \rightarrow$ complete mixture. (28.36)

$\gamma = 1 \rightarrow$ full declared coherence. (28.37)

Intermediate γ represents partial phase retention.

A decay law may be:

$$d\gamma/d\theta = -\kappa_{dec}\gamma. \quad (28.38)$$

Therefore:

$$\gamma(\theta) = \gamma_0 \exp(-\kappa_{dec}\theta). \quad (28.39)$$

The parameter κ_{dec} measures effective decoherence rate.

28.10 Sources of financial decoherence

Candidate phase-destroying processes include:

- public price formation;
- settlement;
- aggregation;
- noisy information;
- heterogeneous clocks;
- liquidity shocks;
- model disagreement;
- regulatory classification;
- irreversible trade execution.

These processes convert:

Phase-Sensitive Alternatives
→ Classical Recorded Mixture. (28.40)

But the analogy is useful only if phase-sensitive alternatives were operationally present before the process.

28.11 Purity

Define purity:

$$\mathcal{P} = \mathrm{Tr}(\rho^2). \quad (28.41)$$

For a pure state:

$$\mathcal{P} = 1. \quad (28.42)$$

For a mixed state:

$$\mathcal{P} < 1. \quad (28.43)$$

A financial purity measure could indicate how much of the effective state is represented as one coherent relational configuration versus an unresolved ensemble.

But low purity may arise from ordinary aggregation.

Thus:

Financial Mixedness ≠ Quantum Decoherence by Default. (28.44)

28.12 Coherence witness

Define l_1 coherence:

$C_{l_1}(\rho) = \Sigma_{n\neq m}|\rho_{nm}|$. (28.45)

Or relative entropy of coherence:

$C_{RE}(\rho) = S(\rho_{diag}) - S(\rho)$. (28.46)

These measures can quantify fitted off-diagonal structure.

A meaningful financial claim requires:

$C(\hat{\rho}) > C_{min}$ (28.47)

and:

PredictiveGain($\hat{\rho}$_full versus classical nulls) > Δ^*. (28.48)

Formal nonzero coherence is not enough.

28.13 The mixture–coherence boundary

Proposition 28.1 — Observable Coherence Criterion

A financial state exhibits operational coherence under protocol P only when:

1. the estimated state contains off-diagonal branch relations;
2. an admissible measurement is sensitive to those relations;
3. the resulting predictions differ from the diagonal mixture;
4. flexible classical interaction models fail to reproduce the effect;
5. the result remains stable under admissible basis transformations.

In compact form:

$\text{Coherence}_{F,P} \Leftrightarrow \text{OffDiagonal} \wedge \text{CrossObservable} \wedge \text{PredictiveDifference} \wedge \text{ClassicalNullFailure} \wedge \text{FrameRobustness}$. (28.49)

This is the minimum standard for advancing from E_5 to E_6 on the entanglement ladder.

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29. From One θ to a Multi-Channel Phase Field

29.1 Why a single phase cannot support interference

The scalar complex state is:

$Z = A\exp(i\theta)$. (29.1)

Multiplying the entire state by a global phase:

$Z' = \exp(i\chi)Z$ (29.2)

does not create relative phase between branches.

Interference requires at least two channels:

$Z_1 = A_1\exp(i\theta_1)$. (29.3)

$Z_2 = A_2\exp(i\theta_2)$. (29.4)

The relevant phase is:

$\Delta\theta_{12} = \theta_1 - \theta_2$. (29.5)

Therefore the Part I angle must be generalized.

29.2 Channelized complex state

Let the secondary world contain N channels:

$Z_n = A_n\exp(i\theta_n)$. (29.6)

Define:

$|\Psi_F\rangle = \Sigma_n c_n Z_n |n\rangle$. (29.7)

The phase field is:

$\Theta_F = (\theta_1,\theta_2,\ldots,\theta_N)$. (29.8)

The relative-phase matrix is:

$\Delta\Theta_{nm} = \theta_n - \theta_m$. (29.9)

Only N−1 independent phases remain after removing one global phase reference.

29.3 Candidate channel types

Channels may represent:

Scenario channels

Different terminal or path outcomes.

Strike channels

Different regions of an option surface.

Maturity channels

Short-, medium-, and long-dated valuation sectors.

Protocol channels

Market, accounting, regulatory, and liquidation views.

Observer channels

Different investor classes or institutions.

Liquidity channels

Continuous, stressed, and frozen market states.

Contract branches

Knock-in, knock-out, exercise, default, and survival paths.

The channel basis must be chosen according to the research question.

29.4 Phase as accumulated valuation action

A branch phase may be defined through an action-like integral:

$\theta_n = \theta_{n,0} + (1/\hbar_F)\int \mathcal{L}_n d\tau$. (29.10)

Where:

$\mathcal{L}_n$ = effective financial Lagrangian for branch n. (29.11)

Possible components include:

$\mathcal{L}_n = \text{Return}_n - \text{RiskCost}_n - \text{FundingCost}_n - \text{LiquidityCost}_n - \text{GatePressure}_n$. (29.12)

Then:

$\Delta\theta_{nm} = (1/\hbar_F)\int(\mathcal{L}_n - \mathcal{L}_m)d\tau + \Delta\theta_{nm,0}$. (29.13)

This gives relative phase an accumulated economic interpretation.

It remains a model hypothesis.

29.5 Phase from filter orientation

Alternatively:

$\theta_n = \arccos(R_n/A_n)$. (29.14)

Then:

$\Delta\theta_{nm} = \arccos(R_n/A_n) - \arccos(R_m/A_m)$. (29.15)

This preserves the original Finance Geometry meaning.

Channels with similar admission ratios are phase-aligned.

Channels with different admission ratios are phase-separated.

The challenge is showing that these phase differences affect cross-channel observables rather than merely restating discount ratios.

29.6 Phase velocity field

Each channel has:

$\omega_n = d\theta_n/dt$. (29.16)

The relative phase evolves as:

$d\Delta\theta_{nm}/dt = \omega_n - \omega_m$. (29.17)

Define phase-locking when:

$\omega_n \approx \omega_m$. (29.18)

Then $\Delta\theta_{nm}$ remains approximately constant.

Define dephasing when:

$|\omega_n - \omega_m|$ is large or unstable. (29.19)

Phase locking may represent:

- synchronized valuation adjustment;
- common funding regime;
- common narrative;
- shared hedge response.

Dephasing may represent:

- market fragmentation;
- protocol disagreement;
- maturity dislocation;
- liquidity separation.

29.7 Kuramoto-style financial synchronization

A stylized phase model is:

$d\theta_n/dt = \omega_n{}^0 + (K/N)\Sigma_m \sin(\theta_m - \theta_n) + \xi_n(t)$. (29.20)

Where:

- ω_n^0 is intrinsic valuation rotation;
- K is coupling strength;
- ξ_n is noise.

Define order parameter:

$re^{i\psi} = (1/N)\Sigma_n e^{i\theta_n}$. (29.21)

Where:

$0 \le r \le 1$. (29.22)

High r means synchronized valuation phases.

Low r means dispersed phases.

This model is classical.

But it can generate phase coherence, collective repricing, and collapse-like synchronization without quantum mechanics.

29.8 Phase dispersion and effective decoherence

Let phase variance be:

$Var_\theta = E[(\theta_n - \psi)^2]$. (29.23)

As dispersion grows:

$r \downarrow$. (29.24)

Cross terms average out:

$E[e^{i\Delta\theta_{nm}}] \to 0$. (29.25)

Thus heterogeneous financial phases can suppress interference-form effects.

This is classical dephasing.

It provides a useful null model for any claimed financial decoherence.

29.9 Gauge freedom

Local channel phases may be redefined:

$|n\rangle \to e^{i\chi_n(x)}|n\rangle$. (29.26)

Then:

$\theta_n \to \theta_n + \chi_n$. (29.27)

Only gauge-invariant combinations should affect observables.

A connection transforms as:

$\mathcal{A}_\mu \to \mathcal{A}_\mu - \partial_\mu \chi$. (29.28)

The covariant derivative is:

$D_\mu = \partial_\mu + i\mathcal{A}_\mu$. (29.29)

This preserves meaningful comparison across local phase conventions.

29.10 Relative phase and path dependence

For channel n transported along path γ:

$\theta_n(\gamma) = \theta_{n,\text{local}} + \int_\gamma \mathcal{A}_\mu dx^\mu$. (29.30)

For two paths γ_1 and γ_2:

$\Delta\theta_{\text{path}} = \int_{\gamma_1}\mathcal{A} - \int_{\gamma_2}\mathcal{A}$. (29.31)

If the connection has curvature:

$\mathbb{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu \neq 0$, (29.32)

then path-dependent phase can remain even when endpoints coincide.

This is geometric phase.

29.11 Ledger contribution to phase

Let each committed event contribute phase increment:

$\delta\theta_{n,k} = \Phi_n(\text{Record}_k)$. (29.33)

Then:

$\theta_{n,k} = \theta_{n,0} + \Sigma_{j\le k}\delta\theta_{n,j}$. (29.34)

Two states with identical visible values but different ledgers may have:

$\theta_n^A \neq \theta_n^B$. (29.35)

The phase therefore carries historical orientation.

This links:

Ledger Memory → Phase Structure. (29.36)

29.12 Phase reset at gates

A gate may reset or randomize phase.

For outcome g:

$\theta_n^+ = R_g(\theta_n^-)$. (29.37)

Examples:

- settlement removes pre-trade uncertainty;
- expiry terminates option phase evolution;
- default changes the relevant valuation frame;
- liquidation destroys the prior going-concern basis.

A complete phase model must include gate-induced discontinuities.

29.13 Non-Abelian extension

If channels possess internal multidimensional frames, phase transport may require matrix-valued connection:

$\mathcal{A}_\mu = \mathcal{A}_\mu^a T_a$. (29.38)

The covariant derivative is:

$D_\mu = \partial_\mu + i\mathcal{A}_\mu$. (29.39)

Curvature becomes:

$\mathbb{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu + i[\mathcal{A}_\mu, \mathcal{A}_\nu]$. (29.40)

The commutator term means transport order matters.

This may model:

- sequential protocol conversion;
- noncommuting risk transformations;
- order-dependent ledger operations;
- multidimensional volatility-mode rotation.

The extension should be introduced only when scalar phase transport fails.

29.14 The multi-phase proposition

Proposition 29.1 — Multi-Channel Phase Requirement

A financial theory can support interference-form dynamics only when:

1. at least two operational channels remain distinct before commitment;
2. each channel carries a phase;
3. relative phase survives admissible transport;

4. a measurement combines channels before projection;

5. cross terms affect observable outcomes.

In compact form:

InterferencePossible ⇔ MultiChannel ∧ RelativePhase ∧ CoherentTransport ∧ PreProjectionCombination ∧ ObservableCrossTerm. (29.41)

A single CAPM angle satisfies none of these requirements by itself.

The multi-channel completion is therefore the minimum structural upgrade.

30. Interference-Form Finance

30.1 From additive scenarios to amplitude combination

Standard finance combines mutually exclusive scenarios through weighted addition:

$P(y) = \Sigma_n q_n P(y|n)$. (30.1)

This is a classical mixture.

Each branch contributes probability or value independently.

An interference-form model instead combines branch amplitudes before producing the observable:

$\mathcal{A}(y) = \Sigma_n \alpha_n \mathcal{A}_n(y)$. (30.2)

The observable weight is then:

$P_F(y) = |\mathcal{A}(y)|^2$. (30.3)

Expanding:

$P_F(y) = \Sigma_n |\alpha_n \mathcal{A}_n(y)|^2 + \Sigma_{n \neq m} \alpha_n \alpha_m \mathcal{A}_n(y) \mathcal{A}_m(y)$. (30.4)

The second sum contains the interference terms.

Classical scenario aggregation gives:

$P_mix(y) = \Sigma_n |\alpha_n \mathcal{A}_n(y)|^2$. (30.5)

Therefore:

InterferenceResidual(y) = P_F(y) − P_mix(y). (30.6)

A financial interference claim requires this residual to be:

- measurable;
- reproducible;
- phase-sensitive;

- not explainable by ordinary nonlinear interaction.

30.2 What counts as a financial branch?

A branch must represent an unresolved alternative that remains operationally distinct before commitment.

Possible branches include:

- exercise versus continuation;
- barrier hit versus not hit;
- default versus survival;
- high-liquidity versus low-liquidity regime;
- alternative valuation protocols;
- different hedge responses;
- competing investor narratives;
- alternative execution paths.

Let branch n be:

$B_n = (State_n, Protocol_n, Path_n, LedgerCondition_n)$. (30.7)

The branch amplitude is:

$\alpha_n = A_n \exp(i\theta_n)$. (30.8)

The observable is produced only after branches are combined through the declared measurement setting.

If the branches are merely sampled separately and averaged afterward, the model remains a mixture.

30.3 Pre-commitment combination

Interference requires:

Branch Combination before Gate. (30.9)

Suppose two channels influence a market price before execution:

$\mathcal{A}_price = \mathcal{A}_buy + \mathcal{A}_sell$. (30.10)

The committed price is:

$PriceOutcome = Gate_price(|\mathcal{A}_price|^2)$. (30.11)

If buy and sell pressures are treated independently and then netted only after execution, the structure is ordinary accounting.

If their phase relationship changes the probability of the committed market outcome before execution, the model has interference form.

The crucial order is:

Branch Preparation
→ Phase Evolution
→ Amplitude Combination
→ Measurement Gate
→ Ledger. (30.12)

Not:

Branch Measurement
→ Classical Result
→ Arithmetic Combination. (30.13)

30.4 Constructive financial interference

For two channels:

$\mathcal{A} = A_1 \exp(i\theta_1) + A_2 \exp(i\theta_2)$. (30.14)

Then:

$|\mathcal{A}|^2 = A_1^2 + A_2^2 + 2A_1A_2\cos(\Delta\theta)$. (30.15)

Where:

$\Delta\theta = \theta_1 - \theta_2$. (30.16)

Constructive interference occurs when:

$\cos(\Delta\theta) > 0$. (30.17)

The combined outcome exceeds the incoherent sum.

Possible financial interpretations include:

- several investor classes reinforcing the same valuation transition;
- hedge flows synchronized with speculative demand;
- multiple maturity sectors rotating into the same regime;
- compatible narratives increasing commitment probability;
- collateral and liquidity conditions jointly amplifying one move.

But ordinary positive feedback can produce the same empirical pattern.

The interference model earns value only if relative phase predicts the amplification more accurately than standard feedback models.

30.5 Destructive financial interference

Destructive interference occurs when:

$\cos(\Delta\theta) < 0$. (30.18)

The combined result is smaller than the incoherent sum.

Possible examples include:

- strong buying interest offset by equally structured hedging pressure;
- optimistic cash-flow expectations opposed by funding deterioration;
- spot momentum opposed by volatility-surface warning;
- two valuation protocols producing mutually cancelling commitment pressure;
- call buying offset by dealer hedge flows in the opposite direction.

Again:

Cancellation ≠ Quantum Interference. (30.19)

Ordinary netting, strategic opposition, and negative feedback can generate cancellation.

The stronger question is whether phase provides a stable hidden coordinate governing the cancellation.

30.6 Interference between valuation protocols

Suppose one primary financial field is projected under two protocols:

$Z_A = A_A \exp(i\theta_A)$. (30.20)

$Z_B = A_B \exp(i\theta_B)$. (30.21)

Examples:

- market-value protocol;
- accounting-value protocol;
- liquidation-value protocol;
- regulatory-capital protocol.

An institution may act only after combining them:

$\mathcal{A}_{commit} = w_A Z_A + w_B Z_B$. (30.22)

The commitment intensity is:

$I_{commit} = |\mathcal{A}_{commit}|^2$. (30.23)

Expanding:

$I_{commit} = w_A^2 A_A^2 + w_B^2 A_B^2 + 2 w_A w_B A_A A_B \cos(\theta_A - \theta_B)$. (30.24)

If the protocols align:

$\theta_A \approx \theta_B$, (30.25)

commitment pressure rises.

If they oppose:

$\theta_A - \theta_B \approx \pi$, (30.26)

commitment may be delayed or cancelled.

This may model institutional hesitation more effectively than averaging scalar valuations.

30.7 Interference in option exercise

An American-option holder may compare:

Exercise Channel. (30.27)

Continuation Channel. (30.28)

Let:

$\mathcal{A}_E = A_E \exp(i\theta_E)$. (30.29)

$\mathcal{A}_C = A_C \exp(i\theta_C)$. (30.30)

The exercise-setting amplitude is:

$\mathcal{A}_{decision} = \mathcal{A}_E + \mathcal{A}_C$. (30.31)

The decision probability may be modelled as:

$P(\text{Exercise}) = |\Pi_E \mathcal{A}_{decision}|^2$. (30.32)

A conventional model compares exercise value and continuation value.

The interference extension asks whether unresolved interaction between these channels creates systematic decision effects not captured by ordinary utility or optimal-stopping models.

This is more naturally a model of bounded decision processes than of market physics.

30.8 Interference in narrative finance

Suppose market participants receive two interpretive channels:

N_1 = “earnings growth.” (30.33)

N_2 = “funding stress.” (30.34)

Each produces a complex valuation contribution:

$Z_{N_1} = A_1 \exp(i\theta_1)$. (30.35)

$Z_{N_2} = A_2 \exp(i\theta_2)$. (30.36)

The resulting price projection may depend on:

$Z_{total} = Z_{N1} + Z_{N2}$. (30.37)

The relative phase determines whether the narratives:

- reinforce;
- cancel;
- rotate the measurement basis;
- create ambiguity.

A purely scalar sentiment score loses this orientation information.

A complex narrative model may therefore be useful even if it remains entirely classical.

30.9 Interference across maturity

Let short- and long-maturity channels be:

$Z_S = A_S \exp(i\theta_S)$. (30.38)

$Z_L = A_L \exp(i\theta_L)$. (30.39)

A maturity-spanning observable may be:

$\mathcal{A}_{term} = w_S Z_S + w_L Z_L$. (30.40)

The cross term is:

$I_{SL} = 2 w_S w_L A_S A_L \cos(\theta_S - \theta_L)$. (30.41)

Possible interpretations:

- aligned short- and long-term risk;
- near-term panic with long-term confidence;
- short-term calm with long-term structural concern;
- maturity inversion.

This creates a phase-based term-structure diagnostic.

30.10 Interference in basket derivatives

For a basket with N underlyings:

$\mathcal{A}_B = \Sigma_i w_i A_i \exp(i\theta_i)$. (30.42)

Then:

$|\mathcal{A}_B|^2 = \Sigma_i w_i^2 A_i^2 + \Sigma_{i \neq j} w_i w_j A_i A_j \cos(\theta_i - \theta_j)$. (30.43)

The cross terms encode phase-adjusted relational contributions.

This resembles covariance decomposition, but the phase model is richer only if:

$\cos(\theta_i - \theta_j)$ (30.44)

captures something not already represented by correlation or copula dependence.

Otherwise phase is merely a different parameterization of covariance.

30.11 Interference and covariance

For real-valued random variables:

$\mathrm{Var}(\Sigma_i w_i X_i) = \Sigma_i w_i^2 \mathrm{Var}(X_i) + \Sigma_\{i \neq j\} w_i w_j \mathrm{Cov}(X_i, X_j)$. (30.45)

The amplitude expression is:

$|\Sigma_i w_i A_i \exp(i\theta_i)|^2 = \Sigma_i w_i^2 A_i^2 + \Sigma_\{i \neq j\} w_i w_j A_i A_j \cos(\Delta\theta_{ij})$. (30.46)

One possible identification is:

$\mathrm{Cov}(X_i, X_j) \approx A_i A_j \cos(\Delta\theta_{ij})$. (30.47)

But this alone does not create a new theory.

It simply writes covariance through amplitude and phase.

The extension becomes useful only when:

1. phase evolves dynamically;
2. phase transports across frames;
3. phase predicts future covariance change;
4. phase holonomy captures path dependence;
5. phase survives out-of-sample testing.

30.12 A phase-implied correlation

Define:

$\rho_{ij}^{\mathrm{phase}} = \cos(\theta_i - \theta_j)$. (30.48)

Then:

$-1 \leq \rho_{ij}^{\mathrm{phase}} \leq 1$. (30.49)

This resembles a correlation coefficient.

However, the matrix:

$R_{ij} = \cos(\theta_i - \theta_j)$ (30.50)

has a restricted low-dimensional structure.

It may fail to represent arbitrary empirical correlation matrices.

A richer state may require:

$\theta_i \in \mathbb{T}^d$, (30.51)

where $\mathbb{T}^d$ is a d-dimensional phase torus.

Then:

ρ_{ij}^phase = $\Sigma_a w_a \cos(\theta_i^a - \theta_j^a)$. (30.52)

This permits multiple latent phase channels.

30.13 Double-path pricing experiment

A clean interference-style experiment would construct two alternative valuation paths leading to one outcome.

Path 1:

Primary State → Protocol A → Outcome y. (30.53)

Path 2:

Primary State → Protocol B → Outcome y. (30.54)

Measure each path separately:

P_A(y). (30.55)

P_B(y). (30.56)

Then allow both protocols to act before commitment:

P_AB(y). (30.57)

Classical mixture prediction:

P_mix(y) = w_AP_A(y) + w_BP_B(y). (30.58)

Interference residual:

I_AB(y) = P_AB(y) − P_mix(y). (30.59)

A phase model predicts:

I_AB(y) = 2√[w_Aw_BP_A(y)P_B(y)]cos Δθ_AB. (30.60)

This can be fitted and tested.

Flexible classical interaction models must be included as nulls.

30.14 Order-dependent interference

Suppose protocols A and B act sequentially.

The outcomes may satisfy:

$$P_{AB}(y) \neq P_{BA}(y). \quad (30.61)$$

An amplitude model may use:

$$\mathcal{A}_{AB} = \langle y|\hat{U}_B\hat{U}_A|\psi\rangle. \quad (30.62)$$

$$\mathcal{A}_{BA} = \langle y|\hat{U}_A\hat{U}_B|\psi\rangle. \quad (30.63)$$

If:

$$[\hat{U}_A,\hat{U}_B] \neq 0, \quad (30.64)$$

the order changes the result.

This may model:

- risk review before trade approval versus after approval;
- collateral call before rating downgrade versus after downgrade;
- volatility recalibration before hedge adjustment versus after;
- accounting recognition before regulatory intervention versus after.

Order effects are common in finance.

The research question is whether a compact operator model predicts them better than standard sequential causal models.

30.15 Interference visibility

Define maximum and minimum outcome intensity over relative phase:

$$I_{max} = A_1^2 + A_2^2 + 2A_1A_2. \quad (30.65)$$

$$I_{min} = A_1^2 + A_2^2 - 2A_1A_2. \quad (30.66)$$

Define visibility:

$$V = (I_{max} - I_{min})/(I_{max} + I_{min}). \quad (30.67)$$

Therefore:

$$V = 2A_1A_2/(A_1^2 + A_2^2). \quad (30.68)$$

Visibility is highest when amplitudes are balanced.

It is low when one channel dominates.

A financial interference experiment should therefore seek regimes where alternative channels have comparable strength.

30.16 Decoherence-adjusted interference

With coherence parameter γ:

$I = A_1^2 + A_2^2 + 2\gamma A_1 A_2 \cos \Delta\theta$. (30.69)

Where:

$0 \le \gamma \le 1$. (30.70)

The interference visibility becomes:

$V_\gamma = \gamma \cdot 2A_1A_2/(A_1^2 + A_2^2)$. (30.71)

This allows gradual suppression rather than an all-or-nothing distinction.

Financial institutions may preserve partial coherence before public commitment and lose it after settlement or broad disclosure.

30.17 Interference versus strategic interaction

A strategic game may produce:

Outcome = F(Action_A,Action_B,Expectations). (30.72)

This can create cross terms without amplitudes.

Therefore an empirical comparison should include:

Model M_0: additive classical mixture. (30.73)

Model M_1: nonlinear classical interaction. (30.74)

Model M_2: latent-state or regime-switching model. (30.75)

Model M_3: complex amplitude model. (30.76)

The complex model is justified only if:

PredictiveGain(M_3) > ComplexityPenalty(M_3). (30.77)

And:

ResidualStructure(M_3) < ResidualStructure(M_0,M_1,M_2). (30.78)

30.18 The interference proposition

Proposition 30.1 — Operational Financial Interference

A financial process exhibits interference form under protocol P when:

1. at least two unresolved channels contribute before commitment;
2. each channel has a complex amplitude;
3. a common measurement combines the channels;

4. the outcome contains phase-sensitive cross terms;

5. the cross terms outperform flexible classical interaction models.

In compact form:

Interference_F,P ⇔ UnresolvedChannels ∧ ComplexAmplitude ∧ PreGateCombination ∧ PhaseCrossTerm ∧ ClassicalNullFailure. (30.79)

Without the fifth condition, interference remains a representation rather than a discovery.

---

31. Decoherence, Clearing, and Settlement

31.1 Why financial possibility does not remain indefinitely coherent

A financial system can maintain multiple unresolved alternatives only while:

- information remains incomplete;
- commitment has not occurred;
- branches remain operationally available;
- observers have not synchronized around one public record;
- the ledger has not eliminated alternatives.

Public quotation, execution, clearing, settlement, and legal recognition progressively reduce the active branch structure.

This suggests a financial decoherence sequence:

Private Possibility
→ Public Measurement
→ Trade
→ Clearing
→ Settlement
→ Stable Ledger. (31.1)

The sequence converts flexible relational possibility into a more classical shared history.

31.2 Environment-induced phase loss

Let the system state be:

$|\Psi_S\rangle = \Sigma_n c_n |s_n\rangle$. (31.2)

Let environmental sectors include:

$\mathscr{H}_E = \mathscr{H}_market \otimes \mathscr{H}_clearing \otimes \mathscr{H}_legal \otimes \mathscr{H}_regulatory$. (31.3)

The initial joint state is:

$|\Psi_SE\rangle = \Sigma_n c_n |s_n\rangle|e0\rangle$. (31.4)

Interaction produces:

$|\Psi_SE'\rangle = \Sigma_n c_n |s_n\rangle|e_n\rangle$. (31.5)

The reduced system state is:

$\rho_S = \Sigma_{n,m} c_n c_m^* \langle e_m|e_n\rangle |s_n\rangle\langle s_m|$. (31.6)

When environmental records become distinguishable:

$\langle e_m|e_n\rangle \to 0$ for $m \neq n$, (31.7)

the off-diagonal terms disappear.

The financial environment has recorded which branch occurred.

31.3 Public price as pointer state

A publicly accepted market price may function as a pointer state.

Before public commitment, several private valuations may coexist:

$|\Psi_private\rangle = \Sigma_n c_n |V_n\rangle$. (31.8)

After execution and dissemination:

$|\Psi_public\rangle \approx |V^*\rangle$. (31.9)

The price V^* is redundantly copied into:

- exchange feeds;
- broker systems;
- risk systems;
- clearing records;
- financial media;
- valuation models.

This redundancy stabilizes V^* as shared operational reality.

The source observer framework similarly treats redundant records as a route from internal outcomes to cross-observer objectivity.

31.4 Trade execution as partial collapse

An indicative quote is not yet a trade.

A trade execution commits:

- price;

- quantity;
- parties;
- time;
- contract identity.

The branch space narrows:

Possible Executions
→ One Executed Trade. (31.10)

But execution may still leave unresolved:

- settlement;
- counterparty failure;
- trade correction;
- legal dispute;
- collateral consequence.

Execution is therefore a partial commitment.

The collapse sequence continues through later gates.

31.5 Clearing as relational compression

Clearing may replace many bilateral obligations with net positions.

Let gross obligation matrix be:

$G = [G_{ij}]$. (31.11)

Netting produces:

$N = Clear(G,P,L)$. (31.12)

Generally:

$\dim(N) < \dim(G)$. (31.13)

Clearing compresses relational distinctions.

Some gross-path information is lost from the active operational state, though it may remain in audit trace.

Thus:

Gross Branch Network
→ Clearing Projection
→ Net Obligation State. (31.14)

This is a financial coarse-graining operation.

31.6 Central counterparty as an environment and observer

A central counterparty:

- measures exposure;
- applies margin rules;
- records obligations;
- mutualizes certain risks;
- changes future admissibility.

It is therefore both:

- an environmental coupling sector;
- an internal observer;
- a gate authority;
- a ledger operator.

The CCP can suppress bilateral branch distinctions by replacing them with standardized cleared states.

This increases shared objectivity while reducing local relational detail.

31.7 Settlement as phase destruction

Before settlement:

Trade State = Executed but Not Final. (31.15)

After settlement:

Trade State = Asset Delivered + Cash Delivered + Final Ledger. (31.16)

Alternative branches such as:

- fail;
- dispute;
- cancellation;
- correction;

are removed from ordinary operational continuation.

Settlement therefore acts like a phase-destroying gate.

Define coherence before settlement:

γ_pre. (31.17)

After finality:

γ_post ≈ 0 for mutually exclusive settlement branches. (31.18)

This does not mean all financial coherence vanishes.

It means that the settled event no longer remains an unresolved live alternative inside the ordinary ledger.

31.8 Legal finality

Legal finality strengthens branch suppression.

Once final settlement is recognized:

Reverse(Event) requires extraordinary procedure. (31.19)

The ledger becomes resistant to ordinary rewriting.

Define finality strength:

F_L = Cost_or_Impossibility_of_Reversal. (31.20)

High F_L corresponds to strong latching.

Legal structure therefore contributes directly to the effective arrow of financial time.

31.9 Decoherence rate

Let off-diagonal coherence evolve as:

$d\rho_{nm}/d\theta = -\kappa_{nm}\rho_{nm}$ for $n \neq m$. (31.21)

Then:

$\rho_{nm}(\theta) = \rho_{nm}(0)\exp(-\kappa_{nm}\theta)$. (31.22)

The rate may depend on:

$\kappa_{nm} = \kappa_info + \kappa_trade + \kappa_clearing + \kappa_settlement + \kappa_legal$. (31.23)

Interpretation:

- κ_info = public information leakage;
- κ_trade = execution-induced differentiation;
- κ_clearing = clearing compression;
- κ_settlement = finality;
- κ_legal = enforceability and irreversibility.

This creates a measurable decomposition of effective financial decoherence.

31.10 Heterogeneous clocks as decoherence

Suppose channels have phase velocities ω_n.

Relative phase becomes:

$$\Delta\theta_{nm}(t) = \Delta\theta_{nm}(0) + \int_0^t [\omega_n(s) - \omega_m(s)]ds. \quad (31.24)$$

If phase velocities vary across observers or desks, the ensemble average satisfies:

$$E[\exp(i\Delta\theta_{nm})] \to 0. \quad (31.25)$$

The channels lose coherent alignment.

No environmental measurement is necessary.

Clock heterogeneity alone can cause dephasing.

31.11 Liquidity as coherence medium

A liquid market rapidly synchronizes:

- prices;
- quotes;
- valuations;
- hedge responses.

This may increase short-run phase alignment.

But it may also accelerate public commitment and destroy private branch coherence.

Liquidity therefore has two opposite effects:

$$\text{Liquidity} \to \text{Faster Synchronization}. \quad (31.26)$$

$$\text{Liquidity} \to \text{Faster Public Decoherence}. \quad (31.27)$$

A frozen market may preserve multiple private valuations longer because no authoritative transaction commits one public price.

Thus:

$$\text{Illiquidity can preserve disagreement while destroying reliable shared phase transport}. \quad (31.28)$$

31.12 Crisis decoherence

During crisis:

- models diverge;
- prices gap;

- liquidity fragments;
- protocols change;
- settlement confidence weakens.

This may produce:

- rapid decoherence within some institutional channels;
- simultaneous fragmentation between channels.

One shared valuation world may split into several locally coherent worlds:

W_market. (31.29)

W_accounting. (31.30)

W_regulatory. (31.31)

W_liquidation. (31.32)

The crisis is not simply loss of coherence.

It may be:

Global Coherence Loss + Local Protocol Locking. (31.33)

31.13 Decoherence-free financial subspaces

Some relational structures may remain stable despite environmental noise.

Examples:

- legally fixed payoff identity;
- collateralized settlement obligation;
- exact offset under perfect netting;
- locked conversion ratio.

Let subspace $\mathcal{K}$ satisfy:

$\mathbb{M}_E(\rho) = \rho$ for all $\rho \in \mathcal{K}$. (31.34)

Then $\mathcal{K}$ is decoherence-resistant under the declared environment.

Financially, this means the relation survives:

- market noise;
- reporting variation;
- local frame changes.

Such structures may serve as robust invariants.

31.14 Recoherence

Financial branches can sometimes become jointly relevant again.

Examples:

- reopened legal dispute;
- trade correction;
- novation;
- contract restructuring;
- reinstated barrier due to data error;
- revised accounting treatment.

A previously suppressed branch re-enters the active state space.

Define recoherence operation:

$\rho_{diag} \to \rho_{recombined}$. (31.35)

However, true reversal is rare because ledger history remains.

Recoherence normally occurs through a new protocol rather than erasure of the old event.

Thus:

Financial Recoherence = New World Built over Old Trace. (31.36)

31.15 Entropy production

Let financial entropy be:

$S_F(\rho) = -\mathrm{Tr}(\rho \ln \rho)$. (31.37)

Environmental coupling can increase reduced-state entropy:

$dS_F/d\theta \geq 0$ (31.38)

under suitable coarse-grained conditions.

But local entropy may decrease when a gate selects one outcome.

The total process separates:

Global Information Redistribution. (31.39)

Local Uncertainty Reduction. (31.40)

Environmental Record Growth. (31.41)

Ledger entropy may increase even while one observer's uncertainty falls.

31.16 Information balance

A schematic information balance is:

$\Delta I_{system} + \Delta I_{environment} + \Delta I_{ledger} = ResidualInformation.$ (31.42)

More explicitly:

$InitialBranchInformation = LocalOutcomeInformation + EnvironmentalRecord + LostResidual.$ (31.43)

A mature financial world should account for where information goes when branches are compressed.

If the model simply deletes residual, it creates false certainty.

31.17 Clearing and settlement as classicalization

The sequence:

Private Complex State
→ Public Measurement
→ Clearing
→ Settlement
→ Redundant Record (31.44)

produces a more classical operational world because:

- one outcome becomes shared;
- alternatives become counterfactual;
- records become redundant;
- future action conditions on the committed result.

This is not necessarily physical classicalization.

It is institutional classicalization.

31.18 The financial decoherence proposition

Proposition 31.1 — Ledgered Decoherence Principle

A financial state undergoes effective decoherence when:

1. alternative branches become correlated with distinguishable environmental records;
2. off-diagonal branch relations cease to affect admissible observables;
3. one operational outcome becomes redundantly recorded;
4. future policies condition on that record;

5. reversal requires a new exceptional protocol.

In compact form:

Decoh_F ⇔ EnvironmentalRecord ∧ CrossTermSuppression ∧ RedundantOutcome ∧ TraceConditioning ∧ IrreversibleRevisionCost. (31.45)

---

32. Geometric Phase and Financial Holonomy

32.1 Returning to the same price does not restore the same state

Suppose an asset begins at:

U0. (32.1)

It follows a path through:

- volatility shock;
- funding stress;
- margin call;
- hedge adjustment;
- recovery.

It later returns to:

U_T = U0. (32.2)

A scalar price description says the asset returned to its starting coordinate.

But the derivative world may now contain:

- different implied volatility;
- different open interest;
- different hedge inventory;
- different collateral;
- different trust;
- different ledger history.

Therefore:

Same Price Endpoint ≠ Same Composite State. (32.3)

This is the basic intuition of financial holonomy.

32.2 Parallel transport

Let the financial state move along path γ in manifold $\mathcal{M}_F$.

A local state $|\psi(x)\rangle$ is transported using connection $\mathcal{A}_\mu$:

$D_\mu|\psi\rangle = (\partial_\mu + i\mathcal{A}_\mu)|\psi\rangle$. (32.4)

Parallel transport satisfies:

$D_\mu|\psi\rangle dx^\mu = 0$. (32.5)

The transported state is:

$|\psi(x_f)\rangle = \mathcal{P}\exp[-i\int_\gamma \mathcal{A}_\mu dx^\mu]|\psi(x_i)\rangle$. (32.6)

Where $\mathcal{P}$ denotes path ordering when required.

The connection determines how local valuation orientation is compared across changing frames.

32.3 Closed-loop transport

For closed loop γ:

$x_f = x_i$. (32.7)

But:

$|\psi_f\rangle = \mathrm{Hol}(\gamma)|\psi_i\rangle$. (32.8)

Where:

$\mathrm{Hol}(\gamma) = \mathcal{P}\exp[-i\oint_\gamma \mathcal{A}_\mu dx^\mu]$. (32.9)

If:

$\mathrm{Hol}(\gamma) \neq I$, (32.10)

the state does not return to its original orientation.

This is holonomy.

In an Abelian case:

$\mathrm{Hol}(\gamma) = \exp(i\Phi_\gamma)$. (32.11)

32.4 Curvature and holonomy

The gauge curvature is:

$\mathbb{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu$. (32.12)

For a small loop enclosing area S:

$\Phi_\gamma \approx \iint_S \mathbb{F}_{\mu\nu} dS^{\{\mu\nu\}}$. (32.13)

Nonzero curvature implies path-dependent phase.

Financially, this may correspond to accumulated residual generated by moving through:

- funding state;
- volatility state;
- collateral state;
- liquidity state;
- regulatory state.

The order of movement matters.

32.5 A simple financial loop

Consider coordinates:

x^1 = volatility σ. (32.14)

x^2 = funding stress f. (32.15)

Path A:

Low σ,Low f
→ High σ,Low f
→ High σ,High f
→ Low σ,High f
→ Low σ,Low f. (32.16)

The visible coordinates return to their starting values.

But the portfolio may retain:

- realized hedge loss;
- changed collateral;
- altered dealer inventory;
- new risk limit;
- changed trust.

The loop leaves a ledgered residue.

32.6 Ledger holonomy

Define ledger transport operator:

$\mathcal{T}_L(\gamma)$. (32.17)

After a loop:

$$L_f = \mathcal{T}_L(\gamma) L_i. \quad (32.18)$$

Generally:

$$L_f \neq L_i. \quad (32.19)$$

Even if:

$$x_f = x_i. \quad (32.20)$$

Define ledger holonomy:

$$H_L(\gamma) = \mathrm{Difference}(L_f, L_i). \quad (32.21)$$

This is one of the strongest financial holonomy concepts because it is directly observable.

Examples include:

- cumulative transaction cost;
- realized tax;
- legal commitment;
- margin history;
- credit deterioration;
- loss-limit activation.

32.7 Phase holonomy versus ledger holonomy

Two kinds of holonomy should be separated.

Phase holonomy

The complex state returns with changed orientation:

$$|\psi_f\rangle = \exp(i\Phi_\gamma)|\psi_i\rangle. \quad (32.22)$$

Ledger holonomy

The committed history changes:

$$L_f \neq L_i. \quad (32.23)$$

They may be linked:

$$\Phi_\gamma = \Phi_{\mathrm{state}}(\gamma) + \Phi_{\mathrm{ledger}}(H_L). \quad (32.24)$$

But they are not identical.

Phase holonomy may exist before irreversible commitment.

Ledger holonomy records historical consequence.

32.8 Option-cycle holonomy

Consider an option portfolio through one volatility cycle.

Initial state:

$(U_0, \sigma_0, \Delta_0, \Gamma_0, H_0, L_0)$. (32.25)

Final visible state:

$U_T = U_0$. (32.26)

$\sigma_T = \sigma_0$. (32.27)

Yet:

$H_T \neq H_0$. (32.28)

$L_T \neq L_0$. (32.29)

$P\&L_T \neq 0$. (32.30)

The loop has produced:

- gamma scalping gain or loss;
- transaction cost;
- funding cost;
- hedge-path dependence.

This is a mature financial example of path-dependent loop residue.

The new theory asks whether that residue can be represented as geometric phase or curvature more economically than by standard path-dependent accounting.

32.9 Gamma as local curvature source

A delta-hedged option has approximate incremental P&L:

$d\Pi \approx \frac{1}{2}\Gamma(dU)^2 + \text{Theta}_D dt + \text{Vega} \cdot d\sigma + \dots$. (32.31)

Over a closed price loop:

$\oint dU = 0$. (32.32)

But:

$\oint \frac{1}{2}\Gamma(dU)^2 \neq 0$. (32.33)

The path can produce nonzero cumulative result.

This is a direct financial analogue of:

Zero Net Displacement + Nonzero Integrated Effect. (32.34)

The effect is already explained by gamma and path variation.

Calling it holonomy is useful only if the geometric formulation unifies multiple path-dependent effects.

32.10 Funding holonomy

Suppose a position moves through:

Normal Funding
→ Stress Funding
→ Collateral Call
→ Recovery Funding
→ Normal Funding. (32.35)

The final funding spread may equal the initial spread.

But cumulative funding cost remains:

$$Cost_f = \oint FundingRate \cdot Exposure \cdot dt. \quad (32.36)$$

And collateral availability may differ.

Thus:

Funding Coordinate Closed
but Funding Ledger Open. (32.37)

This is funding holonomy.

32.11 Regulatory holonomy

A position may pass through:

Low Risk Weight
→ Downgrade
→ Capital Breach
→ Position Reduction
→ Upgrade. (32.38)

The original rating may later return.

But the institution has already:

- raised capital;
- reduced assets;
- changed strategy;
- incurred loss.

The regulatory path has permanently altered the world.

This is protocol-mediated holonomy.

32.12 Non-Abelian holonomy

Suppose two transformations do not commute:

T_A = accounting reclassification. (32.39)

T_B = collateral liquidation. (32.40)

Then:

$T_B T_A \neq T_A T_B$. (32.41)

A closed sequence:

$T_A \to T_B \to T_A^{-1} \to T_B^{-1}$ (32.42)

may fail to restore the original state.

The loop holonomy is:

$H_{AB} = T_A^{-1} T_B^{-1} T_A T_B$. (32.43)

If:

$H_{AB} \neq I$, (32.44)

the order of institutional operations matters.

This is a non-Abelian financial holonomy.

32.13 Geometric phase in multi-channel valuation

Let channel n acquire phase:

$\theta_n = \theta_{n,dyn} + \theta_{n,geom}$. (32.45)

Where:

$\theta_{n,dyn} = -(1/\hbar_F)\int E_n d\tau$. (32.46)

And:

$\theta_{n,geom} = i\int \langle n(x) | \nabla_x n(x) \rangle \cdot dx$. (32.47)

The second term depends on the path of the local basis itself.

Financially, the valuation frame may rotate as:

- volatility surface changes;
- benchmark changes;
- funding frame changes;
- regulatory classification changes.

Even if the local economic amplitude returns, basis transport may leave a geometric phase.

32.14 Berry-like connection

Define local channel basis $|n(x)\rangle$.

The Berry-like financial connection is:

$\mathcal{A}_n(x) = i\langle n(x)|\nabla_x n(x)\rangle$. (32.48)

The geometric phase around loop γ is:

$\Phi_n^{geom} = \oint_\gamma \mathcal{A}_n \cdot dx$. (32.49)

This phase is invariant under admissible local rephasing up to integer winding:

$|n(x)\rangle \to \exp[i\chi_n(x)]|n(x)\rangle$. (32.50)

The observable consequence must depend only on relative or loop phase.

32.15 Winding number

For phase field θ around closed loop:

$\nu = (1/2\pi)\oint_\gamma d\theta$. (32.51)

If:

$\nu \in \mathbb{Z}$, (32.52)

the loop has integer winding.

A financial winding may represent repeated circulation around:

- leverage cycle;
- liquidity cycle;
- risk-on/risk-off cycle;
- volatility cycle.

But topological language is justified only if the winding remains stable under small perturbations.

32.16 Topological robustness

A genuinely topological financial feature should satisfy:

$\nu(\gamma + \text{small deformation}) = \nu(\gamma)$. (32.53)

Unless the path crosses a singularity or regime boundary.

Candidate singularities include:

- default;
- insolvency;
- market closure;
- legal invalidity;
- zero liquidity;
- contract expiry.

This creates a potential topology of financial regimes.

But such claims require careful empirical construction.

32.17 Holonomy measurement protocol

A financial holonomy experiment may proceed:

1. define starting state x_0 and ledger L_0;
2. define a closed path γ in observable state space;
3. transport the valuation state using a declared connection;
4. return to x_0;
5. compare final orientation and ledger with the initial state;
6. compare against standard path-dependent models.

Define measured holonomy:

$$\hat{H}_\gamma = \text{StateDifference}(x_f=x_0, L_f, L_0). \quad (32.54)$$

The geometric model is useful when:

$$\text{PredictionError}_{geom} < \text{PredictionError}_{standard}. \quad (32.55)$$

After complexity adjustment.

32.18 Holonomy as hidden state detection

Suppose visible coordinates return:

$$x_f = x_i. \quad (32.56)$$

But future response differs:

$$\text{Response}_f(u) \neq \text{Response}_i(u). \quad (32.57)$$

Then the visible state description is incomplete.

Holonomy reveals hidden retained structure.

This may indicate:

- omitted ledger;
- omitted phase;
- omitted inventory;
- omitted institutional memory;
- omitted curvature.

Thus:

Loop Test = Diagnostic for Missing State Variables. (32.58)

This may be one of the most practically valuable outcomes of the geometric framework.

32.19 Geometric phase versus hysteresis

Classical hysteresis also produces path dependence.

Let output y depend on path history:

$y_t = F(x_t,History_t)$. (32.59)

A closed input loop can produce:

$y_f \neq y_i$. (32.60)

Therefore:

Holonomy-Like Residue ≠ Quantum Geometric Phase. (32.61)

The geometric formulation must be compared against:

- hysteresis models;
- hidden-state models;
- path-dependent differential equations;
- Preisach-type systems;
- agent memory.

The term geometric phase is earned only if connection and curvature provide a parsimonious invariant description.

32.20 The holonomy proposition

Proposition 32.1 — Financial Holonomy Principle

A financial process exhibits effective holonomy under protocol P when:

1. visible state coordinates traverse a closed loop;
2. the full transported state or ledger does not return;
3. the residual depends systematically on path order;
4. the effect is invariant under admissible local frame changes;
5. a connection-curvature model predicts the residue better than endpoint-only models.

In compact form:

$\mathrm{Hol}_{F,P} \Leftrightarrow \mathrm{ClosedVisibleLoop} \wedge \mathrm{NonClosedFullState} \wedge \mathrm{PathOrderDependence} \wedge \mathrm{FrameInvariant} \wedge \mathrm{PredictiveGain}.$ (32.62)

Part VI — The Layered QM–SR–GR Financial Architecture

33. The GR-Like Base Manifold

33.1 Why the base manifold must be declared separately from the state fibre

The composite financial state lives in a complex fibre.

But the fibre must be attached to a location in a larger financial state space.

Let:

$\boldsymbol{\mathcal{M}}_F$ = financial base manifold. (33.1)

Let:

$\pi: \boldsymbol{\mathcal{H}}_F \to \boldsymbol{\mathcal{M}}_F$. (33.2)

For every $x \in \boldsymbol{\mathcal{M}}_F$:

$\pi^{-1}(x) = \mathscr{H}_x$. (33.3)

The base point x records the effective financial environment.

The fibre $\mathscr{H}_x$ records the local complex or composite valuation state.

This separation prevents the theory from using one object simultaneously as:

- financial location;
- quantum-like state;
- observer frame;
- global geometry.

33.2 Candidate base coordinates

A practical coordinate vector may be:

$x^\mu = (\rho_D,\sigma,\lambda_L,\ell,c,f,m,\tau_L,\ldots)$. (33.4)

Where:

- ρ_D = discount-frame rapidity;
- σ = volatility state;
- λ_L = liquidity coordinate;
- ℓ = leverage coordinate;
- c = collateral state;
- f = funding stress;
- m = market-depth or margin coordinate;
- τ_L = ledger-derived historical coordinate.

Different applications require different manifolds.

The base manifold is protocol-bound:

$\boldsymbol{\mathcal{M}}_F = \boldsymbol{\mathcal{M}}_F(P)$. (33.5)

33.3 Metric from financial distinguishability

One metric candidate is based on statistical distinguishability.

For probability model $p(y|x)$:

$g^F_{\mu\nu}(x) = E[(\partial_\mu \ln p)(\partial_\nu \ln p)]$. (33.6)

This is an information-geometric metric.

A small displacement dx produces:

$ds_F^2 = g^F_{\mu\nu}dx^\mu dx^\nu$. (33.7)

States are far apart when their observable distributions are easily distinguishable.

This metric is empirically estimable.

33.4 Metric from stress and intervention cost

Another metric candidate measures the cost of moving between financial states.

Let $C(x,dx)$ be intervention or transition cost.

For small dx:

C(x,dx) ≈ ½g^F_{μν}(x)dx^μdx^ν. (33.8)

The metric may encode:

- transaction cost;
- liquidity cost;
- capital cost;
- legal friction;
- collateral cost;
- time-to-execute.

Then financial distance measures operational difficulty rather than statistical distinguishability.

These metrics need not coincide.

33.5 Metric from covariance

A local covariance metric may be:

g^F_{μν} = [Cov(dx)]⁻¹_{μν}. (33.9)

Directions with low natural variance become geometrically expensive.

Directions with high variance become easier to traverse.

This resembles a Mahalanobis distance.

But a covariance metric alone does not create GR-like backreaction.

It supplies only a state-dependent geometry.

33.6 Metric from control energy

Suppose dynamics are:

dx/dt = F(x) + B(x)u. (33.10)

Define control cost:

J = ½∫u^TRu dt. (33.11)

The minimum cost to move locally may induce:

ds_F² = dx^TG_control(x)dx. (33.12)

This metric measures how difficult it is to steer the system.

It may be especially useful for:

- liquidity management;

- hedge execution;
- capital intervention;
- crisis stabilization.

33.7 Choosing the metric

The metric must be selected by purpose.

Purpose	Candidate metric
Statistical discrimination	Fisher information
Portfolio state distance	covariance inverse
Execution difficulty	liquidity-cost metric
Institutional transition	intervention-cost metric
Risk propagation	sensitivity or stress metric
Observer reconciliation	gauge-corrected discrepancy metric

No unique universal financial metric is assumed.

The architecture requires only that the chosen metric be:

- declared;
- measurable;
- locally nondegenerate where used;
- compatible with reduction limits;
- useful for prediction or control.

33.8 Signature

The metric may be:

- positive definite;
- pseudo-Riemannian;
- degenerate near constraints;
- piecewise defined across regimes.

A positive-definite metric measures financial distance.

A Lorentz-like signature may be introduced when one coordinate plays a distinguished temporal or valuation role:

ds_F² = −dτ_F² + g_ijdx^idx^j. (33.13)

But a pseudo-Riemannian signature should not be adopted merely to imitate spacetime.

It requires a clear invariant and causal interpretation.

33.9 The local-flat limit

At point x_0 choose local coordinates such that:

$$g^F_{\mu\nu}(x_0) = \eta_{\mu\nu}. \quad (33.14)$$

And:

$$\partial_\lambda g^F_{\mu\nu}(x_0) = 0. \quad (33.15)$$

Then, locally:

$$ds_F^2 \approx \eta_{\mu\nu} dx^\mu dx^\nu. \quad (33.16)$$

This is the financial analogue of a locally inertial frame.

CAPM and local derivative sensitivities may be applied within this neighbourhood.

33.10 Curvature as failure of global frame extension

If:

$$R^\rho_{\sigma\mu\nu} \neq 0, \quad (33.17)$$

no single flat frame extends consistently across the region.

Financially:

- one beta cannot remain valid everywhere;
- one discount frame cannot reconcile all states;
- one volatility surface cannot be transported without residual;
- one local hedge rule cannot remain globally accurate.

Curvature measures the obstruction to globalizing local valuation laws.

33.11 Singular and boundary regions

The manifold may contain boundaries where:

- liquidity vanishes;
- leverage diverges;
- contract expires;
- default occurs;
- legal enforceability fails;
- margin buffer reaches zero.

At such regions, the metric may become:

- singular;
- discontinuous;
- degenerate;
- topologically altered.

A crisis may therefore represent:

Movement toward a boundary of the declared financial manifold. (33.18)

Or:

Failure of the previous manifold to remain valid. (33.19)

33.12 Atlas of financial charts

No single coordinate system may cover the whole market.

Use local charts:

$\{(U_\alpha,\varphi_\alpha)\}$. (33.20)

Where U_α are regions and φ_α coordinate maps.

Examples:

- normal-liquidity chart;
- stressed-liquidity chart;
- pre-default chart;
- post-default chart;
- going-concern chart;
- liquidation chart.

On overlaps:

$\varphi_\beta\circ\varphi_\alpha^{-1}$ (33.21)

provides transition maps.

A protocol transition may require leaving one chart and entering another.

33.13 Ledger-indexed manifold

Because trace changes future geometry, define:

$\mathcal{M}_F(L_k)$. (33.22)

After a gate:

$L_k \to L_{k+1}$. (33.23)

The admissible manifold may change:

$\mathcal{M}_F(L_{k+1}) \neq \mathcal{M}_F(L_k)$. (33.24)

Examples:

- barrier hit changes contract state space;
- default changes legal and settlement space;
- margin breach changes admissible actions;
- regulatory intervention changes capital constraints.

Thus the financial manifold is self-revising.

33.14 Dynamic metric

The metric evolves:

$\partial g^F_{\mu\nu}/\partial\theta \neq 0$. (33.25)

A general law is:

$dg^F_{\mu\nu}/d\theta = \mathcal{G}_{\mu\nu}[g^F,T^F,P,L] + \varepsilon^g_{\mu\nu}$. (33.26)

Where:

- T^F is effective financial stress;
- $\mathcal{G}$ is the geometry-update operator;
- ε^g is unexplained metric residual.

This is the beginning of a GR-like source–geometry theory.

33.15 Effective financial stress tensor

A candidate stress tensor may contain:

$T^F_{\mu\nu} = T^{cashflow}_{\mu\nu} + T^{leverage}_{\mu\nu} + T^{liquidity}_{\mu\nu} + T^{derivative}_{\mu\nu} + T^{ledger}_{\mu\nu}$. (33.27)

Interpretively:

- diagonal terms measure local stress density;
- off-diagonal terms measure directional coupling or stress flow.

For example:

$T^F_{liquidity,collateral}$ (33.28)

may represent how collateral demand creates liquidity pressure.

This object is model-dependent.

It is not derived merely from CAPM.

33.16 Schematic source–geometry equation

A GR-like toy equation may be:

$$G^F_{\mu\nu} + \Lambda_F g^F_{\mu\nu} = \kappa_F T^F_{\mu\nu} + \Xi^F_{\mu\nu}. \quad (33.29)$$

Where:

- $G^F_{\mu\nu}$ is constructed from financial curvature;
- Λ_F is a background geometry term;
- κ_F is calibrated coupling;
- $\Xi^F_{\mu\nu}$ is residual or omitted structure.

The equation is a research template.

It becomes meaningful only if:

1. every tensor has an operational definition;
2. data estimate both sides;
3. the relation predicts out of sample;
4. local CAPM and ordinary finance are recovered in weak-curvature limits.

33.17 Conservation-like condition

A consistent source–geometry model may require:

$$\nabla_\mu T_F^{\mu\nu} = J_{residual}^\nu. \quad (33.30)$$

If:

$$J_{residual}^\nu = 0, \quad (33.31)$$

effective financial stress is covariantly conserved.

If:

$$J_{residual}^\nu \neq 0, \quad (33.32)$$

the declared boundary omits flows.

Examples include:

- off-balance-sheet exposure;

- external funding;
- regulatory transfer;
- hidden liquidity support;
- accounting leakage.

Residual current is therefore a boundary diagnostic.

33.18 The base-manifold proposition

Proposition 33.1 — Curved Financial Base Principle

A GR-like financial base is justified when:

1. local valuation laws remain useful;
2. their parameters vary systematically across state;
3. path transport produces nontrivial residual;
4. a state-dependent metric improves prediction or control;
5. stress and ledger history measurably alter that metric.

In compact form:

GRLike_F ⇔ LocalLaw ∧ StateDependentParameters ∧ TransportResidual ∧ MetricGain ∧ SourceBackreaction. (33.33)

This gives the global environment required by the local SR-like frames and QM-like complex fibres.

34. The SR-Like Local Frame Layer

34.1 Why local financial observers need frames

A financial observer never measures from a completely neutral position.

Its valuation depends on:

- numeraire;
- currency;
- benchmark;
- funding curve;
- collateral agreement;
- accounting regime;
- legal boundary;

- risk horizon;
- admissible instruments;
- accessible ledger.

Define observer frame a as:

$F_a = (N_a, B_a, r_a, h_a, P_a, L_a, \mathcal{A}_a)$. (34.1)

Where:

- N_a is the numeraire;
- B_a is the benchmark;
- r_a is the funding or discount structure;
- h_a is the horizon;
- P_a is the valuation protocol;
- L_a is the accessible ledger;
- $\mathcal{A}_a$ is the measurement algebra.

The same primary economic field may therefore generate different local descriptions:

$\rho_a = \text{Projection}(X \mid F_a)$. (34.2)

$\rho_\beta = \text{Projection}(X \mid F_\beta)$. (34.3)

The SR-like layer asks:

> Which differences between ρ_a and ρ_β are genuine economic differences, and which arise only because the observers use different local frames?

34.2 The local-frame principle

At each point x of the curved financial manifold $\mathcal{M}$_F, define a local frame:

$e^a_\mu(x)$. (34.4)

The global metric is:

$g^F_\mu\nu(x) = e^a_\mu(x) e^b_\nu(x) \eta_{a\beta}$. (34.5)

Here:

- μ,ν index global financial coordinates;
- a,b index local frame coordinates;
- $\eta_{a\beta}$ is the local flat-frame metric.

A local observer uses:

$dx^a = e^a_\mu dx^\mu$. (34.6)

The same displacement expressed globally is:

$dx^\mu = e_a{}^\mu dx^a$. (34.7)

The tetrad-like map translates between:

Global Curved Financial State
↔ Local Approximately Flat Valuation Frame. (34.8)

This is the correct role of the SR-like layer.

It does not replace the GR-like geometry.

It provides the local coordinates in which mature valuation laws such as CAPM can remain approximately valid.

34.3 CAPM inside the local frame

In local frame F_a:

$E_a[r_i] = r_{f,a} + \beta_{i,a} ERP_a$. (34.9)

The local coefficients are:

$\beta_{i,a} = Cov_a(r_i, r_m)/Var_a(r_m)$. (34.10)

$ERP_a = E_a[r_m] - r_{f,a}$. (34.11)

The observer may then construct:

$R_{i,a}$ = admitted value under F_a. (34.12)

$Q_{i,a}$ = retained pressure under F_a. (34.13)

$Z_{i,a} = R_{i,a} + iQ_{i,a}$. (34.14)

The same asset in frame F_β has:

$Z_{i,\beta} = R_{i,\beta} + iQ_{i,\beta}$. (34.15)

A valid theory must specify the transport map:

$Z_{i,\beta} = U_{\beta a} Z_{i,a} + \varepsilon_{\beta a}$. (34.16)

Where $\varepsilon_{\beta a}$ records failure of the declared frame transformation.

34.4 Relative discount rapidity

Let the same financial claim have admitted values R_a and R_β in two discount frames.

Define:

$\rho_{\beta a} = \ln(R_a/R_\beta)$. (34.17)

Then:

$R_\beta = \exp(-\rho_{\beta a})R_a$. (34.18)

For three frames:

$\rho c_a = \rho c_\beta + \rho_{\beta a}$. (34.19)

This additive law is exact for multiplicative discount transformations.

The corresponding bounded coordinate is:

$u_{\beta a} = \tanh \rho_{\beta a}$. (34.20)

The composition rule is:

$uc_a = (uc_\beta + u_{\beta a})/(1 + uc_\beta u_{\beta a})$. (34.21)

This resembles relativistic velocity addition.

But:

$u_{\beta a}$ is a normalized valuation-frame coordinate. (34.22)

$u_{\beta a}$ is not physical velocity. (34.23)

β_CAPM is not $u_{\beta a}$. (34.24)

No universal financial speed limit has been derived.

34.5 Local financial boosts

In hyperbolic coordinates:

$A = R \cosh \rho_D$. (34.25)

$Q_H = A \sinh \rho_D$. (34.26)

A relative boost between frames a and b is:

$A_\beta = A_a \cosh \rho_{\beta a} + Q_H,_a \sinh \rho_{\beta a}$. (34.27)

$Q_H,_\beta = A_a \sinh \rho_{\beta a} + Q_H,_a \cosh \rho_{\beta a}$. (34.28)

The invariant is:

$A^2 - Q_H^2 = \text{constant}$. (34.29)

This representation is useful only when the same governed financial object is preserved.

If the contract, boundary, or protocol changes, the transformation is not merely a boost.

34.6 Frame change versus world change

Suppose observer A values under market continuation.

Observer B values under forced liquidation.

Their values are:

R_market. (34.30)

R_liquidation. (34.31)

One might write:

R_liquidation = exp(−ρ_LM)R_market. (34.32)

But the two observers may not be related by a reversible coordinate transformation.

Liquidation introduces:

- market impact;
- execution sequence;
- legal priority;
- collateral seizure;
- gate activation;
- new ledger consequences.

Therefore:

Market Frame → Liquidation Frame (34.33)

may be a protocol transition rather than a Lorentz-like frame change.

The distinction is:

Same World, Different Coordinates → Frame Transformation. (34.34)

Different Admissibility or Causal Rules → World Transition. (34.35)

34.7 Proper frame transformations

A proper local frame transformation should preserve the declared invariant structure.

Let local state be:

$|\Psi\rangle_a$. (34.36)

Let:

$|\Psi\rangle_\beta = U_{\beta a}|\Psi\rangle_a$. (34.37)

A valid transformation requires:

$U_{\beta a}^\dagger G_\beta U_{\beta a} = G_a$. (34.38)

Where G_a and G_β define the local financial inner products.

Then:

$\langle \Psi|G|\Psi\rangle_a = \langle \Psi|G|\Psi\rangle_\beta$. (34.39)

This preserves the declared financial norm.

Without such a condition, the two observers may not be describing the same state.

34.8 Entanglement invariance under local frame change

For an option–underlying state:

ρ_UD. (34.40)

A local frame transformation acts as:

$\rho'_UD = (U_U \otimes U_D)\rho_UD(U_U^\dagger \otimes U_D^\dagger)$. (34.41)

A valid nonseparability measure E must satisfy:

$E(\rho'_UD) = E(\rho_UD)$. (34.42)

Thus derivative entanglement-like structure should survive:

- change of units;
- equivalent numeraire conversion;
- local basis rotation;
- admissible desk-to-desk mapping.

If the apparent nonseparability disappears under such transformations, it was likely a coordinate artefact.

34.9 Local proper time

A local financial observer may possess an internal clock τ_a.

Define:

$d\tau_a = \Omega_a^{-1} G_a |d\tilde{\theta}_a|$. (34.43)

Where:

- $d\tilde{\theta}_a$ is gauge-corrected phase change;
- G_a is local gate or observability strength;
- Ω_a is local processing or transition capacity.

Different observers may accumulate:

$d\tau_a \neq d\tau_\beta$ (34.44)

over the same calendar interval dt.

This may reflect:

- different update frequency;
- different gate thresholds;
- different information access;
- different processing capacity;
- different ledger commitments.

The analogy is structural:

Same Primary Duration → Different Internal Effective Time. (34.45)

It is not yet a physical relativistic time-dilation law.

34.10 Relative simultaneity

Observer A may group events E_1 and E_2 into one episode:

$k_A(E_1) = k_A(E_2)$. (34.46)

Observer B may record:

$k_B(E_1) < k_B(E_2)$. (34.47)

The difference may arise from:

- latency;
- batching;
- settlement convention;
- observation horizon;
- gate authority.

Thus financial simultaneity is frame-relative.

However, cross-observer agreement requires a consistent causal and ledger map.

Frame relativity does not permit arbitrary event reversal.

34.11 Local inertial observers

A local observer is approximately inertial when:

$d\rho_{\beta a}/d\theta \approx 0$. (34.48)

$\partial_\lambda g^F_{\mu\nu} \approx 0$. (34.49)

$dP/d\theta \approx 0$. (34.50)

$dL_{regime}/d\theta \approx 0$. (34.51)

In this regime:

- funding frame is stable;
- protocol does not change;
- metric variation is small;
- local CAPM coefficients remain usable;
- Greeks remain approximately reliable.

The observer can then treat its local world as flat.

34.12 Accelerated financial observers

When:

$a_F = d\rho_D/d\theta \neq 0$, (34.52)

the observer's frame accelerates.

Examples include:

- rapidly worsening funding;
- changing margin rules;
- emergency regulatory intervention;
- collapsing benchmark reliability;
- accelerating risk aversion.

The observed derivative of R includes a frame term:

$dR'/d\theta = \exp(-\rho_D)[dR/d\theta - R(d\rho_D/d\theta)]$. (34.53)

Thus apparent asset deterioration may partly reflect observer-frame acceleration.

34.13 Equivalence-like financial principle

A local observer may be unable to distinguish immediately between:

1. genuine movement of the financial object;
2. acceleration of its valuation frame.

For a short interval:

$\text{ObservedRepricing} \approx \text{ObjectChange} + \text{FrameAcceleration}$. (34.54)

Only broader cross-frame comparison or invariant measurement can separate them.

This resembles an equivalence principle in a limited structural sense.

It does not establish physical equivalence between gravity and acceleration.

34.14 The SR-like layer proposition

Proposition 34.1 — Local Financial Relativity

An SR-like financial layer is justified when:

1. valuation laws are locally stable;
2. observers use different but mappable frames;
3. multiplicative valuation changes admit additive rapidity coordinates;
4. declared invariants survive proper frame transformation;
5. global inconsistencies require a curved extension.

In compact form:

SRLike_F ⇔ LocalLaw ∧ RelativeFrames ∧ AdditiveRapidity ∧ InvariantTransport ∧ NeedForCurvature. (34.55)

The SR-like layer therefore connects the local CAPM kernel to the curved financial manifold without equating CAPM beta with physical velocity.

---

35. The QM-Like Complex Fibre Layer

35.1 The role of the fibre

At each financial base point x:

x ∈ $\mathcal{M}$_F, (35.1)

there is a local complex state space:

$\mathcal{H}$_x. (35.2)

The collection forms the total bundle:

π: $\mathcal{H}$_F → $\mathcal{M}$_F. (35.3)

The base describes:

- liquidity;
- leverage;
- volatility;
- collateral;

- funding;
- ledger regime.

The fibre describes:

- valuation amplitudes;
- retained pressure;
- derivative branches;
- observer-accessible state relations;
- composite financial possibilities.

Thus:

Base Manifold = Where the Financial World Is. (35.4)

Complex Fibre = What Valuation State Exists There. (35.5)

35.2 Local asset state

For one asset:

$|\psi_U\rangle = Z_U|U\rangle$. (35.6)

Where:

$Z_U = A_U\exp(i\theta_U)$. (35.7)

For multiple asset channels:

$|\psi_U\rangle = \Sigma_i c_i A_i \exp(i\theta_i)|u_i\rangle$. (35.8)

The basis $|u_i\rangle$ may represent:

- scenarios;
- regimes;
- maturities;
- liquidity states;
- investor frames;
- path classes.

The phase relation matters only if admissible observables combine these channels before commitment.

35.3 Local derivative state

For one derivative:

$$|\psi_D\rangle = \Sigma_j d_j A_{D,j} \exp(i\theta_{D,j})|d_j\rangle. \quad (35.9)$$

The basis may represent:

- exercise branches;
- barrier branches;
- payoff regimes;
- volatility modes;
- settlement states;
- hedge states.

A derivative state is therefore richer than one premium.

35.4 Composite derivative state

The option–underlying composite space is:

$$\mathscr{H}_{UD} = \mathscr{H}_U \otimes \mathscr{H}_D. \quad (35.10)$$

The state is:

$$|\Psi_{UD}\rangle = \Sigma_{i,j} c_{ij} |u_i, d_j\rangle. \quad (35.11)$$

A product state satisfies:

$$c_{ij} = \alpha_i \beta_j. \quad (35.12)$$

A nonfactorizable state does not admit this decomposition.

The density-like state is:

$$\rho_{UD} = |\Psi_{UD}\rangle\langle\Psi_{UD}| \quad (35.13)$$

for a pure effective state.

More generally:

$$\rho_{UD} = \Sigma_a p_a |\Psi_a\rangle\langle\Psi_a|. \quad (35.14)$$

35.5 The financial inner product

A local inner product may be:

$$\langle\varphi|\psi\rangle_G = \langle\varphi|G_F|\psi\rangle. \quad (35.15)$$

Where G_F is a positive metric operator.

Possible choices encode:

- risk-neutral weighting;

- state prices;
- liquidity weighting;
- capital weighting;
- information geometry.

Normalization is:

$\langle \psi | G_F | \psi \rangle = 1.$ (35.16)

The financial state norm is therefore declared, not automatically universal.

35.6 Observable operators

An observable $\hat{O}$ acts in the local fibre.

Examples:

$\hat{O}_{price}.$ (35.17)

$\hat{O}_{delta}.$ (35.18)

$\hat{O}_{margin}.$ (35.19)

$\hat{O}_{exercise}.$ (35.20)

$\hat{O}_{payoff}.$ (35.21)

The expected outcome is:

$\langle \hat{O} \rangle = \mathrm{Tr}(\rho G_F \hat{O}).$ (35.22)

The operator family must be tied to actual measurement procedures.

Otherwise the algebra is merely formal.

35.7 Projective measurements and generalized instruments

A simple projective measurement uses:

$\Pi_y^2 = \Pi_y.$ (35.23)

$\Pi_y^\dagger = \Pi_y.$ (35.24)

$\Sigma_y \Pi_y = I.$ (35.25)

The outcome probability is:

$p(y) = \mathrm{Tr}(\rho \Pi_y).$ (35.26)

A more realistic financial instrument uses positive effects:

$E_y \geq 0$. (35.27)

$\Sigma_y E_y = I$. (35.28)

Then:

$p(y) = \mathrm{Tr}(\rho E_y)$. (35.29)

The state update is determined by an instrument map:

$\rho \to \mathbb{M}_y(\rho)/p(y)$. (35.30)

This accommodates:

- noisy measurement;
- partial observation;
- discretionary classification;
- performative intervention;
- settlement gates.

35.8 Density operators for heterogeneous financial worlds

A density operator is especially useful when the effective state aggregates:

- multiple observers;
- uncertain scenarios;
- hidden regimes;
- incomplete records;
- mixed protocols.

Let:

$\rho_F = \Sigma_a p_a |\psi_a\rangle\langle\psi_a|$. (35.31)

The diagonal structure represents classical mixture in the chosen decomposition.

Off-diagonal structure represents declared coherence.

The model must test whether the off-diagonal terms add predictive value.

35.9 Open-system evolution

Finance is not generally closed or unitary.

The local state interacts with:

- market environment;
- clearing system;
- regulators;
- information channels;
- balance sheets;
- legal institutions.

A density-state evolution may be:

$$d\rho/d\theta = -i[\hat{H}_F,\rho] + \Sigma_r \gamma_r (L_r \rho L_r^\dagger - \tfrac{1}{2}\{L_r^\dagger L_r,\rho\}) + \mathscr{R}_F. \quad (35.32)$$

Where:

- $\hat{H}_F$ governs coherent effective evolution;
- L_r are environment or gate operators;
- γ_r are coupling rates;
- $\mathscr{R}_F$ is residual.

This is a Lindblad-like formalism.

It is a useful open-system template, not a claim that markets physically obey quantum master equations.

35.10 Non-Hermitian effective dynamics

Financial systems contain:

- loss;
- default;
- leakage;
- absorption;
- irreversible gates.

An effective non-Hermitian generator may be:

$$\hat{H}_{eff} = \hat{H}_F - i\Gamma_F. \quad (35.33)$$

Then:

$$i\hbar_F d|\psi\rangle/d\theta = \hat{H}_{eff}|\psi\rangle. \quad (35.34)$$

The anti-Hermitian part Γ_F may encode:

- dissipation;
- default absorption;

- liquidity leakage;
- settlement finality;
- branch extinction.

Norm loss must be interpreted and balanced through explicit source or outcome channels.

35.11 Composite local dynamics

For option and underlying:

$\hat{H}_{UD} = \hat{H}_U \otimes I_D + I_U \otimes \hat{H}_D + \hat{H}_{contract} + \hat{H}_{hedge}$. (35.35)

The local evolution is:

$i\hbar_{FD}\partial_\theta|\Psi_{UD}\rangle = \hat{H}_{UD}|\Psi_{UD}\rangle + |\varepsilon_{UD}\rangle$. (35.36)

The interaction terms may generate nonfactorization.

If:

$\hat{H}_{contract} = \hat{H}_{hedge} = 0$, (35.37)

local product evolution becomes possible.

35.12 Fibre dimension and model complexity

The scalar Part I model had:

$\dim \mathscr{H} = 1$ complex channel. (35.38)

A derivative pair requires:

$\dim \mathscr{H}_{UD} = \dim \mathscr{H}_U \times \dim \mathscr{H}_D$. (35.39)

A many-body derivative world may grow exponentially.

Therefore practical modelling requires:

- low-rank approximations;
- tensor networks;
- factorized null models;
- sparse interaction graphs;
- coarse-grained channels.

The mathematical possibility of a large Hilbert-like space does not justify using all of it.

35.13 Tensor-network compression

A many-body state may be approximated as a matrix-product state:

$|\Psi\rangle = \Sigma_{i_1 \ldots i_N} \mathrm{Tr}(A_1^{i_1} A_2^{i_2} \ldots A_N^{i_N}) |i_1 \ldots i_N\rangle$. (35.40)

The bond dimension χ controls relational complexity.

Low χ indicates limited effective entanglement.

High χ indicates stronger nonlocal relational structure within the model.

A financial tensor network may map:

- contract chains;
- collateral chains;
- maturity chains;
- counterparty networks.

This could provide computational value even if the system remains classical.

35.14 Entanglement entropy as model-complexity diagnostic

For bipartition A|B:

$S_A = -\mathrm{Tr}(\rho_A \ln \rho_A)$. (35.41)

In the effective model, high S_A indicates that a simple independent decomposition loses substantial joint information.

This can guide:

- model partitioning;
- risk aggregation;
- desk boundaries;
- data-sharing requirements;
- simulation compression.

The quantity may be useful without implying physical quantum entanglement.

35.15 The fibre-layer proposition

Proposition 35.1 — Complex Fibre Principle

A QM-like financial fibre is justified when:

1. complex amplitudes encode operationally meaningful state relations;
2. composite contracts require tensor structure;

3. observables are explicitly tied to instruments;

4. open-system evolution captures gates and environmental coupling;

5. the representation outperforms simpler classical state models.

In compact form:

QMLike_F ⇔ ComplexState ∧ CompositeStructure ∧ InstrumentAlgebra ∧ OpenDynamics ∧ EmpiricalGain. (35.42)

The complex fibre provides the state grammar.

It does not determine the global geometry or local observer frame.

---

36. Gauge Connection and Covariant Transport

36.1 Why ordinary derivatives are insufficient

Suppose a state $|\psi(x)\rangle$ is defined in a local fibre $\mathscr{H}_x$.

At nearby point x + dx, the state belongs to a different fibre:

$|\psi(x + dx)\rangle \in \mathscr{H}_{x+dx}$. (36.1)

The raw difference:

$|\psi(x + dx)\rangle - |\psi(x)\rangle$ (36.2)

is not meaningful until the local frames are aligned.

A connection supplies the comparison rule.

36.2 Abelian financial connection

For one phase degree of freedom:

$D_\mu = \partial_\mu + i\mathcal{A}_\mu$. (36.3)

Where $\mathcal{A}_\mu$ is the financial gauge connection.

The covariant state derivative is:

$D_\mu|\psi\rangle = \partial_\mu|\psi\rangle + i\mathcal{A}_\mu|\psi\rangle$. (36.4)

Under local rephasing:

$|\psi\rangle \to \exp[i\chi(x)]|\psi\rangle$, (36.5)

the connection transforms:

$\mathcal{A}_\mu \to \mathcal{A}_\mu - \partial_\mu\chi$. (36.6)

Then:

$D_\mu|\psi\rangle \to \exp[i\chi(x)]D_\mu|\psi\rangle$. (36.7)

This preserves covariant comparison.

36.3 Financial meaning of gauge freedom

Local rephasing may correspond to:

- choice of benchmark origin;
- local valuation convention;
- phase-zero convention;
- re-expression of the same admissible state;
- local narrative or basis orientation.

The observable should not depend on arbitrary local phase origin.

Only:

- relative phase;
- loop phase;
- gauge-corrected derivative;
- curvature;

should affect predictions.

Thus:

Absolute Local Phase Convention ≠ Observable Financial Fact. (36.8)

36.4 Connection components

Possible components include:

$\mathcal{A}_\rho$ = transport across discount frames. (36.9)

$\mathcal{A}_\sigma$ = transport across volatility states. (36.10)

$\mathcal{A}_\lambda$ = transport across liquidity states. (36.11)

$\mathcal{A}_L$ = transport across ledger-conditioned states. (36.12)

The connection may encode how valuation orientation changes when moving through each financial direction.

36.5 Gauge-corrected phase velocity

The raw phase derivative is:

$\omega_{raw} = d\theta/dt$. (36.13)

The gauge-corrected phase derivative along path x(t) is:

$\tilde{\omega} = d\theta/dt + \mathcal{A}_\mu(dx^\mu/dt)$. (36.14)

This separates:

- phase change of the state;
- phase change induced by movement of the local frame.

A price movement caused only by changing convention should disappear from gauge-corrected dynamics.

36.6 Covariant valuation change

For complex state Z:

$DZ/dt = dZ/dt + i\mathcal{A}_t Z$. (36.15)

A covariant growth law is:

$DZ/dt = [g_A + i\tilde{\omega}_F]Z + \varepsilon_{cov}$. (36.16)

The real component becomes:

$dR/dt = g_A R - \tilde{\omega}_F Q + \mathrm{Re}(\varepsilon_{cov})$. (36.17)

The imaginary component becomes:

$dQ/dt = g_A Q + \tilde{\omega}_F R + \mathrm{Im}(\varepsilon_{cov})$. (36.18)

This extends the Part I dynamics by correcting phase rotation for frame transport.

36.7 Non-Abelian financial connection

If the local state contains several channels:

$|\psi\rangle \in \mathbb{C}^n$, (36.19)

the connection may be matrix-valued:

$\mathcal{A}_\mu = \mathcal{A}_\mu^a T_a$. (36.20)

The covariant derivative is:

$D_\mu = \partial_\mu + i\mathcal{A}_\mu$. (36.21)

The curvature is:

$\mathbb{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu + i[\mathcal{A}_\mu, \mathcal{A}_\nu]$. (36.22)

The commutator term means transport order matters.

This may describe:

- order-dependent protocol conversion;
- sequential risk transformations;
- noncommuting ledger updates;
- basis mixing across derivative channels.

36.8 Metric connection versus gauge connection

The Levi-Civita connection $\Gamma^\mu_{\alpha\beta}$ transports base-manifold vectors.

The gauge connection $\mathcal{A}_\mu$ transports fibre orientation.

They perform different roles.

Base transport:

$$\nabla_\mu v^\nu = \partial_\mu v^\nu + \Gamma^\nu_{\mu\lambda} v^\lambda. \tag{36.23}$$

Fibre transport:

$$D_\mu |\psi\rangle = \partial_\mu |\psi\rangle + i\mathcal{A}_\mu |\psi\rangle. \tag{36.24}$$

The combined derivative is:

$$\mathcal{D}_\mu = \nabla_\mu + i\mathcal{A}_\mu. \tag{36.25}$$

The metric controls financial distance.

The gauge field controls local state orientation and phase.

They must not be double-counted.

36.9 Gauge curvature

In the Abelian case:

$$\mathbb{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu. \tag{36.26}$$

Nonzero curvature means that local frame alignment depends on path.

For a closed loop:

$$\Phi_\gamma = \oint_\gamma \mathcal{A}_\mu dx^\mu. \tag{36.27}$$

And:

$$\Phi_\gamma = \tfrac{1}{2} \iint_S \mathbb{F}_{\mu\nu} dS^{\mu\nu}. \tag{36.28}$$

This phase is gauge-invariant modulo appropriate winding.

36.10 Electric-like and magnetic-like financial fields

If θ-like time is separated from state coordinates x^i, define:

E_i^F = 𝔽_θi. (36.29)

B_ij^F = 𝔽_ij. (36.30)

Interpretively:

- E_i^F represents valuation pressure gradient across one financial direction;
- B_ij^F represents circulation or path dependence between two state directions.

Examples:

E_liquidity^F may represent pressure generated by changing liquidity. (36.31)

B_funding,collateral^F may represent loop effects between funding and collateral. (36.32)

These names are analogical and should be retained only if they improve diagnosis.

36.11 Parallel transport of derivative entanglement

A composite state transported from x_a to x_β is:

$\rho_UD(x_\beta) = U_\gamma \rho_UD(x_a) U_\gamma^\dagger$. (36.33)

Where:

$U_\gamma = \mathcal{P}\exp[-i\int_\gamma \mathcal{A}_\mu dx^\mu]$. (36.34)

If U_γ is local:

$U_\gamma = U_U,\gamma \otimes U_D,\gamma$, (36.35)

then a valid entanglement measure is preserved:

$E[\rho_UD(x_\beta)] = E[\rho_UD(x_a)]$. (36.36)

If the transport includes interaction between sectors, entanglement may change.

Thus the connection distinguishes:

Local Frame Transport. (36.37)

Joint Dynamical Coupling. (36.38)

36.12 Gauge failure

A gauge-like system fails when local re-description changes the governed object.

Define reconciliation residual:

ε_gauge = Observable_B − Transport_BA(Observable_A). (36.39)

Persistent ε_gauge may indicate:

- hidden exposure;
- incorrect numeraire transformation;
- incompatible protocol;
- data inconsistency;
- missing ledger;
- non-equivalent legal state.

Gauge failure is therefore diagnostically valuable.

36.13 Gauge fixing

To compute, one may choose a convenient gauge.

Examples:

- one benchmark phase set to zero;
- one numeraire fixed;
- one maturity channel used as reference;
- one observer frame designated as anchor.

Let:

$\chi(x)$ chosen such that $\text{Condition}[\mathcal{A}'] = 0$. (36.40)

Gauge fixing simplifies representation.

It must not change gauge-invariant predictions.

36.14 The gauge-layer proposition

Proposition 36.1 — Financial Gauge Transport

A gauge layer is justified when:

1. local valuation states have arbitrary representational orientation;
2. cross-frame comparison requires a connection;
3. relative phase or transported exposure affects observables;
4. closed-loop transport produces reproducible residual;
5. gauge-corrected variables improve reconciliation.

In compact form:

$\text{GaugeLike}_F \Leftrightarrow \text{LocalFreedom} \wedge \text{ConnectionNeed} \wedge \text{InvariantRelativeState} \wedge \text{Holonomy} \wedge$

---

37. The Integrated Effective-State Equation

37.1 Why one equation is useful

The layered architecture contains:

- local CAPM valuation;
- complex state evolution;
- derivative contracts;
- hedge backreaction;
- gauge transport;
- curved geometry;
- gate and ledger effects;
- environmental dissipation.

One schematic equation helps assign each term a role.

It should not be mistaken for a final derived financial law.

37.2 Covariant state equation

Let the global financial state be:

$|\Psi_F(x,\theta,L,P)\rangle$. (37.1)

Define the combined covariant derivative:

$\mathcal{D}_\theta = \partial_\theta + \dot{x}^\mu \nabla_\mu + i\dot{x}^\mu \mathcal{A}_\mu$. (37.2)

The effective evolution is:

$i\hbar_F \mathcal{D}_\theta |\Psi_F\rangle = \hat{H}_F[g^F,\mathcal{A},P,L]|\Psi_F\rangle + |\varepsilon_F\rangle$. (37.3)

This is the central toy equation.

37.3 Generator decomposition

Write:

$\hat{H}_F = \hat{H}_{local} + \hat{H}_{contract} + \hat{H}_{hedge} + \hat{H}_{ledger} + \hat{H}_{environment} + \hat{H}_{intervention}$. (37.4)

Where:

Ĥ_local = local asset and CAPM-seeded dynamics. (37.5)

Ĥ_contract = derivative binding and payoff structure. (37.6)

Ĥ_hedge = derivative-to-underlying feedback. (37.7)

Ĥ_ledger = history-conditioned state modification. (37.8)

Ĥ_environment = liquidity, funding, volatility, information, and institutional coupling. (37.9)

Ĥ_intervention = trading, regulation, margin, and other active controls. (37.10)

37.4 Local CAPM term

A schematic CAPM generator may act diagonally in a local asset basis:

Ĥ_CAPM|i⟩ = ℏ_Fω_i^CAPM|i⟩. (37.11)

Where:

ω_i^CAPM = F[r_f + β_iERP]. (37.12)

The exact map F depends on how expected return is converted into phase evolution.

For example:

dθ_i/dt = κ_θ[r_f + β_iERP]. (37.13)

Then:

Ĥ_CAPM = Σ_iℏ_Fω_i^CAPM|i⟩⟨i|. (37.14)

This is a model choice, not a derivation from standard CAPM.

37.5 Contract interaction term

For underlying branches u_n and derivative branches d_m:

Ĥ_contract = $\Sigma_{n,m} g_{nm}|u_n\rangle\langle u_n| \otimes |d_m\rangle\langle o_D|$ + h.c. (37.15)

Where h.c. denotes the Hermitian-conjugate term when a reversible enlarged-state model is intended.

A simpler controlled interaction is:

Ĥ_contract = $\Sigma_n|u_n\rangle\langle u_n| \otimes$ Ĥ_D,n. (37.16)

The derivative dynamics depend on the underlying branch.

This term prepares joint states.

37.6 Hedge term

Let H be hedge state.

A schematic three-sector coupling is:

$\hat{H}_{hedge} = g_H \hat{O}_D \otimes \hat{O}_H \otimes \hat{O}_U$. (37.17)

This represents:

Derivative Exposure
→ Hedge Adjustment
→ Underlying Market Effect. (37.18)

The coefficient g_H may depend on:

- aggregate gamma;
- liquidity;
- position concentration;
- execution speed;
- ledger state.

Thus:

$g_H = g_H(\Gamma_{market}, \lambda_L, L, P)$. (37.19)

37.7 Ledger term

The ledger modifies future dynamics.

Represent:

$\hat{H}_{ledger} = \hat{H}_{ledger}[L_k]$. (37.20)

After event k:

$L_k \to L_{k+1}$. (37.21)

Therefore:

$\hat{H}_{ledger}[L_{k+1}] \neq \hat{H}_{ledger}[L_k]$. (37.22)

Examples:

- barrier hit;
- margin breach;
- default;
- exercise;
- regulatory restriction.

The generator itself changes after commitment.

This makes the theory self-revising.

37.8 Environmental term

Open-system interaction may be represented through dissipators rather than only Hamiltonian terms.

The density-state equation is:

$d\rho_F/d\theta = -i[\hat{H}_F,\rho_F] + \mathcal{D}_{env}(\rho_F) + \mathcal{D}_{gate}(\rho_F) + \mathscr{R}_F$. (37.23)

Where:

$\mathcal{D}_{env}(\rho) = \Sigma_r \gamma_r(L_r \rho L_r^\dagger - \tfrac{1}{2}\{L_r^\dagger L_r,\rho\})$. (37.24)

$\mathcal{D}_{gate}$ describes commitment and branch suppression.

$\mathscr{R}_F$ is residual governance.

37.9 Gate term

A gate with outcomes g has instruments:

$\{\mathbb{M}_g\}$. (37.25)

The unconditional update is:

$\rho' = \Sigma_g \mathbb{M}_g(\rho)$. (37.26)

The conditional update is:

$\rho_g = \mathbb{M}_g(\rho)/\mathrm{Tr}[\mathbb{M}_g(\rho)]$. (37.27)

After authorization and commitment:

$L_{k+1} = L_k \sqcup \mathrm{Record}_g$. (37.28)

The post-ledger state is:

$\rho_{k+1} = \mathcal{K}_{g,L}[\rho_g]$. (37.29)

The complete runtime alternates continuous evolution and discrete commitment.

37.10 Hybrid continuous–discrete dynamics

Between gates:

$i\hbar_F \mathcal{D}_\theta|\Psi\rangle = \hat{H}_F|\Psi\rangle + |\varepsilon\rangle$. (37.30)

At gate θ_k:

$|\Psi(\theta_k^+)\rangle = G_k[|\Psi(\theta_k^-)\rangle]$. (37.31)

The ledger updates:

$L_{k+1} = L_k \sqcup \mathrm{Record}_k$. (37.32)

The protocol may revise:

$P_{k+1} = \mathcal{U}(P_k, L_{k+1}, \varepsilon_k)$. (37.33)

Thus the complete system is:

Flow → Gate → Ledger → Revision → New Flow. (37.34)

37.11 Backreaction on geometry

The state contributes effective stress:

$T^F_{\mu\nu} = T^F_{\mu\nu}[\rho_F, L, P]$. (37.35)

The metric evolves:

$G^F_{\mu\nu} + \Lambda_F g^F_{\mu\nu} = \kappa_F T^F_{\mu\nu} + \Xi^F_{\mu\nu}$. (37.36)

The changed metric alters the state generator:

$\hat{H}_F = \hat{H}_F[g^F]$. (37.37)

Therefore:

State → Stress → Geometry → State Dynamics. (37.38)

This is the GR-like feedback loop.

37.12 Backreaction on gauge connection

Financial flows may also alter phase transport:

$\mathcal{A}_\mu = \mathcal{A}_\mu[\rho_F, L, P]$. (37.39)

Then:

State → Connection → Relative Phase → State Interference. (37.40)

A derivative concentration may change:

- hedge transport;
- funding orientation;
- liquidity-phase alignment;
- cross-frame reconciliation.

This produces a self-coupled gauge structure.

37.13 Residual equation

The residual is not one scalar noise term.

Decompose:

$|\varepsilon_F\rangle = |\varepsilon_{model}\rangle + |\varepsilon_{frame}\rangle + |\varepsilon_{protocol}\rangle + |\varepsilon_{data}\rangle + |\varepsilon_{novel}\rangle$. (37.41)

Where:

- ε_{model} = dynamic model error;
- ε_{frame} = transport or basis mismatch;
- $\varepsilon_{protocol}$ = declaration failure;
- ε_{data} = measurement failure;
- ε_{novel} = genuinely new regime or unresolved structure.

Residual governance determines whether to:

- continue;
- widen the state space;
- revise the metric;
- change the protocol;
- suspend commitment;
- escalate.

37.14 Effective action form

A schematic action is:

$S_F = \int d\theta \sqrt{|g_F|}[\mathscr{L}_{state} + \mathscr{L}_{gauge} + \mathscr{L}_{geometry} + \mathscr{L}_{contract} + \mathscr{L}_{ledger} + \mathscr{L}_{residual}]$. (37.42)

Possible terms include:

$\mathscr{L}_{state} = i\hbar_F \langle \Psi | \mathcal{D}_\theta \Psi \rangle - \langle \Psi | \hat{H}_F | \Psi \rangle$. (37.43)

$\mathscr{L}_{gauge} = -\frac{1}{4}\mathbb{F}_{\mu\nu}\mathbb{F}^{\mu\nu}$. (37.44)

$\mathscr{L}_{geometry} = (1/2\kappa_F)(R_F - 2\Lambda_F)$. (37.45)

$\mathscr{L}_{contract}$ = coupling between underlying and derivative fibres. (37.46)

$\mathscr{L}_{ledger}$ = gate and history constraint terms. (37.47)

This action is a formal architecture for organizing terms.

It is not asserted as a mature financial field theory.

37.15 The integrated architecture

The full runtime is:

Primary Economic Field
→ Declaration P
→ Curved Base Point x
→ Local Frame e^a_μ
→ Complex Fibre State $|\Psi_F\rangle$
→ Contractual Coupling
→ Covariant Evolution
→ Measurement Setting
→ Gate
→ Ledger L
→ Hedge and Institutional Backreaction
→ Updated Geometry, Connection, and Protocol. (37.48)

This is the intended layered unification.

Unity comes from role separation and coupling.

It does not come from forcing one symbol to represent everything.

---

38. Reduction and Compatibility Limits

38.1 Why reduction limits are necessary

A larger theory is credible only when it recovers the simpler theories under declared conditions.

The architecture must recover:

- ordinary CAPM;
- Part I scalar Finance Geometry;
- standard derivative pricing;
- local SR-like valuation frames;
- flat financial geometry;
- classical mixtures.

Without these limits, the framework becomes disconnected from mature finance.

38.2 Ordinary CAPM limit

Require:

$\dim \mathscr{H}_F = 1$ asset channel. (38.1)

$\hat{H}_{contract} = 0$. (38.2)

$\hat{H}_{hedge} = 0$. (38.3)

$\mathcal{A}_\mu = 0$. (38.4)

$g^{F}_{\mu\nu} = \eta_{\mu\nu}$. (38.5)

$\hat{H}_{ledger} \approx 0$. (38.6)

Then:

$E[r_i] = r_f + \beta_i ERP$. (38.7)

This is the local mature-finance limit.

38.3 Part I scalar complex-state limit

Retain one complex channel:

$Z = R + iQ = Ae^{i\theta}$. (38.8)

Set:

$\dim \mathscr{H}_F = 1$. (38.9)

No tensor product. (38.10)

No gauge curvature. (38.11)

Stable declaration. (38.12)

Then:

$dZ/d\theta = iZ$. (38.13)

And:

$d^2R/d\theta^2 = -R$. (38.14)

$d^2Q/d\theta^2 = -Q$. (38.15)

This recovers the classical rotational model.

38.4 Standard option-pricing limit

Use a classical diagonal scenario state:

$\rho_{mix} = \Sigma_n q_n |u_n, d_n\rangle\langle u_n, d_n|$. (38.16)

Set:

$\rho_{off} = 0$. (38.17)

No phase-sensitive observable. (38.18)

No hedge backreaction. (38.19)

Then:

$D_0 = e^{-rT}\Sigma_n q_n f(u_n)$. (38.20)

This recovers standard risk-neutral valuation.

38.5 Classical coupled-system limit

Allow:

$\hat{H}_{contract} \neq 0$. (38.21)

$\hat{H}_{hedge} \neq 0$. (38.22)

But restrict the state to a separable classical distribution:

$\rho \in Sep(\mathscr{H}_U \otimes \mathscr{H}_D)$. (38.23)

Then the model describes:

- contractual coupling;
- conditional dependence;
- dynamical feedback;
- no-arbitrage constraints;
- ledger backreaction;

without coherent entanglement.

This may be the appropriate description for most derivative markets.

38.6 Coherent effective-state limit

Allow:

$\rho_{off} \neq 0$. (38.24)

Require a cross-branch observable:

$\hat{O}_{cross} \neq$ diagonal. (38.25)

Then:

$Tr(\rho_{coh}\hat{O}_{cross}) \neq Tr(\rho_{mix}\hat{O}_{cross})$. (38.26)

This is the minimum coherent-finance extension.

It remains an empirical hypothesis.

38.7 Local SR-like limit

Let the global metric vary slowly:

$\partial_\lambda g^F_{\mu\nu} \approx 0$. (38.27)

Then:

$\Gamma^\mu_{\alpha\beta} \approx 0$. (38.28)

A local frame relation is sufficient:

$R_\beta = \exp(-\rho_{\beta a})R_a$. (38.29)

This recovers the relative valuation-frame model.

38.8 Flat global limit

Set:

$R^\rho_{\sigma\mu\nu} = 0$. (38.30)

And:

$\mathbb{F}_{\mu\nu} = 0$. (38.31)

Then transport is path-independent.

Closed loops produce:

$\mathrm{Hol}(\gamma) = I$. (38.32)

No geometric phase or gauge holonomy remains.

38.9 Zero-ledger limit

Suppose:

$L_{k+1} = L_k$. (38.33)

Then:

$\hat{H}_{\text{ledger}} = 0$. (38.34)

No gate creates historical latching.

The world becomes reversible or memoryless relative to the declared variables.

This limit may approximate:

- purely indicative pricing;
- simulation without commitment;
- pre-trade analytics.

It cannot represent:

- settlement;
- default;
- barrier history;
- margin;

- institutional time.

38.10 No-backreaction limit

Set:

$\partial F_U/\partial D = 0$. (38.35)

$\partial F_{geometry}/\partial \rho_D = 0$. (38.36)

Then derivative states do not alter:

- underlying price;
- liquidity;
- funding;
- geometry.

The derivative becomes a passive claim on an external process.

This recovers traditional one-way pricing.

38.11 Classical observer limit

Let all measurement operators commute:

$[\hat{O}_a,\hat{O}_b] = 0$. (38.37)

Let all instruments be non-disturbing:

$\mathbb{M}_a \circ \mathbb{M}_b = \mathbb{M}_b \circ \mathbb{M}_a$. (38.38)

Let one global joint distribution exist.

Then the observer world is classically compatible.

Order and contextuality effects disappear.

38.12 No-signalling limit

For local settings:

$p(u|a,b) = p(u|a)$. (38.39)

$p(d|a,b) = p(d|b)$. (38.40)

This limit must be tested separately.

It does not follow from nonfactorization alone.

Most active financial settings will not satisfy it.

38.13 Bell-classical limit

Suppose a local hidden-state model exists:

$$p(u,d|a,b) = \int d\lambda p(\lambda)p(u|a,\lambda)p(d|b,\lambda). \quad (38.41)$$

Then:

$$|S_F| \leq 2 \quad (38.42)$$

under the standard CHSH assumptions.

The financial system remains Bell-classical even if:

- it is contractually bound;
- its state is globally complex;
- it exhibits strong correlation;
- local observers experience conditional collapse.

This is likely the default financial regime.

38.14 Compatibility hierarchy

The full theory should satisfy:

$$\begin{aligned}&\text{Ordinary CAPM}\\ &\subset \text{Scalar Complex Finance}\\ &\subset \text{Classical Composite Derivative Finance}\\ &\subset \text{Curved and Gauge-Connected Finance}\\ &\subset \text{Coherent Effective Financial State Model.}\end{aligned} \quad (38.43)$$

Each extension adds assumptions.

Each must show measurable gain.

The hierarchy must not be reversed rhetorically.

38.15 Contradictions to avoid

The following identifications are prohibited unless separately derived:

$$\theta = \eta_Q. \quad (38.44)$$

$$\eta_Q = \rho_D. \quad (38.45)$$

$$\beta_{CAPM} = \text{relativistic velocity}. \quad (38.46)$$

$$Q_E = Q_H. \quad (38.47)$$

$$\text{Gate Commitment} = \text{Born Collapse}. \quad (38.48)$$

$$\text{Contractual Coupling} = \text{Quantum Entanglement}. \quad (38.49)$$

State-Dependent Metric = Einstein Field Equation. (38.50)

Conditional Update = No-Signalling Nonlocality. (38.51)

Path Dependence = Quantum Geometric Phase. (38.52)

Risk-Neutral Weight = Born Probability. (38.53)

These distinctions protect the architecture from analogy collapse.

38.16 The compatibility proposition

Proposition 38.1 — Layered Reduction Principle

The QM–SR–GR financial architecture is internally compatible only when:

1. each mathematical layer has a distinct role;
2. local laws are recovered in weak-coupling limits;
3. frame, metric, gauge, and protocol transformations are not conflated;
4. stronger quantum-like claims require stronger operational evidence;
5. every extension can be removed when it adds no measurable value.

In compact form:

Compatibility_F ⇔ RoleSeparation ∧ ReductionRecovery ∧ TransformationDiscipline ∧ EvidenceGradient ∧ Removability. (38.54)

The layered architecture is now mathematically assembled.

The next stage is to repeat the quantum subtraction more carefully.

Part I asked which apparent quantum mysteries could already arise from projection, phase, gates, trace, and observer-bounded world formation.

Part II can now ask a stronger question:

> After adding composite derivative states, internal observers, relative valuation frames, curved geometry, gauge transport, conditional collapse, and ledger backreaction, which features of quantum mechanics still resist reconstruction in a non-quantum financial world?

Part VII — Revising the Quantum Subtraction

39. What Part I Already Reproduced

39.1 The purpose of quantum subtraction

The purpose of the financial construction is not to prove that markets are quantum systems.

Its purpose is methodological.

Before treating a strange phenomenon as uniquely quantum, construct the strongest non-quantum observer-bearing control world capable of reproducing similar operational structure.

Then subtract.

The original subtraction was:

Quantum Phenomenon = Generic Observer-Bound Structure + Irreducibly Quantum Residue. (39.1)

Part I supplied a first control world based on:

- declared projection;
- complex valuation geometry;
- phase progression;
- contextual measurement;
- gate commitment;
- ledger trace;
- observer backreaction;
- internal time.

Part II has now enlarged that control world.

Before repeating the subtraction, the achievements of Part I should be stated precisely.

39.2 Complex representation

Part I represented admitted value and retained pressure as:

$Z = R + iQ$. (39.2)

With:

$A^2 = R^2 + Q^2$. (39.3)

$R = A \cos \theta$. (39.4)

$Q = A \sin \theta$. (39.5)

$Z = A \exp(i\theta)$. (39.6)

This demonstrated that a complex coordinate can arise from a mature non-quantum valuation filter.

Therefore:

Use of Complex Numbers ≠ Evidence of Quantum Ontology. (39.7)

Complex numbers may encode:

- orthogonal financial dimensions;
- phase;
- rotation;
- pressure retention;
- hidden coordinate structure.

39.3 Phase evolution

With stable amplitude:

$dZ/d\theta = iZ.$ (39.8)

And:

$d^2Z/d\theta^2 = -Z.$ (39.9)

The real and imaginary coordinates obey:

$dR/d\theta = -Q.$ (39.10)

$dQ/d\theta = R.$ (39.11)

This reproduced oscillatory and phase-bearing behaviour in a classical system.

Therefore:

Oscillation + Complex Phase ≠ Quantum Mechanics. (39.12)

The presence of phase must be distinguished from specifically quantum coherence.

39.4 Rotating measurement settings

Part I reinterpreted valuation rotation as movement of the measurement setting rather than literal rotation of the asset.

Let primary field be X.

Protocol P selects observable:

$\hat{O}_P.$ (39.13)

The observed state is:

$Z_P = \hat{O}_P[X].$ (39.14)

Changing protocol gives:

$Z_P' = \hat{O}_P'[X].$ (39.15)

Thus:

ObservedChange = PrimaryChange + Measurement-FrameChange. (39.16)

This reproduces an important element of quantum measurement language:

The observable outcome depends upon which basis or projection is applied.

But:

Basis Dependence ≠ Quantum Contextuality. (39.17)

A classical system can also generate setting-dependent observations.

39.5 Order effects

Suppose two financial operations A and B are applied sequentially.

Then:

B∘A(X) may differ from A∘B(X). (39.18)

Equivalently:

[A,B]X ≠ 0. (39.19)

Examples include:

- downgrade before collateral posting;
- liquidation before accounting recognition;
- hedge adjustment before volatility recalibration;
- legal declaration before settlement.

This reproduces operational noncommutativity.

Therefore:

Order Dependence ≠ Automatically Quantum Noncommutativity. (39.20)

Classical gates, path dependence, memory, and institutional sequencing can generate the same formal pattern.

39.6 Contextual projection

The same primary field may generate different outputs under different declarations:

Visible_P(X) ≠ Visible_P′(X). (39.21)

The result depends on:

- boundary;
- observable;
- horizon;
- protocol;

- ledger;
- intervention authority.

This reproduces observer-relative effective reality.

But:

Protocol Relativity ≠ Subjective Arbitrariness. (39.22)

A projection remains constrained by:

- declared rules;
- empirical data;
- cross-frame consistency;
- residual;
- auditability.

39.7 Gate commitment

A projected state becomes historical only after passing a gate:

State → Gate → Record. (39.23)

The ledger updates:

$L_{k+1} = L_k \sqcup \text{Record}_k$. (39.24)

This reproduces a collapse-like transition:

Possibility
→ Selection
→ Committed Outcome. (39.25)

But:

Financial Gate Commitment ≠ Born-Rule Collapse. (39.26)

The financial gate may be:

- deterministic;
- threshold-based;
- legally defined;
- institutionally authorized;
- statistically estimated.

No universal amplitude-squared probability law follows.

39.8 Internal certainty

Once an outcome enters an observer's filtration:

$P_O(\text{PastOutcome} = y_k \mid \mathscr{F}_{O,k}) = 1.$ (39.27)

The observer experiences its recorded past as fixed.

This reproduces:

- internal definiteness;
- trace-based certainty;
- history-dependent policy;
- branch latching.

It shows that definite past experience can arise from trace-bearing observer dynamics.

It does not yet explain why quantum outcomes obey the Born rule.

39.9 Cross-observer agreement

Agreement becomes possible when observers possess:

1. a valid frame map;
2. compatible measurement effects;
3. access to a shared record;
4. sufficient redundancy.

Thus:

Objectivity = Frame Reconciliation + Compatible Effects + Accessible Trace + Redundancy. (39.28)

This shows that observer-relative measurement need not destroy objectivity.

Objectivity can emerge through stable agreement structures rather than through an observer-free view from nowhere.

39.10 Internal time

Part I distinguished:

t = calendar time. (39.29)

θ = phase order. (39.30)

k = ledger order. (39.31)

It proposed:

Effective Time = Phase Progression + Gate + Trace. (39.32)

This reproduced a world in which temporal order is generated internally through state change and committed history.

Therefore:

Internal Time Need Not Be Fundamental External Time. (39.33)

A system can carry its own effective clock through recursive disclosure and retained consequence.

39.11 Backreaction

Part I showed that financial observations alter the world they describe:

Valuation
→ Margin
→ Funding
→ Trading
→ New Valuation. (39.34)

Therefore:

Observer Output → Primary-State Change. (39.35)

This reproduces an observer effect without quantum mechanics.

A measurement can be performative because the institution acts on the result.

39.12 Residual governance

Part I retained:

ε = Observation − Declared Model. (39.36)

The residual was not treated automatically as noise.

It could indicate:

- model failure;
- frame mismatch;
- omitted variable;
- protocol failure;
- new regime;
- unresolved structure.

This prevents the financial world from pretending to be complete.

Residual is the boundary through which the larger primary universe remains visible.

39.13 Part I control-world grammar

The first control world therefore contained:

G_world = Declaration + Projection + Phase + Gate + Trace + Backreaction + Residual. (39.37)

This grammar already reproduced:

- complex phase;
- measurement-setting dependence;
- order sensitivity;
- internal collapse-like commitment;
- local observer certainty;
- cross-observer agreement;
- internal time;
- observer backreaction.

The remaining quantum residue after Part I was therefore smaller than a naïve list of quantum mysteries.

But the control world still lacked composite state structure.

That is what Part II adds.

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40. What Part II Adds

40.1 From one complex coordinate to composite state space

Part II introduced:

$\mathscr{H}_{UD} = \mathscr{H}_{U} \otimes \mathscr{H}_{D}$. (40.1)

This is the minimum structure needed to distinguish:

- local underlying state;
- local derivative state;
- global joint state;
- local observables;
- joint observables;
- reduced states;

- product states;
- separable mixtures;
- nonfactorizable states.

Part I could not ask whether a state was entangled because it lacked a subsystem partition.

Part II can ask the question formally.

40.2 Contractual state preparation

The derivative contract became a preparation operator:

$\hat{U}_C(|u_n\rangle|0_D\rangle) = |u_n,d_n\rangle$. (40.2)

For a multi-branch underlying state:

$|\psi_U\rangle = \Sigma_n c_n|u_n\rangle$, (40.3)

the prepared composite state is:

$|\Psi_{UD}\rangle = \Sigma_n c_n|u_n,d_n\rangle$. (40.4)

Thus a mature financial contract supplies:

- subsystem identity;
- joint admissibility;
- branch pairing;
- shared gate structure;
- relational state preparation.

This is substantially stronger than ordinary correlation.

40.3 Global completeness and local mixedness

The global state may be pure:

$\rho_{UD} = |\Psi_{UD}\rangle\langle\Psi_{UD}|$. (40.5)

Yet local observers receive:

$\rho_U = Tr_D(\rho_{UD})$. (40.6)

$\rho_D = Tr_U(\rho_{UD})$. (40.7)

The local states may be mixed.

Thus:

Global Completeness + Local Incompleteness (40.8)

can arise inside a non-quantum effective-world construction.

This reproduces an important part of the operational strangeness associated with entanglement.

But formal purification of a classical distribution remains possible.

Local mixedness alone does not prove quantum coherence.

40.4 Conditional joint update

Measurement of the underlying sector produces:

$$\rho_{D|u,a} = \mathrm{Tr}_U[(\mathbb{M}^U_{a,u} \otimes I_D)(\rho_{UD})] / p(u|a). \quad (40.9)$$

Measurement of the derivative sector produces:

$$\rho_{U|d,b} = \mathrm{Tr}_D[(I_U \otimes \mathbb{M}^D_{b,d})(\rho_{UD})] / p(d|b). \quad (40.10)$$

This reproduces:

Local Measurement
→ Conditional State of Remote Sector. (40.11)

Inside the secondary θ-world, no explicit messenger appears in the conditional state formula.

The correlation belongs to the global state.

However, the primary financial universe may still contain ordinary information and causal channels.

40.5 Entanglement from the internal viewpoint

Part II established three levels:

Primary Constructor Universe
→ Secondary Valuation World
→ Internal Observer. (40.12)

From the primary universe:

The contract and market mechanism explain the option–underlying relation.

From the secondary world:

The relation is given as one prepared composite state.

From the local observer:

Only one reduced sector is directly accessible.

Thus:

External Constructibility ≠ Internal Separability. (40.13)

This is the main perspectival advance of Part II.

40.6 Multi-channel phase

Part II replaced one global phase with:

$\Theta_F = \{\theta_1,\theta_2,\ldots,\theta_N\}$. (40.14)

Relative phase is:

$\Delta\theta_{nm} = \theta_n - \theta_m$. (40.15)

This supplies the minimum structure for interference-form dynamics.

The state becomes:

$|\Psi_F\rangle = \Sigma_n c_n A_n \exp(i\theta_n)|n\rangle$. (40.16)

A global phase is unobservable.

Only relative phase or loop phase can affect joint outcomes.

40.7 Interference-form algebra

When channels combine before commitment:

$\mathcal{A}(y) = \Sigma_n \alpha_n \mathcal{A}_n(y)$. (40.17)

Then:

$|\mathcal{A}(y)|^2 = \Sigma_n |\alpha_n \mathcal{A}_n|^2 + \Sigma_{n\neq m} \alpha_n \alpha_m \mathcal{A}_n \mathcal{A}_m$. (40.18)

The cross terms permit:

- constructive interaction;
- destructive interaction;
- phase-sensitive outcome changes.

Finance already contains nonlinear interaction, netting, and feedback.

Therefore Part II introduced a demanding criterion:

Interference becomes operationally meaningful only when phase-sensitive cross terms outperform flexible classical alternatives.

40.8 Decoherence-like processes

Part II modelled environment-induced branch suppression:

$|\Psi_SE\rangle = \Sigma_n c_n |s_n\rangle|e_n\rangle$. (40.19)

The reduced state is:

$\rho_S = \Sigma_{n,m} c_n c_m^* \langle e_m|e_n\rangle |s_n\rangle\langle s_m|$. (40.20)

When environmental records become distinguishable:

$\langle e_m | e_n \rangle \to 0$ for $m \neq n$. (40.21)

Off-diagonal terms disappear.

Financial environments capable of recording branch distinctions include:

- public prices;
- trades;
- clearing;
- settlement;
- legal records;
- regulatory reports;
- margin systems.

This produces institutional classicalization.

40.9 Geometric phase and holonomy

Part II added gauge transport:

$D_\mu = \partial_\mu + i\mathcal{A}_\mu$. (40.22)

The loop phase is:

$\Phi_\gamma = \oint_\gamma \mathcal{A}_\mu dx^\mu$. (40.23)

A closed visible loop may produce:

$x_{final} = x_{initial}$, (40.24)

while:

$|\psi_{final}\rangle \neq |\psi_{initial}\rangle$, (40.25)

or:

$L_{final} \neq L_{initial}$. (40.26)

This formalizes:

Same Visible Coordinate + Different Path → Different Full State. (40.27)

Finance already contains hysteresis and path dependence.

The geometric formulation is justified only when it supplies invariant predictive structure beyond standard history-dependent models.

40.10 SR-like local frames

Part II distinguished:

θ = circular valuation phase. (40.28)

η_Q = Q-preserving hyperbolic rapidity. (40.29)

ρ_D = additive discount rapidity. (40.30)

It used:

ρ_D = ln(A/R). (40.31)

And relative frame composition:

ρ_CA = ρ_CB + ρ_BA. (40.32)

This supplies an exact additive frame coordinate for multiplicative valuation transformations.

But:

β_CAPM ≠ tanh ρ_D. (40.33)

No financial speed of light has been derived.

40.11 GR-like curved base

Part II introduced:

ds_F² = g^F_μν(x,L,P)dx^μdx^ν. (40.34)

The metric may depend on:

- liquidity;
- leverage;
- volatility;
- funding;
- collateral;
- market depth;
- ledger state.

Local flatness gives:

g^F_μν(x0) ≈ η_μν. (40.35)

Global curvature gives:

R^ρ_σμν ≠ 0. (40.36)

This permits local CAPM laws to remain useful while failing globally.

40.12 Gauge connection

The gauge connection transports local valuation orientation:

$\mathcal{D}_\mu = \nabla_\mu + i\mathcal{A}_\mu$. (40.37)

The metric connection and gauge connection remain distinct:

$\Gamma^\mu_{\alpha\beta}$ transports base-space vectors. (40.38)

$\mathcal{A}_\mu$ transports fibre phase and orientation. (40.39)

This avoids double-counting.

40.13 Backreaction on geometry

The state generates effective stress:

$T^F_{\mu\nu} = T^F_{\mu\nu}[\rho_F,L,P]$. (40.40)

A schematic geometry equation is:

$G^F_{\mu\nu} + \Lambda_F g^F_{\mu\nu} = \kappa_F T^F_{\mu\nu} + \Xi^F_{\mu\nu}$. (40.41)

This does not establish a financial Einstein equation.

It defines the stronger requirement for a genuine GR-like architecture:

The state must measurably alter the geometry through which later states move.

40.14 Part II composite grammar

The additional control-world grammar is:

G_composite = Partition + Joint Preparation + Local Restriction + Conditional Update + Relative Phase + Covariant Transport + Shared Gate. (40.42)

The total non-quantum control architecture becomes:

G_control = G_world + G_composite + G_geometry. (40.43)

Where:

G_geometry = Local Frames + Metric + Connection + Curvature + Holonomy. (40.44)

The quantum subtraction must now be repeated against this stronger control world.

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41. The Entanglement Ladder

41.1 Why entanglement must be decomposed

The word entanglement is often used too broadly.

It may refer to:

- correlation;
- interdependence;
- contractual binding;
- inability to analyse components independently;
- tensor nonfactorization;
- coherent nonseparability;
- Bell-nonclassicality.

These levels are not equivalent.

Part II therefore defines a nine-level ladder.

41.2 E1 — Statistical correlation

Definition:

$p(a,b) \neq p(a)p(b)$. (41.1)

Examples:

- two stocks exposed to the same sector;
- an option and its underlying;
- two institutions sharing funding conditions.

E1 requires no composite ontology.

A latent common cause may explain the relationship.

Finance clearly realizes E1.

41.3 E2 — Functional dependence

Definition:

$B = F(A,X)$. (41.2)

For an option:

$D = V(U,\sigma,r,q,K,T-t,P,L)$. (41.3)

The state of B depends functionally on A and other variables.

This is stronger than raw correlation because the relation has structural direction.

Finance clearly realizes E2.

41.4 E3 — Contractual coupling

The identity of the derivative is constituted through the relation:

$I_D = (U,K,T,Payoff,Exercise,Settlement,Law)$. (41.4)

Changing the underlying or payoff creates a different derivative.

Thus:

Derivative Identity Is Relational. (41.5)

Finance clearly realizes E3.

41.5 E4 — Dynamical binding

The joint dynamics contain cross-sector effects:

$\partial F_U/\partial D \neq 0$. (41.6)

$\partial F_D/\partial U \neq 0$. (41.7)

Examples include:

- delta hedging;
- volatility recalibration;
- margin feedback;
- collateral pressure;
- market impact.

Finance clearly realizes E4.

41.6 E5 — Effective-world nonfactorization

The state satisfies:

$\rho_{AB} \neq \rho_A \otimes \rho_B$. (41.8)

A stronger claim is:

$\rho_{AB} \notin Sep(\mathscr{H}_A \otimes \mathscr{H}_B)$. (41.9)

The state cannot be reconstructed as a convex mixture of independent local states under the declared representation.

Finance can construct E5 formally.

But the claim depends on:

- partition;

- state estimation;
- measurement algebra;
- classical null family.

41.7 E_6 — Observable coherent composite state

The state contains operationally relevant off-diagonal terms:

$\rho_{off} \neq 0$. (41.10)

There exists an observable $\hat{O}_{cross}$ such that:

$\mathrm{Tr}(\rho_{coh}\hat{O}_{cross}) \neq \mathrm{Tr}(\rho_{diag}\hat{O}_{cross})$. (41.11)

Relative phase affects an outcome.

Finance has not yet established E_6 as a general market property.

Some decision and contextual-probability models may use this structure effectively.

But such success would first establish model utility, not physical quantum coherence.

41.8 E_7 — Irreducible contextual joint measurement

No single noncontextual value assignment reproduces all compatible measurement contexts.

Formally:

No map v exists satisfying all context constraints. (41.12)

Finance commonly contains:

- path dependence;
- adaptive policies;
- measurement disturbance;
- institutional context;
- signalling.

These can mimic contextuality.

E_7 therefore requires rejection of rich classical contextual and causal models.

It remains unestablished.

41.9 E_8 — No-signalling entanglement

Strong correlations coexist with:

$p(a|x,y) = p(a|x)$. (41.13)

$p(b|x,y) = p(b|y)$. (41.14)

The distant setting does not alter the local marginal.

Financial markets usually contain ordinary signalling and backreaction.

E_8 is therefore not established in ordinary derivative systems.

41.10 E_9 — Bell-nonclassical entanglement

The joint statistics violate a Bell inequality under valid experimental assumptions.

For CHSH:

$|S| > 2$. (41.15)

A credible claim must control:

- setting independence;
- signalling;
- common causes;
- memory;
- sampling;
- detection;
- post-selection;
- adaptive behaviour.

No Bell-nonclassical financial result is claimed here.

41.11 The ladder table

Level	Structure	Finance status
E_1	Statistical correlation	Established
E_2	Functional dependence	Established
E_3	Contractual coupling	Established
E_4	Dynamical binding	Established
E_5	Effective-state nonfactorization	Formally constructible
E_6	Observable coherent state	Open empirical question
E_7	Irreducible contextuality	Not established
E_8	No-signalling entanglement	Not established
E_9	Bell-nonclassicality	Not established

41.12 Why options remain central

Options reach farther up the ladder than ordinary correlated assets because they provide:

- explicit relational identity;
- natural subsystem partition;
- contractual preparation;
- joint admissibility constraints;
- conditional local states;
- exercise and settlement gates;
- hedge backreaction;
- shared ledger history.

Thus derivatives are not merely correlated securities.

They are the natural financial platform for testing composite-state structure.

The strongest currently defensible claim is:

Options clearly realize E_1–E_4 and provide a principled formal route toward testing E_5. (41.16)

41.13 Observer perspective and the ladder

Observer restriction is relevant beginning at E_5.

For E_1–E_4, ordinary external analysis may suffice.

At E_5 and above:

- global state matters;
- local reductions matter;
- measurement algebra matters;
- internal accessibility matters;
- frame transformation matters.

Thus:

Entanglement Strangeness Begins When Local Completeness Fails. (41.17)

The primary universe may explain how the relation was prepared.

The internal observer experiences the relational state after preparation.

---

42. What Still Remains Quantum

42.1 The residue is smaller, but not empty

After constructing:

- complex phase;
- observer-relative projection;
- internal time;
- composite financial states;
- conditional collapse;
- noncommuting gates;
- curved geometry;
- gauge transport;
- ledger backreaction;

many features commonly described as quantum-like become reproducible in a non-quantum effective world.

But several central features remain unresolved.

The purpose of the stronger financial control world is not to eliminate the quantum residue by verbal analogy.

It is to define that residue more sharply.

42.2 Born-rule necessity

The financial amplitude lift defines:

$\alpha_n = \sqrt{q_n}\exp(i\varphi_n)$. (42.1)

But this is imposed.

The mature financial model begins with q_n.

It does not derive:

$q_n = |\alpha_n|^2$ (42.2)

as a necessary outcome law from more primitive financial assumptions.

A quantum theory must explain why measurement frequencies are governed by squared amplitude rather than:

$|\alpha|$, $|\alpha|^4$, or another function.

Thus the residue includes:

Why Probability = Squared Complex Magnitude. (42.3)

The financial construction reproduces the form only by declaration.

42.3 Empirically mandatory coherence

Finance can introduce relative phase and off-diagonal density terms.

But standard derivative markets are already modelled successfully using:

- stochastic processes;
- state-price densities;
- stochastic volatility;
- jumps;
- copulas;
- agent interactions;
- path dependence.

The stronger question is:

Are there reproducible financial observations that require coherent amplitude addition and cannot be explained by any adequate classical model?

No such general result is established.

Quantum experiments, by contrast, require interference-capable amplitude structure across a wide range of phenomena.

Therefore the residue includes:

Empirical Necessity of Coherent Amplitudes. (42.4)

42.4 Genuine tensor nonseparability

A financial contract may be written in tensor form.

A classical correlated distribution can also be purified formally into a nonfactorizable vector.

Therefore:

Tensor Representation ≠ Physical Nonseparability. (42.5)

A stronger claim requires showing that no separable classical state reproduces the complete measurement statistics.

The financial construction has not done this.

Thus the residue includes:

Nonseparability That Cannot Be Removed by Classical Completion. (42.6)

42.5 Bell inequality violation

Bell-nonclassicality excludes a broad class of local hidden-variable models under explicit assumptions.

Financial markets contain:

- common causes;
- communication;
- adaptive settings;
- memory;
- signalling;
- strategic behaviour.

These features make apparent Bell-style violations comparatively easy to produce without quantum structure.

The financial model has not derived:

$|S_F| > 2$ (42.7)

under loophole-controlled conditions.

Thus Bell nonclassicality remains a major quantum residue.

42.6 No-signalling with strong nonlocal correlation

Derivative markets readily produce strong correlations.

But active measurement settings often change:

- price;
- liquidity;
- information;
- hedge flow.

Therefore:

$p(u|a,b)$ generally depends on b. (42.8)

Quantum entanglement combines strong correlation with no controllable superluminal communication.

The financial construction does not yet reproduce this combination.

Thus the residue includes:

Strong Correlation without Controllable Signalling. (42.9)

42.7 Irreducible contextuality

Financial outcomes are contextual because:

- measurement protocols differ;
- sequence matters;
- agents adapt;
- interventions alter the state;
- ledgers preserve history.

But classical contextual systems can exhibit all these features.

Quantum contextuality is stronger.

It concerns the impossibility of assigning globally consistent values to observables while preserving their contextual compatibility relations.

The financial model has not excluded:

- contextual hidden variables;
- memory models;
- causal models;
- adaptive models.

Thus irreducible contextuality remains unresolved.

42.8 No-cloning

Financial information can generally be copied.

A price, contract description, valuation vector, or portfolio file may be duplicated.

Copying may be imperfect because:

- state changes;
- markets move;
- data are incomplete;
- observer access differs.

But these are practical limitations.

Quantum no-cloning is algebraic:

No universal operation can map:

$|\psi\rangle|0\rangle \rightarrow |\psi\rangle|\psi\rangle$ (42.10)

for every unknown quantum state $|\psi\rangle$.

The financial framework has not derived an equivalent impossibility.

Thus no-cloning remains a specifically quantum residue.

42.9 Specifically quantum disturbance

Financial measurement may disturb the object because:

- trades have market impact;
- disclosures change expectations;
- margin rules change behaviour;
- observers adapt.

This is causal and institutional disturbance.

Quantum measurement disturbance can arise from noncommuting observable structure even without ordinary mechanical market impact.

The financial model has not shown that disturbance is fundamentally unavoidable rather than technologically or institutionally contingent.

Thus:

Practical Observer Disturbance ≠ Fundamental Quantum Disturbance. (42.11)

42.10 Monogamy of entanglement

Quantum entanglement obeys distinctive distribution constraints.

If two systems are maximally entangled, they cannot be equally maximally entangled with arbitrary third systems.

Financial dependencies do not generally obey this restriction.

An underlying can be strongly coupled contractually to many derivatives simultaneously.

Thus:

Financial Shared Dependence ≠ Quantum Entanglement Monogamy. (42.12)

A financial analogue would require a conserved or bounded relational resource.

No such universal law has been derived.

42.11 Tsirelson-type bounds

Quantum correlations can exceed classical Bell bounds but remain below quantum limits.

For CHSH:

$|S_{quantum}| \leq 2\sqrt{2}$. (42.13)

General signalling or unconstrained contextual systems can exceed this.

A financial system can produce arbitrary apparent correlation patterns through:

- communication;
- common control;
- post-selection;
- adaptive sampling.

The existence of specifically quantum upper bounds is therefore part of the residue.

42.12 Quantum teleportation

Financial state information can be transmitted through communication, copying, contractual transfer, or replication.

Quantum teleportation transfers an unknown quantum state using:

- shared entanglement;
- classical communication;
- local measurement;

without physically transporting the original carrier state.

No equivalent necessity has been demonstrated in finance.

Any financial teleportation analogy would presently remain structural metaphor.

42.13 Quantum indistinguishability

Financial instruments are normally distinguishable by:

- contract identifier;
- owner;
- issuer;
- maturity;
- ledger history;
- legal rights.

Even economically identical instruments may remain distinguishable institutionally.

Quantum particles of one species possess much stronger indistinguishability, leading to Bose–Einstein or Fermi–Dirac statistics.

The financial framework has not reproduced this ontological indistinguishability.

42.14 Exchange statistics

Quantum systems can acquire:

- symmetric exchange structure;
- antisymmetric exchange structure;
- more general anyonic phases in suitable dimensions.

Financial object exchange does not naturally generate universal statistics of this kind.

Contract role, ownership, and ledger identity remain relevant.

Thus exchange statistics remain outside the current control-world reconstruction.

42.15 The refined quantum residue

The surviving residue can be summarized as:

Q_residue = Born Necessity

- Empirically Mandatory Coherence
- Irreducible Nonseparability
- No-Signalling Correlation
- Bell Violation
- Fundamental Contextuality
- No-Cloning
- Quantum Disturbance
- Monogamy
- Quantum Statistical Structure. (42.15)

Some elements may later admit broader non-quantum reconstructions.

But Part II has not derived them.

42.16 The epistemic lesson

The control-world analysis changes the question.

Instead of asking:

“Why is quantum mechanics strange?”

ask:

“Which part of the strangeness comes from being a bounded observer inside a generated relational world, and which part requires specifically quantum probability and algebra?”

The answer after Part II is:

A substantial part of the experienced strangeness may arise from:

- global preparation;
- local access;
- contextual projection;
- trace-bearing commitment;
- internal time;
- hidden constructor;
- curved and gauge-connected transport.

But the specifically quantum residue remains in the empirically constrained probability and algebraic structure.

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43. The Revised Subtraction Equation

43.1 First subtraction

Part I used:

Quantum Phenomenon = G_world + Q_residue[1]. (43.1)

Where:

G_world = Declaration + Projection + Phase + Gate + Trace + Backreaction. (43.2)

This removed generic observer-world structure.

43.2 Second subtraction

Part II adds:

G_composite = Partition + Joint Preparation + Local Restriction + Conditional Update + Relative Phase + Joint Gate. (43.3)

It also adds:

G_geometry = Local Frame + Metric + Connection + Curvature + Holonomy. (43.4)

The revised equation is:

Quantum Observation = G_world + G_composite + G_geometry + Q_residue[2]. (43.5)

Where:

$Q_residue^2 \subset Q_residue^1$. (43.6)

The second residue is more precise.

43.3 Expanded form

Write:

Quantum Observation
= Declaration

- Projection
- Complex State
- Internal Phase
- Composite Preparation
- Local Observer Restriction
- Conditional State Update
- Gate Commitment
- Ledger Trace
- Frame Transport
- Curved Geometry
- Backreaction
- Specifically Quantum Residue. (43.7)

This equation is conceptual.

It is not additive in a simple numerical sense.

It describes layers of explanatory structure.

43.4 Observer-strangeness decomposition

Define apparent entanglement strangeness:

$S_ent = S_partition + S_global + S_local + S_temporal + S_residual$. (43.8)

Where:

$S_partition$ = distinct local subsystem identities. (43.9)

S_global = global state contains irreducible relational information. (43.10)

S_local = observers access only reduced states. (43.11)

S_temporal = one internal episode commits correlated outcomes. (43.12)

S_residual = remaining nonclassical probability structure. (43.13)

The first four components may be reproduced in non-quantum effective worlds.

The final component requires further analysis.

43.5 Finance as the stronger control world

The financial control world now contains:

- mature scalar valuation;
- complex completion;
- multi-channel phase;
- derivative tensor structure;
- internal observers;
- adaptive measurement;
- gate and ledger;
- local frame relativity;
- curved geometry;
- gauge transport;
- source–geometry backreaction.

This is a much stronger control than an ordinary classical particle model.

It contains observer-dependent world formation and recursive institutional reality.

Therefore any quantum feature surviving this subtraction deserves sharper attention.

43.6 The one-way implication that must be rejected

The following inference is invalid:

Finance reproduces a structure associated with quantum theory
→ therefore finance is physically quantum. (43.14)

The valid inference is:

Finance reproduces the structure
→ therefore that structure alone is insufficient to identify specifically quantum ontology. (43.15)

This is the methodological value of the analogy.

43.7 The reverse implication

A second inference becomes possible:

A structure arises naturally in finance from observer restriction, composite preparation, gates, and ledgered history
→ a similar structure in physics may also contain a generic observer-world component. (43.16)

This does not prove that quantum mechanics has a hidden classical financial-like constructor.

It identifies a research possibility:

Some quantum strangeness may belong to the architecture of bounded observers inside generated worlds rather than to the ultimate ontology alone.

43.8 Functional homology versus material identity

The final comparison rule is:

Functional Homology ≠ Material Identity. (43.17)

Two systems may share:

- the same state grammar;
- the same observer restriction;
- the same gate structure;
- the same trace logic;
- the same geometric transport pattern;

while differing entirely in physical substrate.

Therefore:

Same Mathematics does not imply Same Material Reality. (43.18)

But:

Same Mathematics may reveal the same control problem. (43.19)

This is the legitimate role of the cross-domain comparison.

43.9 The refined thesis

The second subtraction supports the following thesis:

> A substantial portion of entanglement's experienced strangeness can be reconstructed as the perspective of locally bounded observers inhabiting a secondary world whose globally prepared composite states are disclosed through contextual measurements and fixed through trace-bearing events. Derivative finance provides a non-quantum control world in which this observer-relative structure can be modelled explicitly. What remains specifically quantum is not jointness, context, collapse-like

commitment, phase, curvature, or local incompleteness by themselves, but the empirically constrained probability and algebraic structure that resists adequate classical reconstruction.

Part VIII — Measurement, Simulation, and Falsification

44. Operational Variables

44.1 Why operationalization is decisive

The architecture now contains many powerful concepts:

- retained pressure;
- internal phase;
- frame rapidity;
- metric curvature;
- gauge connection;
- composite state;
- local mixedness;
- interference;
- holonomy;
- entanglement witness.

Without measurable proxies, these terms can become decorative.

Each proposed variable must specify:

OperationalVariable = Definition + Data + Estimator + Intervention + Failure Condition. (44.1)

44.2 Estimating admitted value R

R may be defined as:

- discounted cash-flow value;
- CAPM-filtered value;
- market-admitted price;
- accounting carrying value;
- liquidation value;

- collateral-recognized value.

The protocol must state:

R_P = Value admitted under protocol P. (44.2)

Data may include:

- price;
- cash flow;
- discount rate;
- beta;
- risk premium;
- liquidity adjustment;
- legal constraint.

Different protocols produce different R.

The index P must remain visible.

44.3 Estimating amplitude A

A is the declared pre-filter amplitude.

Candidate definitions include:

A = undiscounted terminal amount. (44.3)

A = pre-risk-adjusted economic value. (44.4)

A = scenario-weighted gross possibility scale. (44.5)

A must not be selected merely to force:

$A^2 = R^2 + Q^2$. (44.6)

Its economic meaning must be independent of the complex completion.

Otherwise Q becomes circularly manufactured.

44.4 Estimating retained pressure Q

Given A and R:

$Q = \sqrt{(A^2 - R^2)}$. (44.7)

This requires:

$A \geq |R|$. (44.8)

When the condition fails, the chosen amplitude or geometry is invalid.

Candidate interpretation:

Q = valuation pressure retained outside admitted value under the declared Euclidean completion. (44.9)

The model should test whether Q predicts:

- repricing;
- volatility;
- gate proximity;
- residual;
- future protocol revision.

44.5 Estimating θ

The local phase is:

θ = arccos(R/A). (44.10)

For signed orientation, use:

θ = atan2(Q,R). (44.11)

The estimator must address:

- branch choice;
- phase wrapping;
- discontinuity;
- sign convention;
- regime change.

A continuous unwrapped phase may be:

θ̃_t = Unwrap[atan2(Q_t,R_t)]. (44.12)

The phase becomes useful when:

dθ̃/dt (44.13)

predicts observable transition better than ordinary return or spread variables.

44.6 Estimating discount rapidity

Define:

ρ_D = ln(A/R). (44.14)

For relative frames a and b:

$\rho_{BA} = \ln(R_A/R_B)$. (44.15)

The composition test is:

$\rho_{CA} \approx \rho_{CB} + \rho_{BA}$. (44.16)

Residual:

$\varepsilon_\rho = \rho_{CA} - \rho_{CB} - \rho_{BA}$. (44.17)

Persistent ε_ρ indicates:

- non-equivalent frames;
- hidden protocol change;
- data inconsistency;
- nonlinear path dependence.

44.7 Estimating the financial metric

Candidate metric estimators include:

Fisher metric:

$g^F_{\mu\nu} = E[(\partial_\mu \ln p)(\partial_\nu \ln p)]$. (44.18)

Inverse covariance metric:

$g^F = \Sigma^{-1}$. (44.19)

Execution-cost metric:

$\text{Cost}(dx) \approx \frac{1}{2} dx^T g^F dx$. (44.20)

Control-energy metric:

$g^F = G_{control}$. (44.21)

The metric should be selected before testing.

Comparing several metrics is legitimate, but post hoc selection must be penalized.

44.8 Estimating curvature

Given estimated metric $\hat{g}$:

$\hat{\Gamma}^\mu_{\alpha\beta} = \frac{1}{2}\hat{g}^{\mu\nu}(\partial_\alpha \hat{g}_{\nu\beta} + \partial_\beta \hat{g}_{\nu\alpha} - \partial_\nu \hat{g}_{\alpha\beta})$. (44.22)

Then estimate:

$\hat{R}^\rho_{\sigma\mu\nu}$. (44.23)

Curvature should predict:

- transport residual;
- path-dependent repricing;
- stress propagation;
- breakdown of local CAPM coefficients.

A nonzero estimated curvature tensor is not enough.

It must improve observable modelling.

44.9 Estimating the gauge connection

Given local states $|\psi(x)\rangle$, estimate connection through parallel alignment:

$$\mathcal{A}_\mu = i\langle\psi(x)|\partial_\mu\psi(x)\rangle. \quad (44.24)$$

Or estimate the transport matrix between neighbouring states:

$$U_{x\to x+dx} \approx \exp[-i\mathcal{A}_\mu dx^\mu]. \quad (44.25)$$

The connection should reduce cross-frame mismatch:

$$\varepsilon_{gauge} = State_B - U_{BA}State_A. \quad (44.26)$$

A successful connection makes:

$$\|\varepsilon_{gauge}\| < \|\varepsilon_{raw}\|. \quad (44.27)$$

44.10 Estimating phase holonomy

For closed loop γ:

$$\hat{\Phi}_\gamma = \oint_\gamma \mathcal{A}_\mu dx^\mu. \quad (44.28)$$

Compare predicted final state:

$$|\psi_{pred,f}\rangle = \exp(i\hat{\Phi}_\gamma)|\psi_i\rangle. \quad (44.29)$$

With observed final state.

A useful holonomy model should predict:

- hedge residue;
- volatility-surface shift;
- ledger difference;
- future response asymmetry.

44.11 Estimating composite states

Choose basis:

{|u_i,d_j⟩}. (44.30)

Estimate joint state ρ_UD from multiple measurement settings.

Constraints:

ρ_UD ≥ 0. (44.31)

Tr ρ_UD = 1. (44.32)

Compare:

M_sep = separable-state family. (44.33)

M_full = unrestricted positive joint-state family. (44.34)

Use:

- held-out likelihood;
- predictive scoring;
- complexity penalties;
- bootstrap uncertainty.

44.12 Estimating local mixedness

Compute:

ρ_U = Tr_D(ρ_UD). (44.35)

ρ_D = Tr_U(ρ_UD). (44.36)

Candidate mixedness measures:

Purity_U = Tr(ρ_U²). (44.37)

Entropy_U = −Tr(ρ_Ulnρ_U). (44.38)

But low purity may reflect:

- aggregation;
- incomplete observation;
- classical hidden regimes.

Therefore mixedness should be interpreted together with separability tests.

44.13 Estimating coherence

In declared branch basis:

$C_{l_1}(\rho) = \Sigma_{i\neq j}|\rho_{ij}|$. (44.39)

Or:

$C_{RE}(\rho) = S(\rho_{diag}) - S(\rho)$. (44.40)

Compare the full model with diagonal mixture:

$\Delta\text{PredictiveGain} = \text{Score}(\rho_{full}) - \text{Score}(\rho_{diag})$. (44.41)

Operational coherence requires:

$\Delta\text{PredictiveGain} >$ threshold after complexity adjustment. (44.42)

44.14 Estimating entanglement-form nonseparability

For low-dimensional systems, candidate tests include:

- positive partial transpose;
- entanglement witnesses;
- distance from separable set;
- fitted Schmidt rank for pure-state models.

Define:

$D_{sep}(\rho) = \min_{\sigma\in Sep}\text{Distance}(\rho,\sigma)$. (44.43)

A nonzero estimate requires uncertainty bounds.

The empirical question is:

Does allowing nonseparability improve prediction beyond classical latent-state models?

44.15 Estimating ledger backreaction

Let event Record_k occur.

Estimate causal effect:

$\Delta Y_{back} = E[Y_{after} \mid Commit] - E[Y_{after} \mid CounterfactualNoCommit]$. (44.44)

Possible outcomes:

- price;
- liquidity;
- margin;
- funding;

- volatility;
- future gate probability.

Use:

- matched controls;
- natural experiments;
- instrumental variables;
- structural causal models;
- agent simulations.

Ledger backreaction is one of the most directly testable components of the architecture.

44.16 Operational declaration template

Every empirical study should declare:

System Boundary: B. (44.45)

Primary Variables: X. (44.46)

Protocol: P. (44.47)

Local Frames: {F_a}. (44.48)

Phase Definition: θ. (44.49)

Composite Partition: Π. (44.50)

Measurement Settings: {a,b}. (44.51)

Gate Rule: G. (44.52)

Ledger Rule: L. (44.53)

Null Models: $\mathcal{N}$. (44.54)

Failure Threshold: ε*. (44.55)

Without this declaration, the theory cannot be falsified cleanly.

45. Candidate Experiments

45.1 Why the first experiments should be modest

The architecture contains several increasingly strong hypotheses.

They should not be tested all at once.

The first experiments should ask whether the framework improves:

- state reconstruction;
- cross-frame reconciliation;
- path-residual prediction;
- gate forecasting;
- backreaction attribution.

Only after these lower-level claims survive should experiments attempt to identify:

- operational coherence;
- effective nonseparability;
- contextuality;
- Bell-style structure.

The testing sequence should therefore follow the entanglement ladder:

E_1–E_4 Validation
→ E_5 State Test
→ E_6 Coherence Test
→ E_7 Contextuality Test
→ E_8–E_9 Boundary Test. (45.1)

The design principle is:

Test the Weakest New Claim Capable of Failing. (45.2)

A protocol-first test must declare its boundary, observation rule, window, admissible interventions, frame-equivalence conditions, and residual handling before interpreting any quantum-style result.

45.2 Experiment 1 — Does retained pressure Q predict future repricing?

Claim

The imaginary coordinate:

$$Q_t = \sqrt{(A_t^2 - R_t^2)} \quad (45.3)$$

contains predictive information not already contained in R, ordinary return, volatility, spread, and standard risk variables.

Protocol

Choose:

- one asset or asset class;
- one economically justified amplitude A_t;

- one admitted value R_t;
- one horizon h;
- one fixed valuation protocol P.

Estimate:

$$\theta_t = \operatorname{atan2}(Q_t, R_t). \quad (45.4)$$

$$\omega_t = \Delta\theta_t / \Delta t. \quad (45.5)$$

$$\Lambda_{F,t} = Q_t \omega_t. \quad (45.6)$$

Test whether:

$$\text{FutureReturn}_{t \to t+h} \quad (45.7)$$

or:

$$\text{FutureVolatility}_{t \to t+h} \quad (45.8)$$

depends on Q_t, ω_t, or $\Lambda_{F,t}$ after controlling for standard predictors.

Baseline model

$$Y_{t+h} = \alpha + \beta_1 R_t + \beta_2 \text{Vol}_t + \beta_3 \text{Spread}_t + \beta_4 \text{Factor}_t + \varepsilon_t. \quad (45.9)$$

Extended model

$$Y_{t+h} = \text{Baseline}_t + \gamma_1 Q_t + \gamma_2 \omega_t + \gamma_3 \Lambda_{F,t} + \varepsilon'_t. \quad (45.10)$$

Pass condition

The extension produces stable out-of-sample gain:

$$\Delta\text{Score}_Q = \text{Score}_{\text{extended}} - \text{Score}_{\text{baseline}} > \delta_Q. \quad (45.11)$$

Failure condition

If the gain disappears under:

- alternative windows;
- plausible definitions of A;
- transaction-cost adjustment;
- regime controls;
- held-out periods;

then Q is not yet an operationally useful coordinate.

45.3 Experiment 2 — Radial change versus frame

rotation

Claim

Observed repricing can be decomposed into:

$dR/dt = g_{AR} - Q\omega_F + Re(\varepsilon_{dyn})$. (45.12)

The angular term explains changes otherwise attributed entirely to economic amplitude.

Test

Estimate:

$RadialComponent_t = g_{A,t}R_t$. (45.13)

$AngularComponent_t = -Q_t\omega_{F,t}$. (45.14)

$Residual_t = \Delta R_t - RadialComponent_t - AngularComponent_t$. (45.15)

Compare against a conventional decomposition based on:

- cash-flow news;
- beta change;
- risk-premium change;
- rate change;
- volatility change.

Prediction

During valuation-frame shocks:

$|AngularComponent| / |\Delta R|$ (45.16)

should rise materially.

Candidate episodes include:

- monetary-policy repricing;
- liquidity shock;
- rating transition;
- regulatory change;
- sudden volatility repricing.

Failure condition

If angular decomposition merely restates ordinary discount-rate changes without improving:

- attribution;

- forecast;
- cross-frame consistency;
- residual reduction;

then the complex geometry adds no new explanatory layer.

45.4 Experiment 3 — Relative-frame composition

Claim

Discount-frame rapidities compose additively:

$$\rho_{CA} = \rho_{CB} + \rho_{BA}. \quad (45.17)$$

Where:

$$\rho_{BA} = \ln(R_A/R_B). \quad (45.18)$$

Candidate frames

- risk-free versus CAPM valuation;
- unsecured versus collateralized funding;
- market versus treasury frame;
- one numeraire versus another;
- desk versus enterprise risk frame.

Test statistic

$$\varepsilon_{comp} = \rho_{CA} - \rho_{CB} - \rho_{BA}. \quad (45.19)$$

Pass condition

For genuinely equivalent frame transformations:

$$|\varepsilon_{comp}| \le \varepsilon_{frame}. \quad (45.20)$$

Interpretation of failure

Persistent residual may indicate:

- hidden fees;
- non-equivalent contracts;
- liquidity cost;
- protocol transition;
- missing ledger state;

- path dependence.

This test is valuable even if the Lorentz analogy is later removed.

It becomes a diagnostic for whether a claimed frame change actually preserves the governed object.

45.5 Experiment 4 — Cross-frame gauge robustness

Claim

Equivalent financial descriptions should preserve control-relevant conclusions after valid transport.

Let:

Decision_A = G(State_A|F_A). (45.21)

Decision_B = G(State_B|F_B). (45.22)

After transport:

State_{A→B} = U_BAState_A. (45.23)

Define gauge residual:

ε_gauge = Distance[G(State_{A→B}|F_B),G(State_B|F_B)]. (45.24)

Candidate comparisons

- trading versus funding exposure;
- trading versus collateral exposure;
- accounting versus economic hedge;
- local desk versus enterprise risk;
- currency or numeraire conversion.

Pass condition

Transported and native conclusions agree within tolerance:

ε_gauge ≤ ε_G. (45.25)

Failure diagnosis

A large residual may reveal:

- hidden funding exposure;
- invalid netting assumption;
- omitted collateral;

- incompatible timing;
- protocol confusion.

Gauge Grammar treats this kind of residual as frame failure rather than ordinary noise and requires explicit cross-frame equivalence testing.

45.6 Experiment 5 — Local CAPM breakdown and curvature

Claim

CAPM performs as a local tangent law, while a curved model explains systematic variation across regimes.

Procedure

Estimate local CAPM parameters in rolling neighbourhoods:

$\beta_i(x), ERP(x), r_f(x)$. (45.26)

Construct a financial metric:

$g^F_{\mu\nu}(x)$. (45.27)

Estimate local curvature proxy:

$\kappa_F(x)$. (45.28)

Test whether CAPM residual increases with curvature:

$E[|\varepsilon_{CAPM}| \mid \kappa_F \text{ high}] > E[|\varepsilon_{CAPM}| \mid \kappa_F \text{ low}]$. (45.29)

Stronger prediction

Transporting local CAPM estimates covariantly should outperform direct extrapolation:

$Error_{covariant} < Error_{flat}$. (45.30)

Failure condition

If curvature measures do not predict:

- parameter instability;
- residual increase;
- regime transition;
- transport failure;

then the GR-like layer has not earned operational status.

45.7 Experiment 6 — Closed-loop financial holonomy

Claim

A closed loop in visible coordinates may leave a systematic state or ledger residual.

Design

Select a loop in:

- spot–volatility space;
- funding–collateral space;
- leverage–liquidity space;
- strike–maturity space.

Require:

$x_{final} \approx x_{initial}$. (45.31)

Measure full-state difference:

$H_\gamma = \text{Distance}[(x_{final}, L_{final}, \Psi_{final}), (x_{initial}, L_{initial}, \Psi_{initial})]$. (45.32)

Null hypothesis

The final response depends only on endpoint coordinates:

$\text{Response}_{final} = F(x_{final})$. (45.33)

Holonomy hypothesis

The final response depends on loop path:

$\text{Response}_{final} = F(x_{final}, \gamma, L_\gamma)$. (45.34)

Pass condition

A connection-curvature model predicts:

- hedge residue;
- transaction cost;
- volatility shift;
- altered future response;

better than endpoint and conventional path-history models.

Failure condition

If ordinary cumulative-cost or hysteresis models explain the effect equally well, geometric phase language should be treated as optional compression rather than discovery.

45.8 Experiment 7 — Barrier gate and latching

Claim

A barrier event creates a persistent internal-world state change not reconstructible from current price alone.

Matched states

Find observations A and B satisfying:

$U_t^A \approx U_t^B$. (45.35)

$\sigma_t^A \approx \sigma_t^B$. (45.36)

$T-t^A \approx T-t^B$. (45.37)

But:

$\text{BarrierHit}_A = 1$. (45.38)

$\text{BarrierHit}_B = 0$. (45.39)

Test whether:

$\text{FutureDerivativeDynamics}_A \neq \text{FutureDerivativeDynamics}_B$. (45.40)

Purpose

This validates:

Same Visible Coordinates + Different Ledger → Different World. (45.41)

It does not test quantum entanglement.

It tests the gate–trace foundation of secondary financial time.

45.9 Experiment 8 — Ledger backreaction

Claim

A committed financial record causally changes later primary-world dynamics.

Candidate events

- margin call;
- credit downgrade;
- default declaration;
- barrier activation;

- forced liquidation;
- regulatory breach.

Define treatment:

T_k = 1 if event is committed. (45.42)

Outcome:

Y_{k+h} = future liquidity, price impact, volatility, funding, or gate probability. (45.43)

Estimate:

ATE_h = E[Y_{k+h}(1) – Y_{k+h}(0)]. (45.44)

Required controls

- pre-event state;
- event anticipation;
- common market shock;
- selection into treatment;
- reverse causality.

Pass condition

The committed event has a stable causal effect after controlling for the underlying economic deterioration that preceded it.

This separates:

Primary Stress Effect (45.45)

from:

Ledger Commitment Effect. (45.46)

45.10 Experiment 9 — Contract preparation versus independent-state model

Claim

An option–underlying joint-state model predicts observations better than separate local-state models.

Independent model

p(u,d) = p_U(u)p_D(d). (45.47)

Classical coupled model

$$p(u,d) = \int p(\lambda)p(u|\lambda)p(d|\lambda)d\lambda. \quad (45.48)$$

Contract-state model

$$\rho_{UD} = \text{Prepare}_C(\rho_U \otimes \rho_{D^0}). \quad (45.49)$$

Test

Compare held-out prediction of:

- joint price movements;
- Greek transitions;
- exercise outcomes;
- hedge errors;
- margin events.

Interpretation

Improvement over independent local models establishes relational state value.

Improvement over strong classical coupled models is required before making an E_5 nonseparability claim.

45.11 Experiment 10 — Reduced-state incompleteness

Claim

A local underlying or derivative state is insufficient to predict certain outcomes that become predictable from the global joint state.

Compare:

$$\text{Model}_U: Y \leftarrow \rho_U. \quad (45.50)$$

$$\text{Model}_D: Y \leftarrow \rho_D. \quad (45.51)$$

$$\text{Model}_{UD}: Y \leftarrow \rho_{UD}. \quad (45.52)$$

Define relational gain:

$$\Delta\text{Rel} = \text{Score}_{UD} - \max(\text{Score}_U, \text{Score}_D). \quad (45.53)$$

A positive ΔRel establishes that joint information matters.

But it does not distinguish classical correlation from entanglement.

The test validates:

$$\text{Global State Information} > \text{Local State Information}. \quad (45.54)$$

45.12 Experiment 11 — Classical separability test

Claim

The observed joint state cannot be adequately represented by a separable classical mixture under the declared basis and instrument family.

Estimate:

$\hat{\rho}_{UD}$. (45.55)

Compute distance to separable set:

$D_{sep} = \min_{\sigma \in Sep} D(\hat{\rho}_{UD}, \sigma)$. (45.56)

Bootstrap uncertainty:

$CI_{sep} = ConfidenceInterval(D_{sep})$. (45.57)

Pass condition

The lower confidence bound is positive:

$LowerBound(CI_{sep}) > 0$. (45.58)

And the result survives:

- basis variation;
- instrument-error correction;
- alternative state dimensions;
- classical latent-factor models;
- out-of-sample testing.

Caution

Because the Hilbert representation itself is constructed, the finding is initially:

Representation-Relative Nonseparability. (45.59)

It is not yet physical quantum entanglement.

45.13 Experiment 12 — Coherent state versus classical mixture

Claim

Relative phase affects observable outcomes.

Estimate two models:

ρ_{mix} = diagonal joint state. (45.60)

ρ_{coh} = full state with off-diagonal terms. (45.61)

Use measurements in more than one basis.

Define:

$\Delta LL_{coh} = LL_{test}(\rho_{coh}) - LL_{test}(\rho_{mix})$. (45.62)

And:

$\Delta BIC_{coh} = BIC_{mix} - BIC_{coh}$. (45.63)

Strong pass condition

The full model:

1. improves held-out likelihood;
2. retains stable phase estimates;
3. predicts a controlled phase-shift experiment;
4. outperforms nonlinear classical alternatives.

Failure condition

If only in-sample fit improves, coherence is over-parameterization.

45.14 Experiment 13 — Phase-shift intervention

Claim

Changing relative phase while keeping branch magnitudes approximately fixed changes the joint outcome.

Let:

$|\alpha_1|, |\alpha_2|$ remain approximately constant. (45.64)

Intervene on:

$\Delta\theta \to \Delta\theta + \delta$. (45.65)

Predict:

$\Delta P(y) = 2|\alpha_1\alpha_2|[\cos(\Delta\theta+\delta) - \cos\Delta\theta]\cdot K_y$. (45.66)

Where K_y represents measurement overlap.

Candidate interventions

- controlled timing shift between two execution channels;
- controlled order of two valuation protocols;
- synchronization or desynchronization of hedge channels;

- simulated gauge-loop phase insertion.

Requirement

The intervention must not materially change:

- branch magnitude;
- liquidity;
- information;
- underlying regime.

Otherwise phase and ordinary causal change cannot be separated.

45.15 Experiment 14 — Decoherence and revival

Claim

A phase-sensitive effect declines with environmental recording and may partially recover after controlled realignment.

Model:

$$\gamma(\theta) = \gamma_0 \exp(-\kappa_{dec}\theta). \quad (45.67)$$

Outcome cross term:

$$I_{cross}(\theta) = 2\gamma(\theta)A_1A_2\cos\Delta\theta. \quad (45.68)$$

Test stages

1. prepare phase-sensitive channels;
2. measure baseline interference visibility;
3. introduce controlled public disclosure, aggregation, or timing jitter;
4. measure suppression;
5. realign channels where operationally possible;
6. test recovery.

Strong signature

Suppression follows environmental coupling rather than simple amplitude decline.

Partial revival follows phase realignment.

Caution

Classical synchronization systems can exhibit the same pattern.

This tests coherent process modelling, not quantum substrate.

45.16 Experiment 15 — Operational incompatibility

Claim

Two measurement procedures cannot be jointly implemented without material disturbance.

Let instruments be:

$\mathbb{M}_A$ and $\mathbb{M}_B$. (45.69)

Define order residual:

$\delta_{AB} = \text{Distance}[(\mathbb{M}_B \circ \mathbb{M}_A)(\rho),(\mathbb{M}_A \circ \mathbb{M}_B)(\rho)]$. (45.70)

Candidate pairs

- liquidation measurement versus going-concern valuation;
- executable liquidity probe versus undisturbed price estimate;
- collateral seizure versus credit assessment;
- public disclosure versus private implied-state estimation.

Pass condition

δ_{AB} remains materially nonzero after controlling for ordinary time evolution.

Interpretation

This establishes operational incompatibility.

It does not establish specifically quantum uncertainty.

45.17 Experiment 16 — Cross-observer fixedness

Claim

A committed event becomes delta-certain across observers only when:

- contexts are mapped;
- effects are compatible;
- the record is accessible.

Design

Observer A records event φ.

Observer B receives either:

1. no frame map;
2. frame map but incompatible effect;
3. compatible effect but no record;
4. all three conditions.

Measure B's posterior certainty.

Prediction:

$P_B[T_\Phi(\varphi)|\mathscr{F}_B] \to 1$ (45.71)

only in the fourth condition.

This experiment directly transfers the internal-observer logic into a financial runtime.

The source observer model specifies the same three-part structure—frame transformation, compatibility or joint measurability, and accessible record—as the basis of cross-observer fixedness.

45.18 Experiment 17 — Redundancy and financial objectivity

Claim

Independent redundant records increase consensus reliability.

Let N observers read fragments of one committed event.

Define disagreement probability:

$\varepsilon_N = P(\text{Consensus}_N \neq \text{TrueRecord})$. (45.72)

Prediction:

ε_N decreases with N (45.73)

under:

- sufficiently distinguishable records;
- bounded individual error;
- partial independence;
- compatible readout frames.

Candidate records

- trade confirmation;
- settlement event;
- barrier observation;

- default event;
- margin breach.

Failure condition

If additional records merely duplicate one common error source, redundancy will not improve truth.

Therefore record independence and provenance must be logged.

45.19 Experiment 18 — Bell-style secondary-world stress test

Purpose

The aim is not initially to prove quantum finance.

The aim is to test whether one secondary-world joint model can reproduce the observed correlations under explicitly declared assumptions.

Settings

Underlying side:

a_0 = price-direction basis. (45.74)

a_1 = volatility or risk-state basis. (45.75)

Derivative side:

b_0 = premium or payoff basis. (45.76)

b_1 = delta–vega or replication basis. (45.77)

Binary outcomes are assigned under pre-registered thresholds.

Compute:

$E(a,b) = \Sigma_{u,d} ud \cdot p(u,d|a,b)$. (45.78)

Then:

$S_F = E(a_0,b_0) + E(a_0,b_1) + E(a_1,b_0) - E(a_1,b_1)$. (45.79)

Required diagnostics

Before interpreting S_F, test:

Measurement Independence Gap:

$MI = D[p(a,b,\lambda), p(a,b)p(\lambda)]$. (45.80)

No-Signalling Gap:

NS = max{|p(u|a,b0)−p(u|a,b1)|,|p(d|a0,b)−p(d|a1,b)|}. (45.81)

Memory Gap:

MG = Dependence[(u_k,d_k),(u_{<k},d_{<k}) | settings]. (45.82)

Selection Gap:

SG = Difference[IncludedTrials,AllPreparedTrials]. (45.83)

Interpretation ladder

Result	Interpretation
	S_F
	S_F
	S_F
	S_F

Even the final row would not immediately prove that financial markets are physical quantum systems.

It would show that the declared secondary-world statistics resist the tested classical model family.

45.20 Experiment 19 — No-signalling test

Estimate:

δ_{B→A} = max_{u,a,b0,b1}|p(u|a,b0)−p(u|a,b1)|. (45.84)

δ_{A→B} = max_{d,b,a0,a1}|p(d|a0,b)−p(d|a1,b)|. (45.85)

Define:

NS_F = max(δ_{B→A},δ_{A→B}). (45.86)

A no-signalling-compatible result requires:

NS_F ≤ ε_NS. (45.87)

The experiment should preferably use:

- passive measurements;
- independently randomized settings;
- concealed outcomes;
- no trading intervention during the trial;
- simulated or archived prepared states.

Active market measurements are likely to violate no-signalling by design.

45.21 Experiment 20 — Monogamy stress test

Claim

A quantum-like entanglement resource would impose restrictions on how strongly one subsystem can share nonseparable relations with several others.

Let U be one underlying and D_1,D_2 several derivatives.

Estimate pairwise effective entanglement:

$E(U:D_1)$. (45.88)

$E(U:D_2)$. (45.89)

Test candidate inequality:

$E(U:D_1)^2 + E(U:D_2)^2 \leq E(U:D_1D_2)^2$. (45.90)

Expected financial result

Ordinary contractual dependence may violate any naïve monogamy analogue because one underlying can support many derivatives.

Failure would clarify that the financial relation is not quantum entanglement.

A stable restricted inequality would instead indicate a bounded relational resource such as:

- liquidity;
- hedge capacity;
- collateral;
- information;
- risk budget.

45.22 Simulation before market experiment

Many strong hypotheses should first be tested in controlled simulation.

A minimum simulation loop is:

1. initialize global state ρ_0;
2. evolve between ticks;
3. select measurement setting from trace;
4. sample or generate outcome;
5. update state;
6. append record;

7. apply gate and backreaction;
8. update metric, connection, or protocol;
9. repeat.

This mirrors the minimal observer simulation structure already proposed for adaptive instruments, latching, cross-observer records, and redundancy tests.

Simulation permits explicit control of:

- hidden state;
- setting independence;
- signalling channels;
- environmental coupling;
- branch coherence;
- ledger access.

Only structures that can be recovered reliably in simulation should proceed to market-data testing.

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46. Null Models

46.1 Why sophisticated null models are essential

A weak null model makes an exotic theory look successful.

A complex-amplitude model should not be compared only against:

- independence;
- linear regression;
- constant volatility;
- simple Gaussian noise.

Financial systems already possess mature models of:

- dependence;
- path history;
- latent regimes;
- nonlinear feedback;
- endogenous volatility;
- strategic interaction.

The correct standard is:

46.2 Null family N_0 — Independent local states

The simplest null is:

$$p(u,d) = p_U(u)p_D(d). \quad (46.2)$$

This tests only whether relational dependence exists.

Options will almost always reject this null.

Rejecting N_0 has little relevance to entanglement.

46.3 Null family N_1 — Classical joint distribution

Use an unrestricted classical joint distribution:

$$p(u,d). \quad (46.3)$$

This captures correlation without phase.

For multiple settings:

$$p(u,d|a,b). \quad (46.4)$$

The model may be contextual in the ordinary statistical sense.

Rejecting N_1 requires demonstrating constraints across settings, not merely dependence within each setting.

46.4 Null family N_2 — Latent-factor model

Let:

$$\lambda_t = \text{latent market state}. \quad (46.5)$$

Then:

$$p(u,d|a,b) = \int p(\lambda)p(u|a,\lambda)p(d|b,\lambda)d\lambda. \quad (46.6)$$

Candidate latent factors include:

- volatility;
- liquidity;
- funding;
- market regime;
- dealer inventory;
- macro news;

- sentiment.

This model can reproduce strong apparent jointness.

46.5 Null family N3 — Dynamic state-space model

Let:

$\lambda_{t+1} = F(\lambda_t,\xi_t)$. (46.7)

Observations:

$u_t = G_U(\lambda_t,a_t,\varepsilon_U,t)$. (46.8)

$d_t = G_D(\lambda_t,b_t,\varepsilon_D,t)$. (46.9)

This captures:

- temporal memory;
- latent regime evolution;
- adaptive measurement;
- common cause.

It should be a default null for sequential financial experiments.

46.6 Null family N4 — Stochastic-volatility model

Underlying:

$dU_t/U_t = \mu_t dt + \sqrt{v_t} dW_t^U$. (46.10)

Variance:

$dv_t = \kappa_v(\theta_v - v_t)dt + \xi\sqrt{v_t} dW_t^v$. (46.11)

With:

$dW_t^U dW_t^v = \rho_v dt$. (46.12)

This model produces:

- skew;
- volatility clustering;
- option–underlying dependence;
- changing Greeks;
- apparent state mixing.

Any complex-phase model of options should outperform stochastic volatility before claiming new structure.

46.7 Null family N_5 — Jump and regime-switching models

Jump diffusion:

$dU_t/U_{t^-} = \mu dt + \sigma dW_t + (J-1)dN_t$. (46.13)

Regime switching:

$S_t \in \{1,\ldots,K\}$. (46.14)

$P(S_{t+1}|S_t) = \Pi$. (46.15)

These models can produce:

- discontinuous gate-like changes;
- apparent collapse;
- path-dependent distributions;
- nonlinear option response.

Rejecting smooth Gaussian models is therefore insufficient.

46.8 Null family N_6 — Copula models

Let marginal distributions be:

$F_U(u)$. (46.16)

$F_D(d)$. (46.17)

Joint distribution:

$F_{UD}(u,d) = C[F_U(u),F_D(d)]$. (46.18)

Use:

- Gaussian copula;
- t-copula;
- Archimedean copulas;
- vine copulas;
- dynamic copulas.

Copulas can model:

- asymmetric dependence;
- tail dependence;

- multi-asset structure.

A claimed nonfactorizable state must outperform sufficiently flexible dependence models.

46.9 Null family N7 — Hawkes and order-flow models

For event type i:

$$\lambda_i(t) = \mu_i + \Sigma_j \int_0^t \varphi_{ij}(t-s)dN_j(s). \quad (46.19)$$

This captures self-excitation and cross-excitation.

It can explain:

- synchronized trading;
- hedge cascades;
- clustered volatility;
- apparent phase locking;
- path dependence.

Hawkes models are particularly important nulls for interference-like order-flow patterns.

46.10 Null family N8 — Agent-based models

Agents may possess:

- heterogeneous beliefs;
- adaptive policies;
- position constraints;
- feedback;
- market impact;
- learning;
- memory.

The aggregate system can produce:

- nonlinearity;
- synchronization;
- regime switching;
- hysteresis;
- apparent contextuality;

- sudden collapse.

A complex-state model should be compared against agent-based mechanisms capable of producing similar macroscopic phenomena.

46.11 Null family N9 — Classical wave and oscillator models

Classical coupled oscillators:

$d^2x_i/dt^2 + \omega_i^2 x_i = \Sigma_j K_{ij} x_j$. (46.20)

Phase models:

$d\theta_i/dt = \omega_i + \Sigma_j K_{ij} \sin(\theta_j - \theta_i)$. (46.21)

These generate:

- relative phase;
- constructive synchronization;
- destructive cancellation;
- dephasing;
- revival.

Therefore:

Phase Behaviour ≠ Quantum Coherence. (46.22)

Classical wave nulls are mandatory for any financial interference claim.

46.12 Null family N10 — Hysteresis models

A path-dependent output may be:

$Y_t = \mathcal{H}[X_{0:t}]$. (46.23)

Preisach-like representation:

$Y_t = \iint \mu(\alpha,\beta) \gamma_{\alpha\beta}[X]_t d\alpha d\beta$. (46.24)

Such models can produce:

- closed-loop residual;
- memory;
- gate thresholds;
- irreversible-looking paths;
- minor-loop structure.

They are strong nulls for financial holonomy.

46.13 Null family N_{11} — Causal feedback models

Use structural equations:

$$D_t = F_D(U_t,\sigma_t,L_t,\varepsilon_D,t). \quad (46.25)$$

$$H_t = F_H(D_t,U_t,L_t,\varepsilon_H,t). \quad (46.26)$$

$$U_{t+1} = F_U(U_t,H_t,News_t,\varepsilon_U,t). \quad (46.27)$$

This explicitly models derivative backreaction.

It can produce global joint-state effects while remaining entirely classical.

46.14 Null family N_{12} — Classical contextual model

Allow outcome to depend on:

- setting;
- previous setting;
- previous outcome;
- hidden state;
- measurement disturbance.

For example:

$$u_k = F_U(a_k,b_{<k},u_{<k},d_{<k},\lambda_k). \quad (46.28)$$

$$d_k = F_D(b_k,a_{<k},u_{<k},d_{<k},\lambda_k). \quad (46.29)$$

Such models can violate simple noncontextual inequalities.

A strong contextuality claim must reject these richer classical explanations under appropriate causal assumptions.

46.15 Null family N_{13} — Signalling model

Allow:

$$p(u|a,b) \neq p(u|a). \quad (46.30)$$

$$p(d|a,b) \neq p(d|b). \quad (46.31)$$

Because finance contains communication and market impact, this may be the natural default.

A Bell-style financial analysis that omits signalling nulls is invalid.

46.16 Null family N_{14} — Measurement-dependent hidden state

Allow:

$p(a,b|\lambda) \neq p(a,b)$. (46.32)

This captures adaptive setting choice.

In markets, agents choose measurements and trades because of the state.

A Bell-style result must quantify rather than ignore this dependence.

46.17 Null family N_{15} — Post-selection and survivor bias

Observed sample:

$\mathcal{D}_obs = Select(\mathcal{D}_full|S=1)$. (46.33)

Selection may depend on:

- liquidity;
- successful settlement;
- available quote;
- surviving firm;
- chosen option series;
- realized trade.

Apparent nonclassical correlation may arise from conditioning on a collider.

All prepared or eligible trials must be logged.

46.18 Nested model comparison

Use a hierarchy:

M_0 = independent. (46.34)

M_1 = classical joint. (46.35)

M_2 = latent dynamic. (46.36)

M_3 = nonlinear feedback. (46.37)

M_4 = classical phase. (46.38)

M_5 = complex coherent state. (46.39)

M_6 = nonseparable contextual state. (46.40)

The new layer is justified only when it improves:

- held-out prediction;
- intervention response;
- compression;
- robustness;
- residual structure.

46.19 Model score

A general score may be:

Score(M) = PredictiveAccuracy(M) − λ_CComplexity(M) − λ_RInstability(M) − λ_AAssumptionCost(M). (46.41)

The winning theory is not merely the best fit.

It must also survive:

- perturbation;
- frame change;
- time shift;
- regime shift;
- parameter uncertainty.

46.20 Residual comparison

For model M:

ε_M,t = Y_t − Ŷ_M,t. (46.42)

Evaluate:

ResidualBias_M. (46.43)

ResidualAutocorrelation_M. (46.44)

ResidualFrameSensitivity_M. (46.45)

ResidualRegimeDependence_M. (46.46)

A theory that reduces average error while leaving structured residual may still be incomplete.

46.21 Null-model adequacy rule

A null family is adequate only when it is allowed to express the mature classical mechanisms already known in the domain.

The rule is:

Do Not Call a Classical Model Refuted after Testing Only a Weak Classical Model. (46.47)

This is especially important because the proposed financial architecture itself is intended as a strong non-quantum control world.

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47. Failure Conditions

47.1 Why explicit failure conditions protect the theory

A broad framework can absorb almost any outcome by redefining:

- protocol;
- phase;
- metric;
- connection;
- residual;
- subsystem partition.

Without strict failure rules, the theory becomes unfalsifiable.

Each layer must therefore possess a removal condition.

The central rule is:

No Measurable Gain → No Theoretical Upgrade. (47.1)

47.2 F1 — Amplitude ambiguity

The framework fails at its first layer if A has no independent economic definition.

If A is chosen only to satisfy:

$A^2 = R^2 + Q^2$, (47.2)

then Q is algebraically manufactured.

Failure condition:

A cannot be estimated independently of R and Q. (47.3)

Required response:

Remove or redefine the complex completion.

47.3 F2 — Q is non-predictive

If Q does not improve:

- forecast;
- diagnosis;
- gate prediction;
- risk attribution;
- residual reduction;

then retained pressure is not operationally useful.

Failure condition:

$\Delta Score_Q \leq 0$ across held-out datasets. (47.4)

Required response:

Treat Q as a visualization coordinate only, or remove it.

47.4 F3 — θ is only a monotonic transform of an existing variable

If:

$\theta = f(StandardDiscountRatio)$ (47.5)

and contributes no independent dynamical or cross-channel information, then phase language may be redundant.

Failure condition:

$ConditionalInformation(\theta;Y|StandardVariables) \approx 0$. (47.6)

Required response:

Retain θ only where phase transport or multi-channel relations add value.

47.5 F4 — Internal time is unstable

θ cannot serve as internal time when:

- it is non-monotonic without branch labels;
- phase unwrapping is arbitrary;
- protocol changes destroy continuity;
- different observers cannot synchronize it.

Failure condition:

$$\mathrm{OrderError}_\theta > \varepsilon_{\mathrm{time}}. \quad (47.7)$$

Required response:

Use piecewise phase clocks, ledger time, or ordinary calendar time.

47.6 F5 — Frame rapidity does not compose

If equivalent valuation transforms fail:

$$\rho_{CA} \approx \rho_{CB} + \rho_{BA}, \quad (47.8)$$

then the additive rapidity model is invalid for those frames.

Required response:

Classify the change as:

- protocol transition;
- path-dependent transport;
- non-multiplicative transformation;
- model error.

Do not preserve Lorentz language by force.

47.7 F6 — Invariant object is undefined

A frame theory fails if it cannot state what remains invariant.

Failure condition:

$$\text{No operational } I \text{ satisfies } I(F_A) \approx I(F_B). \quad (47.9)$$

Without an invariant:

- frame transformation cannot be tested;
- gauge freedom becomes arbitrary;
- cross-observer agreement loses content.

Required response:

Return to protocol declaration.

47.8 F7 — Metric choice is arbitrary

A curved geometry fails when many unrelated metrics produce equally flexible post hoc stories.

Failure condition:

Metric estimates are unstable under minor data or feature changes. (47.10)

Or:

Metric model does not outperform ordinary state-space models. (47.11)

Required response:

Use the simpler statistical or control geometry directly.

47.9 F8 — Curvature is non-predictive

Estimated curvature must explain:

- transport residual;
- local-law breakdown;
- path dependence;
- stress propagation.

Failure condition:

κ_F has no stable relation to any target after controls. (47.12)

Required response:

Remove the GR-like interpretation.

A state-dependent covariance matrix alone does not justify curvature language.

47.10 F9 — Gauge connection does not improve transport

Failure condition:

$\|\varepsilon_{gauge}\| \geq \|\varepsilon_{raw}\|$. (47.13)

If gauge correction makes reconciliation no better—or worse—the chosen connection is not useful.

Required response:

Revise or remove the connection.

47.11 F10 — Holonomy reduces to known accumulated cost

Suppose loop residue is completely explained by:

- transaction cost;
- funding cost;
- realized gamma;
- taxes;
- known hysteresis.

Then:

HolonomyModelGain ≤ 0. (47.14)

Required response:

Use standard path accounting.

Geometric phase should not rename an already adequate explanation.

47.12 F11 — Tensor notation adds no joint predictive value

If:

Score_UD ≤ max(Score_U,Score_D,Score_classicaljoint), (47.15)

then the composite Hilbert representation adds no benefit.

Required response:

Return to an ordinary joint probability or state-space model.

47.13 F12 — Nonfactorization is basis artefact

A claimed nonfactorizable state fails when it disappears under an admissible local transformation or alternative equivalent partition.

Failure condition:

E(ρ) varies materially under proper local frame changes. (47.16)

Required response:

Reject the entanglement interpretation.

47.14 F13 — Coherence is not observable

If:

Tr(ρ_cohÔ) = Tr(ρ_mixÔ) (47.17)

for every admissible financial observable, then coherence has no operational content.

Required response:

Use the mixture.

Formal off-diagonal terms should not be retained without measurable consequences.

47.15 F14 — Phase estimates are not stable

Failure condition:

Estimated $\Delta\theta$ changes radically under:

- small sample perturbation;
- equivalent basis choice;
- minor model variation;
- out-of-sample period.

Required response:

Treat phase as unidentified.

47.16 F15 — Classical nonlinear models explain the interference residual

If a flexible classical model satisfies:

$$\text{Score}_{\text{classical}} \geq \text{Score}_{\text{coherent}}, \quad (47.18)$$

then phase interference has not been established.

Required response:

Prefer the classical model unless the complex representation offers major compression or control advantages.

47.17 F16 — Apparent entanglement is separable

If a separable state σ reproduces observations within tolerance:

$$\text{Distance}(\text{DataPredictions}_{\rho}, \text{DataPredictions}_{\sigma}) \leq \varepsilon_{\text{sep}}, \quad (47.19)$$

then effective entanglement is unsupported.

Required response:

Classify the relation as classical correlation, contractual coupling, or dynamical binding.

47.18 F17 — No-signalling fails

If:

NS_F > ε_NS, (47.20)

then the system contains detectable setting influence.

Required response:

Do not compare the observed correlation directly with Bell entanglement.

Model the signalling channel.

47.19 F18 — Measurement independence fails

If:

MI > ε_MI, (47.21)

settings depend on the hidden or observed market state.

Required response:

Use causal measurement-dependent models.

Do not interpret a CHSH-like value under standard Bell assumptions.

47.20 F19 — Memory explains the result

If a sequential model reproduces the observed setting correlations:

Score_memory ≥ Score_nonclassical, (47.22)

then latching and adaptive policy are sufficient.

Required response:

Retain the internal-observer model without claiming irreducible contextuality.

47.21 F20 — Ledger does not change future dynamics

The gate–trace theory fails for a domain when commitment has no effect beyond underlying state change.

Failure condition:

ATE_ledger ≈ 0. (47.23)

Required response:

Treat the ledger as passive record rather than causal trace in that domain.

47.22 F21 — Cross-observer agreement lacks mapping

If observers cannot define:

- context map;
- outcome map;
- compatible effects;
- accessible record;

then disagreement cannot be interpreted as frame relativity or quantum-like observer difference.

Required response:

The comparison is under-specified.

47.23 F22 — Residual is used as an escape clause

A model fails scientifically when every contradiction is assigned to residual without revision.

Define unexplained residual ratio:

RR = ‖Residual‖/‖ObservedChange‖. (47.24)

Failure condition:

RR remains persistently above RR*. (47.25)

Or residual repeatedly changes type after the result is known.

Required response:

Revise or reject the declaration.

Residual must be logged, typed, or escalated rather than used to protect a preferred theory.

47.24 F23 — The architecture cannot recover mature limits

Failure occurs if the unified model cannot recover:

- CAPM;
- ordinary option valuation;
- classical joint distributions;
- local frame transformations;
- one-way pricing;
- zero-ledger analytics.

Required response:

The theory is not a unification.

It is an incompatible replacement.

47.25 F24 — Over-engineering

A model may be mathematically coherent but operationally inferior.

Define total implementation value:

V_total = PredictiveGain + DiagnosticGain + ControlGain − ComplexityCost − DataCost − GovernanceCost. (47.26)

Failure condition:

V_total ≤ 0. (47.27)

Required response:

Use the simpler model.

This may be the most common outcome for everyday finance.

47.26 F25 — Physics vocabulary adds no role clarity

If replacing:

- gauge;
- fibre;
- curvature;
- entanglement;
- collapse;

with ordinary financial language leaves the model equally clear and effective, the physics terminology is unnecessary.

Required response:

Remove the analogy from the operational layer.

The Gauge Grammar's governing standard is that cross-domain physics language earns its place only through improved explanation, diagnosis, stability, or intervention.

47.27 Falsification matrix

Layer	Required evidence	Failure response
R+iQ	independent A and useful Q	remove complex completion
θ-time	stable ordering and prediction	use t or k

Layer	Required evidence	Failure response
SR-like frame	invariant and composable map	classify as protocol change
GR-like metric	predictive state-dependent geometry	use ordinary state model
Gauge connection	reduced transport residual	remove connection
Composite state	joint predictive gain	use classical joint distribution
Coherence	phase-sensitive observable gain	use diagonal mixture
Entanglement	separable null rejection	classify as correlation/coupling
No-signalling	stable marginal independence	model causal signalling
Bell-like claim	loophole-controlled violation	reject quantum interpretation

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47.28 The rejection principle

The theory should become smaller when evidence fails.

The correct scientific response is:

Failure of One Layer Does Not Require Abandoning Every Lower Layer. (47.28)

For example:

- coherence may fail while joint-state modelling remains useful;
- curvature may fail while local frame rapidity remains useful;
- Bell structure may fail while internal observer analysis remains useful;
- Q may remain descriptive even if not predictive.

This modularity is a strength.

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48. The Research Programme

48.1 The programme should proceed by claim strength

The proposed architecture should be developed through four research tiers.

Tier A — Mature-finance reconstruction

Use established finance to construct:

- R;
- A;
- Q;
- θ;
- local frames;
- derivative composite partitions;

- gates;
- ledger;
- backreaction.

Tier B — Operational effective-world testing

Test:

- predictive value;
- cross-frame invariance;
- path residual;
- joint-state gain;
- ledger causality.

Tier C — Strong quantum-form testing

Test:

- coherence;
- nonseparability;
- incompatibility;
- no-signalling;
- Bell-style constraints.

Tier D — Physics-foundations interpretation

Only after Tiers A–C should the framework be used to make claims about:

- the origin of quantum strangeness;
- hidden constructors;
- observer-relative worlds;
- possible statistical substrate layers.

48.2 Programme Stage 1 — Reproduce Part I

The first implementation should reproduce:

$Z = R + iQ.$ (48.1)

$\theta = \operatorname{atan2}(Q,R).$ (48.2)

$dZ/dt = (g_A + i\omega_F)Z + \varepsilon_{dyn}.$ (48.3)

Validate:

- amplitude definition;
- phase continuity;
- radial/angular decomposition;
- residual classification.

No tensor or quantum terminology is required at this stage.

48.3 Programme Stage 2 — Build the derivative composite world

Declare:

$\mathscr{H}_{UD} = \mathscr{H}_{U} \otimes \mathscr{H}_{D}$. (48.4)

Specify:

- subsystem basis;
- contract preparation;
- local observables;
- joint observables;
- gate rules;
- ledger schema.

Begin with a small finite-state option model.

For example:

$\mathscr{H}_{U} = \mathrm{span}\{|\mathrm{Down}\rangle, |\mathrm{Up}\rangle\}$. (48.5)

$\mathscr{H}_{D} = \mathrm{span}\{|\mathrm{LowPayoff}\rangle, |\mathrm{HighPayoff}\rangle\}$. (48.6)

Contract state:

$|\Psi_{UD}\rangle = c_{D}|\mathrm{Down,LowPayoff}\rangle + c_{U}|\mathrm{Up,HighPayoff}\rangle$. (48.7)

First compare this with the corresponding classical mixture.

48.4 Programme Stage 3 — Implement adaptive internal observers

Define observer:

$O = (\mathcal{A}_{O}, \mathbb{M}_{O}, F_{O}, L_{O}, \pi_{O})$. (48.8)

At tick k:

$a_k = \pi_O(L_{k-1})$. (48.9)

Outcome:

$y_k \sim \mathbb{M}_{a_k}(\rho_{k^-})$. (48.10)

Update:

$\rho_k = \mathbb{M}_{a_k,y_k}(\rho_{k^-})/p(y_k)$. (48.11)

Trace:

$L_k = L_{k-1} \sqcup \text{Record}_k$. (48.12)

Simulate:

- delta-certainty;
- latching;
- adaptive setting changes;
- cross-observer record access;
- redundancy consensus.

The self-referential observer framework supplies an explicit simulation pattern for these processes and identifies which assumptions fail when measurability, compatibility, record accessibility, redundancy, or frame invariance are removed.

48.5 Programme Stage 4 — Add local frames

Implement at least three frames:

- market frame;
- funding frame;
- collateral or accounting frame.

Define transport maps:

T_{BA}. (48.13)

Declare invariants:

$I_{contract}$. (48.14)

$I_{exposure}$. (48.15)

$I_{settlement}$. (48.16)

Measure:

ε_{gauge}. (48.17)

The primary research question is practical:

Does frame-aware transport detect risks hidden by one local description?

48.6 Programme Stage 5 — Estimate curved geometry

Choose one interpretable metric.

A strong first candidate is an intervention-cost metric:

$$\text{Cost}(dx) \approx \tfrac{1}{2} dx^{T} g_F(x) dx. \quad (48.18)$$

Estimate how the metric changes with:

- liquidity;
- leverage;
- funding;
- margin proximity.

Test whether local CAPM or Greek errors rise with estimated curvature.

Avoid beginning with a full Einstein-like field equation.

The first goal is only:

State-Dependent Geometry with Predictive Transport. (48.19)

48.7 Programme Stage 6 — Loop and holonomy tests

Construct controlled loops in:

- spot–volatility;
- funding–collateral;
- leverage–liquidity;
- strike–maturity.

Record:

- endpoint state;
- cumulative costs;
- ledger changes;
- future response.

Compare:

Endpoint Model. (48.20)

History Model. (48.21)

Connection–Curvature Model. (48.22)

The geometric model advances only if it offers:

- better compression;
- stable invariant;
- superior prediction.

48.8 Programme Stage 7 — Multi-channel phase

Define channels:

n = 1,…,N. (48.23)

Estimate:

Z_n = A_nexp(iθ_n). (48.24)

Relative phases:

Δθ_nm = θ_n − θ_m. (48.25)

Candidate channels should first be economically natural:

- maturity buckets;
- strike buckets;
- hedge regimes;
- investor classes;
- valuation protocols.

Test whether relative phase predicts:

- correlation change;
- synchronization;
- gate probability;
- cross-channel residual.

Use classical phase-oscillator models as the initial interpretation.

48.9 Programme Stage 8 — Coherence testing

Only after stable multi-channel phase has been established should the programme introduce:

ρ_coh = ρ_diag + ρ_off. (48.26)

Required evidence:

1. multiple measurement bases;
2. stable off-diagonal estimates;
3. controlled phase intervention;
4. predictive improvement;
5. classical null rejection.

A coherence claim should remain labelled:

Effective Financial Coherence under Protocol P. (48.27)

Not:

Physical Quantum Coherence. (48.28)

48.10 Programme Stage 9 — Nonseparability testing

Estimate:

D_sep(ρ_UD). (48.29)

Use:

- separability witnesses;
- partial-transpose tests where suitable;
- latent classical models;
- basis robustness;
- held-out prediction.

The programme should report:

Representation. (48.30)

Partition. (48.31)

Measurement algebra. (48.32)

Null family. (48.33)

Confidence interval. (48.34)

Without these fields, a nonseparability result is uninterpretable.

48.11 Programme Stage 10 — Contextuality and Bell-style tests

These are late-stage stress tests.

They require:

- independently controlled settings;
- trial isolation;
- no-signalling analysis;
- memory correction;
- complete sampling;
- causal modelling;
- replication.

A Bell-like result should initially be described as:

Failure of Declared Classical Joint Model under Test Assumptions. (48.35)

Only a much stronger cross-domain experimental programme could justify more.

48.12 The assumption ledger

Every experiment should maintain an assumption ledger.

Component	Status
CAPM equation	mature finance
option payoff	mature finance
risk-neutral weights	mature finance
R+iQ completion	coordinate extension
θ as orientation	derived representation
θ as internal time	conditional hypothesis
tensor partition	model extension
contract as operator	effective representation
phase amplitudes	modelling hypothesis
off-diagonal coherence	empirical hypothesis
financial metric	protocol-selected model
gauge connection	estimated transport model
Born-form probability	not derived
Bell nonclassicality	not established

The assumption ledger prevents later sections from quietly treating hypotheses as findings.

48.13 The evidence ladder

Evidence should be labelled:

E0 — Definition

A quantity can be written.

E1 — Fit

It describes existing data.

E2 — Out-of-sample prediction

It predicts held-out observations.

E3 — Intervention response

It predicts controlled change.

E4 — Cross-frame robustness

It survives equivalent representation changes.

E5 — Cross-domain replication

It works in different markets or systems.

E6 — Classical-null rejection

Mature alternatives fail.

E7 — Physical interpretation

A substrate claim is justified.

Most concepts in the present article are currently at E0 or E1.

The research programme aims first at E2–E4.

48.14 The minimum publishable result

A first publishable empirical paper need not establish entanglement.

A sufficient initial result would show:

1. independently defined A and R;
2. reproducible Q and θ;
3. stable frame maps;
4. reduced out-of-sample residual;
5. successful loop-residual prediction;
6. complete protocol and code disclosure.

This would validate the geometry as a financial engineering tool.

Quantum comparison could remain interpretive.

48.15 The minimum coherent-finance result

A stronger paper would require:

1. two or more stable channels;
2. identifiable relative phase;
3. a phase-sensitive measurement;
4. controlled phase shift;
5. observed interference change;
6. rejection of classical mixture and nonlinear-interaction nulls;
7. replication.

Only then should the phrase:

Operational Financial Coherence (48.36)

be used as a result rather than a proposal.

48.16 The minimum derivative-entanglement result

A defensible E5 result requires:

1. natural subsystem partition;
2. multiple local measurement settings;
3. estimated joint state;
4. nonzero distance from separable set;
5. robust entanglement witness;
6. classical latent-model rejection;
7. local-frame invariance;
8. held-out replication.

Even this would establish:

Secondary-World Operational Nonseparability. (48.37)

It would not by itself establish physical quantum entanglement.

48.17 The minimum physics-facing result

A physics-facing claim requires a much higher threshold.

At minimum:

- a clearly specified correspondence;
- no-signalling-compatible correlations;
- measurement independence;
- controlled memory;
- strong classical-null rejection;
- independent replication;
- a reason the same structure should illuminate physical quantum systems.

Without these, the financial model remains a useful non-quantum control world rather than a model of quantum ontology.

48.18 Reproducibility footer

Every experiment should publish:

VerifyTrace_F = [dataset_id][time_window][seed][code_hash][protocol_P][boundary_B][feature_map_φ][amplitude_A][frame_map_T][metric_g][connection_$\mathcal{A}$][partition_Π][settings][gate][ledger_rule][null_models][residual][decision][timestamp]. (48.38)

This footer makes the claim reproducible and auditable.

The wider project's verification grammar similarly requires publishable claims to retain their baseline, feature map, thresholds, decisions, trace, and residual rather than reporting only a final score.

48.19 Governance of theory expansion

A new layer should be admitted only when:

Admission_NewLayer = EmpiricalGain ∧ RoleClarity ∧ ReductionCompatibility ∧ Auditability. (48.39)

A layer should be removed when:

Removal_NewLayer = NoGain ∨ Instability ∨ Redundancy ∨ Unfalsifiability. (48.40)

The theory therefore revises itself through evidence.

It should not grow only by accumulating analogies.

48.20 Final research-programme statement

The practical programme can be summarized as:

Declare
→ Reconstruct
→ Measure

→ Compare
→ Intervene
→ Transport
→ Close the Loop
→ Carry Residual
→ Revise. (48.41)

Or in compact form:

Research_F = Protocol + StrongNulls + ControlledIntervention + FrameRobustness + ResidualHonesty. (48.42)

The immediate scientific aim is not to announce quantum finance.

It is to determine how far a mature non-quantum financial world can reproduce the operational grammar of:

- complex state;
- internal time;
- composite identity;
- conditional collapse;
- curved transport;
- observer-relative strangeness.

Only what survives that reconstruction should remain in the specifically quantum residue.

Part IX — Conclusion

49. From One Complex Price to One Composite Financial World

49.1 The point of departure

This investigation began with an ordinary feature of mature finance.

A future economic possibility does not enter a usable ledger without filtration.

It is:

- discounted;
- risk-adjusted;
- liquidity-adjusted;
- credit-adjusted;
- certainty-adjusted;
- capital-constrained;

- legally classified;
- institutionally admitted.

The visible result is usually one scalar value:

R. (49.1)

Finance Geometry proposed retaining the orthogonal component suppressed by that scalar admission:

$Z = R + iQ$. (49.2)

With:

$A^2 = R^2 + Q^2$. (49.3)

$R = A \cos \theta$. (49.4)

$Q = A \sin \theta$. (49.5)

$Z = A \exp(i\theta)$. (49.6)

The original move was therefore not an attempt to replace mature finance.

It was a completion of its declared filtration geometry.

R represented what the filter admitted.

Q represented pressure implied by the same declared amplitude but hidden when the output was compressed into one real number.

The earlier article then made the geometry dynamic:

$Z(t) = A(t)\exp[i\theta(t)]$. (49.7)

$dZ/dt = [g_A + i\omega_F]Z + \varepsilon_dyn$. (49.8)

This separated visible repricing into:

$dR/dt = g_AR - Q\omega_F + Re(\varepsilon_dyn)$. (49.9)

The three terms represented:

1. radial change in economic amplitude;
2. angular change in the valuation frame;
3. residual not captured by the declared model.

This already showed why identical visible price changes can result from different underlying processes. One decline may arise from deteriorating economic amplitude, while another may arise from a tightening valuation frame. Part I explicitly developed this distinction and treated CAPM as one declared valuation world rather than one universal ontology.

49.2 From coordinate to world

A complex coordinate does not yet constitute a world.

The decisive extension was:

Primary Field
→ Declaration
→ Projection
→ R + iQ
→ Phase
→ Gate
→ Ledger
→ Backreaction
→ Revision. (49.10)

A financial representation became world-like when it acquired:

- effective states;
- approximately closed dynamics;
- admissible observations;
- commitment gates;
- trace;
- intervention;
- backreaction;
- residual governance;
- frame reconciliation.

The world was not required to be fundamentally independent of the primary economy.

It became an effective world because its records changed what happened next.

A margin calculation altered collateral.

A rating altered funding.

A settlement record altered ownership.

A regulatory classification altered admissibility.

Thus:

Representation + Consequence = Effective World. (49.11)

The financial world did not merely describe the primary field.

It participated in constructing its next state.

49.3 CAPM's corrected architectural role

The central correction of Part II is that CAPM should not be treated as the entire financial universe.

CAPM remains:

$$E[r_i] = r_f + \beta_i ERP. \quad (49.12)$$

But its parameters depend on:

- regime;
- benchmark;
- horizon;
- funding;
- leverage;
- liquidity;
- protocol;
- ledger.

Therefore:

$$\beta_i = \beta_i(x,P,L). \quad (49.13)$$

$$ERP = ERP(x,P,L). \quad (49.14)$$

$$r_f = r_f(x,P,L). \quad (49.15)$$

CAPM is most defensibly interpreted as:

$$CAPM = \text{Local Valuation Law inside a Declared Frame}. \quad (49.16)$$

On a curved financial manifold $\mathcal{M}_F$, CAPM operates approximately in a local tangent region:

$$E[r_i]_{local} = r_{f,local} + \beta_{i,local} ERP_{local}. \quad (49.17)$$

The global theory may contain:

- changing metrics;
- changing frames;
- derivatives;
- hedge feedback;
- margin gates;
- ledger history;
- institutional intervention.

Therefore:

$$\hat{H}_{CAPM} \neq \hat{H}_{total}. \quad (49.18)$$

CAPM supplies one local generator.

It does not independently generate the entire QM-like, SR-like, and GR-like architecture.

49.4 Why derivatives were the necessary next step

The scalar complex state:

$Z \in \mathbb{C}$ (49.19)

contains two coordinates but not two subsystems.

R and Q are components of one valuation state.

They do not define:

$\mathscr{H}_A \otimes \mathscr{H}_B$. (49.20)

Without a subsystem decomposition, one cannot rigorously define:

- local states;
- joint states;
- partial traces;
- product states;
- separable mixtures;
- nonfactorizable states;
- entanglement witnesses.

Derivatives supplied the missing composite grammar.

For an underlying sector U and derivative sector D:

$\mathscr{H}_{UD} = \mathscr{H}_U \otimes \mathscr{H}_D$. (49.21)

The contract acts as a preparation or binding operator:

$\hat{U}_C|u_n,0_D\rangle = |u_n,d_n\rangle$. (49.22)

Applied to:

$|\psi_U\rangle = \Sigma_n c_n|u_n\rangle$, (49.23)

it prepares:

$|\Psi_{UD}\rangle = \Sigma_n c_n|u_n,d_n\rangle$. (49.24)

This state is generally not expressible as:

$|\psi_U\rangle \otimes |\psi_D\rangle$. (49.25)

Derivative finance therefore adds something scalar CAPM lacked:

Composite Identity. (49.26)

The derivative is not merely another price correlated with the underlying.

Its effective identity is constituted through:

- underlying;
- payoff;
- strike;
- maturity;
- exercise rule;
- settlement rule;
- legal contract;
- valuation protocol.

49.5 The observer-perspective correction

The most important conceptual correction is that derivative entanglement must not be assessed only from the primary economic universe.

From the primary universe, the option appears ordinary:

Underlying State
→ Pricing Rule
→ Derivative Value. (49.27)

The primary observer sees:

- contract creation;
- pricing machinery;
- data transmission;
- arbitrage;
- hedge execution;
- clearing;
- settlement.

It possesses Constructor Privilege.

From this external position, the option–underlying relationship is explainable.

But the corresponding quantum observer does not stand outside the physical universe with full access to its constructor.

It is an observer inside the world.

The correct financial comparison therefore requires three levels:

Primary Constructor Universe
→ Secondary θ-Time Valuation World
→ Internal Protocol-Bounded Observer. (49.28)

The secondary world is compiled through:

$$\rho_F(\theta) = \mathcal{C}_{P,L}[X(t)]. \quad (49.29)$$

The internal observer accesses only:

$$\text{Visible}_O(\theta) = \hat{O}_{O,P,L}[\rho_F(\theta)]. \quad (49.30)$$

It does not necessarily possess:

- the full preparation process;
- the complete primary state;
- every counterparty position;
- every valuation branch;
- the complete inverse compilation map.

From inside W_θ, the option and underlying may therefore appear as locally incomplete parts of one globally prepared relation.

This yields the central distinction:

$$\text{External Constructibility} \neq \text{Internal Separability}. \quad (49.31)$$

49.6 Entanglement is global; strangeness is local

Suppose the global state is:

$$\rho_{UD}. \quad (49.32)$$

The local states are:

$$\rho_U = \mathrm{Tr}_D(\rho_{UD}). \quad (49.33)$$

$$\rho_D = \mathrm{Tr}_U(\rho_{UD}). \quad (49.34)$$

The global state may contain relational information absent from either local state.

Define:

$$\chi_{UD} = \rho_{UD} - \rho_U \otimes \rho_D. \quad (49.35)$$

When:

$$\chi_{UD} \neq 0, \quad (49.36)$$

the local states do not reconstruct the full joint state.

However, correlation alone is insufficient.

A classically separable state may be:

$\rho_{sep} = \Sigma_j p_j \rho_j^U \otimes \rho_j^D$. (49.37)

The stronger effective nonseparability condition is:

$\rho_{UD} \notin Sep(\mathscr{H}_U \otimes \mathscr{H}_D)$. (49.38)

Observer ignorance does not create entanglement.

The observer's bounded access explains why a global relation appears strange locally, but the global relation must still possess objective nonfactorization relative to the declared partition.

Thus:

Entanglement Is Global Structure; Strangeness Is Local Access. (49.39)

49.7 Three forms of financial time

The completed architecture distinguishes:

t = calendar time. (49.40)

θ = internal phase order. (49.41)

k = ledger-event order. (49.42)

Calendar time belongs to primary economic duration.

Phase time orders state movement inside the secondary valuation world.

Ledger time advances when an event becomes committed trace.

Their relation is:

$t \to \theta(t) \to Gate_k \to Record_k \to L_{k+1}$. (49.43)

A state may evolve continuously in θ without creating a new historical fact.

A historical transition requires:

Phase + Gate + Trace. (49.44)

This explains why two paths can return to the same visible price while remaining different financial worlds:

$R_{final}^A = R_{final}^B$, (49.45)

but:

$L_{final}^A \neq L_{final}^B$. (49.46)

A barrier hit, margin call, default, settlement, or legal recognition changes the future even after the visible coordinate returns.

Thus:

Same Coordinate ≠ Same History. (49.47)

And:

49.8 The role of internal observers

An internal observer was defined as:

$O = (\mathcal{A}_O, \mathbb{M}_O, F_O, L_O, \pi_O)$. (49.49)

It possesses:

- an accessible algebra;
- a measurement family;
- a local frame;
- a trace;
- an adaptive policy.

At episode k:

$a_k = \pi_O(L_{O,k-1})$. (49.50)

The observer receives outcome y_k:

$y_k \sim \mathbb{M}_{a_k}[\rho_F(\theta_k)]$. (49.51)

The record updates:

$L_{O,k} = L_{O,k-1} \sqcup \mathrm{Record}(y_k, a_k)$. (49.52)

The next setting changes:

$a_{k+1} = \pi_O(L_{O,k})$. (49.53)

This produces latching:

Different Outcome
→ Different Trace
→ Different Later Measurement
→ Different Future World. (49.54)

The related observer framework formalizes precisely this filtration-based internal certainty and shows that cross-observer fixedness requires a frame map, compatible effects, and an accessible record.

Thus financial collapse was not defined as magical disappearance of possibility.

It was defined operationally as:

Instrument
→ Conditional Outcome
→ Authorized Gate
→ Persistent Record
→ Trace-Dependent Future. (49.55)

49.9 Why objectivity does not require an observer-free view

Observer relativity does not imply arbitrary subjectivity.

Cross-observer objectivity requires:

1. mapped contexts;
2. mapped outcomes;
3. compatible observations;
4. accessible records;
5. redundant encoding;
6. invariant transport.

In compact form:

Objectivity_P = FrameMap + Compatibility + Record + Redundancy + Invariance. (49.56)

A financial event becomes increasingly objective when it is redundantly encoded across:

- exchange;
- broker;
- clearinghouse;
- custodian;
- bank;
- counterparty;
- regulator.

The observer theory likewise treats redundancy as the mechanism through which high-probability agreement emerges, while warning that geometry alone cannot secure agreement without compatible algebra and accessible records.

Therefore:

Observer-Relative Disclosure + Redundant Trace → Operational Objectivity. (49.57)

49.10 The layered QM–SR–GR architecture

The completed toy architecture assigns each layer a distinct role.

QM-like layer

The complex fibre carries:

- amplitudes;

- phase;
- composite states;
- local and joint observables;
- conditional measurement;
- density operators;
- possible coherence.

SR-like layer

Local valuation frames carry:

- relative benchmarks;
- numeraires;
- funding structures;
- local rapidities;
- local-flat valuation laws;
- invariant transformations.

GR-like layer

The global base manifold carries:

- state-dependent distance;
- liquidity curvature;
- leverage curvature;
- funding curvature;
- collateral curvature;
- path transport;
- source–geometry backreaction.

Gauge layer

The connection carries:

- local phase comparison;
- frame alignment;
- covariant transport;
- loop phase;

- holonomy.

Ledger layer

Trace carries:

- commitment;
- historical order;
- latching;
- institutional finality;
- future admissibility.

The layers combine through:

$$i\hbar_F \mathcal{D}_\theta |\Psi_F\rangle = \hat{H}_F[g^F, \mathcal{A}, P, L]|\Psi_F\rangle + |\varepsilon_F\rangle. \quad (49.58)$$

Where:

$$\mathcal{D}_\theta = \partial_\theta + \dot{x}^\mu \nabla_\mu + i\dot{x}^\mu \mathcal{A}_\mu. \quad (49.59)$$

And:

$$\hat{H}_F = \hat{H}_{CAPM} + \hat{H}_{contract} + \hat{H}_{hedge} + \hat{H}_{ledger} + \hat{H}_{environment} + \hat{H}_{intervention}. \quad (49.60)$$

This is not a physical unification equation.

It is a role-separated effective architecture.

The earlier article's Appendix M explicitly insisted that CAPM's strongest defensible quantum-like relation was a classical complex state with contextual projection, gate, trace, and backreaction—not a literal quantum system.

Part II retains that boundary while showing how additional layers enlarge the available operational grammar.

49.11 Can CAPM now simulate more quantum characteristics?

The answer is:

Yes—but not through CAPM alone.

Scalar CAPM plus R+iQ directly supports:

- complex representation;
- phase-like orientation;
- classical rotational evolution;
- measurement-frame dependence;

- radial versus angular repricing.

It does not directly support:

- composite state spaces;
- derivative entanglement;
- reduced mixed states;
- coherent branch interference;
- contextual joint measurements;
- geometric phase;
- decoherence;
- Bell structure.

Those additional features become formally available only after CAPM is embedded inside a larger architecture:

CAPM Local Kernel

- Multi-Channel Complex Completion
- Derivative Tensor Structure
- Contract Preparation
- Internal Observers
- Gates and Ledger
- Gauge Transport
- Curved Geometry
- Backreaction. (49.61)

The correct statement is therefore:

CAPM Alone Simulates Classical Phase Rotation. (49.62)

CAPM-Seeded Effective Financial Field Theory Simulates a Broader Quantum-Like Operational Grammar. (49.63)

This distinction prevents the entire architecture from being misattributed to one local pricing equation.

49.12 Quantum-like characteristics newly reproducible after Part II

The completed architecture can now reproduce or formally model the following characteristics more strongly than Part I.

Superposition-like unresolved alternatives

A multi-channel state may be:

$|\Psi_F\rangle = \Sigma_n c_{nA}\exp(i\theta_n)|n\rangle$. (49.64)

This represents unresolved alternatives before commitment.

It remains classical unless relative phase has operational consequences.

Entanglement-form composite states

An option–underlying state may satisfy:

$|\Psi_{UD}\rangle \neq |\psi_U\rangle \otimes |\psi_D\rangle$. (49.65)

This creates global completeness with local incompleteness.

Reduced mixed states

Local observers receive:

$\rho_U = Tr_D(\rho_{UD})$. (49.66)

$\rho_D = Tr_U(\rho_{UD})$. (49.67)

Conditional collapse

One local outcome changes the conditional state assigned to the other sector:

$\rho_{D|u} = Update_D(\rho_{UD},u)$. (49.68)

Measurement-basis dependence

The same financial object may be measured through:

- price;
- risk;
- payoff;
- replication;
- collateral;
- liquidation;
- accounting;
- regulatory bases.

Operational noncommutativity

Two gates or instruments may satisfy:

$\mathbb{M}_A \circ \mathbb{M}_B \neq \mathbb{M}_B \circ \mathbb{M}_A$. (49.69)

Decoherence-like classicalization

Environmental records suppress active branch alternatives:

Private Possibility
→ Public Quote
→ Trade
→ Clearing
→ Settlement
→ Redundant Ledger. (49.70)

Geometric phase and holonomy

A closed coordinate loop may leave:

$|\Psi_final\rangle \neq |\Psi_initial\rangle$, (49.71)

or:

$L_final \neq L_initial$. (49.72)

Tunnelling-like gate transition

A state may cross a region of low ordinary accessibility through:

- barrier activation;
- liquidity gap;
- rare intervention;
- regime switch.

But such movement remains classical barrier crossing unless an amplitude law supplies a distinct tunnelling probability.

Uncertainty-like trade-offs

One measurement may disrupt another operational quantity:

Exact Liquidity Probe
↔ Undisturbed Market State. (49.73)

But this remains an engineered measurement–disturbance trade-off unless a universal commutator-bound relation is established.

Thus Part II greatly enlarges the simulation repertoire while keeping the evidence hierarchy explicit.

49.13 Which quantum characteristics remain unsimulated?

The architecture still does not derive:

- Born-rule necessity;

- experimentally mandatory amplitude coherence;
- irreducible tensor nonseparability;
- no-signalling entanglement;
- Bell inequality violation;
- Kochen–Specker contextuality;
- no-cloning;
- entanglement monogamy;
- Tsirelson bounds;
- quantum exchange statistics;
- fundamental measurement disturbance.

The remaining residue can be summarized as:

Q_residue
= Born Necessity

- Mandatory Coherence
- Irreducible Nonseparability
- No-Signalling Correlation
- Bell Nonclassicality
- Fundamental Contextuality
- Quantum Information Constraints. (49.74)

This residue is smaller and more precise than the original undifferentiated category of “quantum strangeness.”

49.14 Does a classical constructor dissolve quantum mystery?

Not automatically.

A deeper primary constructor may explain how a secondary world is generated.

But two questions remain distinct:

1. Is the secondary theory operationally complete for its internal observers?
2. Can the internal statistics be reproduced by a classical joint model?

If all internal statistics admit:

$$p(a,b|x,y) = \int p(\lambda)p(a|x,\lambda)p(b|y,\lambda)d\lambda, \quad (49.75)$$

then the apparent entanglement may have a classical completion.

If no such model survives appropriate tests, the deeper-constructor hypothesis becomes more constrained.

Therefore:

Deeper Constructor ≠ Guaranteed Classical Reducibility. (49.76)

And:

Measurement Constraint ≠ Sufficient Explanation of Bell Nonclassicality. (49.77)

The financial world shows that hidden construction and observer restriction can explain much of the phenomenology.

It does not show that they explain the complete quantum probability structure.

49.15 What finance contributes to quantum foundations

Finance contributes a powerful control example.

It demonstrates that the following can arise in an explicitly non-quantum domain:

- complex states;
- phase;
- contextual projection;
- local observer restriction;
- contract-prepared composite relations;
- collapse-like commitment;
- path-dependent records;
- internal time;
- frame relativity;
- curved geometry;
- gauge transport;
- observer backreaction;
- emergent objectivity.

Therefore none of these features, by itself, proves quantum ontology.

The correct foundations question becomes:

> Which empirically observed quantum structures remain after the strongest observer-bound, composite, trace-bearing, non-quantum world has been subtracted?

This is a more demanding and more informative question than asking why quantum mechanics appears mysterious in the first place.

49.16 What physics contributes back to finance

The correspondence also runs in the opposite direction.

Quantum, relativistic, gauge, and geometric structures provide finance with:

- a checklist of missing state variables;
- a language for local versus global law;
- tools for composite-state modelling;
- tests of basis and frame invariance;
- methods for detecting path-dependent transport;
- stronger distinction between mixture and coherence;
- explicit no-signalling and contextuality boundaries;
- disciplined reduction limits.

The physics architecture therefore acts as:

1. a diagnostic checklist;
2. a benchmark of structural completeness;
3. a pathway toward deeper extensions.

Its value does not depend on markets being physically quantum.

49.17 The practical engineering result

The most immediate financial value may not lie in quantum simulation.

It may lie in detecting hidden incompleteness.

The framework asks:

- Is the observed move radial or frame-induced?
- Which pressure coordinate was suppressed?
- Is the claimed frame transformation truly invariant?
- Did the metric change?
- Did a gate alter the state space?
- Did the ledger alter future dynamics?
- Did a closed loop leave residual?

- Is the local desk state incomplete without the global contract state?
- Does the tensor representation improve prediction?
- Does phase add anything beyond covariance and feedback?

These questions convert the architecture into an engineering audit.

The Gauge Grammar similarly treats mature intelligence as the extraction of stable structure together with explicit residual governance, trace, gates, and invariance—not merely greater scale or richer terminology.

49.18 The final architecture

The complete Part II runtime is:

X_t
→ Declare_P,L(X_t)
→ x ∈ $\mathcal{M}$_F
→ Local Frame e^a_μ
→ Complex Fibre $\mathscr{H}$_x
→ Composite State ρ_UD
→ Contract and Hedge Coupling
→ Covariant θ-Evolution
→ Internal Measurement
→ Gate
→ Ledger k
→ Primary Backreaction
→ Geometry and Protocol Revision. (49.78)

In compact form:

World_{k+1} = Revise[Backreact[Ledger[Gate[Measure[Evolve[Compile(X_k)]]]]]]. (49.79)

The architecture is recursive.

Its effective laws help create the conditions under which their next application occurs.

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50. Final Propositions

50.1 Proposition 1 — Scalar valuation is a projection, not the whole financial state

A scalar value R is the output of a declared filter.

When the amplitude A is independently defined, the pressure-preserving completion is:

Z = R + iQ. (50.1)

Therefore:

Scalar Value = Admitted Coordinate, Not Complete Possibility Field. (50.2)

This proposition does not imply that Q is a second market price.

Q is a retained state coordinate under the declared completion.

50.2 Proposition 2 — Visible repricing has radial, angular, and residual components

For:

$$Z(t) = A(t)\exp[i\theta(t)], \quad (50.3)$$

the real coordinate obeys:

$$dR/dt = g_{AR} - Q\omega_F + \mathrm{Re}(\varepsilon_{dyn}). \quad (50.4)$$

Therefore identical visible price moves may arise from different combinations of:

- economic-amplitude change;
- valuation-frame rotation;
- model residual.

This distinction is empirically testable.

50.3 Proposition 3 — CAPM is most defensibly a local law

CAPM should be written:

$$E[r_i|x,P,L] = r_f(x,P,L) + \beta_i(x,P,L)\mathrm{ERP}(x,P,L). \quad (50.5)$$

Its parameters are local to:

- state;
- protocol;
- frame;
- ledger;
- regime.

Therefore:

$$\text{CAPM} \approx \text{Local Flat-Frame Valuation Law}. \quad (50.6)$$

Its global failure need not invalidate its local usefulness.

50.4 Proposition 4 — CAPM alone does not contain entanglement

The scalar state:

$Z \in \mathbb{C}$ (50.7)

does not supply:

$\mathscr{H}_U \otimes \mathscr{H}_D$. (50.8)

Therefore CAPM plus one complex coordinate cannot independently derive:

- tensor nonseparability;
- reduced mixed states;
- entanglement witnesses;
- Bell structure.

The necessary composite grammar enters through derivatives and other relational financial objects.

50.5 Proposition 5 — Derivatives are natural composite-state constructors

A derivative contract creates a stable distinction between:

- underlying sector;
- derivative sector;

while binding them through one payoff and settlement relation.

The natural state space is:

$\mathscr{H}_{UD} = \mathscr{H}_U \otimes \mathscr{H}_D$. (50.9)

The contract acts as:

$\hat{U}_C|u_n,o_D\rangle = |u_n,d_n\rangle$. (50.10)

Thus derivatives provide the missing architecture required to formulate financial nonseparability.

50.6 Proposition 6 — Contractual coupling is stronger than correlation but weaker than quantum entanglement

Finance clearly realizes:

E_1 = statistical correlation. (50.11)

E_2 = functional dependence. (50.12)

E_3 = contractual coupling. (50.13)

E_4 = dynamical binding. (50.14)

It may formally construct:

E_5 = effective-state nonfactorization. (50.15)

It has not established:

E_8 = no-signalling entanglement. (50.16)

E_9 = Bell-nonclassical entanglement. (50.17)

Therefore the word entanglement must always be qualified by level.

50.7 Proposition 7 — The correct comparison is internal to the secondary world

The primary universe may transparently construct the contract.

But an internal observer receives only the compiled effective state:

$\rho_F(\theta) = \mathcal{C}_{P,L}[X(t)]$. (50.18)

Its visible world is:

$\hat{O}_{O,P,L}[\rho_F(\theta)]$. (50.19)

Therefore:

Primary Transparency Does Not Eliminate Secondary Nonfactorization. (50.20)

And:

Secondary Nonfactorization Does Not Prove Primary Quantum Ontology. (50.21)

Both levels must remain visible.

50.8 Proposition 8 — Entanglement strangeness arises from local access to global structure

A global state may be complete while its local reductions are mixed:

$\rho_U = Tr_D(\rho_{UD})$. (50.22)

$\rho_D = Tr_U(\rho_{UD})$. (50.23)

The local observer does not possess the full relational state.

Thus:

Entanglement Is Global Structure; Strangeness Is Local Access. (50.24)

Observer restriction does not create the global relation.

It determines how that relation is experienced.

50.9 Proposition 9 — Observer ignorance alone is insufficient

A classically correlated state may be:

$\rho_{sep} = \Sigma_j p_j \rho_j^U \otimes \rho_j^D$. (50.25)

Strong correlation and local uncertainty can exist within this separable form.

Therefore:

Ignorance + Correlation ≠ Entanglement. (50.26)

A stronger claim requires:

$\rho_{UD} \notin \mathrm{Sep}(\mathcal{H}_U \otimes \mathcal{H}_D)$, (50.27)

together with operational observables and classical-null rejection.

50.10 Proposition 10 — θ-time and ledger time perform different roles

Phase θ orders internal state progression.

Ledger index k orders committed history.

The relation is:

$t \to \theta(t) \to \mathrm{Gate}_k \to L_{k+1}$. (50.28)

Thus:

Continuous Evolution ≠ Historical Commitment. (50.29)

History begins when a gate produces persistent trace.

50.11 Proposition 11 — Collapse is an internal runtime process

Financial collapse-like commitment is:

Instrument
→ Outcome
→ Gate
→ Record
→ Trace-Dependent Future. (50.30)

It produces:

- local certainty;
- branch latching;

- policy divergence;
- institutional consequence.

It does not by itself produce:

- Born probability;
- physical wavefunction collapse;
- quantum nonlocality.

50.12 Proposition 12 — Objectivity is constructed through compatibility and redundancy

Cross-observer agreement requires:

Frame Map + Compatible Effect + Accessible Record. (50.31)

Multi-observer objectivity further requires sufficient reliable redundancy.

Therefore:

Objectivity ≠ Absence of Observers. (50.32)

Objectivity = Stable Agreement across Admissible Observer Frames. (50.33)

50.13 Proposition 13 — Local relativity and global curvature are compatible

Local frames are related through tetrad-like maps:

$g^{F}_{\mu\nu} = e^{a}_{\ \mu} e^{b}_{\ \nu} \eta_{ab}$. (50.34)

CAPM may remain valid locally even when:

$R^{\rho}_{\ \sigma\mu\nu} \neq 0$ (50.35)

globally.

Thus:

SR-Like Frame = Local Description. (50.36)

GR-Like Geometry = Global State-Dependent Environment. (50.37)

The two layers are nested rather than equated.

50.14 Proposition 14 — Gauge transport is distinct from metric geometry

The metric connection transports base-manifold vectors:

$\nabla_\mu v^\nu = \partial_\mu v^\nu + \Gamma^\nu_{\mu\lambda} v^\lambda$. (50.38)

The gauge connection transports fibre orientation:

$D_\mu|\psi\rangle = \partial_\mu|\psi\rangle + i\mathcal{A}_\mu|\psi\rangle$. (50.39)

Therefore:

Metric Curvature ≠ Gauge Curvature. (50.40)

Double-counting the same financial effect through both layers invalidates the architecture.

50.15 Proposition 15 — A closed coordinate loop need not close the financial state

For a loop γ:

$x_{final} = x_{initial}$. (50.41)

But potentially:

$|\Psi_{final}\rangle = Hol(\gamma)|\Psi_{initial}\rangle$. (50.42)

And:

$L_{final} \neq L_{initial}$. (50.43)

Therefore:

Endpoint Equality ≠ State Equality. (50.44)

Loop tests can diagnose omitted:

- ledger;
- phase;
- inventory;
- path memory;
- curvature.

50.16 Proposition 16 — Phase becomes substantive only through relative-phase observables

A single global phase has no operational effect.

Interference requires:

$\Delta\theta_{nm} = \theta_n - \theta_m$. (50.45)

And an observable sensitive to cross terms:

$Tr(\rho_{coh}\hat{O}_{cross}) \neq Tr(\rho_{diag}\hat{O}_{cross})$. (50.46)

Therefore:

Complex Notation without Phase-Sensitive Observation = Redundant Representation. (50.47)

A financial coherence claim must demonstrate controlled, reproducible phase effects.

50.17 Proposition 17 — Standard option probabilities do not derive the Born rule

A formal lift may define:

$\alpha_n = \sqrt{q_n}\exp(i\varphi_n)$. (50.48)

Then:

$q_n = |\alpha_n|^2$. (50.49)

But risk-neutral weights q_n are already supplied by mature finance.

The squared-amplitude rule is inserted rather than derived.

Therefore:

Risk-Neutral Probability ≠ Born Probability. (50.50)

The Born rule remains part of the quantum residue.

50.18 Proposition 18 — Conditional update is not no-signalling

The conditional state:

$\rho_{D|u}$ (50.51)

may change after an underlying outcome is known.

But no-signalling requires:

$p(d|a,b) = p(d|b)$. (50.52)

Financial measurements often transmit information or alter markets.

Therefore:

Conditional State Change ≠ Bell-Type Nonlocality. (50.53)

A no-signalling test must be performed separately.

50.19 Proposition 19 — Bell-style tests in finance are assumption diagnostics

A financial CHSH-like quantity is:

$S_F = E(a_0,b_0) + E(a_0,b_1) + E(a_1,b_0) - E(a_1,b_1)$. (50.54)

An apparent value:

$|S_F| > 2$ (50.55)

does not establish quantum finance unless the experiment controls:

- signalling;
- measurement dependence;
- memory;
- common causes;
- selection;
- nonstationarity.

Its first use is therefore diagnostic:

Bell-Style Stress Test → Identify Which Classical Assumption Fails. (50.56)

50.20 Proposition 20 — The quantum residue is defined only after strong subtraction

The revised subtraction is:

Quantum Observation = G_world + G_composite + G_geometry + Q_residue. (50.57)

Where:

G_world = Declaration + Projection + Gate + Trace + Backreaction. (50.58)

G_composite = Partition + Joint Preparation + Local Restriction + Conditional Update + Relative Phase. (50.59)

G_geometry = Local Frame + Metric + Connection + Curvature + Holonomy. (50.60)

The quantum residue contains the structures that resist this reconstruction.

50.21 Proposition 21 — The theory must shrink when evidence fails

Each layer has a removal condition.

If Q adds no value, remove Q.

If θ does not provide stable order, use t or k.

If frame rapidity does not compose, classify the change as a protocol transition.

If curvature does not improve transport, use a simpler state model.

If off-diagonal terms do not affect observables, use the mixture.

If separable models fit, reject the entanglement claim.

Therefore:

No Measurable Gain → No Theoretical Upgrade. (50.61)

50.22 Proposition 22 — Functional homology does not imply material identity

Finance and physics may share:

- mathematical form;
- observer architecture;
- phase grammar;
- composite-state logic;
- gate and trace structure;
- geometric transport.

But:

Functional Homology ≠ Material Identity. (50.62)

A shared equation may identify a common control problem without identifying a common substrate.

This is the correct standard for the cross-domain correspondence.

50.23 Proposition 23 — The primary scientific value is sharper subtraction

The financial model does not solve quantum foundations.

It improves the question.

It shows that:

- complex phase;
- local incompleteness;
- contextual measurement;
- collapse-like commitment;
- observer backreaction;
- internal time;

- emergent objectivity;
- curved transport;

can arise in a non-quantum effective world.

Therefore the specifically quantum problem begins later than is often assumed.

The remaining question is:

> Why does the physical quantum world impose its particular probability, nonseparability, no-signalling, contextuality, and information-theoretic constraints?

That is the residue deserving fundamental explanation.

50.24 Final synthesis

The entire article can be compressed into one layered statement:

> The primary financial universe constructs contracts, institutions, and market mechanisms in calendar time. A declared compiler projects part of that universe into a secondary valuation world ordered by θ and historicized by ledger time k. CAPM operates locally within that world as a valuation law. Derivative contracts prepare composite states. Internal observers access only bounded measurement algebras. SR-like transformations relate local frames. GR-like geometry governs global financial distance and path transport. Gauge connections preserve relative orientation. Gates convert possibility into committed trace. Hedging and institutional action return the result to the primary field. From within this world, an option and its underlying may appear as locally incomplete components of one globally prepared relation. This reproduces important operational elements of quantum strangeness while leaving Born probability, no-signalling, Bell nonclassicality, no-cloning, and specifically quantum information structure unresolved.

The final architecture is therefore:

Local CAPM
inside Relative Frames
inside Curved Financial Geometry
carrying Complex Composite States
observed through Contextual Instruments
committed through Gates
retained through Ledgers
and revised through Backreaction. (50.63)

Its central lesson is:

Unity Does Not Come from Making One Symbol Mean Everything. (50.64)

It comes from:

Distinct Roles + Explicit Couplings + Valid Reduction Limits + Testable Residual. (50.65)

And its final methodological proposition is:

Build the Strongest Non-Quantum World First.
Then Ask What Quantum Structure Still Cannot Be Removed. (50.66)

Appendices

Appendix A — Notation, Layer Assignment, and Role Discipline

A.1 Why a notation appendix is necessary

The architecture uses mathematical structures borrowed from several mature fields:

- asset pricing;
- derivative valuation;
- complex analysis;
- probability;
- operator theory;
- information geometry;
- differential geometry;
- gauge theory;
- observer theory;
- ledgered world formation.

The same symbol can carry different meanings across those fields.

The theory remains coherent only when every variable is assigned to one declared layer.

The master rule is:

One Mathematical Role → One Declared Symbol. (A.1)

Related variables may be mapped.

They should not be silently identified.

A.2 Primary-universe symbols

Symbol	Meaning
X(t)	Primary economic state in calendar time
t	Calendar time
U(t)	Underlying economic or market state
D(t)	Derivative market or contract state
CF_T	Cash flow at terminal date T
P	Declared valuation or observation protocol
B	Declared system boundary
L	Ledger inherited by the current state

Symbol	Meaning
u	Intervention or control input
ε	Residual not explained by the declared model

The primary state may contain:

X(t) = [U(t),D(t),σ(t),r(t),q(t),H(t),C(t),L(t),Environment(t),...]. (A.2)

The primary universe contains the mechanisms that construct and update the secondary valuation world.

A.3 Scalar Finance Geometry symbols

Symbol	Meaning
A	Declared pre-filter value amplitude
R	Admitted real valuation coordinate
Q_E	Euclidean retained-pressure coordinate
Z	Complex financial state
θ	Circular valuation-filter angle
g_A	Radial amplitude-growth rate
ω_F	Financial phase velocity
Λ_F	Angular repricing contribution

The basic geometry is:

$Z = R + iQ_E$. (A.3)

$A^2 = R^2 + Q_E^2$. (A.4)

$R = A \cos \theta$. (A.5)

$Q_E = A \sin \theta$. (A.6)

$Z = A \exp(i\theta)$. (A.7)

The dynamic decomposition is:

$dZ/dt = (g_A + i\omega_F)Z + \varepsilon_{dyn}$. (A.8)

Where:

$g_A = (1/A)(dA/dt)$. (A.9)

$\omega_F = d\theta/dt$. (A.10)

$\Lambda_F = Q_E\omega_F$. (A.11)

The admitted-value equation is:

$dR/dt = g_AR - \Lambda_F + Re(\varepsilon_{dyn})$. (A.12)

A.4 Distinguishing Q_E from other uses of Q

The symbol Q_E denotes Euclidean retained pressure:

$$Q_E = \sqrt{(A^2 - R^2)}. \quad (A.13)$$

It must not be confused with the risk-neutral measure:

$\mathbb{Q}$ = risk-neutral probability measure. (A.14)

Nor with the hyperbolic light-cone pressure:

$$Q_H = (A^2 - R^2)/(2R). \quad (A.15)$$

The relation between the two pressure coordinates is:

$$Q_H = Q_E^2/(2R). \quad (A.16)$$

Thus:

$$Q_E \neq Q_H. \quad (A.17)$$

And:

$$Q_E \neq \mathbb{Q}. \quad (A.18)$$

A.5 Circular and hyperbolic variables

Symbol	Definition	Role
θ	$\arccos(R/A)$	Circular phase and filter orientation
η_Q	$\operatorname{artanh}(Q_E/A)$	Q-preserving hyperbolic rapidity
ρ_D	$\ln(A/R)$	Additive discount-frame rapidity
u_D	$\tanh \rho_D$	Bounded relative-frame coordinate

The relations are:

$$\sin \theta = Q_E/A. \quad (A.19)$$

$$\cos \theta = R/A. \quad (A.20)$$

$$\tanh \eta_Q = Q_E/A. \quad (A.21)$$

$$\cosh \eta_Q = A/R. \quad (A.22)$$

$$\rho_D = \ln(A/R). \quad (A.23)$$

Therefore:

$$\sin \theta = \tanh \eta_Q. \quad (A.24)$$

$$\cos \theta = \operatorname{sech} \eta_Q. \quad (A.25)$$

$$\rho_D = \ln \cosh \eta_Q. \quad (A.26)$$

$$\rho_D = -\ln \cos \theta. \quad (A.27)$$

These are maps between representations.

They are not identities between mathematical roles:

$\theta \neq \eta_Q \neq \rho_D$. (A.28)

Appendix M of Part I similarly separates circular phase, Q-preserving rapidity, additive discount boosts, curved geometry, and gauge transport into distinct layers rather than forcing them into one coordinate.

A.6 Time variables

Symbol	Meaning
t	Primary calendar time
θ	Internal phase-order coordinate
k	Discrete ledger-event index
τ_L	Ledger-derived historical order
τ_F	Optional effective financial proper-time parameter
$T-t$	Remaining contractual maturity

The causal-disclosure chain is:

$t \rightarrow \theta(t) \rightarrow Gate_k \rightarrow Record_k \rightarrow L_{k+1}$. (A.29)

Ledger time is:

$\tau_L = order(L_0, L_1, L_2, \ldots)$. (A.30)

A ledger tick occurs when:

$L_{k+1} \neq L_k$. (A.31)

Calendar time can pass without a ledger tick.

Many ledger ticks can occur inside one short calendar interval.

Therefore:

$t \neq \theta \neq k$. (A.32)

The original valuation-world article treats ledger time as the order of committed records and emphasizes that a visible coordinate may return while the ledgered world remains changed.

A.7 Option Greek notation

Symbol	Meaning
Δ	Delta, $\partial D/\partial U$
Γ	Gamma, $\partial^2 D/\partial U^2$
Vega	$\partial D/\partial \sigma$
$Theta_D$	$\partial D/\partial t$
Rho_D	$\partial D/\partial r$
Vanna	$\partial^2 D/(\partial U \partial \sigma)$
Volga	$\partial^2 D/\partial \sigma^2$

Symbol	Meaning
Charm	$\partial^2 D/(\partial U \partial t)$

Option theta is written:

$\text{Theta}_D = \partial D/\partial t$. (A.33)

It must not be confused with the phase coordinate θ:

$\text{Theta}_D \neq \theta$. (A.34)

A.8 Composite-state symbols

Symbol	Meaning
$\mathscr{H}_U$	Effective underlying state space
$\mathscr{H}_D$	Effective derivative state space
$\mathscr{H}_{UD}$	Composite option–underlying state space
	$\psi_U\rangle$
	$\psi_D\rangle$
	$\Psi_{UD}\rangle$
ρ_{UD}	Joint density-like state
ρ_U	Reduced underlying state
ρ_D	Reduced derivative state
Sep	Set of separable states
Π_{contract}	Projector onto contract-admissible states

The joint space is:

$\mathscr{H}_{UD} = \mathscr{H}_U \otimes \mathscr{H}_D$. (A.35)

The local reductions are:

$\rho_U = \text{Tr}_D(\rho_{UD})$. (A.36)

$\rho_D = \text{Tr}_U(\rho_{UD})$. (A.37)

A product state is:

$\rho_{UD} = \rho_U \otimes \rho_D$. (A.38)

A separable state is:

$\rho_{\text{sep}} = \Sigma_j p_j \rho_j^U \otimes \rho_j^D$. (A.39)

A nonseparability claim requires:

$\rho_{UD} \notin \text{Sep}(\mathscr{H}_U \otimes \mathscr{H}_D)$. (A.40)

A.9 Observer symbols

Symbol	Meaning

Symbol	Meaning
O	Internal observer
$\mathcal{A}_O$	Observable algebra accessible to O
$\mathbb{M}_O$	Measurement-instrument family
F_O	Local observer frame
L_O	Accessible observer ledger
π_O	Trace-conditioned setting policy
$\mathscr{F}_{O,k}$	Observer information filtration after k records

An observer is:

$$O = (\mathcal{A}_O, \mathbb{M}_O, F_O, L_O, \pi_O). \quad (A.41)$$

The next setting is:

$$a_{k+1} = \pi_O(\mathscr{F}_{O,k}). \quad (A.42)$$

Internal certainty after a recorded outcome y_k is:

$$P_O(y_k \mid \mathscr{F}_{O,k}) = 1. \quad (A.43)$$

This means the observer's own committed past becomes fixed relative to its later trace.

It does not imply complete knowledge of the primary universe.

A.10 Base-manifold symbols

Symbol	Meaning
$\mathcal{M}_F$	Financial base manifold
x^μ	Global financial coordinate
$g^F_{\mu\nu}$	Financial metric
$\Gamma^\rho_{\mu\nu}$	Metric connection
$R^\rho_{\sigma\mu\nu}$	Curvature tensor
e^a_μ	Local tetrad-like frame
η_{ab}	Local flat-frame metric
$T^F_{\mu\nu}$	Effective financial stress tensor

The metric is:

$$ds_F^2 = g^F_{\mu\nu} dx^\mu dx^\nu. \quad (A.44)$$

The local frame relation is:

$$g^F_{\mu\nu} = e^a_\mu e^b_\nu \eta_{ab}. \quad (A.45)$$

The local-flat limit is:

$$g^F_{\mu\nu} \to \eta_{\mu\nu}. \quad (A.46)$$

The curvature is:

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}. \quad (A.47)$$

A.11 Fibre and gauge symbols

Symbol	Meaning
$\boldsymbol{\mathcal{H}}_F$	Total financial state bundle
$\mathcal{H}_x$	Local fibre at x
π	Bundle projection
$\mathcal{A}_\mu$	Gauge connection
D_μ	Fibre covariant derivative
$\mathbb{F}_{\mu\nu}$	Gauge curvature
$\mathrm{Hol}(\gamma)$	Holonomy around path γ

The bundle is:

$\pi: \boldsymbol{\mathcal{H}}_F \to \boldsymbol{\mathcal{M}}_F$. (A.48)

The fibre derivative is:

$D_\mu = \partial_\mu + i\mathcal{A}_\mu$. (A.49)

The gauge curvature is:

$\mathbb{F}_{\mu\nu} = \partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu + i[\mathcal{A}_\mu,\mathcal{A}_\nu]$. (A.50)

For an Abelian connection:

$\mathbb{F}_{\mu\nu} = \partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu$. (A.51)

The loop transport is:

$\mathrm{Hol}(\gamma) = \mathcal{P}\exp[-i\oint_\gamma \mathcal{A}_\mu dx^\mu]$. (A.52)

A.12 Gate and ledger symbols

Symbol	Meaning
G_k	Gate evaluated at episode k
Record_k	Committed event record
L_k	Ledger before episode k
$\mathcal{K}_L$	Ledger-conditioned state-update map
$\mathcal{U}$	Protocol-revision operator

A gate returns:

$G_k \in \{\mathrm{Commit},\mathrm{Reject},\mathrm{Defer},\mathrm{Escalate}\}$. (A.53)

Commitment updates:

$L_{k+1} = L_k \sqcup \mathrm{Record}_k$. (A.54)

The new state is:

$\rho_{k+1} = \mathcal{K}_{\{L_{k+1}\}}(\rho_k)$. (A.55)

The protocol may revise:

$P_{k+1} = \mathcal{U}(P_k, L_{k+1}, \text{Residual}_k)$. (A.56)

A.13 Generator symbols

The integrated generator is:

$\hat{H}_F = \hat{H}_{local} + \hat{H}_{contract} + \hat{H}_{hedge} + \hat{H}_{ledger} + \hat{H}_{environment} + \hat{H}_{intervention}$. (A.57)

The covariant state equation is:

$i\hbar_F \mathcal{D}_\theta |\Psi_F\rangle = \hat{H}_F |\Psi_F\rangle + |\varepsilon_F\rangle$. (A.58)

Where:

$\mathcal{D}_\theta = \partial_\theta + \dot{x}^\mu \nabla_\mu + i\dot{x}^\mu \mathcal{A}_\mu$. (A.59)

The equation is a formal toy architecture.

It is not asserted as a physical law of markets.

A.14 Prohibited identifications

The following identifications are not permitted without separate derivation:

$\beta_{CAPM} = u_D$. (A.60)

$\theta = \eta_Q$. (A.61)

$\eta_Q = \rho_D$. (A.62)

$Q_E = Q_H$. (A.63)

$\mathbb{Q} = Q_E$. (A.64)

$\text{Theta}_D = \theta$. (A.65)

Gate Commitment = Born Collapse. (A.66)

Risk-Neutral Weight = Born Probability. (A.67)

Contractual Coupling = Quantum Entanglement. (A.68)

Conditional Update = No-Signalling Nonlocality. (A.69)

State-Dependent Metric = Einstein Field Equation. (A.70)

Path Dependence = Quantum Geometric Phase. (A.71)

Operational Incompatibility = Fundamental Quantum Uncertainty. (A.72)

The distinction between these objects is not cosmetic.

It is the primary condition of mathematical compatibility.

Appendix B — A Minimal Two-State Option–Underlying World

B.1 Purpose

This appendix constructs the smallest derivative world capable of illustrating:

- contract preparation;
- product versus nonfactorizable states;
- local mixedness;
- conditional update;
- classical mixture;
- coherent phase;
- gate commitment;
- ledger latching.

It does not attempt realistic option calibration.

Its purpose is architectural transparency.

B.2 Underlying state space

Let the underlying possess two terminal states:

$|D_U\rangle$ = underlying ends below strike. (B.1)

$|U_U\rangle$ = underlying ends above strike. (B.2)

The underlying state space is:

$\mathscr{H}_U = \mathrm{span}\{|D_U\rangle, |U_U\rangle\}$. (B.3)

Let the pre-contract underlying state be:

$|\psi_U\rangle = \alpha|D_U\rangle + \beta\exp(i\varphi)|U_U\rangle$. (B.4)

Normalization requires:

$|\alpha|^2 + |\beta|^2 = 1$. (B.5)

For the worked example:

$\alpha = \sqrt{0.6}$. (B.6)

$\beta = \sqrt{0.4}$. (B.7)

Thus:

$|\psi_U\rangle = \sqrt{0.6}|D_U\rangle + \sqrt{0.4}\exp(i\varphi)|U_U\rangle$. (B.8)

B.3 Derivative state space

Let the derivative begin in neutral state:

$|0_D\rangle$. (B.9)

At maturity the derivative has two relevant payoff states:

$|Z_D\rangle$ = zero payoff. (B.10)

$|P_D\rangle$ = positive payoff. (B.11)

The derivative state space is:

$\mathscr{H}_D = \text{span}\{|0_D\rangle, |Z_D\rangle, |P_D\rangle\}$. (B.12)

For the post-preparation analysis, the active payoff subspace is:

$\mathscr{H}_D^{active} = \text{span}\{|Z_D\rangle, |P_D\rangle\}$. (B.13)

B.4 Contract-preparation operator

The call contract acts as:

$\hat{U}_C|D_U,0_D\rangle = |D_U,Z_D\rangle$. (B.14)

$\hat{U}_C|U_U,0_D\rangle = |U_U,P_D\rangle$. (B.15)

By linearity:

$|\Psi_{UD}\rangle = \hat{U}_C(|\psi_U\rangle \otimes |0_D\rangle)$. (B.16)

Therefore:

$|\Psi_{UD}\rangle = \sqrt{0.6}|D_U,Z_D\rangle + \sqrt{0.4}\exp(i\varphi)|U_U,P_D\rangle$. (B.17)

This is the minimal contract-prepared joint state.

B.5 Why the state does not factorize

Assume:

$|\Psi_{UD}\rangle = |\psi_{U'}\rangle \otimes |\psi_{D'}\rangle$. (B.18)

Write:

$|\psi_{U'}\rangle = a|D_U\rangle + b|U_U\rangle$. (B.19)

$|\psi_{D'}\rangle = c|Z_D\rangle + d|P_D\rangle$. (B.20)

Then:

$|\psi_{U'}\rangle \otimes |\psi_{D'}\rangle = ac|D_U,Z_D\rangle + ad|D_U,P_D\rangle + bc|U_U,Z_D\rangle + bd|U_U,P_D\rangle$. (B.21)

To match the contract state, the cross-pair coefficients must vanish:

ad = 0. (B.22)

bc = 0. (B.23)

But both desired coefficients are nonzero:

ac = √0.6. (B.24)

bd = √0.4exp(iφ). (B.25)

These conditions cannot all hold simultaneously.

Therefore:

$|\Psi_{UD}\rangle \neq |\psi_{U'}\rangle \otimes |\psi_{D'}\rangle$. (B.26)

The state is nonfactorizable in this effective representation.

B.6 Joint density state

The joint state is:

$\rho_{UD} = |\Psi_{UD}\rangle\langle\Psi_{UD}|$. (B.27)

Expanding:

$\rho_{UD} = 0.6|D_U,Z_D\rangle\langle D_U,Z_D| + 0.4|U_U,P_D\rangle\langle U_U,P_D| + \sqrt{0.24}\exp(-i\varphi)|D_U,Z_D\rangle\langle U_U,P_D| + \sqrt{0.24}\exp(i\varphi)|U_U,P_D\rangle\langle D_U,Z_D|$. (B.28)

The first two terms are diagonal.

The final two terms are coherent cross-branch terms.

B.7 Reduced underlying state

Trace over the derivative sector.

Since:

$\langle Z_D|P_D\rangle = 0$, (B.29)

the cross terms vanish under partial trace.

Thus:

$\rho_U = 0.6|D_U\rangle\langle D_U| + 0.4|U_U\rangle\langle U_U|$. (B.30)

The underlying local state is mixed.

Its purity is:

$\mathrm{Tr}(\rho_U^2) = 0.6^2 + 0.4^2 = 0.52$. (B.31)

Its entropy is:

$S(\rho_U) = -0.6\ln 0.6 - 0.4\ln 0.4$. (B.32)

Numerically:

$S(\rho_U) \approx 0.6730$ nats. (B.33)

B.8 Reduced derivative state

Tracing over the underlying sector gives:

$\rho_D = 0.6|Z_D\rangle\langle Z_D| + 0.4|P_D\rangle\langle P_D|$. (B.34)

Its purity is also:

$\mathrm{Tr}(\rho_D^2) = 0.52$. (B.35)

And:

$S(\rho_D) \approx 0.6730$ nats. (B.36)

The global state is pure.

Both local states are mixed.

This reproduces the formal pattern:

Global Purity + Local Mixedness. (B.37)

B.9 Classical mixture with the same local marginals

Define the classical mixture:

$\rho_{mix} = 0.6|D_U,Z_D\rangle\langle D_U,Z_D| + 0.4|U_U,P_D\rangle\langle U_U,P_D|$. (B.38)

Its reduced states are identical:

$\mathrm{Tr}_D(\rho_{mix}) = \rho_U$. (B.39)

$\mathrm{Tr}_U(\rho_{mix}) = \rho_D$. (B.40)

Thus local price or payoff measurements cannot distinguish:

ρ_{UD} (B.41)

from:

ρ_{mix}. (B.42)

The difference lies entirely in the off-diagonal terms.

Therefore:

Local Mixedness Alone Does Not Establish Coherence. (B.43)

B.10 Price-basis measurement

Define projectors:

$\Pi_{Down} = |D_U\rangle\langle D_U|$. (B.44)

$\Pi_{Up} = |U_U\rangle\langle U_U|$. (B.45)

The underlying probabilities are:

$p(Down) = 0.6$. (B.46)

$p(Up) = 0.4$. (B.47)

The derivative payoff probabilities are:

$p(ZeroPayoff) = 0.6$. (B.48)

$p(PositivePayoff) = 0.4$. (B.49)

Joint probabilities are:

$p(Down,ZeroPayoff) = 0.6$. (B.50)

$p(Up,PositivePayoff) = 0.4$. (B.51)

The cross outcomes have zero probability:

$p(Down,PositivePayoff) = 0$. (B.52)

$p(Up,ZeroPayoff) = 0$. (B.53)

Both the coherent state and classical mixture produce these results.

B.11 Conditional derivative state

If the underlying outcome is Down:

$\rho_{D|Down} = |Z_D\rangle\langle Z_D|$. (B.54)

If the underlying outcome is Up:

$\rho_{D|Up} = |P_D\rangle\langle P_D|$. (B.55)

Thus the underlying outcome conditionally fixes the derivative payoff state.

From inside the joint θ-world:

One Local Outcome → One Correlated Conditional State. (B.56)

From the primary universe:

The call payoff rule explains the pairing.

Both descriptions are valid at their respective levels.

B.12 Rotated joint basis

Define correlated branch states:

$|0\rangle = |D_U,Z_D\rangle$. (B.57)

$|1\rangle = |U_U,P_D\rangle$. (B.58)

Define rotated projectors:

$|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. (B.59)

$|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. (B.60)

For the coherent state:

$p_{coh}(+) = \frac{1}{2}[1 + 2\sqrt{0.24}\cos\varphi]$. (B.61)

Since:

$2\sqrt{0.24} \approx 0.9798$, (B.62)

we obtain:

$p_{coh}(+) \approx \frac{1}{2}[1 + 0.9798\cos\varphi]$. (B.63)

The mixture gives:

$p_{mix}(+) = 1/2$. (B.64)

Thus the rotated joint measurement distinguishes coherence when:

$\cos\varphi \neq 0$. (B.65)

B.13 Constructive case

Let:

$\varphi = 0$. (B.66)

Then:

$p_{coh}(+) \approx 0.9899$. (B.67)

And:

$p_{coh}(-) \approx 0.0101$. (B.68)

The coherent state is almost entirely aligned with $|+\rangle$.

The mixture remains:

$p_{mix}(+) = p_{mix}(-) = 0.5$. (B.69)

B.14 Destructive case

Let:

φ = π. (B.70)

Then:

p_coh(+⃗) ≈ 0.0101. (B.71)

And:

p_coh(−⃗) ≈ 0.9899. (B.72)

The phase reverses the rotated-basis outcome.

The branch probabilities in the payoff basis remain:

0.6 and 0.4. (B.73)

Thus the same diagonal scenario probabilities can support radically different joint-basis outcomes if coherence is operational.

B.15 Why ordinary option pricing does not perform this measurement

Standard option pricing evaluates observables diagonal in the payoff basis:

$D_0 = e^{-rT}[0.6f(D_U) + 0.4f(U_U)]$. (B.74)

Such observables do not detect φ.

A financial coherence claim therefore requires an instrument implementing the equivalent of the rotated joint basis.

Without it:

φ remains unidentifiable. (B.75)

The coherent state is empirically indistinguishable from the classical mixture.

B.16 Maturity gate

At maturity, the terminal-price gate produces either:

G_T = Down. (B.76)

Or:

G_T = Up. (B.77)

The contract gate maps:

Down → ZeroPayoff. (B.78)

Up → PositivePayoff. (B.79)

The committed record is:

Record_T = (UnderlyingOutcome,DerivativePayoff,ContractID,Frame,Evidence,Residual). (B.80)

The ledger updates:

$L_{T+1} = L_T \sqcup Record_T$. (B.81)

After settlement, the unresolved two-branch state no longer governs the ordinary operational world.

B.17 Ledger latching

Suppose the Up branch is committed.

Then:

L_{T+1} contains PositivePayoff. (B.82)

Future policy may include:

- cash settlement;
- delivery;
- tax recognition;
- hedge closure;
- accounting entry.

If the underlying later declines below strike:

$U_{later} < K$, (B.83)

the prior payoff does not disappear.

The historical record remains:

$L_{later} \supset Record_T$. (B.84)

Thus:

Later Coordinate Reversal ≠ Erasure of Committed Outcome. (B.85)

B.18 Interpretation ladder for the toy model

The model establishes:

1. natural subsystem distinction;
2. contract-prepared joint state;
3. nonfactorizable pure-state representation;

4. local mixedness;
5. conditional fixing;
6. phase-sensitive joint-basis prediction.

It does not establish that real option markets implement:

- coherent branch amplitudes;
- rotated joint measurements;
- no-signalling;
- Bell violation;
- physical quantum entanglement.

The worked example therefore demonstrates mathematical possibility, not empirical market fact.

Appendix C — Classical Mixture, Coherent Lift, and Model Selection

C.1 The central empirical problem

A classical scenario model and coherent amplitude model may share the same diagonal branch weights.

Let:

$q = (q_1,q_2,\ldots,q_N)$. (C.1)

The mixture is:

$\rho_{mix} = \Sigma_n q_n |n\rangle\langle n|$. (C.2)

The coherent lift is:

$|\Psi\rangle = \Sigma_n \sqrt{q_n} \exp(i\varphi_n)|n\rangle$. (C.3)

And:

$\rho_{coh} = |\Psi\rangle\langle\Psi|$. (C.4)

The difference is:

$\rho_{off} = \rho_{coh} - \rho_{mix}$. (C.5)

The scientific question is not whether ρ_{off} can be written.

It is whether ρ_{off} changes a measurable outcome.

C.2 Diagonal observables

Let:

$\hat{O}_{diag} = \Sigma_n O_n |n\rangle\langle n|$. (C.6)

Then:

$\mathrm{Tr}(\rho_{mix}\hat{O}_{diag}) = \Sigma_n q_n O_n$. (C.7)

And:

$\mathrm{Tr}(\rho_{coh}\hat{O}_{diag}) = \Sigma_n q_n O_n$. (C.8)

Therefore:

$\mathrm{Tr}[(\rho_{coh} - \rho_{mix})\hat{O}_{diag}] = 0$. (C.9)

Any dataset containing only diagonal measurements is insufficient to identify coherence.

C.3 Cross-branch observables

Let:

$\hat{O}_{cross} = \Sigma_{n\neq m} O_{nm} |n\rangle\langle m|$. (C.10)

Then:

$\mathrm{Tr}(\rho_{mix}\hat{O}_{cross}) = 0$. (C.11)

While:

$\mathrm{Tr}(\rho_{coh}\hat{O}_{cross}) = \Sigma_{n\neq m}\sqrt{(q_n q_m)}\exp[i(\varphi_n - \varphi_m)]O_{mn}$. (C.12)

Coherence becomes observable only when the measurement protocol contains off-diagonal sensitivity.

C.4 Two-path empirical form

Suppose two branches A and B can each generate outcome y.

Measured separately:

$P_A(y)$. (C.13)

$P_B(y)$. (C.14)

A classical mixture predicts:

$P_{mix}(y) = w_A P_A(y) + w_B P_B(y)$. (C.15)

A coherent model predicts:

$P_{coh}(y) = P_{mix}(y) + 2\gamma\sqrt{[w_A w_B P_A(y) P_B(y)]}\cos\Delta\varphi$. (C.16)

Where:

$\gamma \in [0,1]$ (C.17)

is an effective coherence parameter.

The interference residual is:

$I_{AB}(y) = P_{observed}(y) - P_{mix}(y)$. (C.18)

The implied phase relation is:

$\cos \Delta\varphi = I_{AB}(y)/[2\gamma\sqrt{(w_A w_B P_A(y) P_B(y))}]$. (C.19)

This expression is valid only when the denominator is nonzero and the result lies in $[-1,1]$.

C.5 Immediate falsification check

A proposed two-channel coherent model fails if:

$|I_{AB}(y)| > 2\gamma\sqrt{[w_A w_B P_A(y) P_B(y)]}$. (C.20)

Because then:

$|\cos \Delta\varphi| > 1$. (C.21)

The measured residual cannot be represented by the declared two-channel amplitude model.

One must then:

- add channels;
- revise weights;
- revise γ;
- use a nonlinear classical model;
- reject the amplitude construction.

C.6 Classical nonlinear alternative

A flexible classical model may be:

$P_{class}(y) = \sigma_{logistic}[c_0 + c_1 x_A + c_2 x_B + c_3 x_A x_B + c_4 x_A^2 + c_5 x_B^2]$. (C.22)

The interaction term:

$c_3 x_A x_B$ (C.23)

can produce constructive or destructive effects.

Therefore the comparison is not:

Amplitude Model versus Additive Model. (C.24)

It is:

Amplitude Model versus Adequate Nonlinear Classical Model. (C.25)

C.7 Latent-regime alternative

Let hidden regime s satisfy:

$s \in \{1,\ldots,K\}$. (C.26)

Then:

$P(y|A,B) = \Sigma_s P(s)P(y|A,B,s)$. (C.27)

Regime mixing can produce:

- phase-like reversals;
- cancellation;
- amplification;
- apparent contextuality.

A coherent model must outperform dynamic latent-regime alternatives.

C.8 Model hierarchy

Define:

M_0 = independent additive model. (C.28)

M_1 = classical nonlinear interaction model. (C.29)

M_2 = latent dynamic regime model. (C.30)

M_3 = classical coupled-phase model. (C.31)

M_4 = complex amplitude model. (C.32)

The complex model advances only when:

$\text{Score}(M_4) > \max[\text{Score}(M_0),\text{Score}(M_1),\text{Score}(M_2),\text{Score}(M_3)]$. (C.33)

After complexity, stability, and assumption penalties.

C.9 Scoring rule

One possible score is:

$\text{Score}(M) = LL_{test}(M) - \lambda_{KK_M} - \lambda_{UUncertainty_M} - \lambda_{SInstability_M}$. (C.34)

Where:

- LL_{test} is held-out log likelihood;
- K_M is model complexity;
- Uncertainty_M penalizes weakly identified parameters;

- Instability_M penalizes sensitivity to window or basis.

A coherent model with unstable phases should not win merely through higher in-sample flexibility.

C.10 Phase-identification condition

The phase is identified only if:

FisherInformation(Δφ) > I*. (C.35)

Or operationally:

Var(Δφ^) < V*. (C.36)

If multiple phase values produce almost identical predictions:

Δφ is not identified. (C.37)

The correct report is:

Phase Underdetermined. (C.38)

Not:

Hidden Quantum Phase Discovered. (C.39)

C.11 Controlled phase-shift test

The strongest coherence test intervenes on phase while holding magnitudes approximately fixed.

Before intervention:

P_before(y) = P_mix(y) + 2γK_ycos Δφ. (C.40)

After shift δ:

P_after(y) = P_mix(y) + 2γK_ycos(Δφ+δ). (C.41)

Where:

K_y = √[w_Aw_BP_A(y)P_B(y)]. (C.42)

The predicted change is:

ΔP(y) = 2γK_y[cos(Δφ+δ) − cos Δφ]. (C.43)

A convincing result requires:

- stable magnitudes;
- controlled δ;
- no information shock;
- no liquidity change;

- no hidden intervention;
- out-of-sample prediction.

C.12 Decoherence test

Introduce controlled phase noise ξ:

$\xi \sim$ distribution with $\mathrm{Var}(\xi) = \sigma_\xi^2$. (C.44)

Then:

$E[\cos(\Delta\varphi+\xi)] = \cos\Delta\varphi \cdot E[\cos\xi] - \sin\Delta\varphi \cdot E[\sin\xi]$. (C.45)

For zero-mean Gaussian phase noise:

$E[\exp(i\xi)] = \exp(-\sigma_\xi^2/2)$. (C.46)

Therefore the effective coherence is:

$\gamma_{eff} = \gamma_0 \exp(-\sigma_\xi^2/2)$. (C.47)

The predicted interference becomes:

$I_{noisy} = 2\gamma_{eff} K_y \cos\Delta\varphi$. (C.48)

This provides a quantitative dephasing test.

Classical oscillator models make similar predictions, so the test establishes phase-process utility rather than quantum ontology.

C.13 Revival test

After phase noise, apply a realignment protocol reducing dispersion:

$\sigma_{\xi,after}^2 < \sigma_{\xi,before}^2$. (C.49)

The model predicts:

$\gamma_{after} > \gamma_{before}$. (C.50)

And:

$|I_{after}| > |I_{before}|$ (C.51)

when the mean phase remains aligned.

A repeatable suppression-and-revival pattern is stronger evidence for phase modelling than a one-time nonlinear residual.

C.14 Basis-robustness test

Estimate coherence in basis B:

$C_B(\rho)$. (C.52)

Transform under an admissible unitary U:

$\rho' = U\rho U^\dagger$. (C.53)

The operational predictions should remain invariant when observables transform consistently:

$\hat{O}' = U\hat{O}U^\dagger$. (C.54)

Then:

$\mathrm{Tr}(\rho\hat{O}) = \mathrm{Tr}(\rho'\hat{O}')$. (C.55)

If the result depends on arbitrary basis labelling, it is not a robust physical or financial claim.

C.15 Mixture–coherence decision table

Finding	Classification
Diagonal model fits all admissible measurements	Classical mixture sufficient
Full state improves only in sample	Overfitting likely
Off-diagonal model improves held-out fit	Candidate effective coherence
Controlled phase shift matches prediction	Stronger phase-process evidence
Classical phase model performs equally well	Classical coherence sufficient
Nonlinear classical nulls fail repeatedly	Candidate irreducible amplitude structure
Result survives frame and basis changes	Operationally robust coherence
No-signalling and Bell tests also pass	Much stronger anomaly requiring replication

---

C.16 The proper conclusion rule

The correct conclusion should match the strongest completed test.

Examples:

“Relative phase is a useful predictive latent variable.” (C.56)

“An effective complex-amplitude model outperforms tested classical alternatives.” (C.57)

“The fitted joint state is representation-relative and nonseparable.” (C.58)

These are not equivalent to:

“The market is physically quantum.” (C.59)

Appendix D — Three-Clock Runtime and Ledgered Time

D.1 Minimal runtime

The secondary valuation world uses three distinct orders:

Primary duration:

$t_0 < t_1 < t_2 < \dots$. (D.1)

Phase progression:

$\theta_0 \to \theta_1 \to \theta_2 \to \dots$. (D.2)

Ledger commitment:

$L_0 \to L_1 \to L_2 \to \dots$. (D.3)

The compilation relation is:

$\rho_F(\theta) = \mathcal{C}_{P,L}[X(t)]$. (D.4)

The event runtime is:

$X(t) \to \rho_F(\theta) \to$ Instrument $\to$ Outcome $\to$ Gate $\to$ Record $\to L_{k+1}$. (D.5)

D.2 Calendar-time evolution

Let the primary economic state satisfy:

$dX/dt = F_X(X,u,P,L) + \xi_X$. (D.6)

Calendar time governs:

- cash flow;
- news;
- trading;
- maturity;
- settlement duration;
- institutional operations.

Calendar time may continue even when no valuation phase movement occurs:

$dt > 0$ but $d\theta = 0$. (D.7)

For example:

- a market is closed;
- no new admissible information arrives;
- a fixed valuation frame is maintained.

D.3 Phase-time evolution

Let:

$d\theta/dt = \omega_F(X,P,L)$. (D.8)

If:

$\omega_F \neq 0$, (D.9)

then:

$d/d\theta = (1/\omega_F)d/dt$. (D.10)

For:

$dZ/dt = (g_A + i\omega_F)Z + \varepsilon_{dyn}$, (D.11)

the θ-form is:

$dZ/d\theta = [(g_A/\omega_F) + i]Z + \varepsilon_{dyn}/\omega_F$. (D.12)

When:

$g_A = 0$, (D.13)

and:

$\varepsilon_{dyn} = 0$, (D.14)

we recover:

$dZ/d\theta = iZ$. (D.15)

D.4 Phase reversal

If:

ω_F changes sign, (D.16)

then θ ceases to be globally monotonic.

Suppose:

$\omega_F > 0$ for $t < t^*$. (D.17)

$\omega_F < 0$ for $t > t^*$. (D.18)

At t^*:

$\omega_F(t^*) = 0$. (D.19)

The internal clock requires a branch label:

θ^+ = forward-orientation branch. (D.20)

θ^- = reverse-orientation branch. (D.21)

A global clock can then be represented as:

$\tau_\theta = (\text{branch},\theta)$. (D.22)

Without the branch label, identical θ values may correspond to different histories.

D.5 Multiple local phase clocks

For N channels:

$d\theta_n/dt = \omega_n(X,P,L)$. (D.23)

The relative phase is:

$\Delta\theta_{nm} = \theta_n - \theta_m$. (D.24)

The relative phase velocity is:

$d\Delta\theta_{nm}/dt = \omega_n - \omega_m$. (D.25)

Synchronization occurs when:

$|\omega_n - \omega_m| \le \varepsilon_{sync}$. (D.26)

Dephasing occurs when:

$|\omega_n - \omega_m| > \varepsilon_{sync}$. (D.27)

The channels may share calendar time while accumulating different internal phase.

D.6 Ledger time

A ledger tick requires commitment:

$L_{k+1} = L_k \sqcup Record_k$. (D.28)

If an outcome remains provisional:

$L_{k+1} = L_k$. (D.29)

Thus many phase changes may occur between ledger ticks:

$\theta_k \to \theta_{k+\delta_1} \to \theta_{k+\delta_2} \to \ldots \to Gate$. (D.30)

Conversely, several ledger events may be committed within one small calendar interval.

Ledger time therefore orders consequences rather than duration.

D.7 Same phase, different ledger

Consider two histories A and B:

$\theta_A = \theta_B$. (D.31)

$R_A = R_B$. (D.32)

$Q_A = Q_B$. (D.33)

But:

$L_A \neq L_B$. (D.34)

Then the complete states differ:

$State_A = (R,Q,\theta,L_A)$. (D.35)

$State_B = (R,Q,\theta,L_B)$. (D.36)

Therefore:

$State_A \neq State_B$. (D.37)

The visible complex coordinate is not sufficient to specify a ledger-bearing world.

D.8 Same ledger index, different calendar duration

Suppose event k occurs after:

$\Delta t_A = 1$ second. (D.38)

In another system it occurs after:

$\Delta t_B = 30$ days. (D.39)

Both produce:

$L_k \rightarrow L_{k+1}$. (D.40)

Ledger distance is equal:

$\Delta k = 1$. (D.41)

Calendar duration differs:

$\Delta t_A \neq \Delta t_B$. (D.42)

This is why ledger time is not a substitute for physical duration.

It is a different order structure.

D.9 Internal simultaneity

Two outcomes E_U and E_D are θ-simultaneous when:

$Sim_\theta(E_U,E_D) \Leftrightarrow SameJointInstrument \wedge SameGateEpisode$. (D.43)

They need not possess identical microsecond timestamps.

They belong to one effective disclosure event.

Calendar simultaneity is:

$Sim_t(E_U,E_D) \Leftrightarrow t_U = t_D$ within tolerance. (D.44)

Ledger simultaneity is:

$$\mathrm{Sim}_k(E_U, E_D) \Leftrightarrow k_U = k_D. \quad \text{(D.45)}$$

These three simultaneity relations should not be conflated.

D.10 A worked five-stage trace

Stage 0 — Prepared state

Calendar time:

$$t = 0. \quad \text{(D.46)}$$

Phase:

$$\theta = 0.20. \quad \text{(D.47)}$$

Ledger:

$$L_0. \quad \text{(D.48)}$$

Joint option state:

$$\rho_{UD,0}. \quad \text{(D.49)}$$

Stage 1 — Market movement

Calendar time:

$$t = 1. \quad \text{(D.50)}$$

Phase:

$$\theta = 0.30. \quad \text{(D.51)}$$

No gate passes:

$$L \text{ remains } L_0. \quad \text{(D.52)}$$

Stage 2 — Volatility repricing

Calendar time:

$$t = 2. \quad \text{(D.53)}$$

Phase:

$$\theta = 0.47. \quad \text{(D.54)}$$

The option approaches a margin threshold.

Ledger remains:

$$L_0. \quad \text{(D.55)}$$

Stage 3 — Margin gate

Calendar time:

t = 2.1. (D.56)

Gate condition:

Exposure − Collateral > Threshold. (D.57)

Commit:

Recordo = MarginCall. (D.58)

Ledger:

$L_1 = L_0 \sqcup$ MarginCall. (D.59)

Stage 4 — Backreaction

Calendar time:

t = 2.2. (D.60)

The institution sells assets to post collateral.

The primary state changes:

$X_{2.2} \neq X_{2.1}$. (D.61)

The valuation world recompiles:

$\rho_{F,new} = \mathcal{C}_{\{P,L_1\}}[X_{2.2}]$. (D.62)

The metric and phase velocity may change:

$g_{F,new} \neq g_{F,old}$. (D.63)

$\omega_{F,new} \neq \omega_{F,old}$. (D.64)

The ledger event has changed the conditions of future time.

D.11 Proper-time candidate

A tentative effective proper time may be:

$d\tau_F = N_F(X,P,L)dt$. (D.65)

Where N_F is a financial lapse-like function.

Alternatively:

$d\tau_F = G_F|d\tilde{\theta}|/\Omega_F$. (D.66)

Where:

- G_F is gate or observability intensity;

- $\tilde{\theta}$ is gauge-corrected phase;
- Ω_F is local processing capacity.

This is a modelling option.

It is not derived as a universal financial time law.

D.12 Time-bearing-world criterion

A secondary financial construction becomes time-bearing when:

1. state progression is ordered;
2. some transitions become committed;
3. the record persists;
4. future admissibility depends on that record;
5. observers inherit the trace.

In compact form:

$\text{TimeBearing}_W \Leftrightarrow \text{OrderedChange} \wedge \text{Commitment} \wedge \text{Persistence} \wedge \text{FutureConstraint} \wedge \text{Inheritance}.$ (D.67)

This expresses the wider ledger ontology:

Time is not merely that something changes.

Time becomes historically consequential when selected change enters a ledger and helps generate the future.

Appendix E — Contract Preparation, Joint Instruments, and Ledger Updates

E.1 Purpose

This appendix assembles the complete option–underlying runtime in one formal sequence.

The sequence begins with an underlying possibility state.

A contract prepares a joint state.

Internal observers apply local or joint instruments.

A gate determines whether an outcome becomes operationally committed.

A ledger preserves the event.

Backreaction changes the later primary and secondary worlds.

The complete sequence is:

Underlying Preparation
→ Contract Binding
→ Joint Evolution
→ Instrument Selection
→ Outcome
→ Gate
→ Ledger
→ Backreaction. (E.1)

This structure separates six processes that are often compressed into the single word collapse:

1. preparation;
2. evolution;
3. observation;
4. conditioning;
5. commitment;
6. historical consequence.

E.2 Initial underlying state

Let the underlying state be:

$\rho_U^0 \in \mathcal{D}(\mathscr{H}_U)$. (E.2)

Let the derivative registration sector begin in neutral state:

$\rho_D^0 = |0_D\rangle\langle 0_D|$. (E.3)

The initially independent composite state is:

$\rho_UD^0 = \rho_U^0 \otimes \rho_D^0$. (E.4)

For a pure underlying state:

$|\psi_U\rangle = \Sigma_n c_n |u_n\rangle$. (E.5)

The initial state is:

$|\Psi_UD^0\rangle = \Sigma_n c_n |u_n, 0_D\rangle$. (E.6)

At this stage, the derivative has not yet been relationally bound to the underlying.

E.3 Contract-preparation channel

Let the contract-preparation map be:

$\mathcal{C}_C: \mathcal{D}(\mathscr{H}_U \otimes \mathscr{H}_D) \to \mathcal{D}(\mathscr{H}_U \otimes \mathscr{H}_D)$. (E.7)

A deterministic branch-binding rule is:

$\mathcal{C}_C[|u_n,0_D\rangle\langle u_n,0_D|] = |u_n,d_n\rangle\langle u_n,d_n|$. (E.8)

For pure-state preparation:

$\hat{U}_C|u_n,0_D\rangle = |u_n,d_n\rangle$. (E.9)

Applying the contract:

$\rho_{UD}^C = \mathcal{C}_C(\rho_{UD}^0)$. (E.10)

For the pure case:

$|\Psi_{UD}^C\rangle = \Sigma_n c_n|u_n,d_n\rangle$. (E.11)

The contract has converted an independent registration state into a relationally structured joint state.

E.4 Preparation as a completely positive map

A realistic contract preparation may include:

- model uncertainty;
- liquidity state;
- funding state;
- legal ambiguity;
- contract-registration error.

Represent the preparation through Kraus-like operators K_r:

$\mathcal{C}_C(\rho) = \Sigma_r K_r \rho K_r^\dagger$. (E.12)

The normalization condition is:

$\Sigma_r K_r^\dagger K_r = I$ (E.13)

for a trace-preserving preparation map.

If some branches represent rejection or failed registration, the accepted contract channel may be trace-decreasing:

$\Sigma_r K_r^\dagger K_r \leq I$. (E.14)

The missing trace corresponds to:

- contract rejection;
- incomplete execution;
- invalid legal state;
- discarded preparation branch.

This is more realistic than assuming every contract declaration succeeds.

E.5 Contract-admissibility projector

Let the full tensor-product space contain all formal pairs:

$\{|u_n,d_m\rangle\}$. (E.15)

Only some pairs satisfy the contract.

Define admissibility relation:

$C_P(u_n,d_m) = 1$ if pair (u_n,d_m) is contractually admissible. (E.16)

The contract projector is:

$\Pi_C = \Sigma_{n,m}\, C_P(u_n,d_m)|u_n,d_m\rangle\langle u_n,d_m|$. (E.17)

A fully admissible state satisfies:

$\Pi_C\rho_UD\Pi_C = \rho_UD$. (E.18)

An admissibility residual is:

$\varepsilon_C = \rho_UD - \Pi_C\rho_UD\Pi_C$. (E.19)

A large:

$\|\varepsilon_C\|$ (E.20)

indicates that the declared joint state contains branches inconsistent with the contract.

E.6 Continuous pre-measurement evolution

After preparation, the joint state evolves in θ-time:

$d\rho_UD/d\theta = -i[\hat{H}_UD,\rho_UD] + \mathcal{D}_env(\rho_UD) + \mathscr{R}_UD$. (E.21)

Where:

$\hat{H}_UD = \hat{H}_U \otimes I_D + I_U \otimes \hat{H}_D + \hat{H}_contract + \hat{H}_hedge$. (E.22)

The environmental term is:

$\mathcal{D}_env(\rho) = \Sigma_r\gamma_r[L_r\rho L_r^\dagger - \tfrac{1}{2}L_r^\dagger L_r\rho - \tfrac{1}{2}\rho L_r^\dagger L_r]$. (E.23)

The residual term $\mathscr{R}_UD$ captures:

- model error;
- omitted channels;
- frame mismatch;
- protocol instability;
- unmodelled intervention.

Between gates, the state remains in the possibility-bearing layer.

E.7 Local underlying instrument

Observer A selects underlying setting a.

The instrument family is:

$\mathbb{M}^U_a = \{\mathbb{M}^U_{a,u}\}_u$. (E.24)

Outcome probability:

$p(u|a) = \mathrm{Tr}[(E^U_{a,u} \otimes I_D)\rho_{UD}]$. (E.25)

The unnormalized post-measurement state is:

$\tilde{\rho}_{UD|u,a} = (\mathbb{M}^U_{a,u} \otimes I_D)(\rho_{UD})$. (E.26)

Normalized:

$\rho_{UD|u,a} = \tilde{\rho}_{UD|u,a} / p(u|a)$. (E.27)

The conditional derivative state is:

$\rho_{D|u,a} = \mathrm{Tr}_U(\rho_{UD|u,a})$. (E.28)

This update describes the derivative state relative to the underlying outcome.

It does not imply that the derivative observer has already received the outcome record.

E.8 Local derivative instrument

Observer B selects derivative setting b.

The instrument family is:

$\mathbb{M}^D_b = \{\mathbb{M}^D_{b,d}\}_d$. (E.29)

Outcome probability:

$p(d|b) = \mathrm{Tr}[(I_U \otimes E^D_{b,d})\rho_{UD}]$. (E.30)

The conditional underlying state is:

$\rho_{U|d,b} = \mathrm{Tr}_D[(I_U \otimes \mathbb{M}^D_{b,d})(\rho_{UD})] / p(d|b)$. (E.31)

Examples of derivative settings include:

- premium basis;
- payoff basis;
- delta–gamma basis;
- volatility basis;
- collateral basis;

- exercise basis;
- liquidation basis.

Different settings disclose different relational aspects of the same global contract state.

E.9 Joint instrument

For simultaneous local settings a and b:

$$\mathbb{M}_{a,b}^{u,d} = \mathbb{M}^U_{a,u} \otimes \mathbb{M}^D_{b,d}. \quad (E.32)$$

The joint probability is:

$$p(u,d|a,b) = \mathrm{Tr}[\mathbb{M}_{a,b}^{u,d}(\rho_{UD})]. \quad (E.33)$$

The conditional joint state is:

$$\rho_{UD|u,d,a,b} = \mathbb{M}_{a,b}^{u,d}(\rho_{UD}) / p(u,d|a,b). \quad (E.34)$$

The event pair:

$$(u,d) \quad (E.35)$$

belongs to one internal measurement episode when:

- the settings are jointly declared;
- the instruments are compatible;
- one gate evaluates the pair;
- one record binds the outcome pair.

E.10 Sequential instruments

Suppose underlying instrument a acts before derivative instrument b.

The sequential probability is:

$$p(u \text{ then } d|a,b) = \mathrm{Tr}[(\mathbb{M}^D_{b,d} \circ \mathbb{M}^U_{a,u})(\rho_{UD})]. \quad (E.36)$$

Reversing the order:

$$p(d \text{ then } u|b,a) = \mathrm{Tr}[(\mathbb{M}^U_{a,u} \circ \mathbb{M}^D_{b,d})(\rho_{UD})]. \quad (E.37)$$

Operational noncommutativity occurs when:

$$\mathbb{M}^D_{b,d} \circ \mathbb{M}^U_{a,u} \neq \mathbb{M}^U_{a,u} \circ \mathbb{M}^D_{b,d}. \quad (E.38)$$

This may arise from:

- measurement disturbance;
- trade impact;

- gate activation;
- path dependence;
- ledger update;
- adaptive recalibration.

It does not automatically imply specifically quantum incompatibility.

E.11 Instrument disturbance

Define disturbance caused by instrument a:

$$Dist_a(\rho) = Distance[\rho,\Sigma_u \mathbb{M}_{a,u}(\rho)]. \quad (E.39)$$

A nearly passive instrument satisfies:

$$Dist_a(\rho) \approx 0. \quad (E.40)$$

A strongly performative instrument satisfies:

$$Dist_a(\rho) \gg 0. \quad (E.41)$$

Examples:

Instrument	Typical disturbance
Historical-data classification	Low
Non-executable indicative quote	Low to moderate
Public rating announcement	Moderate to high
Large executable trade	High
Margin call	High
Forced liquidation	Very high

Any contextuality or Bell-style test must quantify this disturbance.

E.12 Provisional outcome

An instrument may produce provisional outcome y:

$$y = (u,d). \quad (E.42)$$

But provisional outcome is not yet a committed event.

Define provisional state:

$$\rho_{prov} = \rho_{UD|u,d,a,b}. \quad (E.43)$$

The gate receives:

$$Input_G = (\rho_{prov},u,d,a,b,P,L_k,Evidence). \quad (E.44)$$

Possible decisions are:

G_k ∈ {Commit,Reject,Defer,Escalate}. (E.45)

This preserves the distinction:

Measurement Outcome ≠ Historical Fact. (E.46)

E.13 Gate conditions

A gate may evaluate:

C_G = C_G(u,d,a,b,P,L_k,Evidence). (E.47)

A deterministic threshold gate commits when:

C_G ≥ 0. (E.48)

A probabilistic gate may commit with:

p_G = σ_logistic(C_G). (E.49)

An authority-bound gate also requires:

AuthorityValid = 1. (E.50)

Thus:

Commit iff ConditionPassed ∧ AuthorityValid ∧ EvidenceSufficient. (E.51)

A financial event may be economically plausible but institutionally uncommitted when any one requirement fails.

E.14 Record schema

A strong event record is:

Record_k = (EventID,ContractID,Outcome,Settings,Frame,Authority,Evidence,Confidence,Residual,t,θ,k). (E.52)

Each field performs a distinct role.

Field	Purpose
EventID	Distinguish event identity
ContractID	Preserve governed object
Outcome	State what was committed
Settings	Record measurement context
Frame	Preserve observer coordinates
Authority	Record commitment power
Evidence	Support later audit
Confidence	Preserve uncertainty
Residual	Preserve unresolved structure
t	Calendar location

Field	Purpose
θ	Internal phase location
k	Ledger order

A record that omits setting or frame cannot support reliable cross-observer reconstruction.

E.15 Ledger update

On commitment:

$L_{k+1} = L_k \sqcup \text{Record}_k$. (E.53)

On rejection:

$L_{k+1} = L_k \sqcup \text{RejectRecord}_k$ (E.54)

when rejection itself is consequential.

On deferral:

$L_{k+1} = L_k \sqcup \text{PendingRecord}_k$ (E.55)

when pending status changes later admissibility.

Thus even non-commitment can enter history.

The ledger should distinguish:

- no event occurred;
- event was rejected;
- event remains unresolved;
- evidence was insufficient;
- authority was absent.

E.16 Post-ledger state update

The committed record changes the effective state:

$\rho_{k+1} = \mathcal{K}_{\text{Record}_k}(\rho_{\text{prov}})$. (E.56)

Examples:

Exercise

$\rho_D \to$ settled payoff state. (E.57)

Barrier hit

Active-state subspace changes. (E.58)

Default

Credit and settlement sectors change. (E.59)

Margin call

Collateral and funding sectors change. (E.60)

Liquidation

Underlying inventory and liquidity sectors change. (E.61)

The ledger update is therefore not merely a database write.

It is a state-transition operator.

E.17 Adaptive observer update

After the record:

$\pi_O,k+1$ = UpdatePolicy(π_O,k,Record_k). (E.62)

The next setting becomes:

$a_k+1 = \pi_O,k+1(L_k+1)$. (E.63)

Thus two different outcomes produce two different future measurement paths:

Record_A ≠ Record_B (E.64)

implies generally:

$\pi_A,k+1 \neq \pi_B,k+1$. (E.65)

This is the formal source of latching.

E.18 Primary-world backreaction

The record induces an intervention:

u_k = PolicyAction(Record_k,P,L_k+1). (E.66)

The primary state changes:

$X_k+1 = \mathscr{B}$(X_k,u_k,Record_k). (E.67)

Examples:

- hedge execution;
- collateral transfer;
- forced asset sale;

- regulatory restriction;
- settlement delivery;
- legal enforcement.

The secondary world then recompiles:

$\rho_{F,k+1} = \mathcal{C}_{P_{k+1},L_{k+1}}[X_{k+1}]$. (E.68)

The loop closes:

Secondary Commitment
→ Primary Backreaction
→ New Secondary State. (E.69)

E.19 Protocol revision

Residual may trigger protocol revision:

$P_{k+1} = \mathcal{U}(P_k,L_{k+1},\text{Residual}_k)$. (E.70)

Possible revisions include:

- widen state space;
- change metric;
- change threshold;
- change frame;
- change contract interpretation;
- suspend automated commitment;
- escalate to human authority.

The world therefore revises its own disclosure rules.

E.20 Full hybrid runtime

The complete one-cycle runtime is:

ρ_k^-
→ Evolve_θ
→ Select(a_k,b_k)
→ Measure
→ Condition
→ Gate
→ Record
→ L_{k+1}
→ PolicyUpdate
→ Intervention
→ X_{k+1}

→ Recompile
→ ρ_{k+1}^{-}. (E.71)

In operator form:

$\rho_{k+1} = \mathcal{C}_{P_{k+1},L_{k+1}} \circ \mathscr{B} \circ \text{Policy} \circ \text{Ledger} \circ \text{Gate} \circ \text{Measure} \circ \text{Evolve}(\rho_k)$. (E.72)

This is the minimal derivative-world kernel.

E.21 Runtime failure points

Stage	Failure
Preparation	Contract state misdeclared
Evolution	Dynamics omit major coupling
Setting	Observer basis inappropriate
Measurement	Instrument noisy or performative
Conditioning	Local update confused with public knowledge
Gate	Threshold or authority invalid
Record	Context or residual omitted
Ledger	Trace corrupted or inaccessible
Policy	Adaptive response destabilizing
Backreaction	Impact misestimated
Recompilation	New world uses obsolete protocol

A credible implementation must log failures by stage rather than flattening them into one model error.

E.22 Kernel proposition

Proposition E.1 — Derivative Runtime Kernel

A derivative world becomes a self-revising observer-bearing runtime when:

1. contracts prepare joint states;
2. observers apply bounded instruments;
3. outcomes remain provisional until gated;
4. records change future policy and admissibility;
5. interventions alter the primary constructor;
6. the next secondary state is recompiled from the changed primary field.

In compact form:

$\text{Runtime}_D = \text{Prepare} + \text{Evolve} + \text{Observe} + \text{Gate} + \text{Ledger} + \text{Backreact} + \text{Recompile}$. (E.73)

Appendix F — The Two Lorentz Embeddings

F.1 Why two embeddings are required

The original Finance Geometry begins with:

$A^2 = R^2 + Q_E^2$. (F.1)

This is Euclidean.

A Lorentz-like representation can be introduced in at least two different ways.

Route 1 — Q-preserving embedding

Preserve the original retained pressure Q_E.

Route 2 — Discount-composition embedding

Preserve multiplicative valuation ratios as additive rapidities.

The two constructions are related.

They are not identical.

F.2 Route 1 — Q-preserving embedding

Rewrite:

$A^2 - Q_E^2 = R^2$. (F.2)

Define rapidity η_Q:

$A = R \cosh \eta_Q$. (F.3)

$Q_E = R \sinh \eta_Q$. (F.4)

Therefore:

$\tanh \eta_Q = Q_E/A$. (F.5)

$\eta_Q = \operatorname{artanh}(Q_E/A)$. (F.6)

The invariant is:

$A^2 - Q_E^2 = R^2$. (F.7)

In this representation:

- R is interval-like;
- A is hyperbolic time-like coordinate;
- Q_E is pressure-like spatial coordinate;
- η_Q parametrizes their relation.

The primary benefit is continuity with the original $R+iQ_E$ geometry.

F.3 Circular–hyperbolic map

Since:

$R = A \cos \theta$, (F.8)

and:

$Q_E = A \sin \theta$, (F.9)

we obtain:

$\cos \theta = \operatorname{sech} \eta_Q$. (F.10)

$\sin \theta = \tanh \eta_Q$. (F.11)

$\tan \theta = \sinh \eta_Q$. (F.12)

Therefore:

$\eta_Q = \operatorname{artanh}(\sin \theta)$. (F.13)

The circular and hyperbolic descriptions encode the same local triangle through different functions.

F.4 Route 2 — Discount-composition embedding

Define discount rapidity:

$\rho_D = \ln(A/R)$. (F.14)

Then:

$R = A \exp(-\rho_D)$. (F.15)

For frames a and b:

$\rho_{BA} = \ln(R_A/R_B)$. (F.16)

Thus:

$R_B = \exp(-\rho_{BA}) R_A$. (F.17)

Composition is additive:

$\rho_{CA} = \rho_{CB} + \rho_{BA}$. (F.18)

This route is designed for relative valuation-frame transport.

F.5 Light-cone coordinates

Define:

$R_+ = A \exp(\rho_D)$. (F.19)

$R_- = A \exp(-\rho_D)$. (F.20)

Identify:

$R_- = R$. (F.21)

Then:

$R_+R_- = A^2$. (F.22)

Define:

$U_H = (R_+ + R_-)/2$. (F.23)

$Q_H = (R_+ - R_-)/2$. (F.24)

Therefore:

$U_H = A \cosh \rho_D$. (F.25)

$Q_H = A \sinh \rho_D$. (F.26)

And:

$U_H^2 - Q_H^2 = A^2$. (F.27)

F.6 Relation between Q_E and Q_H

Using:

$R_+ = A^2/R$, (F.28)

and:

$R_- = R$, (F.29)

we obtain:

$Q_H = \frac{1}{2}(A^2/R - R)$. (F.30)

Therefore:

$Q_H = (A^2 - R^2)/(2R)$. (F.31)

Since:

$Q_E^2 = A^2 - R^2$, (F.32)

we have:

$Q_H = Q_E^2/(2R)$. (F.33)

Thus:

Q_H is quadratic in Q_E near low pressure. (F.34)

They should never be treated as one coordinate.

F.7 Relation between η_Q and ρ_D

From Route 1:

$A/R = \cosh \eta_Q$. (F.35)

From Route 2:

$A/R = \exp(\rho_D)$. (F.36)

Therefore:

$\exp(\rho_D) = \cosh \eta_Q$. (F.37)

And:

$\rho_D = \ln \cosh \eta_Q$. (F.38)

Using the circular angle:

$\rho_D = -\ln \cos \theta$. (F.39)

The three variables form:

$\theta \leftrightarrow \eta_Q \leftrightarrow \rho_D$. (F.40)

But their composition laws differ.

F.8 Small-pressure comparison

Let:

$x = Q_E/A$. (F.41)

For $x \ll 1$:

$\theta \approx x$. (F.42)

$\eta_Q \approx x$. (F.43)

$\rho_D \approx x^2/2$. (F.44)

Therefore:

$\theta \approx \eta_Q$. (F.45)

But:

ρ_D is second-order in pressure. (F.46)

This explains why small-regime calibration may obscure the distinction between the embeddings.

F.9 High-pressure comparison

As:

$R/A \to 0$, (F.47)

we obtain:

$\theta \to \pi/2$. (F.48)

$\eta_Q \to \infty$. (F.49)

$\rho_D \to \infty$. (F.50)

The circular angle saturates.

The rapidities continue to distinguish increasingly extreme compression.

This may be useful near:

- insolvency;
- liquidity collapse;
- severe discounting;
- near-zero admitted certainty.

F.10 Relative-frame matrix

The discount-frame transformation may be written in the hyperbolic coordinate pair (U_H,Q_H):

$U_{H,B} = U_{H,A} \cosh \rho_{BA} + Q_{H,A} \sinh \rho_{BA}$. (F.51)

$Q_{H,B} = U_{H,A} \sinh \rho_{BA} + Q_{H,A} \cosh \rho_{BA}$. (F.52)

The invariant is:

$U_{H,B}^2 - Q_{H,B}^2 = U_{H,A}^2 - Q_{H,A}^2$. (F.53)

In light-cone form:

$R_{+,B} = \exp(\rho_{BA}) R_{+,A}$. (F.54)

$R_{-,B} = \exp(-\rho_{BA}) R_{-,A}$. (F.55)

F.11 Bounded frame variable

Define:

$u_D = \tanh \rho_D$. (F.56)

Relative composition is:

$u_{CA} = (u_{CB} + u_{BA})/(1 + u_{CB}u_{BA})$. (F.57)

The variable is bounded:

$-1 < u_D < 1$. (F.58)

But its financial meaning is:

normalized relative valuation-frame displacement. (F.59)

It is not market velocity.

F.12 CAPM beta remains distinct

CAPM beta is:

$\beta_i = \mathrm{Cov}(r_i, r_m)/\mathrm{Var}(r_m)$. (F.60)

It measures sensitivity to market return.

The bounded frame variable is:

$u_D = \tanh \ln(A/R)$. (F.61)

The two may correlate in some empirical regime.

But:

$\beta_i \neq u_D$. (F.62)

A mapping:

$u_D = f(\beta_i, ERP, r_f, T, \ldots)$ (F.63)

would require an explicit derivation and empirical test.

It cannot be assumed from symbol resemblance.

F.13 Which route should be used?

Use Route 1 when the purpose is:

- preserve Q_E;
- retain the original complex geometry;
- reinterpret pressure hyperbolically;
- compare circular and hyperbolic orientation.

Use Route 2 when the purpose is:

- compare valuation frames;
- compose discount transformations;
- define relative rapidity;
- use light-cone variables;

- construct bounded frame coordinates.

Use neither when:

- the protocol changes;
- the contract changes;
- the transformation is path-dependent;
- no invariant is preserved.

F.14 Mixed use

A model may use both routes if notation remains explicit.

For example:

Local complex state:

$Z = R + iQ_E$. (F.64)

Q-preserving rapidity:

$\eta_Q = \operatorname{artanh}(Q_E/A)$. (F.65)

Relative frame rapidity:

$\rho_{BA} = \ln(R_A/R_B)$. (F.66)

The maps are:

$\rho_D = \ln \cosh \eta_Q$. (F.67)

$Q_H = Q_E^2/(2R)$. (F.68)

This layered use is coherent.

Direct substitution of one pressure or rapidity for another is not.

F.15 Embedding compatibility proposition

Proposition F.1 — Dual Lorentz Embedding

The Q-preserving and discount-composition embeddings are compatible when:

1. both derive from the same declared A and R;
2. Q_E and Q_H remain distinct;
3. η_Q and ρ_D remain distinct;
4. transformations preserve the declared invariant;
5. protocol transitions are excluded from frame composition.

In compact form:

CompatibleLorentz_F ⇔ SharedAR ∧ DistinctPressures ∧ DistinctRapidities ∧ InvariantPreservation ∧ StableProtocol. (F.69)

Appendix G — Base Curvature, Fibre Curvature, and Holonomy

G.1 Three geometric structures

The layered architecture contains three different geometric objects.

Base metric geometry

g^F_μν describes financial distance.

Base connection geometry

Γ^ρ_μν describes transport of tangent vectors.

Fibre gauge geometry

$\mathcal{A}$_μ describes phase and internal-state transport.

The corresponding curvatures are:

R^ρ_σμν for the base. (G.1)

$\mathbb{F}$_μν for the fibre. (G.2)

They should not be identified.

G.2 Base manifold

Let:

x^μ = (ρ_D,σ,λ_L,ℓ,c,f,m,…). (G.3)

The metric is:

ds_F² = g^F_μνdx^μdx^ν. (G.4)

A path is:

γ: s ↦ x^μ(s). (G.5)

Its financial length is:

Length_F(γ) = ∫√|g^F_μν(dx^μ/ds)(dx^ν/ds)|ds. (G.6)

The metric may measure:

- distinguishability;
- execution cost;
- intervention effort;
- risk sensitivity;
- institutional difficulty.

G.3 Base connection

The Levi-Civita-like connection is:

$\Gamma^{\rho}_{\mu\nu} = \tfrac{1}{2} g^{\rho\lambda}(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu})$. (G.7)

A tangent vector V^μ transported along path obeys:

$dV^\mu/ds + \Gamma^\mu_{\alpha\beta} V^\alpha (dx^\beta/ds) = 0$. (G.8)

The base curvature is:

$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}$. (G.9)

Nonzero curvature means:

local-flat frames cannot be extended globally without path dependence.

G.4 Local frame

The tetrad-like map is:

$g^F_{\mu\nu} = e^a_{\ \mu} e^b_{\ \nu} \eta_{ab}$. (G.10)

The local coordinates are:

$dx^a = e^a_{\ \mu} dx^\mu$. (G.11)

At a selected point x_0:

$g^F_{\mu\nu}(x_0) = \eta_{\mu\nu}$. (G.12)

And locally:

$\Gamma^\rho_{\mu\nu}(x_0) \approx 0$. (G.13)

This is where local CAPM and local Greek approximations are most reliable.

G.5 Fibre bundle

At each x:

$\mathscr{H}_x$ = local complex financial state space. (G.14)

The bundle is:

$\pi: \boldsymbol{\mathcal{H}}_F \to \boldsymbol{\mathcal{M}}_F$. (G.15)

A local state is:

$|\psi(x)\rangle \in \mathscr{H}_x$. (G.16)

The local phase convention can vary:

$|\psi(x)\rangle \to \exp[i\chi(x)]|\psi(x)\rangle$. (G.17)

A connection is required to compare states at different x.

G.6 Gauge connection

The fibre covariant derivative is:

$D_\mu = \partial_\mu + i\mathcal{A}_\mu$. (G.18)

Under local rephasing:

$\mathcal{A}_\mu \to \mathcal{A}_\mu - \partial_\mu\chi$. (G.19)

The Abelian curvature is:

$\mathbb{F}_{\mu\nu} = \partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu$. (G.20)

For a matrix-valued connection:

$\mathbb{F}_{\mu\nu} = \partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu + i[\mathcal{A}_\mu,\mathcal{A}_\nu]$. (G.21)

G.7 Combined covariant derivative

For a fibre-valued tensor:

$\mathcal{D}_\mu = \nabla_\mu + i\mathcal{A}_\mu$. (G.22)

The base connection corrects coordinate transport.

The gauge connection corrects internal orientation.

A complete derivative along path x(s) is:

$\mathcal{D}_s = (dx^\mu/ds)\mathcal{D}_\mu$. (G.23)

Parallel transport satisfies:

$\mathcal{D}_s|\psi\rangle = 0$. (G.24)

G.8 Base holonomy

Transport a tangent vector around closed loop γ:

V_final = Hol_base(γ)V_initial. (G.25)

For small loop:

Hol_base(γ) ≈ I + ½R_μνArea^μν. (G.26)

Financial interpretation:

- the meaning of a local risk direction changes around the loop;
- a hedge transported through several regimes does not return unchanged;
- local CAPM exposure may not return after global state cycling.

G.9 Fibre holonomy

Transport a state around closed loop γ:

|ψ_final⟩ = Hol_gauge(γ)|ψ_initial⟩. (G.27)

Where:

Hol_gauge(γ) = $\mathcal{P}$exp[−i∮_γ$\mathcal{A}$_μdx^μ]. (G.28)

In the Abelian case:

Hol_gauge(γ) = exp(iΦ_γ). (G.29)

With:

Φ_γ = ∮_γ$\mathcal{A}$_μdx^μ. (G.30)

The state returns to the same base coordinate with changed internal orientation.

G.10 Ledger holonomy

The ledger update around path γ is:

L_final = $\mathcal{T}$_L(γ)L_initial. (G.31)

Define:

H_L(γ) = Difference(L_final,L_initial). (G.32)

Even when:

x_final = x_initial, (G.33)

one may have:

H_L(γ) ≠ 0. (G.34)

Ledger holonomy is directly observable through:

- realized P&L;

- transaction cost;
- margin history;
- settlement record;
- tax consequence;
- legal obligation.

G.11 Three holonomies

The complete loop output may be represented as:

$$\mathrm{Hol}_{\mathrm{total}}(\gamma) = \mathrm{Hol}_{\mathrm{base}}(\gamma) \times \mathrm{Hol}_{\mathrm{gauge}}(\gamma) \times \mathrm{Hol}_{\mathrm{ledger}}(\gamma). \quad (G.35)$$

The three factors mean:

Factor	Meaning
$\mathrm{Hol}_{\mathrm{base}}$	Tangent and metric transport residue
$\mathrm{Hol}_{\mathrm{gauge}}$	Internal phase or orientation residue
$\mathrm{Hol}_{\mathrm{ledger}}$	Historical commitment residue

They may interact.

They should not be collapsed into one undifferentiated “memory effect.”

G.12 Example: spot–volatility loop

Let coordinates be:

$$x^1 = U. \quad (G.36)$$

$$x^2 = \sigma. \quad (G.37)$$

Consider loop:

$$(U_0,\sigma_0) \to (U_1,\sigma_0) \to (U_1,\sigma_1) \to (U_0,\sigma_1) \to (U_0,\sigma_0). \quad (G.38)$$

The endpoint matches the start.

But the option portfolio may retain:

- hedge P&L;
- transaction cost;
- changed inventory;
- different implied-surface shape;

- altered risk limit.

Thus:

x_final = x_initial. (G.39)

But:

State_final ≠ State_initial. (G.40)

G.13 Example: funding–collateral loop

Coordinates:

x^1 = funding spread f. (G.41)

x^2 = collateral requirement c. (G.42)

Loop:

Low f,Low c
→ High f,Low c
→ High f,High c
→ Low f,High c
→ Low f,Low c. (G.43)

Possible retained effects:

- cumulative funding cost;
- collateral opportunity cost;
- asset liquidation;
- credit deterioration;
- altered trust.

The loop residue may be decomposed:

H_total = H_cost + H_ledger + H_phase + ε_H. (G.44)

The geometric model is useful only if this decomposition improves prediction or control.

G.14 Curvature from noncommuting transport

Let transport operators be:

T_f = move along funding direction. (G.45)

T_c = move along collateral direction. (G.46)

The loop operator is:

$H_{fc} = T_f^{-1} T_c^{-1} T_f T_c$. (G.47)

If:

$H_{fc} \neq I$, (G.48)

the operations do not commute.

For a small loop:

$H_{fc} \approx I + \mathbb{F}_{fc}\Delta f \Delta c$. (G.49)

This provides an empirical route to estimate curvature from order-dependent transition residual.

G.15 Gauge-invariant observables

The absolute phase θ may change under gauge:

$\theta \rightarrow \theta + \chi$. (G.50)

Observable quantities should depend on:

- relative phase;
- loop phase;
- covariant derivative;
- curvature.

Examples:

$\Delta\theta_{nm} = \theta_n - \theta_m$. (G.51)

$\Phi_\gamma = \oint_\gamma \mathcal{A}$. (G.52)

$|D_\mu \psi|^2$. (G.53)

$\mathbb{F}_{\mu\nu}\mathbb{F}^{\mu\nu}$. (G.54)

A financial model that predicts using arbitrary absolute phase origin is not gauge-robust.

G.16 Metric and gauge double-counting test

Suppose path residual is predicted by both:

Metric curvature term C_g. (G.55)

Gauge curvature term C_A. (G.56)

The combined model is:

Residual = $C_g + C_A + \varepsilon$. (G.57)

Double-counting occurs when both terms encode the same empirical feature.

A practical diagnostic is high parameter redundancy:

Corr(C_g,C_A) ≈ ±1. (G.58)

Or unstable coefficient allocation across samples.

Required response:

- redefine the layers;
- constrain one term;
- remove the redundant structure.

G.17 Curvature estimation through transport triangles

Select three nearby states A, B, and C.

Transport A→B→C:

T_CBT_BA. (G.59)

Compare with direct A→C:

T_CA. (G.60)

Define triangle residual:

ε_ABC = T_CBT_BA − T_CA. (G.61)

In flat path-independent transport:

ε_ABC ≈ 0. (G.62)

Systematic nonzero ε_ABC indicates:

- curvature;
- protocol mismatch;
- omitted ledger;
- invalid connection.

The experiment must distinguish among these explanations.

G.18 Holonomy versus hysteresis

A hysteresis model writes:

Y_t = $\mathcal{H}$[X_{0:t}]. (G.63)

A geometric model writes:

Y_final = Hol(γ)Y_initial. (G.64)

The geometric model earns its role when:

1. the loop effect is frame-invariant;
2. one connection summarizes many histories;
3. curvature predicts new loops;
4. the model compresses path dependence.

Otherwise:

Use Hysteresis Directly. (G.65)

G.19 Geometry proposition

Proposition G.1 — Triple Geometry Principle

A financial world may require three distinct geometric structures:

1. a metric geometry of financial distance;
2. a gauge geometry of internal-state orientation;
3. a ledger geometry of historical accessibility.

In compact form:

Geometry_F = MetricDistance + FibreOrientation + LedgeredPath. (G.66)

A complete loop analysis should state which geometry generates each residual.

Appendix H — Separability Witnesses and Bell-Style Stress Tests

H.1 Purpose

This appendix defines a cautious hierarchy for testing whether an option–underlying joint model exceeds classical correlation.

The tests proceed from weakest to strongest:

1. independence;
2. classical joint dependence;
3. separability;
4. coherent nonseparability;
5. contextuality;
6. no-signalling;
7. Bell-type nonclassicality.

Failure at a higher level does not invalidate lower-level joint modelling.

H.2 Independence test

The null hypothesis is:

H_0^{ind}: $p(u,d) = p(u)p(d)$. (H.1)

Rejecting H_0^{ind} establishes correlation.

It does not establish:

- contractual coupling;
- nonfactorization;
- coherence;
- entanglement.

For options, rejection is expected.

H.3 Conditional-independence test

Introduce latent state λ:

H_0^{latent}: $p(u,d|\lambda) = p(u|\lambda)p(d|\lambda)$. (H.2)

Candidate λ includes:

- underlying price;
- volatility;
- liquidity;
- funding;
- time to maturity;
- dealer inventory;
- market regime.

If conditional dependence disappears after adequate λ:

$I(U;D|\lambda) \approx 0$, (H.3)

then common classical state explains the relation.

H.4 Separable-state model

The separable family is:

Sep = {Σ_jp_jρ_j^U ⊗ ρ_j^D}. (H.4)

Estimate unrestricted state:

ρ̂_UD. (H.5)

Estimate nearest separable state:

σ̂_sep = arg min_{σ∈Sep}D(ρ̂_UD,σ). (H.6)

Define separability distance:

D_sep = D(ρ̂_UD,σ̂_sep). (H.7)

A positive point estimate is insufficient.

One needs:

LowerCI(D_sep) > 0. (H.8)

H.5 Partial-transpose test

For a bipartite state, define partial transpose on D:

ρ_UD^{T_D}. (H.9)

If:

ρ_UD^{T_D} has a negative eigenvalue, (H.10)

the state is nonseparable for many low-dimensional cases.

Define negativity:

$\mathcal{N}(\rho) = [\|\rho^{T_D}\|_1 - 1]/2$. (H.11)

A positive:

$\mathcal{N}(\rho) > 0$ (H.12)

is a representation-relative nonseparability witness.

The result depends on the validity of:

- the state reconstruction;
- the tensor partition;
- the measurement basis;
- the positive-state model.

H.6 Entanglement witness

An entanglement witness W satisfies:

$\mathrm{Tr}(W\sigma_{sep}) \geq 0$ for all $\sigma_{sep} \in \mathrm{Sep}$. (H.13)

But:

$\mathrm{Tr}(W\rho_{candidate}) < 0$. (H.14)

A data-driven witness can be built from measured observables:

$W = \Sigma_k c_k \hat{O}_k$. (H.15)

The test statistic is:

$w_{data} = \Sigma_k c_k \langle \hat{O}_k \rangle_{data}$. (H.16)

The separable bound is:

$w_{sep} = \inf_{\sigma \in \mathrm{Sep}} \mathrm{Tr}(W\sigma)$. (H.17)

Witness violation occurs when:

$w_{data} < w_{sep} - \delta_{stat}$. (H.18)

The coefficients c_k must be pre-registered or corrected for search bias.

H.7 Classical latent-model challenge

Before accepting nonseparability, compare against:

- stochastic volatility;
- copula models;
- hidden Markov states;
- Hawkes processes;
- agent-based feedback;
- classical contextual memory;
- nonlinear interaction models.

Define best classical score:

$\mathrm{Score}_{class} = \max_{M \in \mathcal{N}_{classical}} \mathrm{Score}(M)$. (H.19)

Define joint-state score:

$\mathrm{Score}_{joint} = \mathrm{Score}(\rho_{UD})$. (H.20)

The joint-state representation advances only if:

$\mathrm{Score}_{joint} - \mathrm{Score}_{class} > \Delta^*$. (H.21)

H.8 Coherence witness

Let branch basis be $\{|n\rangle\}$.

A coherence witness may be:

$$W_{coh} = |n\rangle\langle m| + |m\rangle\langle n|. \quad (H.22)$$

Its expectation is:

$$\langle W_{coh}\rangle = 2\mathrm{Re}(\rho_{nm}). \quad (H.23)$$

The corresponding quadrature witness is:

$$W_{phase} = -i|n\rangle\langle m| + i|m\rangle\langle n|. \quad (H.24)$$

Its expectation is:

$$\langle W_{phase}\rangle = 2\mathrm{Im}(\rho_{nm}). \quad (H.25)$$

Both are needed to reconstruct complex off-diagonal structure.

A diagonal payoff measurement cannot estimate them.

H.9 Measurement-setting design

Let underlying settings be:

a_0 = price-direction basis. (H.26)

a_1 = volatility or liquidity basis. (H.27)

Let derivative settings be:

b_0 = payoff or premium basis. (H.28)

b_1 = Greek or replication basis. (H.29)

Outcomes are encoded:

$$u,d \in \{-1,+1\}. \quad (H.30)$$

Thresholds must be pre-registered.

Changing thresholds after observing the result introduces selection bias.

H.10 Correlation function

For setting pair (a,b):

$$E(a,b) = \Sigma_{\{u,d\}} ud \cdot p(u,d|a,b). \quad (H.31)$$

The CHSH-like statistic is:

$$S_F = E(a_0,b_0) + E(a_0,b_1) + E(a_1,b_0) - E(a_1,b_1). \quad (H.32)$$

Under standard local hidden-variable assumptions:

$|S_F| \le 2$. (H.33)

But finance usually violates some assumptions before the inequality is tested.

H.11 No-signalling diagnostics

Compute underlying marginal:

$p(u|a,b) = \Sigma_d p(u,d|a,b)$. (H.34)

Derivative-to-underlying signalling gap:

$\delta_{D\to U} = \max_{u,a,b_0,b_1} |p(u|a,b_0) - p(u|a,b_1)|$. (H.35)

Underlying-to-derivative signalling gap:

$\delta_{U\to D} = \max_{d,b,a_0,a_1} |p(d|a_0,b) - p(d|a_1,b)|$. (H.36)

Define:

$NS_F = \max(\delta_{D\to U}, \delta_{U\to D})$. (H.37)

A Bell-like interpretation requires:

$NS_F \le \varepsilon_{NS}$. (H.38)

Otherwise the setting choices alter local marginals through ordinary causal channels.

H.12 Measurement-independence diagnostics

Let λ denote observed or latent market state.

Measurement independence requires:

$p(a,b|\lambda) = p(a,b)$. (H.39)

Define dependence measure:

$MI_F = I[(a,b);\lambda]$. (H.40)

A Bell-like interpretation requires:

$MI_F \le \varepsilon_{MI}$. (H.41)

In ordinary finance:

$a = \pi_A(\lambda, L_A)$. (H.42)

$b = \pi_B(\lambda, L_B)$. (H.43)

Therefore setting independence usually fails.

Randomized passive settings are preferable.

H.13 Memory diagnostics

Let trial history be:

H_k = (a_{<k},b_{<k},u_{<k},d_{<k},L_{<k}). (H.44)

Memory-free trials require approximately:

p(u_k,d_k|a_k,b_k,H_k) ≈ p(u_k,d_k|a_k,b_k). (H.45)

Define memory gap:

MG_F = I[(u_k,d_k);H_k | a_k,b_k]. (H.46)

A large MG_F indicates:

- adaptive policy;
- ledger dependence;
- inventory memory;
- regime persistence.

A sequential classical model must then be used.

H.14 Post-selection diagnostics

Let S=1 indicate inclusion in the analysed sample.

Selection bias arises when:

p(S=1|u,d,a,b,λ) (H.47)

depends on outcomes or hidden state.

Define:

SG_F = Distance[p_data(u,d|a,b,S=1),p_all(u,d|a,b)]. (H.48)

Common financial selection sources include:

- only traded quotes;
- only liquid options;
- only settled trades;
- only surviving institutions;
- only available data windows.

A Bell-style test must log all eligible trials.

H.15 Time-order diagnostics

Let:

t_A = time of setting or outcome A. (H.49)

t_B = time of setting or outcome B. (H.50)

If:

$t_A < t_B$, (H.51)

ordinary information transfer from A to B may be possible.

The experiment must define a trial window Δt_{trial} and causal isolation assumption.

Ledger simultaneity:

$k_A = k_B$ (H.52)

does not imply physical causal isolation.

H.16 Strong classical causal model

A general causal null may be:

$\lambda_{k+1} = F(\lambda_k, a_k, b_k, u_k, d_k, \xi_k)$. (H.53)

$u_k = G_U(a_k, b_k, \lambda_k, H_k, \varepsilon_{U,k})$. (H.54)

$d_k = G_D(a_k, b_k, \lambda_k, H_k, \varepsilon_{D,k})$. (H.55)

This model allows:

- signalling;
- measurement dependence;
- memory;
- common cause;
- feedback.

If such a model reproduces the data, a Bell-nonclassical interpretation is unnecessary.

H.17 Interpretation table

Observation	Defensible conclusion
Joint dependence only	Correlation
Functional constraint	Contractual coupling
Joint model beats local models	Global relational information
Separable state fits	Classical composite state
Separable state rejected	Representation-relative nonseparability
Off-diagonal model predicts new basis	Candidate operational coherence

Observation	Defensible conclusion
No-signalling fails	Ordinary causal interaction present
Setting independence fails	Bell inference invalid
Memory explains result	Adaptive classical process
CHSH exceeds 2 with gaps uncontrolled	Anomalous but non-diagnostic
CHSH exceeds 2 with all major gaps controlled	Strong anomaly requiring replication

---

H.18 Bell-style experiment in simulation

A controlled simulation may proceed:

1. prepare hidden primary state X_k;
2. compile secondary state ρ_k;
3. generate settings a_k and b_k independently;
4. apply passive local instruments;
5. delay record sharing until both outcomes are fixed;
6. estimate S_F;
7. separately introduce signalling, memory, and setting dependence;
8. measure how each changes S_F.

This reveals which classical mechanisms can mimic a violation.

The experiment is more valuable as an assumption audit than as an immediate quantum claim.

H.19 World-relative entanglement

A financial entanglement statement should be written:

Ent_θ(U,D | W_θ,Π,$\mathcal{A}$,P). (H.56)

Where:

- W_θ is the secondary valuation world;
- Π is the subsystem partition;
- $\mathcal{A}$ is the admissible measurement algebra;
- P is the protocol.

This notation prevents the claim from being interpreted as unconditional material identity with physical quantum entanglement.

H.20 Witness proposition

Proposition H.1 — Evidence-Graded Nonseparability

A derivative system may be described as operationally nonseparable only when:

1. a natural subsystem partition exists;
2. the joint state is estimated from multiple settings;
3. separable models fail within uncertainty;
4. rich classical causal nulls fail;
5. the result survives local frame transformations;
6. the evidence level is stated explicitly.

In compact form:

$$\text{OperationalNonSep}_F \Leftrightarrow \text{NaturalPartition} \wedge \text{MultiSettingState} \wedge \text{SepFailure} \wedge \text{ClassicalNullFailure} \wedge \text{FrameRobustness} \wedge \text{EvidenceLabel}. \quad (H.57)$$

The next appendix can then separate assumptions already inherited from mature finance from assumptions introduced only by the layered toy architecture.

Appendix I — Assumption Ledger and Evidence Status

I.1 Why the assumption ledger is indispensable

The architecture combines mature finance with several layers of mathematical extension.

Without an explicit assumption ledger, a concept may silently move through the following stages:

$$\begin{aligned}&\text{Definition}\\ &\rightarrow \text{Convenient Representation}\\ &\rightarrow \text{Model Hypothesis}\\ &\rightarrow \text{Empirical Claim}\\ &\rightarrow \text{Ontological Assertion}. \quad (I.1)\end{aligned}$$

These stages are not equivalent.

The assumption ledger therefore classifies every important component by:

1. source;
2. mathematical status;
3. empirical status;
4. failure condition;
5. strongest presently defensible interpretation.

The governing rule is:

$$\text{No Assumption May Be Promoted without New Evidence}. \quad (I.2)$$

I.2 Evidence-status codes

The article uses the following status codes.

Code	Status
M	Mature finance or established mathematics
D	Derived from declared definitions
E	Engineering extension
H	Empirical hypothesis
T	Toy-theory proposal
U	Unestablished strong claim
R	Residual or open problem

A component may carry more than one code.

For example:

R+iQ completion = D + E. (I.3)

It is derived once A and R are declared.

But using it as a predictive financial coordinate is an engineering extension requiring empirical validation.

I.3 Mature-finance assumptions

I.3.1 CAPM local valuation relation

$E[r_i] = r_f + \beta_i ERP$. (I.4)

Status:

M. (I.5)

Limitations:

- model assumptions;
- empirical instability;
- benchmark dependence;
- horizon dependence;
- regime dependence.

Strongest defensible use:

Local expected-return benchmark under a declared market frame. (I.6)

I.3.2 Risk-neutral valuation

$D_0 = E_{\mathbb{Q}}[\text{DiscountedPayoff}]$. (I.7)

Status:

M. (I.8)

Subject to:

- no-arbitrage;
- tradability;
- measure construction;
- market-completeness assumptions or calibration conventions.

Strongest defensible use:

Pricing relation under the declared valuation measure. (I.9)

It is not a physical measurement-probability law.

I.3.3 Option payoff relation

For a European call:

$D_T = \max(U_T - K, 0)$. (I.10)

Status:

M. (I.11)

This provides a mature example of contractual branch binding.

I.3.4 Greeks

Delta:

$\Delta = \partial D / \partial U$. (I.12)

Gamma:

$\Gamma = \partial^2 D / \partial U^2$. (I.13)

Vega:

$\text{Vega} = \partial D / \partial \sigma$. (I.14)

Status:

M. (I.15)

The geometric interpretation of Greeks as tangent and curvature coordinates is:

E. (I.16)

I.3.5 Hedging backreaction

Derivative positions can generate hedge demand, which can affect underlying order flow.

Status:

M in mechanism. (I.17)

The strength and direction are empirically conditional.

The operator representation:

Ĥ_hedge (I.18)

has status:

T. (I.19)

I.3.6 Margin and collateral feedback

Valuation changes can produce:

Valuation
→ Exposure
→ Margin
→ Funding
→ Trading
→ New Valuation. (I.20)

Status:

M. (I.21)

This is one of the strongest established examples of secondary-world backreaction on the primary economy.

I.4 Complex-completion assumptions

I.4.1 Independent amplitude A

The theory requires an independently meaningful amplitude:

A = declared pre-filter economic scale. (I.22)

Status:

H. (I.23)

The theory fails if A is chosen solely to manufacture Q.

Required condition:

A must be estimated independently of Q. (I.24)

I.4.2 Admitted coordinate R

R is:

R = value admitted by protocol P. (I.25)

Status:

D. (I.26)

Its meaning is protocol-relative.

Examples include:

- market value;
- CAPM-filtered value;
- accounting value;
- liquidation value;
- collateral value.

I.4.3 Retained pressure Q_E

$Q_E = \sqrt{(A^2 - R^2)}$. (I.27)

Status:

D + E. (I.28)

Derived mathematically once A and R are declared.

Its interpretation as economically meaningful retained pressure is:

H. (I.29)

It becomes operational only when it predicts or explains something beyond A and R.

I.4.4 Complex state

$Z = R + iQ_E$. (I.30)

Status:

D + E. (I.31)

This does not imply:

- quantum state;
- physical wavefunction;
- complex probability amplitude.

Strongest defensible use:

Two-coordinate valuation representation preserving admitted value and orthogonal retained

pressure. (I.32)

I.4.5 Circular phase

$\theta = \operatorname{atan2}(Q_E, R)$. (I.33)

Status:

D. (I.34)

The interpretation of θ as internal financial orientation is:

E. (I.35)

The interpretation of θ as time is:

H. (I.36)

I.5 Internal-time assumptions

I.5.1 Phase order

A channel has phase order when:

$\theta_1 \to \theta_2 \to \theta_3$ (I.37)

provides a stable sequence of internal valuation states.

Status:

E + H. (I.38)

I.5.2 Phase velocity

$\omega_F = d\theta/dt$. (I.39)

Status:

D once θ(t) is defined. (I.40)

Its predictive meaning is:

H. (I.41)

I.5.3 Ledger time

$L_{k+1} = L_k \sqcup Record_k$. (I.42)

Status:

M + E. (I.43)

Persistent records and event order are mature institutional structures.

Their interpretation as generated historical time is:

T. (I.44)

I.5.4 Time-bearing world

TimeBearing_W ⇔ OrderedChange ∧ Commitment ∧ Persistence ∧ FutureConstraint. (I.45)

Status:

T. (I.46)

This is a general world-formation proposition.

It is not a standard theorem of finance or physics.

I.6 Composite-state assumptions

I.6.1 Natural subsystem partition

$\mathscr{H}$_UD = $\mathscr{H}$_U ⊗ $\mathscr{H}$_D. (I.47)

Status:

E. (I.48)

The option and underlying are institutionally distinguishable and contractually linked.

The tensor product is an effective modelling choice.

It is not supplied automatically by standard option theory.

I.6.2 Contract-preparation operator

Û_C|u_n,o_D⟩ = |u_n,d_n⟩. (I.49)

Status:

T. (I.50)

This is a formal representation of the contract's branch-binding role.

Its usefulness must be tested against ordinary conditional probability models.

I.6.3 Joint state

|Ψ_UD⟩ = Σ_n c_n|u_n,d_n⟩. (I.51)

Status:

T. (I.52)

If c_n are merely square roots of classical probabilities, this may be a formal purification.

Operational coherence requires more.

I.6.4 Reduced local states

$\rho_U = \mathrm{Tr}_D(\rho_{UD})$. (I.53)

$\rho_D = \mathrm{Tr}_U(\rho_{UD})$. (I.54)

Status:

M mathematically within the declared tensor representation. (I.55)

Their financial meaning depends on the validity of the partition and observable algebra.

I.6.5 Effective nonfactorization

$\rho_{UD} \neq \rho_U \otimes \rho_D$. (I.56)

Status:

E. (I.57)

This establishes correlation in the effective state representation.

It does not establish entanglement.

I.6.6 Strict nonseparability

$\rho_{UD} \notin \mathrm{Sep}(\mathscr{H}_U \otimes \mathscr{H}_D)$. (I.58)

Status:

U. (I.59)

A financial claim requires:

- estimated joint state;
- multiple measurement settings;
- natural partition;
- strong classical-null rejection;
- frame robustness.

I.7 Phase and coherence assumptions

I.7.1 Risk-neutral amplitude lift

$\alpha_n = \sqrt{q_n}\exp(i\varphi_n)$. (I.60)

Status:

D + T. (I.61)

The magnitude is defined from mature pricing weights.

The phase is additional.

I.7.2 Relative phase

$\Delta\varphi_{nm} = \varphi_n - \varphi_m$. (I.62)

Status:

D once channel phases are assigned. (I.63)

Its financial observability is:

H. (I.64)

I.7.3 Off-diagonal coherence

$\rho_{off} \neq 0$. (I.65)

Status:

U. (I.66)

A fitted nonzero off-diagonal term may reflect:

- overfitting;
- basis choice;
- hidden classical state;
- actual effective coherence.

I.7.4 Operational coherence

$\text{Coherence}_F \Leftrightarrow \text{OffDiagonal} \wedge \text{CrossObservable} \wedge \text{PredictiveGain} \wedge \text{ClassicalNullFailure}$. (I.67)

Status:

U. (I.68)

No general market evidence is asserted.

I.7.5 Interference

$P_F(y) = P_{mix}(y) + \text{CrossTerm}(y)$. (I.69)

Status:

T + H. (I.70)

Ordinary nonlinearity is the default null explanation.

I.8 Observer assumptions

I.8.1 Internal observer

O = ($\mathcal{A}$_O,$\mathbb{M}$_O,F_O,L_O,π_O). (I.71)

Status:

E + T. (I.72)

The observer is defined operationally, not psychologically.

I.8.2 Restricted local algebra

$\mathcal{A}$_O ⊂ $\mathcal{A}$_global. (I.73)

Status:

M as an information-access concept. (I.74)

Its use in financial state reduction is:

E. (I.75)

I.8.3 Internal certainty

P_O(y_k | $\mathscr{F}$_O,k) = 1. (I.76)

Status:

D under the assumption that the observer's own committed record is trusted. (I.77)

This is record-relative certainty.

It is not universal truth.

I.8.4 Latching

Record difference causes future policy divergence:

Record_A ≠ Record_B → π_A,next ≠ π_B,next. (I.78)

Status:

M + E. (I.79)

This is common in adaptive financial systems.

I.8.5 Cross-observer fixedness

ABFixed ⇔ FrameMap ∧ CompatibleEffects ∧ AccessibleRecord. (I.80)

Status:

T supported by observer-theory architecture. (I.81)

This is testable in institutional systems.

I.9 SR-like assumptions

I.9.1 Discount rapidity

ρ_D = ln(A/R). (I.82)

Status:

D. (I.83)

Its additive composition follows mathematically:

ρ_CA = ρ_CB + ρ_BA. (I.84)

when the transformations are multiplicative and protocol-preserving.

I.9.2 Bounded relative-frame coordinate

u_D = tanh ρ_D. (I.85)

Status:

D. (I.86)

Its relativistic interpretation is:

T. (I.87)

It is not physical velocity.

I.9.3 Local financial frame

F_a = (Numeraire,Benchmark,Funding,Horizon,Protocol,Ledger,Algebra). (I.88)

Status:

M + E. (I.89)

Finance already uses local valuation conventions.

The tetrad representation is:

T. (I.90)

I.9.4 Local-flat valuation law

CAPM is treated as locally valid:

CAPM_local(x). (I.91)

Status:

H. (I.92)

This becomes useful if local parameter stability correlates with estimated geometric flatness.

I.9.5 Financial equivalence principle

Observed repricing may combine:

Object Change + Frame Acceleration. (I.93)

Status:

T + H. (I.94)

The proposition must be tested through cross-frame invariants.

I.10 GR-like assumptions

I.10.1 Financial manifold

$\mathcal{M}$_F = state space of declared financial coordinates. (I.95)

Status:

E. (I.96)

State manifolds are mathematically standard.

Their specific economic coordinate selection is protocol-dependent.

I.10.2 Financial metric

ds_F² = g^F_μνdx^μdx^ν. (I.97)

Status:

E + H. (I.98)

Candidate definitions include:

- Fisher information;
- inverse covariance;
- execution cost;
- intervention energy.

There is no unique established financial spacetime metric.

I.10.3 Financial curvature

$R^{\rho}_{\sigma\mu\nu} \neq 0$. (I.99)

Status:

D after a metric is estimated. (I.100)

Its interpretation as economically useful curvature is:

H. (I.101)

I.10.4 Effective financial stress tensor

$T^{F}_{\mu\nu} = T^{F}_{\mu\nu}[\rho, L, P]$. (I.102)

Status:

T. (I.103)

No unique mature definition exists.

I.10.5 Financial Einstein-like equation

$G^{F}_{\mu\nu} + \Lambda_{F} g^{F}_{\mu\nu} = \kappa_{F} T^{F}_{\mu\nu} + \Xi^{F}_{\mu\nu}$. (I.104)

Status:

T + U. (I.105)

This is a schematic architecture.

It is not an empirical field equation presently established in finance.

I.11 Gauge assumptions

I.11.1 Gauge freedom

$|\psi(x)\rangle \rightarrow \exp[i\chi(x)]|\psi(x)\rangle$. (I.106)

Status:

M mathematically within complex-state modelling. (I.107)

Its economic meaning depends on what local phase convention represents.

I.11.2 Gauge connection

$D_{\mu} = \partial_{\mu} + i\mathcal{A}_{\mu}$. (I.108)

Status:

T + H. (I.109)

The connection becomes useful only if it improves cross-frame transport.

I.11.3 Gauge curvature

$\mathbb{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu + i[\mathcal{A}_\mu, \mathcal{A}_\nu]$. (I.110)

Status:

D after connection specification. (I.111)

Its financial interpretation is:

H. (I.112)

I.11.4 Holonomy

$\mathrm{Hol}(\gamma) = \mathcal{P}\exp[-i\oint_\gamma \mathcal{A}]$. (I.113)

Status:

D mathematically. (I.114)

Its empirical role in finance is:

H. (I.115)

Known path dependence and hysteresis are mandatory null models.

I.12 Gate and ledger assumptions

I.12.1 Gate

$G \in \{\text{Commit}, \text{Reject}, \text{Defer}, \text{Escalate}\}$. (I.116)

Status:

M + E. (I.117)

Financial institutions already use gates.

The operator interpretation is an engineering abstraction.

I.12.2 Commitment

$L_{k+1} = L_k \sqcup \text{Record}_k$. (I.118)

Status:

M. (I.119)

I.12.3 Collapse-like interpretation

Collapse_F = Instrument + Outcome + Gate + Trace + Latching. (I.120)

Status:

T. (I.121)

This is an operational analogy.

It is not physical wavefunction collapse.

I.12.4 Ledger backreaction

Record_k → Action_k → X_{k+1}. (I.122)

Status:

M + H. (I.123)

The existence of backreaction is mature.

The magnitude must be estimated causally.

I.13 Strong quantum claims

I.13.1 Born rule

$p_n = |\alpha_n|^2$. (I.124)

Status in financial theory:

Inserted form, not derived. (I.125)

Classification:

U. (I.126)

I.13.2 No-signalling

p(u|a,b) = p(u|a). (I.127)

p(d|a,b) = p(d|b). (I.128)

Status in ordinary finance:

Generally false for active settings. (I.129)

Classification:

U. (I.130)

I.13.3 Bell nonclassicality

|S_F| > 2 under valid Bell assumptions. (I.131)

Status:

Not established. (I.132)

Classification:

U. (I.133)

I.13.4 Kochen–Specker contextuality

No globally consistent noncontextual value map exists. (I.134)

Status:

Not established. (I.135)

Classification:

U. (I.136)

I.13.5 No-cloning

No universal operation duplicates every unknown state. (I.137)

Status in finance:

Not derived. (I.138)

Classification:

U. (I.139)

I.13.6 Monogamy

Strong nonseparability cannot be shared arbitrarily across subsystems. (I.140)

Status in finance:

No universal law established. (I.141)

Classification:

U. (I.142)

I.14 Master assumption table

Component	Status	Strongest current interpretation
CAPM	M	Local valuation benchmark

Component	Status	Strongest current interpretation
Risk-neutral pricing	M	Mature derivative valuation
Greeks	M	Sensitivity and curvature measures
R	D	Protocol-admitted value
A	H	Independently defined pre-filter amplitude
Q_E	D+H	Derived pressure coordinate with untested predictive meaning
Z=R+iQ_E	D+E	Complex valuation completion
θ	D+H	Orientation; candidate internal phase
k	M+T	Ledger-event order
Tensor product	E	Effective subsystem representation
Contract operator	T	Branch-binding representation
Reduced states	M within model	Local effective descriptions
Nonfactorization	E	Joint relational structure
Nonseparability	U	Requires strong empirical evidence
Relative phase	H	Candidate cross-channel state variable
Coherence	U	Not generally established
Internal observer	E+T	Instrument–trace–policy runtime
Discount rapidity	D	Additive valuation-frame coordinate
Financial metric	E+H	Protocol-selected geometry
Curvature	H	Candidate transport-residual structure
Gauge connection	T+H	Candidate frame-transport model
Holonomy	H	Candidate path-invariant residue
Ledger backreaction	M+H	Mature mechanism, magnitude testable
Born rule	U	Not derived
No-signalling	U	Usually violated by active finance
Bell nonclassicality	U	Not established

I.15 Assumption-promotion rule

A component may be promoted only through the following ladder:

Definition
→ Identified Estimator
→ Out-of-Sample Prediction
→ Controlled Intervention
→ Frame Robustness
→ Strong Null Rejection
→ Replication. (I.143)

For example:

Q_E begins as D. (I.144)

If Q_E predicts held-out repricing, it becomes:

H with E2 evidence. (I.145)

If controlled intervention confirms its role, it advances further.

It must never be promoted directly from a geometric definition to a physical ontology.

I.16 Assumption-demotion rule

A component must be demoted when:

- its estimator is unstable;
- equivalent frames disagree;
- simpler models perform equally well;
- residual remains structured;
- the claimed invariant cannot be identified;
- the theory survives only through post hoc redefinition.

In compact form:

$$\mathrm{Demote}(C) \Leftrightarrow \mathrm{Instability} \vee \mathrm{Redundancy} \vee \mathrm{NullSurvival} \vee \mathrm{ResidualFailure} \vee \mathrm{InvariantFailure}. \tag{I.146}$$

A healthy theory contains explicit downward paths.

I.17 Assumption-ledger proposition

Proposition I.1 — Evidence-Preserving Expansion

A layered cross-domain theory remains scientifically disciplined only when every extension retains:

1. its source;
2. its current evidence status;
3. its operational estimator;
4. its failure condition;
5. its strongest permitted interpretation.

In compact form:

$$\mathrm{DisciplinedExtension} \Leftrightarrow \mathrm{Source} \wedge \mathrm{Status} \wedge \mathrm{Estimator} \wedge \mathrm{FailureRule} \wedge \mathrm{InterpretationBound}. \tag{I.147}$$

Appendix J — Quantum-Characteristic Capability Matrix

J.1 Purpose

Part I concluded that scalar CAPM Finance Geometry could reproduce some quantum-like characteristics but not others.

Part II added:

- derivative composite states;
- internal observer restriction;
- multi-channel phase;
- relative frames;
- curved geometry;
- gauge transport;
- gate–ledger dynamics;
- backreaction.

This appendix compares the simulation capability before and after the extension.

The question is not:

“Does finance become quantum?”

The question is:

“Which quantum-associated structures can now be represented more faithfully in a non-quantum control world?”

J.2 Capability levels

Code	Meaning
0	Not represented
1	Loose analogy
2	Formal structural representation
3	Operationally testable model
4	Empirically established in finance
Q	Specifically quantum residue remains

The capability score describes structural coverage.

It does not describe physical identity.

J.3 Complex numbers

Part I

$Z = R + iQ$. (J.1)

Capability:

3. (J.2)

The complex coordinate is mathematically explicit and operationally testable.

Part II

The state becomes multi-channel:

$|\Psi_F\rangle = \Sigma_n c_n A_n \exp(i\theta_n)|n\rangle$. (J.3)

Capability:

3. (J.4)

Improvement:

From one complex coordinate to a complex state space.

Quantum residue:

Complex numbers alone are not specifically quantum.

J.4 Phase

Part I

One angle:

$\theta = \operatorname{atan2}(Q,R)$. (J.5)

Capability:

3 for local orientation. (J.6)

Part II

Relative phase:

$\Delta\theta_{nm} = \theta_n - \theta_m$. (J.7)

Capability:

3 for multi-channel phase modelling. (J.8)

Improvement:

Relative phase can now enter cross-channel observables.

Quantum residue:

Empirically mandatory coherence remains unestablished.

J.5 Superposition

Part I

One scalar complex state did not provide multiple unresolved branches.

Capability:

1. (J.9)

Part II

$|\Psi\rangle = \Sigma_n c_n|n\rangle$. (J.10)

Capability:

2–3. (J.11)

Improvement:

Explicit unresolved channel representation.

Quantum residue:

A classical possibility mixture may produce the same diagonal outcomes.

J.6 Interference

Part I

No relative phase channels.

Capability:

0–1. (J.12)

Part II

$P(y) = P_{mix}(y) + CrossTerm(y)$. (J.13)

Capability:

2–3. (J.14)

Improvement:

Phase-sensitive interference is formally representable and experimentally testable.

Quantum residue:

Classical nonlinear and phase-oscillator models remain strong alternatives.

J.7 Composite systems

Part I

No tensor-product decomposition.

Capability:

0. (J.15)

Part II

$\mathscr{H}_{UD} = \mathscr{H}_U \otimes \mathscr{H}_D$. (J.16)

Capability:

3. (J.17)

Improvement:

Derivative and underlying become distinct but contractually bound subsystems.

Quantum residue:

Tensor notation alone does not establish physical subsystem ontology.

J.8 Entanglement-like nonfactorization

Part I

Undefined.

Capability:

0. (J.18)

Part II

$\rho_{UD} \neq \rho_U \otimes \rho_D$. (J.19)

And potentially:

$\rho_{UD} \notin \mathrm{Sep}(\mathscr{H}_U \otimes \mathscr{H}_D)$. (J.20)

Capability:

2–3. (J.21)

Improvement:

Effective nonfactorization can now be defined and tested.

Quantum residue:

Irreducible nonseparability remains unestablished.

J.9 Local mixedness

Part I

No partial trace.

Capability:

0. (J.22)

Part II

$\rho_U = \mathrm{Tr}_D \rho_{UD}$. (J.23)

$\rho_D = \mathrm{Tr}_U \rho_{UD}$. (J.24)

Capability:

3. (J.25)

Improvement:

Global completeness and local incompleteness become representable.

Quantum residue:

Mixedness may arise from classical aggregation or hidden variables.

J.10 Conditional collapse

Part I

Gate commitment was modelled:

State → Gate → Record. (J.26)

Capability:

3 for institutional collapse-like commitment. (J.27)

Part II

Local measurement conditions the remote subsystem state:

$\rho_{D|u}$. (J.28)

Capability:

3. (J.29)

Improvement:

The collapse grammar now acts on a composite state.

Quantum residue:

The update may remain ordinary Bayesian conditioning plus causal feedback.

J.11 Context-dependent measurement

Part I

Different valuation filters produced different outputs.

Capability:

3. (J.30)

Part II

Multiple local and joint measurement bases are explicit.

Capability:

3. (J.31)

Improvement:

Settings can be modelled algebraically and sequentially.

Quantum residue:

Kochen–Specker contextuality is not established.

J.12 Noncommutativity

Part I

Order-dependent gates and protocols.

Capability:

3 operationally. (J.32)

Part II

Instrument composition:

$\mathbb{M}_A \circ \mathbb{M}_B \neq \mathbb{M}_B \circ \mathbb{M}_A$. (J.33)

Capability:

3. (J.34)

Improvement:

Noncommutativity is tied to explicit instrument maps.

Quantum residue:

The effect may arise from ordinary causal disturbance and memory.

J.13 Decoherence

Part I

Ledger commitment suppressed alternatives conceptually.

Capability:

2. (J.35)

Part II

System–environment state:

$|\Psi_{SE}\rangle = \Sigma_n c_n|s_n,e_n\rangle$. (J.36)

Capability:

3. (J.37)

Improvement:

Off-diagonal suppression can be modelled quantitatively.

Quantum residue:

Institutional classicalization is not physical quantum decoherence.

J.14 Geometric phase

Part I

Phase existed but no gauge connection or loop transport was fully developed.

Capability:

1–2. (J.38)

Part II

$\Phi_\gamma = \oint_\gamma \mathcal{A}_\mu dx^\mu$. (J.39)

Capability:

3. (J.40)

Improvement:

Closed-loop phase and holonomy become testable.

Quantum residue:

Classical hysteresis and path dependence remain null explanations.

J.15 Relative frames

Part I

Appendix M hinted at Lorentz-like mappings.

Capability:

2. (J.41)

Part II

ρ_BA = ln(R_A/R_B). (J.42)

ρ_CA = ρ_CB + ρ_BA. (J.43)

Capability:

3. (J.44)

Improvement:

Exact additive frame composition is defined.

Quantum residue:

No physical Lorentz invariance or universal speed limit is derived.

J.16 Curved geometry

Part I

Curvature was proposed schematically.

Capability:

2. (J.45)

Part II

ds_F² = g^F_μνdx^μdx^ν. (J.46)

R^ρ_σμν. (J.47)

Capability:

2–3. (J.48)

Improvement:

Metric, connection, curvature, and local frames are role-separated.

Quantum residue:

No Einstein equation or physical spacetime curvature follows.

J.17 Gauge transport

Part I

Only hinted.

Capability:

1–2. (J.49)

Part II

$D_\mu = \partial_\mu + i\mathcal{A}_\mu$. (J.50)

Capability:

3. (J.51)

Improvement:

Cross-frame phase transport and holonomy become explicit.

Quantum residue:

The financial gauge group and connection remain empirical modelling choices.

J.18 Observer-relative reality

Part I

Strongly represented through protocol, gate, and trace.

Capability:

3–4. (J.52)

Part II

Three-level architecture:

Primary Constructor
→ Secondary World
→ Internal Observer. (J.53)

Capability:

3–4. (J.54)

Improvement:

The location of entanglement strangeness becomes clearer.

Quantum residue:

Observer boundedness alone does not derive quantum probability.

J.19 Internal time

Part I

t, θ, and k were separated.

Capability:

3. (J.55)

Part II

Multiple local phase clocks and frame-relative simultaneity were added.

Capability:

3. (J.56)

Improvement:

The financial world now supports local clocks and global ledger order.

Quantum residue:

No physical time-dilation law is established.

J.20 Tunnelling-like behaviour

Part I

Gate crossing and hidden-pressure release suggested a loose analogy.

Capability:

1–2. (J.57)

Part II

Barrier, liquidity, and regime-transition states are more explicit.

Capability:

2. (J.58)

Improvement:

Transition across low-admissibility regions can be modelled.

Quantum residue:

No quantum amplitude-penetration law is derived.

J.21 Uncertainty relations

Part I

Measurement constraints and observer disturbance were discussed.

Capability:

1–2. (J.59)

Part II

Operational incompatibility is explicit:

$\mathbb{M}_A \circ \mathbb{M}_B \neq \mathbb{M}_B \circ \mathbb{M}_A$. (J.60)

Capability:

2–3. (J.61)

Improvement:

Instrument-level trade-offs can be measured.

Quantum residue:

No universal lower bound such as:

$\Delta A \Delta B \geq \frac{1}{2}|\langle [A,B] \rangle|$ (J.62)

has been established as a fundamental financial law.

J.22 No-signalling

Part I

Not reproduced.

Capability:

0. (J.63)

Part II

Explicitly testable:

$p(u|a,b) = p(u|a)$. (J.64)

Capability:

2 as a test criterion. (J.65)

Empirical status:

Usually violated in active markets.

Quantum residue:

Remains.

J.23 Bell nonclassicality

Part I

Not reproduced.

Capability:

0. (J.66)

Part II

CHSH-like stress test defined:

$S_F = E(a_0,b_0)+E(a_0,b_1)+E(a_1,b_0)-E(a_1,b_1)$. (J.67)

Capability:

2 as an experimental framework. (J.68)

Empirical status:

Not established.

Quantum residue:

Remains strongly.

J.24 Born rule

Part I

Not derived.

Capability:

0. (J.69)

Part II

Formal amplitude lift:

$\alpha_n = \sqrt{q_n}\exp(i\varphi_n)$. (J.70)

Capability:

1–2. (J.71)

Improvement:

The form can be represented.

Quantum residue:

The squared-amplitude probability law remains inserted rather than derived.

J.25 No-cloning

Part I

Not represented.

Capability:

0. (J.72)

Part II

State copying can be discussed in the composite formalism.

Capability:

1. (J.73)

Quantum residue:

Financial states remain copyable in principle.

No universal algebraic no-cloning theorem follows.

J.26 Entanglement monogamy

Part I

Undefined.

Capability:

0. (J.74)

Part II

Multipartite states can be represented and tested.

Capability:

2. (J.75)

Quantum residue:

No universal monogamy law is expected for ordinary contracts.

J.27 Quantum indistinguishability

Part I

Not represented.

Capability:

0. (J.76)

Part II

Still not reproduced.

Financial instruments retain:

- identifiers;
- ownership;
- legal history;
- counterparty identity.

Capability:

0–1. (J.77)

Quantum residue:

Remains.

J.28 Capability summary

Characteristic	Part I	Part II	Status after Part II
Complex state	3	3	Strong non-quantum reconstruction
Phase	3	3	Multi-channel extension added
Superposition form	1	2–3	Formal unresolved-channel model
Interference	0–1	2–3	Testable, not established
Composite systems	0	3	Strong extension
Nonfactorization	0	2–3	Formally testable
Local mixedness	0	3	Formally explicit
Conditional collapse	2–3	3	Composite update added
Context dependence	3	3	Strongly represented
Noncommutativity	2–3	3	Instrument-level form
Decoherence	2	3	Environment and ledger model
Geometric phase	1–2	3	Holonomy formalized

Characteristic	Part I	Part II	Status after Part II
Local relativity	2	3	Exact frame maps defined
Curved geometry	2	2–3	Operational metric required
Gauge transport	1–2	3	Explicit connection added
Internal time	3	3	Multi-clock refinement
Tunnelling-like transition	1–2	2	Still classical barrier crossing
Uncertainty-like trade-off	1–2	2–3	Operational, not fundamental
No-signalling	0	2 test framework	Not established
Bell violation	0	2 test framework	Not established
Born rule	0	1–2 formal lift	Not derived
No-cloning	0	1	Not derived
Monogamy	0	2 test framework	Not established
Quantum statistics	0	0	Remains quantum residue

J.29 The corrected answer to the article's second question

The article asked:

> After adding Appendix M's layered QM–SR–GR architecture, can CAPM simulate more quantum characteristics than Part I originally allowed?

The precise answer is:

Yes at the architectural level. (J.78)

No at the level of CAPM alone. (J.79)

The extended architecture enables simulation of:

- composite states;
- local mixedness;
- entanglement-form nonfactorization;
- relative-phase interference;
- decoherence;
- geometric phase;
- observer-relative collapse;
- frame transport;
- curvature and backreaction.

But these capabilities arise from:

CAPM + Derivatives + Observer Runtime + Geometry + Gauge + Ledger. (J.80)

They do not arise from the CAPM expected-return equation by itself.

J.30 Capability-matrix proposition

Proposition J.1 — Architecture-Dependent Quantum Simulation

The number of quantum-associated structures a financial model can simulate depends not on the presence of one complex formula but on the completeness of its architecture.

In compact form:

QuantumLikeCoverage_F = f(StateSpace,Partition,Phase,Observers,Instruments,Gates,Trace,Frames,Geometry,Backreaction) (J.81)

Therefore:

More Complete Architecture → Smaller Unexplained Quantum-Like Residue. (J.82)

But:

Smaller Residue ≠ Quantum Identity. (J.83)

Appendix K — Minimal Computational Runtime

K.1 Purpose

This appendix translates the conceptual framework into a minimal implementable runtime.

The aim is not computational realism.

The aim is to ensure that every theoretical role corresponds to an executable component.

The runtime contains:

1. primary state;
2. compiler;
3. financial geometry;
4. composite contract state;
5. internal observers;
6. evolution;
7. measurement;
8. gate;
9. ledger;
10. backreaction;
11. residual-driven revision.

K.2 Primary-state schema

Define:

$$X_t = \{\text{Underlying},\text{Volatility},\text{Funding},\text{Liquidity},\text{Collateral},\text{Positions},\text{News},\text{Contracts},\text{Environment}\}. \tag{K.1}$$

A practical record may contain:

Field	Example
underlying_price	Current spot or forward
volatility_state	Realized and implied surface factors
funding_curve	Relevant discount and borrowing rates
liquidity_state	Spread, depth, impact
collateral_state	Posted and required collateral
positions	Underlying, derivative, hedge inventory
contracts	Strike, maturity, payoff, gates
news_state	Exogenous information
protocol_state	Current valuation and gate rules

K.3 Protocol schema

Define protocol:

$$P = \{\text{Boundary},\text{AmplitudeRule},\text{AdmissionRule},\text{Frame},\text{Metric},\text{Connection},\text{Measurements},\text{Gates},\text{Residu} \tag{K.2}$$

A protocol should not be hidden in source code.

It should be a versioned configuration object.

Example conceptual fields:

ProtocolID. (K.3)

BoundaryDefinition. (K.4)

AmplitudeEstimator. (K.5)

RealValueEstimator. (K.6)

PhaseConvention. (K.7)

SubsystemPartition. (K.8)

MeasurementSettings. (K.9)

GateRules. (K.10)

LedgerSchema. (K.11)

NullModels. (K.12)

K.4 Compilation step

The compiler maps:

$\rho_F = \mathcal{C}_{P,L}(X_t)$. (K.13)

A minimal compilation performs:

$A_t = \text{EstimateAmplitude}(X_t,P)$. (K.14)

$R_t = \text{EstimateAdmittedValue}(X_t,P)$. (K.15)

$Q_t = \sqrt{\max(A_t^2 - R_t^2,0)}$. (K.16)

$\theta_t = \text{atan2}(Q_t,R_t)$. (K.17)

$Z_t = R_t + iQ_t$. (K.18)

For multiple channels:

$Z_{n,t} = A_{n,t}\exp(i\theta_{n,t})$. (K.19)

The state vector is normalized:

$|\psi_F,t\rangle = \text{Normalize}_G[\Sigma_n c_n Z_{n,t}|n\rangle]$. (K.20)

K.5 Contract preparation

For each contract C:

$\rho_{UD,C} = \text{Prepare}_C(\rho_U \otimes \rho_{D^0})$. (K.21)

A discrete implementation may map each underlying scenario n to derivative state d_n:

Scenario n
→ Underlying state u_n
→ Payoff or valuation state d_n. (K.22)

The prepared density state may be either:

Classical mixture:

$\rho_{mix} = \Sigma_n q_n|u_n,d_n\rangle\langle u_n,d_n|$. (K.23)

Or coherent candidate:

$\rho_{coh} = |\Sigma_n \sqrt{q_n}\exp(i\varphi_n)|u_n,d_n\rangle|^2$. (K.24)

The runtime should support both.

The diagonal model must remain the default until coherence is justified.

K.6 Local frame construction

For observer a:

F_a = BuildFrame(X_t,P_a,L_a). (K.25)

The frame may specify:

- numeraire;
- benchmark;
- funding curve;
- horizon;
- protocol;
- accessible ledger.

Relative rapidity is:

ρ_BA = ln(R_A/R_B). (K.26)

Transport is:

State_B,pred = T_BA(State_A). (K.27)

Gauge residual:

ε_BA = State_B,observed − State_B,pred. (K.28)

The runtime logs ε_BA.

K.7 Metric and connection estimation

Estimate metric:

g_F,t = EstimateMetric(X_t,L_t,P). (K.29)

Estimate gauge connection:

$\mathcal{A}$_t = EstimateConnection(StateHistory,FrameHistory,P). (K.30)

The covariant phase velocity is:

$\tilde{\omega}$_t = dθ/dt + $\mathcal{A}$_μ(dx^μ/dt). (K.31)

The runtime may begin with:

g_F = identity metric. (K.32)

$\mathcal{A}$_μ = 0. (K.33)

More complex geometry should be introduced only after simpler versions fail.

K.8 Continuous evolution

For state vector:

$i\hbar_F \mathcal{D}_\theta |\Psi\rangle = \hat{H}_F |\Psi\rangle + |\varepsilon\rangle$. (K.34)

A numerical time step may be:

$|\Psi_{t+\Delta t}\rangle = \text{Normalize}[|\Psi_t\rangle - i\Delta\theta \hat{H}_F |\Psi_t\rangle/\hbar_F + \text{Noise} + \text{Residual}]$. (K.35)

For density state:

$\rho_{t+\Delta t} = \rho_t + \Delta\theta[-i[\hat{H}_F,\rho_t] + \mathcal{D}_{env}(\rho_t)] + \text{Residual}$. (K.36)

A classical runtime may instead evolve scenario weights directly.

Both implementations should be compared.

K.9 Observer selection

Each observer has:

$O_a = (\mathcal{A}_a, \mathbb{M}_a, F_a, L_a, \pi_a)$. (K.37)

At episode k:

$\text{Setting}_{a,k} = \pi_a(L_{a,k}, \text{StateEstimate}_{a,k})$. (K.38)

A low-risk observer may select a fixed passive setting.

A high-risk observer may select adaptive stress measurements.

The setting decision itself must be logged.

K.10 Measurement

For local effect E_y:

$p(y) = \text{Tr}(\rho E_y)$. (K.39)

Outcome generation may be:

$y \sim \text{Categorical}[p(y)]$. (K.40)

The provisional post-measurement state is:

$\rho_y = \mathbb{M}_y(\rho)/p(y)$. (K.41)

For deterministic institutional measurements, the runtime may instead calculate outcome directly.

The distinction between stochastic sampling and deterministic classification must remain explicit.

K.11 Gate evaluation

Gate input:

GInput_k = {Outcome,State,Setting,Frame,Evidence,Authority,Ledger,Residual}. (K.42)

Gate output:

Decision_k ∈ {Commit,Reject,Defer,Escalate}. (K.43)

A threshold gate may use:

Score_k = w^{T}Features_k. (K.44)

Commit when:

Score_k ≥ Threshold_k. (K.45)

A governance-aware gate also checks:

EvidenceSufficient. (K.46)

AuthorityValid. (K.47)

ResidualWithinLimit. (K.48)

K.12 Record creation

On any consequential gate result, create:

Record_k = {EventID,Outcome,Decision,Setting,Frame,Evidence,Confidence,Residual,t,θ,k,ProtocolVersion}. (K.49)

The record is immutable under ordinary operation.

Corrections create new records rather than silently rewriting history.

Thus:

Correction ≠ Erasure. (K.50)

Correction = New Record referencing Old Record. (K.51)

K.13 Ledger update

L_{k+1} = Append(L_k,Record_k). (K.52)

The runtime should support:

- local observer ledgers;
- shared institutional ledger;
- public redundant records;
- provenance links;

- delayed access.

Cross-observer objectivity tests require modelling which observer can read which record at which episode.

K.14 Policy update

Observer policy changes:

π_a,k_{+1} = UpdatePolicy(π_a,k,Record_k,Residual_k). (K.53)

Examples:

- increase measurement frequency;
- change hedge threshold;
- suspend automated trading;
- demand more collateral;
- escalate to human review.

This is the latching mechanism.

K.15 Backreaction

The committed decision generates action:

u_k = ActionPolicy(Record_k). (K.54)

Primary state changes:

X_t_{+1} = PrimaryTransition(X_t,u_k,Environment_t). (K.55)

Examples:

- trade;
- hedge;
- collateral transfer;
- liquidation;
- settlement;
- regulatory restriction.

The next financial state is recompiled from X_t_{+1}.

K.16 Geometry update

After backreaction:

$g_{F,t+1} = \mathrm{UpdateMetric}(g_{F,t}, X_{t+1}, L_{k+1})$. (K.56)

$\mathcal{A}_{t+1} = \mathrm{UpdateConnection}(\mathcal{A}_t, \mathrm{StateHistory}, \mathrm{FrameHistory}, L_{k+1})$. (K.57)

The state changes the geometry through which later states move.

This is the minimal GR-like feedback.

K.17 Residual classification

Compute:

$\varepsilon_{\mathrm{model}} = \mathrm{ObservedState} - \mathrm{PredictedState}$. (K.58)

Classify residual:

$\mathrm{ResidualType} \in \{\mathrm{Data}, \mathrm{Frame}, \mathrm{Protocol}, \mathrm{Dynamics}, \mathrm{NovelRegime}, \mathrm{Unknown}\}$. (K.59)

Residual response:

Type	Response
Data	repair measurement or provenance
Frame	revise transport map
Protocol	revise declaration or gate
Dynamics	revise generator
Novel regime	expand state space
Unknown	defer or escalate

Residual should not be automatically absorbed into random noise.

K.18 Revision step

Protocol update:

$P_{k+1} = \mathrm{Revise}(P_k, \mathrm{ResidualHistory}, \mathrm{PerformanceHistory})$. (K.60)

Admission rule:

$\mathrm{AdmitRevision} \iff \mathrm{PredictiveGain} \wedge \mathrm{Stability} \wedge \mathrm{Auditability}$. (K.61)

The revised protocol receives a new version.

Past results remain linked to the version under which they were produced.

K.19 Minimal pseudocode

```
initialize primary state X
initialize protocol P
initialize ledgers L_global and L_observer
initialize metric g and connection A
initialize observer policies π
```

```
for each calendar step t:

    rho = compile_secondary_world(X, P, L_global)

    rho = evolve_state(rho, g, A, P)

    for each observer O:

        frame = build_local_frame(X, P, O.ledger)
        setting = O.policy.select(rho, frame, O.ledger)

        provisional_outcome = measure(rho, setting, frame)
        provisional_state = condition(rho, provisional_outcome)

        decision = gate(
            provisional_outcome,
            provisional_state,
            frame,
            P,
            O.ledger
        )

        record = create_record(
            provisional_outcome,
            decision,
            setting,
            frame,
            residual
        )

        O.ledger = append(O.ledger, record)

        if record is globally consequential:
            L_global = append(L_global, record)

        O.policy = update_policy(O.policy, record)

        action = derive_action(record, O.policy)
        X = apply_backreaction(X, action)

    g = update_metric(X, L_global, P)
    A = update_connection(X, L_global, P)

    residual = compare_prediction_with_observation()

    if residual exceeds threshold:
        P = revise_or_escalate(P, residual)
```

K.20 Classical baseline implementation

The runtime must include a classical baseline.

Replace the density state with:

$p_t(s)$ = classical state distribution. (K.62)

Evolution:

p_{t+1} = TransitionMatrix·p_t. (K.63)

Measurement:

$p(y|a) = \Sigma_s p(y|a,s)p_t(s)$. (K.64)

Gate, ledger, and backreaction remain unchanged.

This baseline tests whether complex amplitudes or tensor structure add value beyond ordinary hidden-state modelling.

K.21 Coherent implementation

The coherent extension adds:

- complex amplitudes;
- off-diagonal state terms;
- multiple measurement bases;
- phase-sensitive observables;
- decoherence operators.

It should be enabled only when the classical baseline is already stable.

The development order is:

Classical Runtime
→ Composite Classical Runtime
→ Complex Phase Runtime
→ Coherent Runtime. (K.65)

Not the reverse.

K.22 Data outputs

Every episode should output:

- primary state snapshot;
- compiled financial state;
- local observer frame;
- selected setting;
- provisional outcome;
- gate decision;
- record;
- ledger hash;
- action;
- metric;
- connection;

- residual;
- protocol version.

This supports:

- audit;
- replay;
- counterfactual analysis;
- frame comparison;
- falsification.

K.23 Replay mode

Given saved records, the runtime should support:

Replay(P_version,Seed,DataSnapshot). (K.66)

A replay succeeds when:

Output_replay = Output_original within tolerance. (K.67)

If not, the system lacks reproducibility.

K.24 Counterfactual mode

The runtime should support replacing one decision:

Record_k → AlternativeRecord_k′. (K.68)

Then recompute:

World_k_{+1}′ = Runtime(World_k,Record_k′). (K.69)

This reveals:

- ledger backreaction;
- policy latching;
- geometric divergence;
- future gate differences.

Counterfactual replay is essential for measuring the causal role of commitment.

K.25 Runtime-kernel proposition

Proposition K.1 — Executable World Criterion

A proposed financial world architecture becomes technically meaningful only when each conceptual role maps to an executable operation with a logged input, output, residual, and revision path.

In compact form:

ExecutableTheory ⇔ State ∧ Compiler ∧ Evolution ∧ Instrument ∧ Gate ∧ Ledger ∧ Backreaction ∧ Residual ∧ Replay. (K.70)

Appendix L — Claim Language and Publication Discipline

L.1 Why claim language matters

The framework operates near several high-risk conceptual boundaries.

Terms such as:

- quantum;
- entanglement;
- collapse;
- relativity;
- curvature;
- gauge;
- field;
- spacetime;

can imply more than the evidence supports.

The publication language must therefore match the achieved evidence level.

L.2 Permitted low-risk claims

The following claims are generally permissible after mathematical construction:

- “The model uses a complex financial state.”
- “The option and underlying are represented as a composite effective system.”
- “The contract acts as a branch-binding operator in the toy representation.”
- “The model contains local observer restrictions.”
- “The framework admits a gauge-like transport connection.”
- “The estimated state exhibits path-dependent residual.”

- “The system has an internal gate–ledger time order.”

These describe the model.

They do not claim that markets are physically quantum.

L.3 Claims requiring empirical validation

The following require out-of-sample evidence:

- “Q predicts future repricing.”
- “θ functions as a stable internal clock.”
- “The financial metric improves transport.”
- “Gauge correction reduces cross-frame error.”
- “Curvature predicts local CAPM failure.”
- “Holonomy predicts loop residue.”
- “The joint-state model outperforms classical latent-state models.”
- “Relative phase improves forecast.”

L.4 Claims requiring strong classical-null rejection

The following require substantially stronger evidence:

- “The financial state is operationally coherent.”
- “The option–underlying state is nonseparable.”
- “Financial measurement is irreducibly contextual.”
- “Classical hidden-state models fail.”
- “The observed interference cannot be represented classically.”

Such claims must disclose:

- partition;
- measurement settings;
- basis;
- state estimator;
- null models;
- uncertainty;
- replication status.

L.5 Claims not presently supported

The article does not support:

- “CAPM is quantum mechanics.”
- “Options are physically entangled with their underlyings.”
- “Financial prices obey the Born rule.”
- “Markets violate Bell inequalities under valid assumptions.”
- “Financial gauge fields are physical gauge fields.”
- “Financial curvature obeys Einstein’s equation.”
- “Market collapse is wavefunction collapse.”
- “Financial internal time is physical proper time.”
- “The model proves a hidden classical substrate beneath quantum physics.”

These may be research questions.

They are not conclusions.

L.6 Recommended terminology ladder

Evidence level	Recommended term
Pure analogy	quantum-inspired
Formal role match	QM-like / SR-like / GR-like
Executable model	effective complex-state model
Predictive phase model	operational phase dynamics
Full-state improvement	composite-state model
Separable-null rejection	representation-relative nonseparability
Phase-sensitive null rejection	operational coherence
No-signalling and Bell tests passed	anomalous nonclassical-style correlation
Physical substrate established	physical quantum behaviour

The final row requires evidence far beyond the present article.

L.7 Required qualification for entanglement

Use:

“entanglement-like contractual structure.” (L.1)

Or:

“secondary-world operational nonseparability.” (L.2)

Or:

“effective nonfactorization under the declared partition.” (L.3)

Avoid the unqualified phrase:

“financial entanglement” (L.4)

unless the level is explicitly stated.

L.8 Required qualification for collapse

Use:

“collapse-like commitment.” (L.5)

Or:

“gate-mediated state commitment.” (L.6)

Or:

“ledgered branch fixation.” (L.7)

Avoid suggesting that institutional commitment is identical to physical quantum collapse.

L.9 Required qualification for relativity

Use:

“Lorentz-like valuation-frame mapping.” (L.8)

Or:

“additive discount rapidity.” (L.9)

Avoid:

“financial object moves at relativistic speed.” (L.10)

Or:

“CAPM beta is velocity.” (L.11)

L.10 Required qualification for gravity

Use:

“effective financial curvature.” (L.12)

Or:

“state-dependent financial geometry.” (L.13)

Or:

“GR-like source–geometry toy architecture.” (L.14)

Avoid claiming that leverage or liquidity is literally spacetime gravity.

L.11 Required qualification for gauge

Use:

“gauge-like transport of local valuation orientation.” (L.15)

The term is justified when:

- local conventions vary;
- an invariant survives;
- a connection improves transport;
- loop phase is measurable.

Without these:

“frame correction” (L.16)

may be the better term.

L.12 Abstract claim template

A defensible abstract should include:

1. the mature-finance foundation;
2. the mathematical extension;
3. the internal-observer perspective;
4. the operational results;
5. the limits;
6. the quantum residue.

A suitable structure is:

> We construct a non-quantum financial control world in which CAPM functions as a local valuation kernel, derivative contracts generate composite effective states, internal observers access restricted measurement algebras, and gates write outcomes into a backreactive ledger. The framework formally reproduces several operational features associated with quantum systems, including local mixedness, conditional state updates, phase-sensitive interference form, decoherence-like record formation, and geometric holonomy. These constructions do not derive Born probability, no-signalling, Bell nonclassicality, or physical quantum ontology.

L.13 Result-reporting template

For every reported result:

Claim

State exactly what was tested.

System boundary

Declare included and excluded sectors.

Protocol

State valuation and observation rules.

Representation

Define state space, basis, and partition.

Data

Provide source, period, and selection rule.

Null models

List mature alternatives.

Result

Report effect size and uncertainty.

Residual

Report unexplained structure.

Failure condition

State what would reject the claim.

Interpretation bound

State the strongest permitted conclusion.

L.14 Reproducibility footer

Each empirical paper should publish:

VerifyTrace = [Data][Code][Seed][Protocol][Boundary][Frames][Metric][Connection] [Partition][Settings][Gate][Ledger][Nulls][Residual][Decision][Version]. (L.17)

This footer is not optional for strong quantum-style claims.

L.15 Negative results

A negative result is scientifically valuable when it shows:

- Q adds no predictive value;
- phase reduces to covariance;
- the joint state is separable;
- holonomy reduces to transaction cost;
- gauge correction fails;
- no-signalling fails;
- Bell-style anomalies vanish under memory correction.

Such results identify where the quantum analogy stops.

The research programme should publish them.

L.16 The publication threshold

A paper should not make a stronger claim merely because the weaker claim appears less exciting.

The publication rule is:

Claim Strength ≤ Evidence Strength. (L.18)

And:

Vocabulary Strength ≤ Operational Specificity. (L.19)

L.17 Final publication proposition

Proposition L.1 — Interpretation-Bounded Reporting

A cross-domain result is trustworthy only when its language never exceeds:

1. the declared mathematical structure;
2. the completed empirical test;
3. the strongest surviving null comparison;
4. the stated evidence level.

In compact form:

TrustworthyClaim ⇔ StructureBound ∧ TestBound ∧ NullBound ∧ EvidenceBound. (L.20)

Appendix M — The Completed Grand-Unification Map

M.1 The architecture in one view

The final layered system is:

Primary Constructor Layer

$X(t) \in \Sigma_{primary}$. (M.1)

Contains:

- economic activity;
- contracts;
- market institutions;
- data transmission;
- balance sheets;
- legal structures;
- physical and computational mechanisms.

Compilation Layer

$\rho_F(\theta) = \mathcal{C}_{P,L}[X(t)]$. (M.2)

Declares:

- boundary;
- valuation rule;
- state basis;
- protocol;
- retained pressure;
- observer interface.

GR-Like Base Layer

$x \in \mathcal{M}_F$. (M.3)

Metric:

$ds_F^2 = g^F_{\mu\nu} dx^\mu dx^\nu$. (M.4)

Carries:

- liquidity;
- leverage;

- funding;
- collateral;
- volatility;
- stress;
- regime;
- ledger-conditioned geometry.

SR-Like Local Frame Layer

$g^{F}_{\mu\nu} = e^{a}_{\mu} e^{b}_{\nu} \eta_{ab}$. (M.5)

Local frame:

$F_a = (\text{Numeraire}, \text{Benchmark}, \text{Funding}, \text{Horizon}, \text{Protocol}, \text{Ledger})$. (M.6)

Carries:

- local CAPM law;
- relative discount rapidity;
- frame transformation;
- local simultaneity;
- invariant comparison.

QM-Like Fibre Layer

$\pi: \boldsymbol{\mathcal{H}}_F \to \boldsymbol{\mathcal{M}}_F$. (M.7)

Local state:

$|\Psi_F\rangle \in \mathscr{H}_x$. (M.8)

Carries:

- complex amplitude;
- relative phase;
- derivative composite states;
- local and joint observables;
- density states;
- conditional measurement.

Gauge Layer

$D_\mu = \partial_\mu + i\mathcal{A}_\mu$. (M.9)

Carries:

- local phase alignment;
- covariant transport;
- path dependence;
- loop phase;
- holonomy.

Observer Layer

$O = (\mathcal{A}_O, \mathbb{M}_O, F_O, L_O, \pi_O)$. (M.10)

Carries:

- bounded measurement;
- context selection;
- local state reduction;
- adaptive policy;
- internal certainty.

Gate–Ledger Layer

$L_{k+1} = L_k \sqcup \mathrm{Record}_k$. (M.11)

Carries:

- commitment;
- historical order;
- latching;
- finality;
- cross-observer records.

Backreaction Layer

$\mathrm{Record}_k \to \mathrm{Action}_k \to X_{k+1}$. (M.12)

Carries:

- hedge;
- margin;
- collateral;
- liquidation;

- settlement;
- regulatory consequence.

M.2 The master loop

The completed loop is:

Primary State
→ Declaration
→ Curved Base
→ Local Frame
→ Complex Fibre
→ Composite Contract State
→ Covariant Evolution
→ Internal Measurement
→ Conditional Update
→ Gate
→ Ledger
→ Action
→ Primary Backreaction
→ Geometry Revision
→ New State. (M.13)

In compact runtime form:

World_k₊₁ = Recompile[Backreact[Ledger[Gate[Measure[Evolve[Compile(World_k)]]]]]]. (M.14)

M.3 The central state equation

The effective-state equation is:

$i\hbar_F \mathcal{D}_\theta |\Psi_F\rangle = \hat{H}_F[g^F, \mathcal{A}, P, L]|\Psi_F\rangle + |\varepsilon_F\rangle$. (M.15)

Where:

$\mathcal{D}_\theta = \partial_\theta + \dot{x}^\mu \nabla_\mu + i\dot{x}^\mu \mathcal{A}_\mu$. (M.16)

And:

$\hat{H}_F = \hat{H}_{CAPM} + \hat{H}_{contract} + \hat{H}_{hedge} + \hat{H}_{ledger} + \hat{H}_{environment} + \hat{H}_{intervention}$. (M.17)

The geometry equation is:

$G^F_{\mu\nu} + \Lambda_F g^F_{\mu\nu} = \kappa_F T^F_{\mu\nu} + \Xi^F_{\mu\nu}$. (M.18)

The gate update is:

$\rho^+ = \mathbb{M}_y(\rho^-)/p(y)$. (M.19)

The ledger update is:

$L_{k+1} = L_k \sqcup Record_y$. (M.20)

The backreaction is:

$X_{k+1} = \mathscr{B}[X_k, \text{Record}_y]$. (M.21)

M.4 Reduction ladder

The architecture reduces as follows.

Full layered model

QM-like fibre

- SR-like local frame
- GR-like base
- gauge transport
- gate–ledger
- backreaction. (M.22)

Remove coherence

$\rho_{\text{off}} \to 0$. (M.23)

Result:

Classical composite derivative system. (M.24)

Remove tensor structure

$\dim \mathscr{H}_{\text{UD}} \to 1$ complex channel. (M.25)

Result:

Scalar Finance Geometry. (M.26)

Remove phase pressure

$Q \to 0$. (M.27)

Result:

Real-valued valuation state. (M.28)

Remove curvature and gauge

$g_F \to \eta$. (M.29)

$\mathcal{A} \to 0$. (M.30)

Result:

Flat local valuation world. (M.31)

Remove backreaction and ledger

$\hat{H}_{ledger} \to 0$. (M.32)

$\hat{H}_{hedge} \to 0$. (M.33)

Result:

Passive CAPM or option-pricing analytics. (M.34)

M.5 What each layer explains

Layer	Explains
CAPM local law	expected-return relation
R+iQ completion	admitted value and retained pressure
Multi-channel phase	relative orientation and interference form
Contract tensor structure	composite identity
Local observer algebra	bounded access
Gate	outcome commitment
Ledger	history and latching
SR-like frame	relative valuation description
GR-like metric	global state-dependent distance
Gauge connection	phase and basis transport
Holonomy	closed-loop residual
Backreaction	valuation-world causality
Residual governance	model incompleteness and revision

M.6 What no layer explains by itself

No single layer independently explains:

- Born probability;
- no-signalling;
- Bell nonclassicality;
- quantum contextuality;
- no-cloning;
- exchange statistics;
- quantum ontology.

These remain outside the completed toy architecture unless separately derived and tested.

M.7 Final unification proposition

Proposition M.1 — Role-Separated Financial Unification

A coherent QM–SR–GR-like financial architecture is possible without immediate mathematical contradiction when:

1. the complex fibre carries state;
2. the local frame carries relative description;
3. the curved base carries global geometry;
4. the gauge connection carries internal transport;
5. the gate and ledger carry commitment and time;
6. backreaction couples the secondary world to its primary constructor;
7. reduction limits recover mature finance.

In compact form:

UnifiedFinanceToy ⇔ FibreState ∧ LocalFrame ∧ CurvedBase ∧ GaugeTransport ∧ LedgerTime ∧ Backreaction ∧ MatureLimits. (M.35)

M.8 Final warning

This architecture is:

- a formal toy unification;
- a control-world construction;
- a research grammar;
- an engineering checklist.

It is not:

- a physical unification of quantum mechanics, special relativity, and general relativity;
- evidence that markets are quantum fields;
- proof that CAPM contains hidden quantum gravity;
- proof that financial derivatives are physically entangled systems.

The final boundary statement is:

Structural Coexistence ≠ Physical Identity. (M.36)

And the final research instruction is:

Preserve the Layers.
Test the Couplings.
Recover the Limits.
Carry the Residual. (M.37)

Appendix N — A Worked End-to-End Numerical Example

N.1 Purpose

This appendix follows one fictional asset and one European call option through the complete layered architecture.

It demonstrates how the same case can be represented through:

- local CAPM;
- scalar complex completion;
- circular and hyperbolic coordinates;
- relative valuation frames;
- derivative composite state;
- internal measurement;
- gate and ledger;
- hedge backreaction;
- revised geometry.

The numbers are illustrative.

They are not calibrated market estimates and are not investment advice.

N.2 Primary economic state

Consider one underlying asset U.

Its current primary-state variables are:

Spot price = 100. (N.1)

Risk-free rate = 5%. (N.2)

Expected market return = 11%. (N.3)

CAPM beta = 1.20. (N.4)

Market risk premium is:

ERP = 11% − 5% = 6%. (N.5)

The CAPM expected return is:

E[r_U] = 5% + 1.20×6%. (N.6)

Therefore:

$E[r_U] = 12.2\%$. (N.7)

This is the local expected-return law under the declared market frame.

It is not yet the complete financial-world state.

N.3 Declared amplitude and admitted value

Assume the protocol declares:

$A = 120$. (N.8)

$R_M = 100$. (N.9)

Where:

- A is the independently estimated pre-filter economic scale;
- R_M is the value admitted in the market frame M.

The Euclidean retained-pressure coordinate is:

$Q_{E,M} = \sqrt{(A^2 - R_M^2)}$. (N.10)

Therefore:

$Q_{E,M} = \sqrt{(120^2 - 100^2)}$. (N.11)

$Q_{E,M} = \sqrt{4400}$. (N.12)

$Q_{E,M} \approx 66.3325$. (N.13)

The local complex state is:

$Z_M = 100 + i66.3325$. (N.14)

Its modulus is:

$|Z_M| = 120$. (N.15)

N.4 Circular valuation phase

The market-frame angle is:

$\theta_M = \operatorname{atan2}(Q_{E,M}, R_M)$. (N.16)

Therefore:

$\theta_M \approx \operatorname{atan2}(66.3325, 100)$. (N.17)

$\theta_M \approx 0.5857$ radians. (N.18)

Equivalently:

$\theta_M \approx 33.56°$. (N.19)

The circular relations are:

$\cos \theta_M = 100/120 = 0.8333$. (N.20)

$\sin \theta_M = 66.3325/120 \approx 0.5528$. (N.21)

Thus:

$Z_M = 120\exp(i0.5857)$. (N.22)

The number 0.5857 is a valuation-orientation coordinate.

It is not calendar time.

N.5 Q-preserving hyperbolic embedding

Define:

$\eta_{Q,M} = \operatorname{artanh}(Q_{E,M}/A)$. (N.23)

Then:

$\eta_{Q,M} = \operatorname{artanh}(0.5528)$. (N.24)

$\eta_{Q,M} \approx 0.6224$. (N.25)

Check:

$A = R_M \cosh \eta_{Q,M}$. (N.26)

$120 \approx 100\cosh(0.6224)$. (N.27)

And:

$Q_{E,M} = R_M \sinh \eta_{Q,M}$. (N.28)

$66.3325 \approx 100\sinh(0.6224)$. (N.29)

The invariant is:

$A^2 - Q_{E,M}^2 = R_M^2$. (N.30)

Therefore:

$14400 - 4400 = 10000$. (N.31)

N.6 Discount-composition rapidity

The discount rapidity is:

$\rho_{D,M} = \ln(A/R_M)$. (N.32)

Therefore:

$\rho_{D,M} = \ln(120/100)$. (N.33)

$\rho_{D,M} = \ln 1.2$. (N.34)

$\rho_{D,M} \approx 0.1823$. (N.35)

The bounded frame coordinate is:

$u_{D,M} = \tanh \rho_{D,M}$. (N.36)

Thus:

$u_{D,M} \approx \tanh(0.1823)$. (N.37)

$u_{D,M} \approx 0.1803$. (N.38)

The corresponding light-cone pressure is:

$Q_{H,M} = Q_{E,M}^2/(2R_M)$. (N.39)

Therefore:

$Q_{H,M} = 4400/200$. (N.40)

$Q_{H,M} = 22$. (N.41)

This example makes the distinctions explicit:

$\theta_M \approx 0.5857$. (N.42)

$\eta_{Q,M} \approx 0.6224$. (N.43)

$\rho_{D,M} \approx 0.1823$. (N.44)

$Q_{E,M} \approx 66.3325$. (N.45)

$Q_{H,M} = 22$. (N.46)

None of these quantities is interchangeable.

N.7 Three valuation frames

Now introduce three local valuation frames.

Market frame M

$R_M = 100$. (N.47)

Collateral frame C

$R_C = 94$. (N.48)

Liquidation frame L

$R_L = 82$. (N.49)

The relative market-to-collateral rapidity is:

$\rho_{CM} = \ln(R_M/R_C)$. (N.50)

Therefore:

$\rho_{CM} = \ln(100/94)$. (N.51)

$\rho_{CM} \approx 0.06188$. (N.52)

The collateral-to-liquidation rapidity is:

$\rho_{LC} = \ln(R_C/R_L)$. (N.53)

Therefore:

$\rho_{LC} = \ln(94/82)$. (N.54)

$\rho_{LC} \approx 0.13658$. (N.55)

The direct market-to-liquidation rapidity is:

$\rho_{LM} = \ln(R_M/R_L)$. (N.56)

Therefore:

$\rho_{LM} = \ln(100/82)$. (N.57)

$\rho_{LM} \approx 0.19845$. (N.58)

The additive composition check is:

$\rho_{LM} = \rho_{LC} + \rho_{CM}$. (N.59)

Numerically:

$0.19845 \approx 0.13658 + 0.06188$. (N.60)

The small rounding residual is:

$\varepsilon_{frame} \approx -0.00001$. (N.61)

This confirms exact multiplicative frame composition for the declared values.

N.8 Why the liquidation frame may not be a pure frame change

Although the rapidities compose numerically, liquidation may activate new mechanisms:

- market impact;
- forced execution;
- legal priority;
- collateral seizure;
- altered timing;

- irreversible ledger consequences.

Therefore the numerical transformation:

$R_M \to R_L$ (N.62)

does not prove that M and L are merely equivalent coordinates.

The model must test whether the governed object and admissible dynamics are preserved.

If liquidation changes the world's causal rules, then:

$M \to L$ = Protocol Transition. (N.63)

Not merely:

$M \to L$ = Reversible Frame Boost. (N.64)

The rapidity remains a useful discrepancy coordinate.

Its interpretation depends on the protocol.

N.9 The derivative contract

Introduce a one-year European call.

Strike:

$K = 100$. (N.65)

Maturity:

$T = 1$ year. (N.66)

Consider two coarse terminal underlying scenarios.

Down branch:

$u_0 = 80$. (N.67)

Up branch:

$u_1 = 130$. (N.68)

The call payoffs are:

$d_0 = \max(80-100,0) = 0$. (N.69)

$d_1 = \max(130-100,0) = 30$. (N.70)

Assume risk-neutral branch weights:

$q_0 = 0.60$. (N.71)

$q_1 = 0.40$. (N.72)

The discounted call value is:

$D_0 = \exp(-0.05)[0.60\times 0 + 0.40\times 30]$. (N.73)

Therefore:

$D_0 = \exp(-0.05)\times 12$. (N.74)

$D_0 \approx 11.4148$. (N.75)

This is standard classical risk-neutral valuation in the two-state toy market.

N.10 The classical joint distribution

The classical contract state is:

$\rho_{mix} = 0.60|80,0\rangle\langle 80,0| + 0.40|130,30\rangle\langle 130,30|$. (N.76)

The option and underlying are perfectly paired within each scenario.

But the state is separable because it is a convex mixture of product branches.

Therefore:

Strong Contract Correlation ≠ Quantum Entanglement. (N.77)

The classical state already reproduces:

- payoff dependence;
- conditional certainty;
- local uncertainty;
- joint scenario prediction.

N.11 Formal amplitude lift

Define branch amplitudes:

$\alpha_0 = \sqrt{0.60}$. (N.78)

$\alpha_1 = \sqrt{0.40}\exp(i\varphi)$. (N.79)

Choose illustrative relative phase:

$\varphi = 0.60$ radians. (N.80)

The formal underlying state is:

$|\psi_U\rangle = \sqrt{0.60}|80\rangle + \sqrt{0.40}\exp(i0.60)|130\rangle$. (N.81)

The contract-preparation operator acts as:

$\hat{U}_C|80,0_D\rangle = |80,0\rangle$. (N.82)

$\hat{U}_C|130,0_D\rangle = |130,30\rangle$. (N.83)

The prepared joint state is:

$|\Psi_{UD}\rangle = \sqrt{0.60}|80,0\rangle + \sqrt{0.40}\exp(i0.60)|130,30\rangle$. (N.84)

This state is formally nonfactorizable.

But its coherence is empirically meaningful only if some admissible measurement detects the relative phase.

N.12 Local reduced states

Assume the branch states are orthogonal.

The reduced underlying state is:

$\rho_U = 0.60|80\rangle\langle 80| + 0.40|130\rangle\langle 130|$. (N.85)

The reduced derivative state is:

$\rho_D = 0.60|0\rangle\langle 0| + 0.40|30\rangle\langle 30|$. (N.86)

The local purity is:

$\mathrm{Tr}(\rho_U^2) = 0.60^2 + 0.40^2$. (N.87)

Therefore:

$\mathrm{Tr}(\rho_U^2) = 0.52$. (N.88)

Similarly:

$\mathrm{Tr}(\rho_D^2) = 0.52$. (N.89)

The global pure-state representation has purity:

$\mathrm{Tr}(\rho_{UD}^2) = 1$. (N.90)

Thus the formal architecture produces:

Global Purity + Local Mixedness. (N.91)

But the classical mixture produces the same local reduced states.

Local mixedness alone cannot distinguish the two models.

N.13 A phase-sensitive joint measurement

Define correlated branch basis:

$|\bar{0}\rangle = |80,0\rangle$. (N.92)

$|\bar{1}\rangle = |130,30\rangle$. (N.93)

Define rotated state:

$|\bar{+}\rangle = (|\bar{0}\rangle + |\bar{1}\rangle)/\sqrt{2}$. (N.94)

The coherent model predicts:

p_coh(+) = ½[1 + 2√(0.60×0.40)cos0.60]. (N.95)

Since:

2√0.24 ≈ 0.9798. (N.96)

And:

cos0.60 ≈ 0.8253, (N.97)

we obtain:

p_coh(+) ≈ ½[1 + 0.8087]. (N.98)

Therefore:

p_coh(+) ≈ 0.9044. (N.99)

The classical mixture predicts:

p_mix(+) = 0.50. (N.100)

The models differ sharply in this rotated joint basis.

However, ordinary option pricing does not implement such a measurement.

The experimental problem is therefore not mathematical.

It is operational:

What real financial instrument combines the two branches before commitment in a manner sensitive to their relative phase? (N.101)

Until such an instrument exists:

The amplitude phase remains a formal extension. (N.102)

N.14 Internal measurement and conditional update

Suppose the terminal underlying measurement records:

$u_1 = 130$. (N.103)

The conditional derivative state becomes:

$\rho_{D|130} = |30\rangle\langle 30|$. (N.104)

The observer now assigns:

P(D_T = 30 | U_T = 130) = 1. (N.105)

From the primary universe, this is explained by the call payoff.

From inside the secondary contract world:

One Global State

→ One Underlying Outcome
→ One Conditionally Definite Derivative Outcome. (N.106)

No separate messenger is required inside the conditional-state formula.

But settlement information must still reach the relevant institutions before they can act on it.

Thus:

Conditional Fixing ≠ Instantaneous Communicable Knowledge. (N.107)

N.15 The maturity gate

The exercise condition is:

Exercise = 1 iff $U_T > 100$. (N.108)

Since:

$U_T = 130$, (N.109)

the gate returns:

Exercise = 1. (N.110)

The committed payoff is:

$D_T = 30$. (N.111)

The event record is:

Record_1 = (ContractID,Underlying=130,Exercise=1,Payoff=30,Frame=M,Evidence,Residual). (N.112)

The ledger updates:

$L_2 = L_1 \sqcup \text{Record}_1$. (N.113)

Before the gate, the derivative was a contingent possibility.

After the gate and record, it becomes a settled obligation.

N.16 Same later spot, different ledger

Suppose the underlying later falls back to:

$U_{later} = 100$. (N.114)

This equals the original spot.

But the world is not restored.

The exercised payoff remains recorded:

$\text{Record}_1 \in L_{later}$. (N.115)

Therefore:

U_later = U_initial. (N.116)

But:

World_later ≠ World_initial. (N.117)

The difference includes:

- settlement;
- realized profit and loss;
- tax consequences;
- hedge closure;
- changed cash ownership;
- contract expiry.

This is ledger holonomy in its simplest form.

N.17 Hedge state before maturity

Assume that before maturity the call has:

Delta = 0.55. (N.118)

Gamma = 0.018 per currency unit. (N.119)

A dealer is short:

N_D = −1000 calls. (N.120)

The delta hedge is:

H = −N_DDelta. (N.121)

Therefore:

H = −(−1000)(0.55). (N.122)

H = 550 underlying units. (N.123)

The dealer holds 550 units of the underlying to hedge the short call position.

N.18 Gamma-induced hedge adjustment

Suppose the underlying rises by:

ΔU = 5. (N.124)

The delta change is approximately:

ΔDelta ≈ Gamma×ΔU. (N.125)

Therefore:

ΔDelta ≈ 0.018×5. (N.126)

ΔDelta ≈ 0.09. (N.127)

The hedge adjustment is:

ΔH = −N_DΔDelta. (N.128)

Thus:

ΔH = −(−1000)(0.09). (N.129)

ΔH = 90. (N.130)

The dealer must buy approximately 90 additional underlying units.

The derivative has now backreacted on the underlying market through hedging.

N.19 Price impact

Assume a simple impact coefficient:

λ_I = 0.002 price units per underlying unit purchased. (N.131)

The hedge-induced impact is:

ΔU_impact = λ_IΔH. (N.132)

Therefore:

ΔU_impact = 0.002×90. (N.133)

ΔU_impact = 0.18. (N.134)

The initial exogenous move was:

5.00. (N.135)

The derivative hedge adds:

0.18. (N.136)

The total simplified move becomes:

5.18. (N.137)

This is a positive feedback loop:

Underlying Rise
→ Delta Increase
→ Hedge Buying
→ Additional Underlying Rise. (N.138)

No quantum mechanism is required.

The loop arises from mature derivative hedging and market impact.

N.20 Loop-gain estimate

Define simplified loop gain:

$\mathcal{G} = \lambda_I(-N_D)Gamma.$ (N.139)

Substitute:

$\mathcal{G} = 0.002 \times 1000 \times 0.018.$ (N.140)

Therefore:

$\mathcal{G} = 0.036.$ (N.141)

Since:

$0 < \mathcal{G} < 1,$ (N.142)

the local feedback is amplifying but not self-explosive in this toy approximation.

A much larger position, gamma, or impact coefficient could move the system toward:

$\mathcal{G} \geq 1.$ (N.143)

At that point the local linear model would become unstable or inadequate.

N.21 Margin gate

Suppose the dealer's post-move derivative exposure is:

Exposure = 1,250,000. (N.144)

Posted collateral is:

Collateral = 1,100,000. (N.145)

The unsecured exposure is:

X_margin = Exposure − Collateral. (N.146)

Thus:

X_margin = 150,000. (N.147)

Suppose the contractual threshold is:

M* = 100,000. (N.148)

The margin condition is:

MarginCall = 1 iff X_margin > M*. (N.149)

Since:

150,000 > 100,000, (N.150)

the gate commits:

MarginCall = 1. (N.151)

The ledger records:

Record$_2$ = MarginCall for 50,000 excess over threshold. (N.152)

N.22 Margin backreaction

Assume the dealer must raise 50,000 by selling a liquid asset.

The secondary valuation record therefore causes:

Margin Record
→ Funding Action
→ Asset Sale
→ Market Impact. (N.153)

Let the sale reduce liquidity metric λ_L:

$\lambda_{L,before} = 1.00$. (N.154)

$\lambda_{L,after} = 0.92$. (N.155)

The financial metric changes:

$g^F_{before} \neq g^F_{after}$. (N.156)

Later movements through the same visible price coordinates may now be more expensive because market depth has deteriorated.

This is a minimal state-to-geometry backreaction.

N.23 Metric update

Suppose the one-dimensional liquidity-cost metric is:

$ds_F^2 = g_{LL}dL^2$. (N.157)

Before the margin event:

$g_{LL,before} = 1/\lambda_{L,before}^2$. (N.158)

Thus:

$g_{LL,before} = 1$. (N.159)

After the event:

$g_{LL,after} = 1/0.92^2$. (N.160)

Therefore:

$g_{LL,after} \approx 1.1815$. (N.161)

The same nominal liquidity displacement dL now has larger financial distance:

$ds_{after}^2 \approx 1.1815 dL^2$. (N.162)

The committed record has altered the effective geometry.

N.24 Phase-velocity update

Suppose financial phase velocity depends on liquidity stress:

$\omega_F = \omega_0 + \kappa_\lambda \lambda_{stress}$. (N.163)

Let:

$\omega_0 = 0.10$. (N.164)

$\kappa_\lambda = 0.50$. (N.165)

Define:

$\lambda_{stress} = 1 - \lambda_L$. (N.166)

Before the event:

$\lambda_{stress,before} = 0$. (N.167)

Therefore:

$\omega_{F,before} = 0.10$. (N.168)

After the event:

$\lambda_{stress,after} = 1 - 0.92 = 0.08$. (N.169)

Thus:

$\omega_{F,after} = 0.10 + 0.50 \times 0.08$. (N.170)

$\omega_{F,after} = 0.14$. (N.171)

The margin event has accelerated the local valuation rotation.

This is a toy model of:

Ledger Event
→ Geometry Change
→ Phase-Velocity Change. (N.172)

N.25 Revised admitted-value dynamics

Using:

$dR/dt = g_{AR} - Q_E \omega_F + \mathrm{Re}(\varepsilon_{dyn})$, (N.173)

assume temporarily:

$g_A = 0.02$. (N.174)

$R = 100$. (N.175)

$Q_E = 66.3325$. (N.176)

Before the margin event:

$\omega_F = 0.10$. (N.177)

Then:

$dR/dt_{before} = 0.02 \times 100 - 66.3325 \times 0.10 + Re(\varepsilon_{dyn})$. (N.178)

Therefore:

$dR/dt_{before} = 2 - 6.6333 + Re(\varepsilon_{dyn})$. (N.179)

$dR/dt_{before} = -4.6333 + Re(\varepsilon_{dyn})$. (N.180)

After the margin event:

$\omega_F = 0.14$. (N.181)

Then:

$dR/dt_{after} = 2 - 66.3325 \times 0.14 + Re(\varepsilon_{dyn})$. (N.182)

Therefore:

$dR/dt_{after} = 2 - 9.2866 + Re(\varepsilon_{dyn})$. (N.183)

$dR/dt_{after} = -7.2866 + Re(\varepsilon_{dyn})$. (N.184)

The primary amplitude-growth term has not changed.

The stronger decline comes from faster valuation-frame rotation induced by the ledgered liquidity event.

This illustrates the intended radial–angular decomposition.

N.26 Full runtime sequence

The example can now be summarized as:

Step 1 — Local valuation

CAPM estimates expected return:

12.2%. (N.185)

Step 2 — Complex completion

$Z_M = 100 + i66.3325$. (N.186)

Step 3 — Internal phase

$\theta_M = 0.5857$. (N.187)

Step 4 — Relative frames

Market, collateral, and liquidation values are linked by additive discount rapidities.

Step 5 — Contract preparation

$|\Psi_{UD}\rangle = \sqrt{0.60}|80,0\rangle + \sqrt{0.40}\exp(i0.60)|130,30\rangle$. (N.188)

Step 6 — Local measurement

Underlying outcome 130 conditionally fixes payoff 30.

Step 7 — Exercise gate

The payoff is committed.

Step 8 — Ledger

The exercise and settlement record becomes persistent.

Step 9 — Hedge backreaction

Gamma causes 90 additional underlying units to be purchased.

Step 10 — Margin gate

A 50,000 collateral shortfall is committed.

Step 11 — Primary intervention

Assets are sold to fund collateral.

Step 12 — Geometry update

Liquidity metric rises from 1.0000 to approximately 1.1815.

Step 13 — Phase update

ω_F rises from 0.10 to 0.14.

Step 14 — New valuation dynamics

The angular repricing contribution becomes more negative.

The secondary world has altered the primary conditions from which its next state is compiled.

N.27 What is mature in this example?

The following elements are mature or conventional:

- CAPM expected-return calculation;
- risk-neutral option valuation;
- derivative payoff;
- delta;
- gamma;
- hedge adjustment;
- margin threshold;
- collateral action;
- market impact;
- settlement record.

N.28 What is a coordinate extension?

The following are mathematical extensions:

- A as declared pre-filter amplitude;
- Q_E as retained-pressure coordinate;
- Z = R+iQ_E;
- θ as valuation orientation;
- η_Q as Q-preserving rapidity;
- ρ_D as discount rapidity;
- Q_H as light-cone pressure coordinate.

N.29 What is a toy-theory hypothesis?

The following are modelling hypotheses:

- the coherent option–underlying amplitude state;
- the phase value φ = 0.60;
- the rotated joint financial measurement;
- the liquidity-derived metric;
- the phase-velocity response to liquidity stress;
- the financial state equation;

- the state-to-geometry coupling.

N.30 What is not demonstrated?

The example does not demonstrate:

- physical quantum coherence;
- Born-rule derivation;
- no-signalling;
- Bell inequality violation;
- irreducible quantum contextuality;
- financial no-cloning;
- physical spacetime curvature;
- physical gauge fields.

It demonstrates only that the layered architecture can be instantiated without forcing its mathematical roles into direct contradiction.

N.31 Reduction checks

Classical option limit

Set:

φ unobservable. (N.189)

ρ_off → 0. (N.190)

Then the model reduces to:

ρ_mix. (N.191)

And standard risk-neutral price:

D0 ≈ 11.4148. (N.192)

Scalar Finance Geometry limit

Remove derivative tensor structure:

$\mathscr{H}$_UD → one channel. (N.193)

Then:

Z = 100 + i66.3325. (N.194)

Real valuation limit

Set:

$Q_E \to 0$. (N.195)

Then:

$Z \to R = 100$. (N.196)

Flat geometry limit

Set:

$g_{LL} \to 1$. (N.197)

$\mathcal{A}_\mu \to 0$. (N.198)

Then transport becomes path-independent.

Passive valuation limit

Set:

Gamma = 0. (N.199)

Margin backreaction = 0. (N.200)

Ledger effect = 0. (N.201)

Then the derivative becomes a passive claim on the underlying.

All major lower-level models are recoverable.

N.32 Numerical architecture table

Quantity	Value	Role
A	120	Pre-filter amplitude
R_M	100	Market-admitted value
$Q_{E,M}$	66.3325	Euclidean retained pressure
θ_M	0.5857	Circular valuation phase
$\eta_{Q,M}$	0.6224	Q-preserving rapidity
$\rho_{D,M}$	0.1823	Discount rapidity
$Q_{H,M}$	22	Hyperbolic pressure coordinate
R_C	94	Collateral-frame value
R_L	82	Liquidation-frame value
D_0	11.4148	Toy risk-neutral call price
Delta	0.55	First-order coupling
Gamma	0.018	Coupling curvature
Hedge before move	550	Underlying units
Hedge adjustment	90	Backreaction order
Margin shortfall	50,000	Gate-triggering deficit
g_{LL} before	1.0000	Liquidity metric
g_{LL} after	1.1815	Revised geometry

Quantity	Value	Role
ω_F before	0.10	Initial phase velocity
ω_F after	0.14	Post-ledger phase velocity

N.33 Worked-example proposition

Proposition N.1 — End-to-End Layer Compatibility

One financial case can be represented simultaneously through:

1. a local mature-finance valuation law;
2. a complex pressure-preserving completion;
3. additive relative-frame coordinates;
4. a derivative composite-state representation;
5. an internal measurement-and-gate process;
6. ledgered historical commitment;
7. derivative and margin backreaction;
8. a state-dependent geometry update;

without identifying any one layer with another.

In compact form:

WorkedArchitecture = CAPM_Local + ComplexState + RelativeFrames + ContractTensor + InternalObserver + GateLedger + Backreaction + GeometryRevision. (N.202)

The example therefore illustrates the article's central design principle:

Mathematical Unification Requires Coupled Roles, Not Collapsed Meanings. (N.203)

Appendix O — A Minimal CHSH-Style Financial Sandbox

O.1 Purpose

This appendix designs a controlled secondary-world experiment for testing joint financial correlations.

Its immediate purpose is not to prove that finance is quantum.

It is to determine whether a declared option–underlying model can distinguish among:

1. ordinary correlation;
2. signalling interaction;
3. adaptive contextuality;

4. representation-relative nonseparability;
5. stronger Bell-style anomaly.

The central discipline is:

Compute the Inequality Only after Auditing Its Assumptions. (O.1)

A large CHSH-like statistic is not informative when:

- settings transmit information;
- trials retain memory;
- samples are selected after outcomes;
- settings depend on market state;
- the two sectors are not operationally local.

O.2 Why a sandbox is preferable to immediate market testing

Live markets contain:

- continuous communication;
- common news;
- strategic reaction;
- market impact;
- changing liquidity;
- adaptive measurement;
- nonstationary state;
- incomplete trial enumeration.

These features violate the clean assumptions needed for a Bell-style interpretation.

A sandbox permits direct control of:

- state preparation;
- setting selection;
- local instruments;
- signalling channels;
- memory;
- outcome disclosure;

- gate timing;
- ledger access.

Thus:

Sandbox First
→ Assumption Recovery
→ Market Approximation
→ Real-World Stress Test. (O.2)

O.3 Two-sector state space

Let the underlying sector be two-dimensional:

$\mathcal{H}_U = \text{span}\{|0_U\rangle, |1_U\rangle\}$. (O.3)

Let the derivative sector be:

$\mathcal{H}_D = \text{span}\{|0_D\rangle, |1_D\rangle\}$. (O.4)

The composite space is:

$\mathcal{H}_{UD} = \mathcal{H}_U \otimes \mathcal{H}_D$. (O.5)

Interpret the basis states as:

$|0_U\rangle$ = lower underlying regime. (O.6)

$|1_U\rangle$ = higher underlying regime. (O.7)

$|0_D\rangle$ = lower derivative payoff or risk regime. (O.8)

$|1_D\rangle$ = higher derivative payoff or risk regime. (O.9)

O.4 Prepared joint state

Use the parameterized state:

$|\Psi(\alpha,\varphi)\rangle = \cos\alpha|0_U,0_D\rangle + \exp(i\varphi)\sin\alpha|1_U,1_D\rangle$. (O.10)

Where:

$0 \le \alpha \le \pi/2$. (O.11)

For:

$\alpha = 0$ or $\alpha = \pi/2$, (O.12)

the state is a product branch.

For:

$0 < \alpha < \pi/2$, (O.13)

the pure-state representation is nonfactorizable.

At:

$\alpha = \pi/4$, (O.14)

the branch magnitudes are balanced.

O.5 Classical separable control state

Construct a classical mixture with identical branch weights:

$\rho_sep = \cos^2\alpha|0,0\rangle\langle 0,0| + \sin^2\alpha|1,1\rangle\langle 1,1|$. (O.15)

The coherent state is:

$\rho_coh = |\Psi(\alpha,\varphi)\rangle\langle\Psi(\alpha,\varphi)|$. (O.16)

Both models produce the same outcome frequencies when measured only in the branch basis.

They differ only under measurements sensitive to the cross terms.

This makes ρ_sep the first essential control.

O.6 Local measurement settings

Observer U selects one of two settings:

$a \in \{a_0,a_1\}$. (O.17)

Observer D selects:

$b \in \{b_0,b_1\}$. (O.18)

Represent a binary local observable by angle ξ:

$\hat{O}_U(\xi) = \cos\xi \cdot Z_U + \sin\xi \cdot X_U$. (O.19)

Similarly:

$\hat{O}_D(\zeta) = \cos\zeta \cdot Z_D + \sin\zeta \cdot X_D$. (O.20)

Where the branch-basis operators are:

$Z|0\rangle = |0\rangle$. (O.21)

$Z|1\rangle = -|1\rangle$. (O.22)

And the basis-mixing operators are:

$X|0\rangle = |1\rangle$. (O.23)

$X|1\rangle = |0\rangle$. (O.24)

The measurement outcomes are:

$u \in \{-1,+1\}$. (O.25)

$d \in \{-1,+1\}$. (O.26)

O.7 Financial interpretation of the settings

Possible underlying settings are:

a_0 — Direction basis

Classifies the underlying as:

- lower regime;
- higher regime.

a_1 — Mixed price–volatility basis

Classifies a rotated combination of:

- directional state;
- volatility or liquidity orientation.

Possible derivative settings are:

b_0 — Payoff basis

Classifies:

- lower payoff or exposure;
- higher payoff or exposure.

b_1 — Mixed premium–Greek basis

Classifies a rotated combination of:

- premium state;
- delta–vega or replication state.

The rotations must correspond to implementable data transformations or instruments.

Otherwise the settings remain formal.

O.8 Correlation function

For state ρ:

$E(\xi,\zeta) = \mathrm{Tr}[\rho \cdot \hat{O}_U(\xi) \otimes \hat{O}_D(\zeta)]$. (O.27)

For the balanced coherent state:

$|\Psi\rangle = [|0,0\rangle + \exp(i\varphi)|1,1\rangle]/\sqrt{2}$, (O.28)

the correlation is:

$$E_{coh}(\xi,\zeta) = \cos\xi\cos\zeta + \cos\varphi\cdot\sin\xi\sin\zeta. \quad (O.29)$$

For:

$$\varphi = 0, \quad (O.30)$$

this becomes:

$$E_{coh}(\xi,\zeta) = \cos(\xi-\zeta). \quad (O.31)$$

For the corresponding separable mixture:

$$E_{sep}(\xi,\zeta) = \cos\xi\cos\zeta. \quad (O.32)$$

The difference is:

$$\Delta E(\xi,\zeta) = \cos\varphi\cdot\sin\xi\sin\zeta. \quad (O.33)$$

This is the phase-sensitive joint term.

O.9 CHSH statistic

Choose settings:

$$a_0 = \xi_0. \quad (O.34)$$

$$a_1 = \xi_1. \quad (O.35)$$

$$b_0 = \zeta_0. \quad (O.36)$$

$$b_1 = \zeta_1. \quad (O.37)$$

Define:

$$S_F = E(\xi_0,\zeta_0) + E(\xi_0,\zeta_1) + E(\xi_1,\zeta_0) - E(\xi_1,\zeta_1). \quad (O.38)$$

A standard local hidden-variable model under its required assumptions satisfies:

$$|S_F| \le 2. \quad (O.39)$$

The coherent mathematical model can reach:

$$|S_F| = 2\sqrt{2} \quad (O.40)$$

for suitable settings and phase.

But the financial interpretation depends entirely on whether the measurement and causal assumptions are satisfied.

O.10 Illustrative settings

Use:

$$\xi_0 = 0. \quad (O.41)$$

$\xi_1 = \pi/2$. (O.42)

$\zeta_0 = \pi/4$. (O.43)

$\zeta_1 = -\pi/4$. (O.44)

For:

$\varphi = 0$, (O.45)

we obtain:

$E(\xi_0,\zeta_0) = 1/\sqrt{2}$. (O.46)

$E(\xi_0,\zeta_1) = 1/\sqrt{2}$. (O.47)

$E(\xi_1,\zeta_0) = 1/\sqrt{2}$. (O.48)

$E(\xi_1,\zeta_1) = -1/\sqrt{2}$. (O.49)

Therefore:

$S_F = 2\sqrt{2}$. (O.50)

For the separable mixture:

$E_{sep}(\xi,\zeta) = \cos\xi \cos\zeta$. (O.51)

Thus:

$E_{sep}(0,\pi/4) = 1/\sqrt{2}$. (O.52)

$E_{sep}(0,-\pi/4) = 1/\sqrt{2}$. (O.53)

$E_{sep}(\pi/2,\pi/4) = 0$. (O.54)

$E_{sep}(\pi/2,-\pi/4) = 0$. (O.55)

Therefore:

$S_{sep} = \sqrt{2}$. (O.56)

The formal coherent and separable models are distinguishable.

The real challenge is constructing legitimate financial instruments corresponding to the rotated settings.

O.11 Trial preparation

Each trial k should begin with:

$\rho_k^0 = \text{Prepare}(P_{seed},\lambda_k)$. (O.57)

Where:

- P_{seed} is the declared preparation protocol;
- λ_k is the controlled hidden primary state.

Preparation must finish before settings are selected.

The sequence is:

Prepare
→ Freeze Preparation
→ Select Settings
→ Measure Locally
→ Seal Outcomes
→ Share Records. (O.58)

This ordering reduces ordinary information leakage.

O.12 Independent setting generation

Settings should be generated by independent random sources:

$a_k \sim Random_A$. (O.59)

$b_k \sim Random_B$. (O.60)

Require:

$I[(a_k,b_k);\lambda_k] \leq \varepsilon_{MI}$. (O.61)

Where I is mutual information.

If settings depend materially on λ_k, measurement independence fails.

For example:

a_k = choose stress basis when volatility is high (O.62)

is adaptive and therefore unsuitable for a clean Bell-style test.

It may still be suitable for an observer-policy experiment.

O.13 Local outcome generation

The local outcome probabilities are:

$p(u,d|a,b) = Tr[\rho(E^U_{a,u} \otimes E^D_{b,d})]$. (O.63)

The local marginal is:

$p(u|a,b) = \Sigma_d p(u,d|a,b)$. (O.64)

And:

$p(d|a,b) = \Sigma_u p(u,d|a,b)$. (O.65)

The outcome-generation process must not use the remote setting except through the declared global state relation.

O.14 Outcome sealing

Observer A records:

$R_{A,k} = \mathrm{Hash}(a_k,u_k,t_{A,k})$. (O.66)

Observer B records:

$R_{B,k} = \mathrm{Hash}(b_k,d_k,t_{B,k})$. (O.67)

The records are sealed before sharing:

$\mathrm{ShareTime} > \max(t_A,t_B)$. (O.68)

This prevents one outcome from being modified after learning the other.

A later reconciliation record contains:

$R_{AB,k} = \mathrm{Join}(R_{A,k},R_{B,k},\mathrm{PreparationID}_k)$. (O.69)

O.15 No-signalling audit

Estimate:

$\delta_{D\to U} = \max_{u,a,b_0,b_1}|p(u|a,b_0)-p(u|a,b_1)|$. (O.70)

Estimate:

$\delta_{U\to D} = \max_{d,b,a_0,a_1}|p(d|a_0,b)-p(d|a_1,b)|$. (O.71)

Define:

$NS_F = \max(\delta_{D\to U},\delta_{U\to D})$. (O.72)

A clean no-signalling result requires:

$NS_F \le \varepsilon_{NS}$. (O.73)

This should be tested before interpreting S_F.

O.16 Controlled signalling injection

The sandbox should deliberately introduce a signalling channel.

For example:

$u_k = \mathrm{BaseOutcome}_U + \kappa_{sig}\cdot b_k$. (O.74)

Or:

$d_k = \mathrm{BaseOutcome}_D + \kappa_{sig}\cdot a_k$. (O.75)

As κ_{sig} increases:

NS_F should increase. (O.76)

The experiment should verify that the diagnostic detects the injected channel.

A test incapable of detecting known signalling cannot establish its absence.

O.17 Memory injection

Introduce ledger-dependent evolution:

$\lambda_{k+1} = F(\lambda_k,u_k,d_k,a_k,b_k)$. (O.77)

Then:

$p(u_k,d_k|a_k,b_k,H_k) \neq p(u_k,d_k|a_k,b_k)$. (O.78)

Where H_k is prior history.

Define:

$MG_F = I[(u_k,d_k);H_k|a_k,b_k]$. (O.79)

The analysis should compare:

- memory-free model;
- first-order Markov model;
- longer-memory model;
- adaptive observer model.

This establishes whether an apparent inequality violation can be generated through sequential latching.

O.18 Post-selection injection

Let a trial be included only when:

$S_k = 1$. (O.80)

Suppose:

$p(S_k=1|u_k,d_k,a_k,b_k)$ (O.81)

depends on the outcomes.

Then:

$p_{obs}(u,d|a,b) \neq p_{all}(u,d|a,b)$. (O.82)

The sandbox should demonstrate how selective retention can distort S_F.

Examples include retaining only:

- completed trades;
- liquid observations;

- successful settlements;
- large moves;
- trials with complete data.

O.19 Common-cause injection

Let hidden variable λ_k influence both outcomes:

$u_k = f_U(a_k,\lambda_k,\varepsilon_U,k)$. (O.83)

$d_k = f_D(b_k,\lambda_k,\varepsilon_D,k)$. (O.84)

This can create strong joint correlation without direct signalling.

The test should evaluate whether latent-state estimation recovers λ_k sufficiently to explain the observed statistics.

A strong classical control family should include:

- discrete hidden states;
- continuous latent factors;
- nonlinear response;
- nonstationary latent distributions.

O.20 Observer-policy mode

A second sandbox mode should intentionally allow adaptive settings:

$a_k = \pi_A(L_A,k_{-1})$. (O.85)

$b_k = \pi_B(L_B,k_{-1})$. (O.86)

This mode tests:

- internal observer latching;
- contextual measurement;
- policy divergence;
- cross-observer agreement.

It should not be interpreted under standard Bell assumptions.

The same runtime can therefore support two distinct experiments:

Isolation mode

Tests joint statistics under controlled assumptions.

Observer mode

Tests trace-conditioned world formation.

These modes must not be mixed.

O.21 Classical hidden-variable benchmark

Define:

$p(u,d|a,b) = \int p(\lambda)p(u|a,\lambda)p(d|b,\lambda)d\lambda$. (O.87)

Fit a flexible model family.

For example:

$\lambda \in \mathbb{R}^m$. (O.88)

$p(u=1|a,\lambda) = \sigma_{logistic}(f_a(\lambda))$. (O.89)

$p(d=1|b,\lambda) = \sigma_{logistic}(g_b(\lambda))$. (O.90)

Use nonlinear f_a and g_b.

The benchmark should be trained on one subset and evaluated on held-out trials.

O.22 Coherent-state benchmark

Fit state parameters:

α. (O.91)

φ. (O.92)

Measurement angles:

$\xi_0,\xi_1,\zeta_0,\zeta_1$. (O.93)

Possible decoherence parameter:

γ. (O.94)

The partially coherent state is:

$\rho_\gamma = \rho_{sep} + \gamma(\rho_{coh}-\rho_{sep})$. (O.95)

Estimate:

$\hat{\gamma} \in [0,1]$. (O.96)

A stable:

$\hat{\gamma} > 0$ (O.97)

is only candidate effective coherence.

The coherent model must outperform the flexible classical hidden-variable benchmark.

O.23 Evidence sequence

The evidence sequence should be reported as:

Stage O1 — Correlation

Reject independence.

Stage O2 — Relational-state gain

Joint state outperforms separate local models.

Stage O3 — Separable-null challenge

A separable representation is tested.

Stage O4 — Phase-sensitive prediction

Off-diagonal terms predict rotated-setting outcomes.

Stage O5 — No-signalling audit

Marginals are setting-independent within tolerance.

Stage O6 — Measurement-independence audit

Settings are independent of preparation state.

Stage O7 — Memory and selection audit

Sequential and sampling loopholes are controlled.

Stage O8 — Bell-style anomaly

The remaining correlation exceeds the tested classical bound.

Each stage requires the previous stages.

O.24 Decision table

Outcome	Correct interpretation
Strong correlation, classical model fits	Classical contractual dependence
Coherent model fits, classical phase model also fits	Classical phase structure
Off-diagonal state improves prediction	Candidate operational coherence
No-signalling fails	Ordinary causal influence
Measurement independence fails	Adaptive setting dependence
Memory model explains result	Ledgered sequential process

Outcome	Correct interpretation
Selection explains result	Sampling artefact
Separable models fail but Bell assumptions fail	Representation-relative nonseparability
All major classical controls fail	Strong anomaly requiring independent replication

---

O.25 Sandbox runtime

The minimal trial loop is:

Prepare ρ_k
→ Randomize a_k,b_k
→ Apply Local Instruments
→ Seal u_k,d_k
→ Update Local Ledgers
→ Reconcile Records
→ Audit NS,MI,MG,SG
→ Estimate S_F
→ Fit Classical and Coherent Models. (O.98)

Where:

NS = no-signalling gap. (O.99)

MI = measurement-independence gap. (O.100)

MG = memory gap. (O.101)

SG = selection gap. (O.102)

The statistic S_F should never be reported without these four diagnostics.

O.26 Falsification conditions

The strong nonclassical-style interpretation fails if any of the following holds:

NS_F > ε_NS. (O.103)

MI_F > ε_MI. (O.104)

MG_F > ε_MG and memory model fits. (O.105)

SG_F > ε_SG. (O.106)

Score_classical ≥ Score_coherent. (O.107)

Phase parameters are unstable. (O.108)

The result disappears under equivalent setting representation. (O.109)

The prepared-state assumption cannot be reproduced. (O.110)

Failure at this level does not invalidate the broader financial-world architecture.

It only limits the entanglement claim.

O.27 Sandbox proposition

Proposition O.1 — Assumption-First Bell Testing

A Bell-style financial statistic has interpretive value only when the runtime separately measures and constrains:

1. ordinary signalling;
2. setting dependence;
3. sequential memory;
4. post-selection;
5. classical latent-state alternatives.

In compact form:

BellMeaning_F ⇔ CHSHResult ∧ NoSignallingAudit ∧ SettingIndependence ∧ MemoryControl ∧ SelectionControl ∧ StrongClassicalNulls. (O.111)

Without this conjunction:

|S_F| is merely a correlation summary. (O.112)

Appendix P — Curvature and Holonomy Estimation Workflow

P.1 Purpose

The GR-like and gauge-like layers become scientifically meaningful only when they support measurable transport.

This appendix provides a practical workflow for estimating:

- local financial metric;
- base connection;
- gauge connection;
- curvature;
- loop holonomy;
- ledger residual.

The goal is not to imitate physical spacetime.

The goal is to determine whether financial state transport is:

- locally regular;

- globally path-dependent;
- frame-sensitive;
- history-bearing.

P.2 Select the state coordinates

Choose a coordinate vector:

$x_t = (x_t^1, x_t^2, \ldots, x_t^m)$. (P.1)

A derivative-risk example might use:

$x_t = (U_t, \sigma_t, Funding_t, Liquidity_t, Collateral_t, Inventory_t)$. (P.2)

Coordinates should satisfy:

1. economic interpretability;
2. measurable variation;
3. sufficient data density;
4. relevance to the target transport problem.

Adding coordinates only to improve in-sample fit should be penalized.

P.3 Select the transported object

The transported object may be:

- beta vector;
- Greek vector;
- option-surface factor;
- hedge portfolio;
- observer state;
- complex phase vector;
- admissible-action set.

Let the object be:

$V(x) \in T_x\mathcal{M}_F$ (P.3)

for a tangent-like risk vector.

Or:

$|\psi(x)\rangle \in \mathcal{H}_x$ (P.4)

for a fibre state.

Metric and gauge transport must not be mixed without declaring the object type.

P.4 Candidate metric from covariance

Estimate local covariance:

$\Sigma(x) = \mathrm{Cov}[\Delta x | \mathrm{Neighbourhood}(x)]$. (P.5)

Define:

$g_cov(x) = [\Sigma(x) + \lambda I]^{-1}$. (P.6)

Where λ regularizes inversion.

The local distance is:

$ds_cov^2 = dx^T g_cov dx$. (P.7)

This metric measures statistical unusualness.

A move in a normally stable direction has greater distance than an equally sized move in a volatile direction.

P.5 Candidate metric from intervention cost

Let $C(x \to x+dx)$ be estimated cost of moving the system.

Fit:

$C(x,dx) = \frac{1}{2} dx^T g_cost(x) dx + O(\|dx\|^3)$. (P.8)

Possible costs include:

- execution cost;
- capital cost;
- collateral cost;
- funding cost;
- liquidation loss;
- control energy.

This metric has a direct engineering interpretation.

P.6 Candidate metric from predictive distinguishability

Suppose outcome distribution is:

$p(y|x)$. (P.9)

The Fisher metric is:

g_Fisher,μν(x) = E[(∂_μln p)(∂_νln p)]. (P.10)

Small displacement dx changes the distribution by approximately:

D_KL[p(y|x)||p(y|x+dx)] ≈ ½g_Fisher,μνdx^μdx^ν. (P.11)

This metric measures observational distinguishability.

P.7 Metric-selection criterion

For candidate metric g_j, evaluate:

Score_metric(j) = TransportGain_j + ForecastGain_j + ControlGain_j – ComplexityPenalty_j. (P.12)

Select the metric before final hypothesis testing.

If several metrics are retained, report results for all of them.

Do not choose the metric only because it produces the desired curvature.

P.8 Estimate local frames

At point x, decompose:

g(x) = E(x)^{T}ηE(x). (P.13)

The frame matrix E contains:

e^{a}_μ(x). (P.14)

Local coordinates are:

dx_local = E(x)dx_global. (P.15)

A local-flat approximation is valid when metric variation across the neighbourhood is small:

‖g(x+dx)−g(x)‖ ≤ ε_flat. (P.16)

P.9 Estimate the base connection

Given the metric, compute:

Γ^ρ_μν = ½g^{ρλ}(∂_μg_λν + ∂_νg_λμ − ∂_λg_μν). (P.17)

In empirical data, derivatives may be estimated using:

- local polynomial regression;
- Gaussian processes;
- neural differential models;

- finite differences on a state grid.

Connection uncertainty must be propagated into curvature estimates.

P.10 Direct transport estimation

A model-free alternative estimates transport directly.

Given nearby states x_A and x_B, fit:

V_B ≈ T_BA V_A. (P.18)

Estimate T_BA from repeated transitions.

For small displacement:

T_BA ≈ I − Γ_μdx^μ. (P.19)

Then:

Γ_μ ≈ −(T_BA−I)/dx^μ. (P.20)

This approach can be more stable than differentiating a noisy metric.

P.11 Triangle transport test

Select nearby states A, B, and C.

Two-step transport gives:

V_C^{A→B→C} = T_CBT_BAV_A. (P.21)

Direct transport gives:

V_C^{A→C} = T_CAV_A. (P.22)

Define:

ε_ABC = V_C^{A→B→C} − V_C^{A→C}. (P.23)

In locally flat, path-independent transport:

ε_ABC ≈ 0. (P.24)

Systematic nonzero residual may indicate:

- curvature;
- protocol mismatch;
- omitted state;
- nonstationarity;
- ledger difference.

P.12 Estimate base curvature

Using the connection:

$R^{\rho}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}$. (P.25)

For a small coordinate loop with area $\Delta S^{\mu\nu}$:

$\Delta V^{\rho} \approx R^{\rho}_{\sigma\mu\nu} V^{\sigma} \Delta S^{\mu\nu}$. (P.26)

Thus curvature predicts the vector-transport residual around a small loop.

This prediction should be tested on held-out loops.

P.13 Estimate the fibre connection

For normalized fibre states $|\psi(x)\rangle$, estimate:

$\mathcal{A}_\mu(x) = i\langle \psi(x) | \partial_\mu \psi(x) \rangle$. (P.27)

If the states are noisy, estimate local unitary alignment:

$U_{BA} = \arg\min_U \| |\psi_B\rangle - U|\psi_A\rangle \|$. (P.28)

For small displacement:

$U_{BA} \approx \exp[-i\mathcal{A}_\mu dx^\mu]$. (P.29)

The connection should improve cross-frame state alignment.

P.14 Gauge-fixing step

Select one reference convention.

For example:

$\theta_{reference}(x_0) = 0$. (P.30)

Or impose:

$\nabla \cdot \mathcal{A} = 0$. (P.31)

The chosen gauge simplifies estimation.

All final predictions should depend only on:

- relative phase;
- covariant derivative;
- loop phase;
- gauge curvature.

If results depend on the arbitrary gauge choice, the model is invalid.

P.15 Estimate gauge curvature

For an Abelian connection:

$\mathbb{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu$. (P.32)

For matrix-valued transport:

$\mathbb{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu + i[\mathcal{A}_\mu, \mathcal{A}_\nu]$. (P.33)

The small-loop phase is:

$\Phi_\gamma \approx \tfrac{1}{2}\mathbb{F}_{\mu\nu}\Delta S^{\mu\nu}$. (P.34)

The measured phase should agree with transport around the loop.

P.16 Construct closed loops

A loop should return approximately to the starting visible coordinates:

$\|x_{final} - x_{initial}\| \le \varepsilon_{loop}$. (P.35)

Candidate loops include:

Spot–volatility loop

(U_0, σ_0)
$\to (U_1, \sigma_0)$
$\to (U_1, \sigma_1)$
$\to (U_0, \sigma_1)$
$\to (U_0, \sigma_0)$. (P.36)

Funding–collateral loop

(f_0, c_0)
$\to (f_1, c_0)$
$\to (f_1, c_1)$
$\to (f_0, c_1)$
$\to (f_0, c_0)$. (P.37)

Leverage–liquidity loop

(ℓ_0, λ_0)
$\to (\ell_1, \lambda_0)$
$\to (\ell_1, \lambda_1)$
$\to (\ell_0, \lambda_1)$
$\to (\ell_0, \lambda_0)$. (P.38)

P.17 Measure full loop output

The visible endpoint residual is:

ε_x = x_final − x_initial. (P.39)

The transported-vector residual is:

ε_V = V_final − V_initial. (P.40)

The fibre-state residual is:

ε_ψ = Distance[|ψ_final⟩,Hol_gauge(γ)|ψ_initial⟩]. (P.41)

The ledger residual is:

ε_L = Difference(L_final,L_initial). (P.42)

The total residual is:

ε_total = (ε_x,ε_V,ε_ψ,ε_L). (P.43)

A closed visible loop may still contain large ε_V, ε_ψ, or ε_L.

P.18 Separate known accumulated costs

Calculate ordinary cumulative effects:

Cost_transaction. (P.44)

Cost_funding. (P.45)

Gamma_P&L. (P.46)

Tax. (P.47)

MarginCost. (P.48)

LegalCost. (P.49)

Define known-cost residual:

ε_known = ObservedLoopEffect − ΣKnownCosts. (P.50)

Only ε_known should be attributed to additional hidden state, curvature, or holonomy.

P.19 Hysteresis benchmark

Fit a classical history model:

Y_t = F(x_t,H_t). (P.51)

Where H_t summarizes:

- prior extrema;
- cumulative volume;
- threshold crossings;

- state duration;
- previous gates.

Compare:

Model_endpoint. (P.52)

Model_hysteresis. (P.53)

Model_geometry. (P.54)

Model_geometry+ledger. (P.55)

The geometric model advances only if it improves held-out loop prediction.

P.20 Order-reversal test

Traverse loop γ:

$A \to B \to C \to D \to A$. (P.56)

Then traverse reverse loop:

γ^{-1}: $A \to D \to C \to B \to A$. (P.57)

For an Abelian geometric phase:

$\Phi_{\gamma^{-1}} = -\Phi_\gamma$. (P.58)

For a purely dissipative cost:

$\text{Cost}_{\gamma^{-1}}$ may remain positive. (P.59)

This helps separate:

- orientation-sensitive phase;
- symmetric accumulated cost;
- irreversible ledger effect.

P.21 Loop-area scaling

For small loops, geometric residual should scale approximately with enclosed area:

$\Phi_\gamma \propto \text{Area}(\gamma)$. (P.60)

If loop size is scaled by factor s:

$\text{Area} \to s^2\text{Area}$. (P.61)

Then:

$\Phi \to s^2\Phi$ (P.62)

in the small-loop regime.

Ordinary threshold hysteresis may instead show:

- discontinuous activation;
- saturation;
- non-area scaling.

This provides a useful diagnostic.

P.22 Locality of curvature

Estimate curvature in several neighbourhoods:

$\kappa_F(x_1),\kappa_F(x_2),\ldots,\kappa_F(x_n)$. (P.63)

Test whether high curvature predicts:

- unstable beta;
- unstable Greeks;
- transport error;
- increased hedge residual;
- regime-boundary proximity.

A curvature estimate that has no relation to any observable failure is not operationally useful.

P.23 Ledger-conditioned geometry

Estimate metric before and after a committed event:

$g_{before} = g(x,L_k)$. (P.64)

$g_{after} = g(x,L_{k+1})$. (P.65)

Holding visible coordinates approximately fixed, test:

$\Delta g_L = g_{after} - g_{before}$. (P.66)

Examples:

- before and after margin breach;
- before and after default declaration;
- before and after barrier activation;
- before and after regulatory intervention.

This directly tests whether the ledger changes effective geometry.

P.24 State-to-geometry causal test

Let treatment be committed event T.

Estimate:

ATE_g = E[g_after(1)−g_after(0)]. (P.67)

Control for:

- pre-event stress;
- anticipation;
- common market movement;
- selection into the event;
- reverse causality.

A significant ATE_g supports:

Record → Geometry Change. (P.68)

This is stronger than correlation between stress and curvature.

P.25 Geometry-to-state feedback test

Having estimated g_after, test whether later dynamics depend on it:

Y_{t+h} = F(X_t,g_t,L_t) + ε. (P.69)

Compare with:

Y_{t+h} = F(X_t,L_t) + ε′. (P.70)

Define:

ΔScore_g = Score_withmetric − Score_withoutmetric. (P.71)

The GR-like loop requires both directions:

State → Geometry. (P.72)

Geometry → Later State Dynamics. (P.73)

One-way correlation is insufficient.

P.26 Gauge-transport validation

For observers A and B:

State_B,pred = U_BAState_A. (P.74)

The raw discrepancy is:

ε_raw = State_B − State_A. (P.75)

The transported discrepancy is:

ε_cov = State_B − U_BAState_A. (P.76)

A useful connection requires:

E[‖ε_cov‖] < E[‖ε_raw‖]. (P.77)

And the gain must survive out of sample.

P.27 Holonomy prediction

Estimate connection on training loops.

For held-out loop γ_test, predict:

Hol_pred(γ_test) = $\mathcal{P}$exp[−i∮_{γ_test}$\mathcal{A}$]. (P.78)

Predict final state:

|ψ_f,pred⟩ = Hol_pred(γ_test)|ψ_i⟩. (P.79)

Compare with observed final state.

The holonomy model advances only if it predicts new loops rather than merely describing old ones.

P.28 Geometry evidence levels

G0 — Coordinate description

A state space is defined.

G1 — Metric fit

A metric summarizes local distances.

G2 — Transport gain

Covariant transport improves prediction.

G3 — Curvature prediction

Loop residual is predicted from curvature.

G4 — Ledger-conditioned geometry

Committed records alter the metric.

G5 — Backreactive geometry

The altered metric changes later state dynamics.

The full GR-like analogy begins only near G4–G5.

P.29 Failure conditions

The geometric interpretation fails when:

Metric estimates are unstable. (P.80)

Curvature depends mainly on smoothing choice. (P.81)

Loop residual is explained by known costs. (P.82)

Hysteresis performs equally well. (P.83)

Gauge correction does not reduce error. (P.84)

Loop orientation has no predicted relation to phase. (P.85)

Geometry does not predict future dynamics. (P.86)

Ledger events do not change geometry after controls. (P.87)

The appropriate response is to retain only the successful lower-level components.

P.30 Workflow summary

The full workflow is:

Declare Coordinates
→ Select Transported Object
→ Estimate Metric
→ Estimate Connection
→ Construct Loops
→ Measure Transport Residual
→ Subtract Known Costs
→ Compare Hysteresis Null
→ Test Gauge Invariance
→ Test Ledger Backreaction
→ Predict Held-Out Loops. (P.88)

P.31 Geometry proposition

Proposition P.1 — Predictive Curvature Requirement

A financial curvature or holonomy claim is justified only when:

1. local state coordinates and transported objects are declared;
2. a metric or connection is estimable;
3. closed-loop residual is reproducible;

4. known costs and classical hysteresis are insufficient;
5. the geometric structure predicts held-out transport;
6. ledger-conditioned backreaction is separately measured where claimed.

In compact form:

OperationalGeometry_F ⇔ DeclaredState ∧ EstimableTransport ∧ ReproducibleLoop ∧ ClassicalNullFailure ∧ HeldOutPrediction ∧ BackreactionAudit. (P.89)

Appendix Q — Cross-Observer Agreement and Redundant Objectivity

Q.1 Purpose

A secondary valuation world may contain several observers with:

- different local frames;
- different accessible variables;
- different measurement settings;
- different ledgers;
- different update times;
- different institutional authority.

Observer relativity does not imply that every observer inhabits an unrelated private world.

Agreement can emerge when observers possess compatible ways to:

1. translate frames;
2. compare outcomes;
3. access records;
4. verify provenance;
5. preserve the same governed object.

This appendix formalizes how locally bounded financial observers may construct shared operational objectivity.

Q.2 Two internal observers

Let observer A be:

$O_A = (\mathcal{A}_A, \mathbb{M}_A, F_A, L_A, \pi_A)$. (Q.1)

Let observer B be:

$O_B = (\mathcal{A}_B, \mathbb{M}_B, F_B, L_B, \pi_B)$. (Q.2)

Their local state estimates are:

$\rho_A = \hat{O}_A[\rho_F]$. (Q.3)

$\rho_B = \hat{O}_B[\rho_F]$. (Q.4)

In general:

$\rho_A \neq \rho_B$. (Q.5)

The difference may arise from:

- frame;
- measurement algebra;
- information delay;
- protocol;
- ledger access;
- local noise.

The first task is to determine whether A and B are observing the same governed financial object.

Q.3 Governed-object identity

Let object identity be:

$I_G = (\text{ContractID}, \text{Underlying}, \text{Payoff}, \text{Maturity}, \text{SettlementRule}, \text{LegalBoundary})$. (Q.6)

Observers refer to the same governed object only when:

$I_G^A = I_G^B$. (Q.7)

A mismatch means the disagreement may concern different objects.

Examples include:

- spot asset versus forward;
- secured claim versus unsecured claim;
- pre-default contract versus post-default recovery claim;
- market value versus liquidation proceeds.

Before comparing measurements:

ObjectIdentity must be established. (Q.8)

Q.4 Frame map

Let the map from A's frame to B's frame be:

$T_{BA}: F_A \to F_B$. (Q.9)

The mapped state is:

$\rho_{A\to B} = T_{BA}(\rho_A)$. (Q.10)

A successful frame map requires:

$\text{Distance}[\rho_{A\to B}, \rho_B] \le \varepsilon_{frame}$. (Q.11)

The residual is:

$\varepsilon_{BA}^{frame} = \rho_B - T_{BA}(\rho_A)$. (Q.12)

Persistent residual may indicate:

- incorrect conversion;
- hidden exposure;
- protocol mismatch;
- delayed information;
- different legal state;
- genuinely different observations.

Q.5 Outcome map

Observer A may label an outcome:

$y_A \in \mathcal{Y}_A$. (Q.13)

Observer B may use:

$y_B \in \mathcal{Y}_B$. (Q.14)

A comparison requires an outcome translation:

$\tau_{BA}: \mathcal{Y}_A \to \mathcal{Y}_B$. (Q.15)

Agreement occurs when:

$\tau_{BA}(y_A) = y_B$. (Q.16)

Examples:

- "barrier activated" ↔ "knock-in state entered";
- "margin deficit" ↔ "variation-margin call";
- local-currency value ↔ base-currency value;
- desk exposure ↔ enterprise-equivalent exposure.

Without τ_BA, apparent disagreement may be terminological rather than substantive.

Q.6 Measurement-effect compatibility

Let A's effect for outcome y_A be:

E_A,y. (Q.17)

Let B's corresponding effect be:

E_B,τ(y). (Q.18)

Compatibility under frame transport requires:

E_B,τ(y) ≈ U_BAE_A,yU_BA†. (Q.19)

Define effect mismatch:

ε_effect = ‖E_B,τ(y) − U_BAE_A,yU_BA†‖. (Q.20)

A comparison is admissible only when:

ε_effect ≤ ε_E. (Q.21)

Otherwise A and B are applying materially different questions to the same object.

Q.7 Record accessibility

Suppose A commits event record:

R_A,k. (Q.22)

Observer B can treat the event as fixed only when:

R_A,k ∈ AccessibleRecords_B. (Q.23)

Define access variable:

χ_B(R_A,k) = 1 if B can verify the record. (Q.24)

χ_B(R_A,k) = 0 otherwise. (Q.25)

Record access may require:

- network delivery;
- permission;
- format compatibility;
- identity authentication;
- timestamp verification;
- provenance validation.

Without record access, B may retain uncertainty even when A is internally certain.

Q.8 Cross-observer fixedness

Let φ be the committed proposition.

A is internally fixed when:

$P_A(\varphi|\mathscr{F}_A,k) = 1.$ (Q.26)

B becomes fixed when:

$P_B[T_{BA}(\varphi)|\mathscr{F}_B,j] = 1.$ (Q.27)

A sufficient operational condition is:

$Fixed_{AB}(\varphi) \Leftrightarrow ObjectMatch \wedge FrameMap \wedge EffectCompatibility \wedge AccessibleRecord.$ (Q.28)

This is a conjunction.

Failure of any term can prevent shared fixedness.

Q.9 Delay to agreement

Let A commit at calendar time:

$t_A^{commit}.$ (Q.29)

Let B gain verified access at:

$t_B^{access}.$ (Q.30)

Agreement delay is:

$\Delta t_{AB} = t_B^{access} - t_A^{commit}.$ (Q.31)

In ledger order:

$\Delta k_{AB} = k_B^{fixed} - k_A^{commit}.$ (Q.32)

The two delays differ.

A short calendar delay may still span several ledger events.

A long calendar delay may produce no intervening ledger event.

Q.10 Agreement probability

When records and measurements are noisy, define:

$P_{agree}(A,B) = P[\tau_{BA}(y_A)=y_B].$ (Q.33)

Conditioned on a common verified record R:

$P_{agree}(A,B|R)$. (Q.34)

A strong objectivity regime requires:

$P_{agree}(A,B|R) \geq 1 - \varepsilon_{obj}$. (Q.35)

The tolerance ε_{obj} includes:

- measurement error;
- frame-conversion error;
- record corruption;
- timing mismatch;
- protocol disagreement.

Q.11 Redundant records

Suppose event φ is recorded independently by N channels:

$R_1, R_2, \ldots, R_N$. (Q.36)

Let each channel have error probability:

$p_i = P(R_i \text{ incorrect})$. (Q.37)

For independent equal-quality channels:

$p_i = p$. (Q.38)

A majority-vote consensus with odd N has error:

$p_{majority} = \Sigma_{j=(N+1)/2}^{N} C(N,j) p^j (1-p)^{N-j}$. (Q.39)

For:

$p < 1/2$, (Q.40)

majority error decreases as N grows.

This is one route to operational objectivity.

Q.12 Correlated record errors

Record channels may share one upstream source.

Let correlation between errors i and j be:

$\mathrm{Corr}(\mathrm{Error}_i, \mathrm{Error}_j) = \rho_{ij}$. (Q.41)

If:

$\rho_{ij} \approx 1$, (Q.42)

additional records add little independent evidence.

Examples include:

- several systems copying one faulty market feed;
- several databases inheriting one incorrect contract identifier;
- multiple reports generated from one mistaken model.

Therefore:

Record Count ≠ Independent Redundancy. (Q.43)

Effective redundancy depends on source diversity and provenance.

Q.13 Effective redundancy

Define effective record number:

$N_{eff} = (\Sigma_i w_i)^2 / \Sigma_{i,j} w_i w_j \rho_{ij}$. (Q.44)

Where:

- w_i is reliability weight;
- $\rho_{ii} = 1$;
- ρ_{ij} captures dependence.

When records are independent:

$\rho_{ij} = 0$ for $i \neq j$. (Q.45)

Then N_{eff} approaches the ordinary weighted count.

When all records are perfectly correlated:

$N_{eff} \approx 1$. (Q.46)

This quantity is more informative than raw replication count.

Q.14 Record provenance graph

Let record provenance be a directed graph:

$\mathcal{G}_R = (V_R, E_R)$. (Q.47)

Vertices represent:

- data sources;
- transformations;
- observers;

- ledgers;
- publication systems.

An edge:

$R_i \rightarrow R_j$ (Q.48)

means record j derives from record i.

Independent confirmation requires paths whose earliest common ancestor is sufficiently remote.

Define provenance overlap:

$\Omega_{ij} = \text{SharedAncestry}(R_i,R_j)$. (Q.49)

High Ω_{ij} reduces effective redundancy.

Q.15 Record distinguishability

Two possible outcomes φ and ψ must leave distinguishable records.

Let record distributions be:

$p(R|\varphi)$. (Q.50)

$p(R|\psi)$. (Q.51)

Define distinguishability:

$D_{record}(\varphi,\psi) = D[p(R|\varphi),p(R|\psi)]$. (Q.52)

Operational objectivity requires:

$D_{record}(\varphi,\psi) \geq D^*$. (Q.53)

If different outcomes produce nearly indistinguishable records, later observers cannot reconstruct which event occurred reliably.

Q.16 Record persistence

Let survival probability of record R after ledger interval n be:

$S_R(n) = P(R \text{ remains accessible and valid after } n \text{ ticks})$. (Q.54)

A persistent record satisfies:

$S_R(n) \geq 1 - \varepsilon_{persist}$ (Q.55)

over the relevant horizon.

Threats include:

- deletion;
- format obsolescence;

- permission loss;
- cryptographic failure;
- institutional failure;
- silent revision.

Objectivity requires not only initial redundancy but continuing retrievability.

Q.17 Revision records

A record may later be corrected.

Let initial record be:

R_k. (Q.56)

Correction creates:

R_k′ = Corrects(R_k). (Q.57)

The ledger should preserve both:

L_j ⊃ {R_k,R_k′}. (Q.58)

The correct current proposition is determined by a revision rule:

CurrentTruth(L_j) = ResolveRevisionChain(R_k,R_k′,…). (Q.59)

Silent deletion would destroy auditability.

Therefore:

Correction = New Trace, Not Erasure of Old Trace. (Q.60)

Q.18 Conflicting authoritative records

Suppose two authorized systems commit:

R_A = φ. (Q.61)

R_B = ¬φ. (Q.62)

The world contains a record conflict.

Define conflict state:

C_R = (R_A,R_B,Authorities,Frames,Evidence). (Q.63)

A higher-order gate must decide:

G_conflict ∈ {Prefer_A,Prefer_B,Merge,Defer,Litigate}. (Q.64)

Until resolved, shared objectivity remains incomplete.

The conflict itself should enter the ledger:

$L \to L \sqcup \text{ConflictRecord}$. (Q.65)

Q.19 Authority hierarchy

Let authority ordering be:

$\text{Authority}_1 \succ \text{Authority}_2 \succ \dots$. (Q.66)

Examples:

- clearinghouse over local desk;
- court judgment over internal interpretation;
- official fixing over indicative quote;
- settlement record over pre-trade estimate.

A resolution rule may be:

Select record from highest valid authority. (Q.67)

But authority is protocol-dependent.

Different worlds may recognize different hierarchies.

Q.20 Consensus versus truth

Consensus is:

$\text{Consensus}(\varphi)$ = many observers record φ. (Q.68)

Truth relative to the declared primary event is:

$\text{Truth}_P(\varphi)$ = φ matches governed event under protocol P. (Q.69)

Consensus may fail to track truth when:

- all observers share one corrupted source;
- institutional incentives align around error;
- measurement protocol is invalid;
- the governed object was misidentified.

Therefore:

Consensus ≠ Truth. (Q.70)

Operational objectivity requires:

Consensus + Provenance + Compatibility + Residual Audit. (Q.71)

Q.21 Observer partitions

Observers may access different parts of one global state.

Let:

$\mathcal{A}$_A ∩ $\mathcal{A}$_B = shared observable algebra. (Q.72)

Agreement can be tested only on the shared subalgebra.

For observable Ô:

Ô ∈ $\mathcal{A}$_A ∩ $\mathcal{A}$_B. (Q.73)

If:

Ô ∉ $\mathcal{A}$_B, (Q.74)

B cannot directly verify A's claim.

The proper statement is:

Unverifiable by B under current access. (Q.75)

Not:

False for B. (Q.76)

Q.22 Partial agreement

Observers may agree on one invariant while disagreeing on local decomposition.

For example:

TotalExposure_A = TotalExposure_B. (Q.77)

But:

FundingComponent_A ≠ FundingComponent_B. (Q.78)

CollateralComponent_A ≠ CollateralComponent_B. (Q.79)

The invariant may be:

I_total = Funding + Collateral + MarketValueAdjustment. (Q.80)

Thus:

Local Decomposition Disagreement + Invariant Agreement (Q.81)

can coexist.

This is a normal consequence of frame-relative representation.

Q.23 Agreement matrix

For N observers, define:

$A_{ij} = P[\tau_{ji}(y_i)=y_j]$. (Q.82)

The agreement matrix is:

$\mathfrak{A} = [A_{ij}]$. (Q.83)

A highly coherent observer network has:

$A_{ij} \approx 1$ (Q.84)

for all compatible pairs.

Clusters of high internal but low external agreement may indicate:

- protocol fragmentation;
- delayed information;
- local worldview separation;
- incompatible legal or accounting frames.

Q.24 Observer-network phases

Each observer may carry local phase:

θ_i. (Q.85)

Define network order parameter:

$re^{i\psi} = (1/N)\Sigma_i e^{i\theta_i}$. (Q.86)

High r indicates phase-aligned valuation orientation.

Low r indicates disagreement or dephasing.

But high r does not guarantee correctness.

A synchronized network can converge on a shared error.

Thus phase alignment should be combined with record and residual audits.

Q.25 Consensus gate

A multi-observer gate may commit when:

$\Sigma_i w_i Vote_i \geq Threshold$. (Q.87)

Where:

$Vote_i \in \{-1,+1\}$. (Q.88)

The weight may depend on:

- authority;
- evidence quality;
- independence;
- frame relevance;
- historical reliability.

A better gate uses effective evidence rather than raw observer count.

Q.26 Delta-certainty and record trust

Suppose B receives a trusted record from A.

Then:

$P_B(\varphi|R_A,Trust_A) = 1$. (Q.89)

This delta-certainty depends on:

- record authentication;
- authority;
- mapping;
- no unresolved conflict.

If trust is probabilistic:

$P_B(\varphi|R_A) = 1 - p_{corrupt}$. (Q.90)

Therefore internal certainty is protocol-relative rather than metaphysically absolute.

Q.27 Objectivity emergence sequence

The emergence sequence is:

Local Outcome
→ Local Record
→ Authenticated Sharing
→ Frame Translation
→ Effect Compatibility
→ Redundant Confirmation
→ Stable Consensus
→ Future Policy Alignment. (Q.91)

The last step matters.

A fact becomes world-forming when multiple observers condition future action on the same record.

Thus:

Q.28 Objectivity failure modes

Failure	Consequence
Object mismatch	observers discuss different entities
Frame mismatch	numerical disagreement without substantive conflict
Effect mismatch	observers ask different questions
Record inaccessibility	local certainty remains private
Provenance collapse	false redundancy
Authority conflict	competing committed worlds
Revision erasure	audit history lost
Timing mismatch	stale agreement
Protocol drift	old translation no longer valid
Residual suppression	consensus becomes overconfident

Q.29 Cross-observer experiment

A controlled experiment may use four conditions.

Condition 1 — No frame map

B receives A's value without transformation.

Condition 2 — Frame map only

B receives a mapped value but not the measurement definition.

Condition 3 — Frame map and effect compatibility

B can interpret the result but lacks the committed record.

Condition 4 — Full structure

B receives:

- frame map;
- compatible effect;
- authenticated record;
- provenance.

Prediction:

Agreement and certainty increase across the four conditions.

This directly tests the architecture's claim that objectivity requires more than numerical similarity.

Q.30 Redundancy experiment

Prepare an event φ.

Generate N records with controlled:

- error rate;
- source correlation;
- authority weight;
- transmission delay.

Measure:

ConsensusAccuracy(N). (Q.93)

Compare raw count with N_eff.

Prediction:

Consensus accuracy tracks effective redundancy more closely than raw record count.

Q.31 Observer-objectivity proposition

Proposition Q.1 — Ledgered Objectivity Principle

A financial event becomes operationally objective across bounded observers when:

1. they refer to the same governed object;
2. their local frames can be reconciled;
3. their measurement effects are compatible;
4. an authenticated record is accessible;
5. the record is redundantly preserved with sufficient independence;
6. future observer policies condition on the same resolved trace.

In compact form:

Objectivity_F ⇔ ObjectIdentity ∧ FrameMap ∧ EffectCompatibility ∧ RecordAccess ∧ IndependentRedundancy ∧ SharedBackreaction. (Q.94)

Objectivity is therefore not the absence of observers.

It is the stable alignment of observer-bound worlds around a governed and reproducible trace.

Appendix R — Falsification Dashboard and Theory-Reduction Decision Tree

R.1 Purpose

The layered architecture contains many optional components.

A practical research programme requires one dashboard showing:

- what is being claimed;
- what evidence supports it;
- what would falsify it;
- which lower-level model remains after failure.

The theory should not behave as one indivisible package.

Each layer should be independently testable and removable.

R.2 Claim hierarchy

Define twelve claim levels.

Level	Claim
C_1	R is a meaningful protocol-admitted value
C_2	A is independently definable
C_3	Q carries useful retained-pressure information
C_4	θ supports stable phase dynamics
C_5	relative frames admit invariant transport
C_6	state-dependent geometry improves modelling
C_7	derivative joint states outperform local states
C_8	off-diagonal coherence affects observables
C_9	separable classical states are inadequate
C_{10}	contextuality is irreducible to memory and disturbance
C_{11}	no-signalling-compatible correlation exists
C_{12}	Bell-classical models fail under controlled assumptions

The levels are ordered approximately by evidential strength.

R.3 Layer-dependence graph

The claim dependencies are:

$C_2 \rightarrow C_3$. (R.1)

$C_3 \rightarrow C_4$. (R.2)

C_5 and C_6 may be tested independently of C_8–C_{12}. (R.3)

$C_7 \rightarrow C_8 \rightarrow C_9 \rightarrow C_{10} \rightarrow C_{11} \rightarrow C_{12}$. (R.4)

Failure of C_{12} does not imply failure of C_7.

Failure of C8 does not imply that composite-state modelling is useless.

This prevents all-or-nothing evaluation.

R.4 Dashboard fields

Each claim card should contain:

ClaimID. (R.5)

OperationalDefinition. (R.6)

Estimator. (R.7)

Dataset. (R.8)

Baseline. (R.9)

NullModels. (R.10)

Threshold. (R.11)

Result. (R.12)

Uncertainty. (R.13)

Residual. (R.14)

Status. (R.15)

NextAction. (R.16)

The status values are:

{Untested,Supported,Weak,Failed,Inconclusive,Deprecated}. (R.17)

R.5 C1 dashboard — Admitted value

Claim

R is the output of a declared and reproducible valuation protocol.

Test

Repeated implementation of P on identical inputs produces:

R_replay ≈ R_original. (R.18)

Metric

ε_R = |R_replay − R_original|. (R.19)

Pass condition

$\varepsilon_R \le \varepsilon_{replay}$. (R.20)

Failure response

Repair protocol definition before adding any complex geometry.

R.6 C_2 dashboard — Independent amplitude

Claim

A has an economic definition independent of Q.

Test

Estimate A using data or model inputs not derived from:

$Q = \sqrt{(A^2 - R^2)}$. (R.21)

Pass condition

Amplitude estimator is reproducible and economically interpretable.

Failure response

Stop the R+iQ construction.

Without independent A, Q is circular.

R.7 C_3 dashboard — Retained pressure

Claim

Q adds information beyond A and R.

Metric

$\Delta\text{Score}_Q = \text{Score}(Y|A,R,Q) - \text{Score}(Y|A,R)$. (R.22)

Pass condition

$\Delta\text{Score}_Q > \delta_Q$ out of sample. (R.23)

Failure response

Retain Q only as a visualization coordinate or remove it.

R.8 C_4 dashboard — Phase dynamics

Claim

θ or relative phase predicts transition structure.

Metrics

Phase stability:

Var(θ^ across resamples). (R.24)

Predictive gain:

ΔScore_θ. (R.25)

Ordering error:

OrderError_θ. (R.26)

Pass condition

Stable phase + predictive gain + acceptable ordering.

Failure response

Return to real-valued dynamics or ordinary state variables.

R.9 C5 dashboard — Frame transport

Claim

Different financial descriptions are related by an invariant transformation.

Metrics

Composition residual:

ε_comp = ρ_CA − ρ_CB − ρ_BA. (R.27)

Transport residual:

ε_cov = State_B − T_BAState_A. (R.28)

Pass condition

Both residuals remain within declared tolerance.

Failure response

Classify the difference as protocol transition, hidden state, or non-equivalent object.

R.10 C6 dashboard — Curved geometry

Claim

A state-dependent metric improves local-to-global transport.

Metrics

ΔScore_metric. (R.29)

Loop prediction error:

Error_loop. (R.30)

Curvature–failure association:

Corr(κ_F,|ε_local|). (R.31)

Pass condition

Metric and curvature predict held-out transport failure.

Failure response

Use ordinary state-space, covariance, or hysteresis models.

R.11 C7 dashboard — Composite-state value

Claim

The option–underlying joint state improves prediction.

Metric

ΔRel = Score_UD − max(Score_U,Score_D,Score_classical_baseline). (R.32)

Pass condition

ΔRel > δ_rel. (R.33)

Failure response

Use local or ordinary classical joint models.

R.12 C8 dashboard — Operational coherence

Claim

Off-diagonal state terms affect admissible observables.

Metrics

ΔLL_coh. (R.34)

Phase intervention accuracy. (R.35)

Coherence stability. (R.36)

Pass condition

The coherent model predicts held-out rotated-setting or phase-shift outcomes better than diagonal and nonlinear classical models.

Failure response

Set:

$\rho_{off} = 0$. (R.37)

Use the classical mixture.

R.13 C9 dashboard — Nonseparability

Claim

No separable state adequately reproduces the joint observations.

Metrics

D_{sep}. (R.38)

Negativity:

$\mathcal{N}(\rho)$. (R.39)

Witness value:

$\mathrm{Tr}(W\rho)$. (R.40)

Pass condition

Separable-state confidence bound is violated and strong classical latent models fail.

Failure response

Classify the state as correlated, contractually coupled, or classically composite.

R.14 C10 dashboard — Irreducible contextuality

Claim

Order and context effects cannot be explained by:

- measurement disturbance;
- signalling;
- memory;

- adaptive setting;
- hidden variables.

Metric

ContextualityResidual = ObservedEffect − BestClassicalContextualModel. (R.41)

Pass condition

Residual remains stable across pre-registered contexts and compatibility conditions.

Failure response

Use adaptive classical observer dynamics.

R.15 C_{11} dashboard — No-signalling compatibility

Claim

Local marginals do not depend on the remote setting.

Metric

$NS_F = \max(\delta_{D\to U}, \delta_{U\to D})$. (R.42)

Pass condition

$NS_F \le \varepsilon_{NS}$. (R.43)

Failure response

Model the ordinary causal channel.

Do not interpret the correlation as Bell-style entanglement.

R.16 C_{12} dashboard — Bell-classical failure

Claim

A Bell-style inequality is violated under controlled assumptions.

Metrics

S_F. (R.44)

MI_F. (R.45)

NS_F. (R.46)

MG_F. (R.47)

SG_F. (R.48)

Pass condition

|S_F| > 2 + δ_S (R.49)

while:

MI_F ≤ ε_MI. (R.50)

NS_F ≤ ε_NS. (R.51)

MG_F ≤ ε_MG. (R.52)

SG_F ≤ ε_SG. (R.53)

and strong classical models fail.

Failure response

Report the violated assumption rather than claiming nonclassicality.

R.17 Evidence vector

For each claim C_j, define evidence vector:

E_j = (Fit,Prediction,Intervention,FrameRobustness,NullRejection,Replication). (R.54)

Each component may be scored from 0 to 1.

A weighted evidence score is:

Score_E,j = w·E_j. (R.55)

But no scalar score should conceal a zero in a mandatory component.

For example, a Bell-style claim requires nonzero evidence in:

- intervention control;
- no-signalling;
- setting independence;
- null rejection.

R.18 Mandatory-condition mask

Let m_j be a binary requirement vector.

A claim is admissible only when:

E_j,k ≥ Threshold_j,k for every k with m_j,k = 1. (R.56)

This prevents high fit from compensating for absent causal control.

R.19 Interpretation ceiling

Define interpretation ceiling:

I_max(C_j) = strongest statement permitted by evidence. (R.57)

Examples:

C_3 supported permits:

“Q is a useful predictor.” (R.58)

It does not permit:

“Q is a physical imaginary dimension.” (R.59)

C_8 supported permits:

“Operational coherence under protocol P.” (R.60)

It does not permit:

“Markets are quantum.” (R.61)

C_{12} supported would permit:

“Tested local classical models failed under declared assumptions.” (R.62)

Physical interpretation would still require further work.

R.20 Theory-reduction tree

The reduction decision begins with:

Node 1 — Is A independently meaningful?

If no:

Use ordinary R-only finance. (R.63)

If yes:

Proceed to Q.

Node 2 — Does Q add value?

If no:

Use A and R without complex completion. (R.64)

If yes:

Proceed to phase.

Node 3 — Does θ add stable dynamics?

If no:

Use R+iQ as static geometry only. (R.65)

If yes:

Proceed to multi-channel phase.

Node 4 — Do joint states add value?

If no:

Use scalar or multivariate classical finance. (R.66)

If yes:

Proceed to separability testing.

Node 5 — Does coherence add value?

If no:

Use classical joint mixture. (R.67)

If yes:

Proceed to strong classical-null tests.

Node 6 — Are separable models rejected?

If no:

Use coherent or phase-bearing but separable model if still useful. (R.68)

If yes:

Report representation-relative nonseparability.

Node 7 — Does no-signalling hold?

If no:

Model causal interaction. (R.69)

If yes:

Proceed to Bell-style testing.

Node 8 — Are Bell-classical models rejected?

If no:

Retain nonseparable or contextual effective model without stronger claim. (R.70)

If yes:

Escalate to independent replication and foundations analysis.

R.21 Geometry-reduction branch

Does frame transport compose?

If no:

Use protocol-transition modelling. (R.71)

Does metric improve prediction?

If no:

Remove curved geometry. (R.72)

Does connection improve alignment?

If no:

Remove gauge structure. (R.73)

Does loop residual remain after known costs and hysteresis?

If no:

Use ordinary path accounting. (R.74)

Does ledger alter the metric causally?

If no:

Use static geometry. (R.75)

Does geometry alter later dynamics?

If no:

Do not claim GR-like backreaction. (R.76)

R.22 Residual escalation rule

Let normalized residual be:

$RR_j = \|\varepsilon_j\| / \|Y_j\|$. (R.77)

Define thresholds:

$RR_j < r_1 \rightarrow$ Accept. (R.78)

$r_1 \leq RR_j < r_2 \rightarrow$ Monitor. (R.79)

$r_2 \leq RR_j < r_3 \rightarrow$ Revise. (R.80)

$RR_j \geq r_3 \rightarrow$ Suspend or reject. (R.81)

Thresholds must be declared before the result.

R.23 Stability score

Let parameter estimate under perturbation p be:

$\hat{\theta}_p$. (R.82)

Define relative instability:

$\text{Instability}(\hat{\theta}) = \text{SD}_p(\hat{\theta}_p)/|\text{Mean}_p(\hat{\theta}_p)|$. (R.83)

A model with high predictive fit but unstable parameters may be unsuitable for interpretation or control.

R.24 Complexity cost

Let:

K_{model} = parameter count or effective complexity. (R.84)

D_{data} = data and calibration burden. (R.85)

G_{cost} = governance and audit burden. (R.86)

Total cost is:

$C_{total} = \lambda_K K_{model} + \lambda_D D_{data} + \lambda_G G_{cost}$. (R.87)

Net model value is:

$V_{net} = \text{PredictiveGain} + \text{DiagnosticGain} + \text{ControlGain} - C_{total}$. (R.88)

Adopt only if:

$V_{net} > 0$. (R.89)

R.25 Dashboard example

Claim	Status	Evidence	Residual	Decision
C_1 reproducible R	Supported	replay error low	minor	retain
C_2 independent A	Weak	economic proxy unstable	moderate	revise
C_3 useful Q	Untested	none	unknown	test
C_5 frame transport	Supported locally	composition good	liquidation mismatch	split frame/ protocol
C_6 curvature	Inconclusive	in-sample gain	unstable out of sample	simplify

Claim	Status	Evidence	Residual	Decision
C_7 joint state	Supported	relational prediction gain	low	retain
C_8 coherence	Failed	no phase-intervention gain	high	use mixture
C_9 nonseparability	Not tested	insufficient settings	unknown	defer
C_{11} no-signalling	Failed in live data	setting impact visible	high	model signalling
C_{12} Bell anomaly	Inadmissible	assumptions fail	—	do not claim

This is a scientifically healthy outcome.

Several lower layers may survive even when stronger quantum-style layers fail.

R.26 Research-stop rule

A programme should stop expanding a branch when:

ExpectedInformationGain < MarginalResearchCost. (R.90)

Or when:

Three successive stronger models fail to improve held-out performance. (R.91)

The correct response is not to add more vocabulary.

It is to consolidate the successful lower-level model.

R.27 Falsification-dashboard proposition

Proposition R.1 — Modular Falsifiability Principle

A layered theory remains scientifically useful only when every component has:

1. a measurable claim;
2. an explicit threshold;
3. a strong null comparison;
4. a failure response;
5. a lower-level model to which the theory can retreat.

In compact form:

ModularScience ⇔ MeasurableClaim ∧ Threshold ∧ StrongNull ∧ FailureResponse ∧ ReductionPath. (R.92)

Appendix S — Compact Glossary and One-Page

Architecture

S.1 Primary universe

The domain containing the economic, legal, institutional, computational, and physical mechanisms that construct financial contracts and observations.

Symbol:

$X(t)$. (S.1)

S.2 Secondary valuation world

The effective world compiled from the primary universe under protocol P and ledger L.

Symbol:

$\rho_F(\theta) = \mathcal{C}_{P,L}[X(t)]$. (S.2)

S.3 Admitted value

The value recognized by a declared financial filter.

Symbol:

R. (S.3)

S.4 Pre-filter amplitude

An independently defined economic scale before the declared admission filter.

Symbol:

A. (S.4)

S.5 Retained pressure

The orthogonal coordinate completing A and R under the Euclidean construction.

Symbol:

$Q_E = \sqrt{(A^2 - R^2)}$. (S.5)

S.6 Complex financial state

The two-coordinate valuation state:

$Z = R + iQ_E$. (S.6)

S.7 Circular phase

The orientation of the complex state:

$$\theta = \operatorname{atan2}(Q_E, R). \tag{S.7}$$

S.8 Discount rapidity

The additive logarithmic coordinate relating multiplicative valuation frames:

$$\rho_D = \ln(A/R). \tag{S.8}$$

S.9 Local financial frame

A valuation context containing:

- numeraire;
- benchmark;
- funding;
- horizon;
- protocol;
- ledger access;
- observable algebra.

Symbol:

$$F_a. \tag{S.9}$$

S.10 Financial manifold

The global state space of declared financial coordinates.

Symbol:

$$\mathcal{M}_F. \tag{S.10}$$

S.11 Financial metric

A rule measuring local financial distance, distinguishability, or transition cost.

Symbol:

$$ds_F^2 = g^F_{\mu\nu} dx^\mu dx^\nu. \tag{S.11}$$

S.12 Local tetrad

The map between a curved global financial coordinate system and one local approximately flat frame.

Symbol:

$g^{F}_{\mu\nu} = e^{a}_{\mu} e^{b}_{\nu} \eta_{ab}$. (S.12)

S.13 Complex fibre

The local complex state space attached to one base-manifold point.

Symbol:

$\mathscr{H}_x$. (S.13)

S.14 Financial bundle

The collection of local fibres over the financial manifold.

Symbol:

$\pi: \boldsymbol{\mathcal{H}}_F \to \boldsymbol{\mathcal{M}}_F$. (S.14)

S.15 Gauge connection

The rule for comparing phase and internal orientation across nearby local fibres.

Symbol:

$D_\mu = \partial_\mu + i\mathcal{A}_\mu$. (S.15)

S.16 Gauge curvature

The obstruction to path-independent internal-state transport.

Symbol:

$\mathbb{F}_{\mu\nu}$. (S.16)

S.17 Holonomy

The residual transformation after transporting a state around a closed loop.

Symbol:

$\mathrm{Hol}(\gamma)$. (S.17)

S.18 Contract preparation

The map binding underlying and derivative branches into one joint state.

Symbol:

$\mathcal{C}_C$ or $\hat{U}_C$. (S.18)

S.19 Composite financial state

A state defined on:

$\mathscr{H}_U \otimes \mathscr{H}_D$. (S.19)

S.20 Product state

A joint state containing no relation beyond independent local states:

$\rho_{UD} = \rho_U \otimes \rho_D$. (S.20)

S.21 Separable state

A classical mixture of product states:

$\rho_{sep} = \Sigma_j p_j \rho_j^U \otimes \rho_j^D$. (S.21)

S.22 Effective nonseparability

Failure of the joint state to belong to the separable set under the declared representation:

$\rho_{UD} \notin Sep$. (S.22)

S.23 Local reduced state

The state accessible to one subsystem after tracing over the other:

$\rho_U = Tr_D \rho_{UD}$. (S.23)

S.24 Operational coherence

Off-diagonal state relations that affect admissible observables and outperform classical alternatives.

S.25 Internal observer

A bounded runtime possessing:

- observable algebra;
- instrument family;
- local frame;
- ledger;
- adaptive policy.

Symbol:

$O = (\mathcal{A}_O, \mathbb{M}_O, F_O, L_O, \pi_O)$. (S.24)

S.26 Instrument

A measurement operation producing an outcome and a conditional state update.

Symbol:

$\mathbb{M}_y$. (S.25)

S.27 Gate

The operation deciding whether a provisional outcome becomes committed, rejected, deferred, or escalated.

Symbol:

G. (S.26)

S.28 Ledger

The persistent ordered record of committed or otherwise consequential events.

Symbol:

L_k. (S.27)

S.29 Latching

The process through which one record changes later measurement policy or admissibility.

S.30 Backreaction

The causal effect of a secondary-world record or decision on the primary economic state.

S.31 Internal phase time

The ordering supplied by phase progression θ.

S.32 Ledger time

The ordering supplied by committed records k.

S.33 Calendar time

The external duration coordinate t.

S.34 Operational objectivity

Stable cross-observer agreement based on:

- shared object identity;
- frame maps;
- compatible effects;
- accessible records;
- independent redundancy;
- shared consequence.

S.35 Quantum subtraction

The method of constructing the strongest non-quantum observer-bearing control world and identifying which quantum features remain unexplained.

S.36 Quantum residue

The structures not yet reconstructed by the financial control world, including:

- Born-rule necessity;
- mandatory coherent amplitudes;
- irreducible nonseparability;
- no-signalling entanglement;
- Bell nonclassicality;
- fundamental contextuality;

- quantum information constraints.

S.37 One-page architecture

The complete architecture is:

Primary Universe X(t)
↓ declaration and compilation
Secondary Financial World ρ_F(θ)
↓ placement on curved base $\mathcal{M}$_F
Local Financial Frame F_a
↓ attachment of complex fibre $\mathcal{H}$_x
Underlying–Derivative Composite State ρ_UD
↓ covariant evolution through g_F and $\mathcal{A}$
Internal Observer Instrument $\mathbb{M}$
↓ provisional outcome
Gate G
↓ commitment
Ledger L_{k+1}
↓ policy and institutional action
Primary Backreaction X(t+Δt)
↓ metric, connection, and protocol revision
New Secondary World. (S.28)

S.38 One-page master equation

The state evolves through:

$$i\hbar_F \mathcal{D}_\theta |\Psi_F\rangle = \hat{H}_F[g^F,\mathcal{A},P,L]|\Psi_F\rangle + |\varepsilon_F\rangle. \quad (S.29)$$

Where:

$$\mathcal{D}_\theta = \partial_\theta + \dot{x}^\mu \nabla_\mu + i\dot{x}^\mu \mathcal{A}_\mu. \quad (S.30)$$

The generator is:

$$\hat{H}_F = \hat{H}_{CAPM} + \hat{H}_{contract} + \hat{H}_{hedge} + \hat{H}_{ledger} + \hat{H}_{environment} + \hat{H}_{intervention}. \quad (S.31)$$

Measurement produces:

$$\rho_y = \mathbb{M}_y(\rho)/Tr[\mathbb{M}_y(\rho)]. \quad (S.32)$$

Commitment produces:

$$L_{k+1} = L_k \sqcup Record_y. \quad (S.33)$$

Backreaction produces:

$$X_{k+1} = \mathcal{B}(X_k,Record_y). \quad (S.34)$$

The geometry may update through:

$$g^F_{k+1} = \mathcal{G}[g^F_k,X_{k+1},L_{k+1},P_k]. \quad (S.35)$$

The protocol may update through:

$P_{k+1} = \mathcal{U}[P_k, \text{Residual}_k, L_{k+1}]$. (S.36)

S.39 One-page reduction ladder

Full layered architecture:

Complex Fibre

- Composite State
- Internal Observer
- Local Frames
- Curved Geometry
- Gauge Transport
- Gate–Ledger
- Backreaction. (S.37)

Remove coherence:

Classical Composite Finance. (S.38)

Remove tensor structure:

Scalar Complex Finance. (S.39)

Remove Q:

Real-Valued Finance. (S.40)

Remove curvature:

Flat Local Finance. (S.41)

Remove ledger and backreaction:

Passive Valuation Analytics. (S.42)

At the base remains:

Mature CAPM and Derivative Pricing. (S.43)

S.40 One-page conclusion

The strongest defensible conclusion is:

> CAPM is not a quantum theory. It can, however, serve as a local valuation kernel inside a larger observer-bearing financial world that contains complex phase, derivative composite states, local observer restrictions, contextual instruments, curved geometry, gauge transport, commitment gates, historical ledgers, and

institutional backreaction. From inside that secondary world, an option and its underlying may appear as locally incomplete parts of one globally prepared relation. This reconstructs a substantial portion of quantum-like operational strangeness without deriving the specifically quantum constraints of Born probability, no-signalling, Bell nonclassicality, no-cloning, or quantum statistics.

The architecture's final discipline is:

Do Not Confuse the Kernel with the World. (S.44)

Do Not Confuse the Representation with the Substrate. (S.45)

Do Not Confuse Local Incompleteness with Quantum Proof. (S.46)

Do Not Confuse Mathematical Beauty with Empirical Necessity. (S.47)

Preserve the Layers. (S.48)

Test the Couplings. (S.49)

Recover the Limits. (S.50)

Publish the Residual. (S.51)

Yes. The **main argument, formal architecture, conclusion, worked example, testing programme, failure conditions, and Appendices A–S are complete**.

The main remaining improvement is editorial rather than conceptual: add several comparative tables so readers can quickly distinguish layers, claims, limits, and evidence levels. The following appendix can serve that purpose.

Appendix T — Comparative Reference Tables

T.1 Purpose

The architecture deliberately separates structures that are often collapsed into one analogy.

These tables provide quick comparisons among:

- Part I and Part II;
- CAPM, QM-like, SR-like, and GR-like roles;
- primary and secondary worlds;
- correlation and entanglement levels;
- classical mixtures and coherent states;
- phase, rapidity, and time variables;
- geometry, gauge, and ledger structures;
- reproducible and unreproduced quantum characteristics.

The central reading rule is:

T.2 Part I versus Part II

Dimension	Part I	Part II
Core object	One complex valuation coordinate	Composite observer-bearing financial world
Basic state	$Z = R + iQ$	ρ_{UD} on $\mathscr{H}_U \otimes \mathscr{H}_D$
Main financial kernel	CAPM-filtered valuation	CAPM embedded inside derivative and geometric architecture
Phase structure	One circular phase θ	Multiple relative phases $\theta_n - \theta_m$
Subsystem partition	Absent	Underlying and derivative sectors
Entanglement grammar	Undefined	Formally definable
Observer model	Contextual valuation observer	Internal observer with algebra, instrument, frame, ledger, and policy
Collapse analogue	Gate and ledger commitment	Conditional joint update followed by gate and ledger
Time structure	Calendar time, θ-time, ledger time	Local phase clocks, frame-relative simultaneity, and shared ledger order
Relativity	Lorentz-like analogy hinted	Local frame maps and additive discount rapidity formalized
Geometry	Effective curvature proposed	Base metric, connection, curvature, and local tetrads separated
Gauge structure	Hinted	Explicit fibre connection and holonomy
Decoherence	Branch suppression through commitment	Environment–record interaction and off-diagonal suppression
Backreaction	Valuation affects primary state	Derivative, hedge, ledger, metric, and protocol backreaction
Quantum subtraction	First subtraction	Stronger subtraction against a composite curved control world
Main unresolved residue	Broadly identified quantum remainder	Born rule, Bell nonclassicality, no-signalling, no-cloning, and quantum information constraints

The transition is:

Scalar Complex Coordinate
→ Composite Time-Bearing Financial World. (T.2)

T.3 Mathematical role assignment

Layer	Mathematical object	Financial role	Must not be confused with
CAPM layer	$E[r_i] = r_f + \beta_i ERP$	Local expected-return law	Complete market dynamics
Complex layer	$Z = R + iQ_E$	Admitted value plus retained pressure	Physical wavefunction

Layer	Mathematical object	Financial role	Must not be confused with
Phase layer	θ_n	Local valuation orientation	Ledger time or option Theta
Composite layer	$\mathscr{H}_U \otimes \mathscr{H}_D$	Underlying–derivative joint state	Mere scalar correlation
Observer layer	$O = (\mathcal{A},\mathbb{M},F,L,\pi)$	Bounded measurement runtime	External omniscient analyst
SR-like layer	ρ_BA, e^a_μ	Relative local valuation frames	Physical velocity or CAPM beta
GR-like layer	$g^{F}_{\mu\nu}$, $R^{\rho}_{\sigma\mu\nu}$	State-dependent global geometry	Physical spacetime curvature
Gauge layer	$\mathcal{A}_\mu$, $\mathbb{F}_{\mu\nu}$	Fibre orientation and phase transport	Metric connection
Gate layer	G	Commitment decision	Born-rule measurement
Ledger layer	L_k	Historical trace and latching	Mere timestamp list
Backreaction layer	$X_{k+1} = \mathscr{B}(X_k,\mathrm{Record}_k)$	Secondary-world effects on primary state	Passive observation
Residual layer	ε	Omitted structure and revision signal	Disposable noise

T.4 Primary universe, secondary world, and internal observer

Question	Primary constructor universe	Secondary θ-time world	Internal observer
Main object	Economic and institutional mechanisms	Compiled valuation state	Accessible local state
Time	Calendar time t	Phase order θ and ledger order k	Observer filtration $\mathscr{F}_{O,k}$
Knowledge	May include contract construction and market mechanism	Contains effective joint relations	Accesses only bounded algebra
Option–underlying relation	Pricing, payoff, hedge, legal construction	One prepared composite state	Local reduced state and conditional updates
Causality	Ordinary communication, trading, hedge flow	Effective joint evolution	Protocol-bounded observation
Measurement	Data production and institutional action	State disclosure through instruments	Local setting selection
Collapse	External institutional event	Gate-mediated branch commitment	Internally experienced definite outcome
Record	Databases, settlement, law, balance sheet	Shared ledger	Accessible trace subset
Main limitation	Too external to reproduce internal strangeness	Effective, not necessarily fundamental	Locally incomplete
Main insight	Explains preparation	Contains global relational structure	Experiences local incompleteness

The perspectival relation is:

Primary Constructibility
≠ Secondary Separability
≠ Local Completeness. (T.3)

T.5 CAPM's role before and after extension

Question	CAPM alone	CAPM inside Part II
Expected return	Yes	Yes, locally
Complex completion	Only after adding R+iQ	Retained as local state coordinate
Phase	One scalar orientation	Multiple channel phases
Tensor products	No	Supplied by derivatives
Composite identity	No	Supplied by contracts
Conditional local state	No	Supplied by joint-state formalism
Observer algebra	No	Explicit
Gauge transport	No	Explicit
Curved geometry	No	Global embedding
Ledger time	No	Explicit
Backreaction	Not intrinsic	Added through hedge, margin, and institutional response
Entanglement-like modelling	No	Formally possible
Bell nonclassicality	No	Testable only as an open boundary
Physical quantum interpretation	Unsupported	Still unsupported

Therefore:

CAPM Is the Local Kernel.
The Architecture Supplies the World. (T.4)

T.6 Scalar complex state versus composite derivative state

Property	Scalar state Z = R+iQ	Composite state ρ_UD
Number of sectors	One	At least two
State space	$\mathbb{C}$ or one complex channel	$\mathscr{H}_U \otimes \mathscr{H}_D$
Local states	Undefined	ρ_U and ρ_D
Partial trace	Not available	Available
Product state	Not meaningful	$\rho_U \otimes \rho_D$
Separability	Not meaningful	Formally definable
Relative phase	Requires channel extension	Natural across branches
Entanglement witness	Impossible	Formally possible

Property	Scalar state Z = R+iQ	Composite state ρ_UD
Contract preparation	External to state	Built into joint-state construction
Observer-local incompleteness	Limited	Explicit
Main use	Radial–angular valuation decomposition	Relational state and observer analysis

A two-coordinate scalar state is not a two-subsystem state.

Therefore:

R and Q Are Coordinates of One State.
U and D Are Candidate Subsystems. (T.5)

T.7 Correlation, coupling, and entanglement ladder

Level	Name	Formal signature	Financial example	Current status
E_1	Statistical correlation	$p(u,d) \neq p(u)p(d)$	Option price moves with underlying	Established
E_2	Functional dependence	$D = V(U,\sigma,r,\ldots)$	Payoff and pricing dependence	Established
E_3	Contractual coupling	Derivative identity contains underlying relation	Strike, maturity, payoff, settlement	Established
E_4	Dynamical binding	$\partial F_U/\partial D \neq 0$ and $\partial F_D/\partial U \neq 0$	Hedging and volatility feedback	Established conditionally
E_5	Effective nonfactorization	$\rho_{UD} \neq \rho_U \otimes \rho_D$	Joint contract-state representation	Formally constructible
E_6	Coherent composite state	Observable $\rho_{off} \neq 0$	Phase-sensitive joint branch interaction	Open
E_7	Irreducible contextuality	No global noncontextual value map	Context- and order-dependent measurement	Not established
E_8	No-signalling entanglement	Local marginals independent of remote settings	Passive isolated joint test	Not established
E_9	Bell nonclassicality		S	> 2 under valid assumptions

The strongest presently defensible general statement is:

Derivative Finance Clearly Realizes E_1–E_4.
It Provides a Formal Test Platform for E_5–E_9. (T.6)

T.8 Classical mixture versus coherent state

Property	Classical mixture	Coherent state
State	$\rho_{mix} = \Sigma_n q_n$	$n\rangle\langle n$
Branch weights	q_n	

Property	Classical mixture	Coherent state
Relative phase	Absent or irrelevant	Present
Off-diagonal terms	Zero	Potentially nonzero
Payoff-basis predictions	Weighted expectation	Usually identical
Rotated-basis predictions	No interference cross term	Phase-sensitive cross term
Local marginals	May be mixed	May be identical to mixture
Required measurement	Ordinary branch measurement	Cross-branch observable
Strong classical nulls needed	No	Yes
Financial status	Mature default	Open empirical hypothesis
Physical quantum implication	None	Still none by itself

The empirical distinction is:

$\mathrm{Tr}(\rho_{coh}\hat{O}_{cross}) \neq \mathrm{Tr}(\rho_{mix}\hat{O}_{cross})$. (T.7)

Without an admissible $\hat{O}_{cross}$:

Coherent State and Mixture Are Operationally Indistinguishable. (T.8)

T.9 Circular phase, hyperbolic rapidity, and time variables

Variable	Definition	Mathematical role	Financial interpretation
θ	$\mathrm{atan2}(Q_E,R)$	Circular angle	Valuation orientation
η_Q	$\mathrm{artanh}(Q_E/A)$	Q-preserving rapidity	Hyperbolic pressure coordinate
ρ_D	$\ln(A/R)$	Additive discount rapidity	Relative valuation-frame displacement
u_D	$\tanh \rho_D$	Bounded rapidity coordinate	Normalized frame relation
t	External calendar coordinate	Duration	Market and contractual time
k	Ledger-event index	Discrete causal order	Number and order of committed events
τ_F	Optional effective proper time	Local process clock	Gate- and phase-weighted internal duration
Theta_D	$\partial D/\partial t$	Option Greek	Time decay of derivative value

Relations:

$\sin\theta = \tanh\eta_Q$. (T.9)

$\rho_D = \ln\cosh\eta_Q$. (T.10)

$\rho_D = -\ln\cos\theta$. (T.11)

But:

$\theta \neq \eta_Q \neq \rho_D \neq t \neq k$. (T.12)

T.10 Euclidean and Lorentz-like embeddings

Feature	Euclidean completion	Q-preserving hyperbolic embedding	Discount-composition embedding
Core relation	$A^2 = R^2 + Q_E^2$	$A^2 - Q_E^2 = R^2$	$U_H^2 - Q_H^2 = A^2$
Main pressure coordinate	Q_E	Q_E	Q_H
Main parameter	θ	η_Q	ρ_D
Parameter definition	$\cos\theta = R/A$	$\tanh\eta_Q = Q_E/A$	$\rho_D = \ln(A/R)$
Preserved structure	Euclidean amplitude	Original Q_E	Multiplicative valuation ratio
Composition benefit	Circular rotation	Hyperbolic pressure map	Additive frame rapidity
Best use	Local complex valuation	Reinterpret retained pressure	Cross-frame valuation transport
Main warning	Not a tensor state	η_Q is not θ	Q_H is not Q_E

The pressure relation is:

$$Q_H = Q_E^2/(2R). \quad (T.13)$$

The rapidity relation is:

$$\rho_D = \ln\cosh\eta_Q. \quad (T.14)$$

T.11 Metric, gauge, and ledger geometry

Structure	Metric geometry	Gauge geometry	Ledger geometry
Main object	$g^{F}_{\mu\nu}$	$\mathcal{A}_\mu$	L_k
What it measures	Distance or transition cost	Internal orientation transport	Historical accessibility
Connection	$\Gamma^{\rho}_{\mu\nu}$	$\mathcal{A}_\mu$	Ledger update rule
Curvature/ residual	$R^{\rho}_{\sigma\mu\nu}$	$\mathbb{F}_{\mu\nu}$	$H_L(\gamma)$
Closed-loop effect	Tangent vector changes	Phase/orientation changes	History changes
Example	Liquidity stress alters distance	Funding–collateral loop alters phase	Margin call remains recorded
Local invariant	ds_F^2	Relative phase or holonomy	Record identity and order
Main null model	State-space or covariance model	Frame correction	Ordinary database history
Main danger	Calling covariance “gravity”	Calling arbitrary phase “gauge”	Treating every timestamp as causal trace

The total loop structure is:

$$\mathrm{Hol}_{\mathrm{total}}(\gamma) = \mathrm{Hol}_{\mathrm{metric}}(\gamma) \times \mathrm{Hol}_{\mathrm{gauge}}(\gamma) \times \mathrm{Hol}_{\mathrm{ledger}}(\gamma). \quad (T.15)$$

T.12 Measurement, conditioning, gate, and ledger

Stage	Input	Output	Does it create a historical fact?
Preparation	Primary state and contract	Joint possibility state	No
Evolution	Current state	Updated possibility state	No
Measurement	State and setting	Provisional outcome	Not necessarily
Conditioning	Outcome	Conditional state	No by itself
Gate	Outcome, evidence, authority	Commit, reject, defer, escalate	Potentially
Record creation	Gate decision	Structured event record	Yes when consequential
Ledger update	Prior ledger and record	New ledger	Yes
Policy update	New ledger	Revised observer policy	Changes future
Backreaction	Record and policy	Primary-world intervention	Changes future world

Therefore:

Measurement Outcome ≠ Commitment. (T.16)

Conditioning ≠ Public Knowledge. (T.17)

Commitment + Trace + Future Constraint = Historical Event. (T.18)

T.13 Primary causation versus secondary conditional update

Question	Primary-universe explanation	Secondary-world description
Why option follows underlying	Pricing rule and payoff contract	Prepared joint state
Why derivative state changes after underlying result	Information transmission and contract evaluation	Conditional state update
Why underlying may react to derivative	Hedge and market impact	Backreaction term
Why outcomes appear simultaneous	Shared computation and event protocol	One joint θ-episode
Why local observer lacks full explanation	Incomplete information access	Reduced local state
Why record becomes definite	Institutional authorization and settlement	Gate and ledger latching
Why this is not Bell nonlocality	Ordinary causal channels remain	No-signalling not guaranteed

The two descriptions are complementary.

Primary Causation Explains Construction.
Secondary State Explains Internal Relational Form. (T.19)

T.14 Decoherence-like finance versus quantum decoherence

Feature	Financial institutional classicalization	Quantum decoherence
Environment	Market, clearing, law, settlement, records	Physical environment
Branch distinction	Different contractual or institutional outcomes	Quantum alternatives
Off-diagonal interpretation	Effective model relation	Physical quantum coherence
Record mechanism	Trade, settlement, legal and database trace	Environment correlations
Pointer-like state	Publicly accepted price or settlement result	Stable quantum pointer basis
Irreversibility source	Cost, law, authority, institutional finality	Effective environmental entanglement
Recoherence	Reopened or revised protocol	Physical phase restoration
Main similarity	Branch distinguishability suppresses cross terms	Same formal suppression structure
Main difference	May be wholly classical	Empirically quantum

Thus:

Institutional Classicalization
Is a Functional Homologue of Decoherence,
Not Proof of Physical Decoherence. (T.20)

T.15 Financial holonomy versus ordinary path dependence

Test	Ordinary accumulated cost	Classical hysteresis	Gauge-like holonomy	Ledger holonomy
Endpoint returns	Possible	Possible	Required for loop test	Possible
Path matters	Yes	Yes	Yes	Yes
Orientation reversal changes sign	Usually no	Sometimes	Often expected for phase	Generally no
Small-loop area scaling	Not necessary	Not necessary	Candidate signature	Not necessary
State frame invariance	Not required	Not required	Required	Record-based
Persistent history	Through cumulative cost	Through hidden memory	Through transported orientation	Through explicit trace
Example	Funding expense	Leverage cycle	Phase after protocol loop	Margin event remains recorded

Test	Ordinary accumulated cost	Classical hysteresis	Gauge-like holonomy	Ledger holonomy
Strongest null	Direct accounting	Hysteresis model	Classical transport model	Ordinary event history

A geometric-phase claim is justified only when it predicts more than:

Known Cost + Known Memory. (T.21)

T.16 Local flatness versus global curvature

Feature	Local flat financial regime	Globally curved financial regime
CAPM parameters	Approximately stable	State-dependent
Greeks	Reliable locally	Drift across paths and regimes
Metric	Nearly constant	Varies with state and ledger
Connection	Approximately removable	Path-dependent
Transport	Endpoint-dominated	Path-sensitive
Frame map	One local mapping sufficient	Multiple charts required
Typical condition	Stable liquidity and funding	Stress, leverage, collateral interaction
Model choice	Local CAPM or standard derivatives	State-dependent geometry
Diagnostic	Small transport residual	Persistent triangle or loop residual

The local-to-global relation is:

CAPM_local

- Transport
- Curvature Correction
  = Global Financial Architecture. (T.22)

T.17 Quantum-characteristic comparison matrix

Characteristic	Reproduced by Part II?	Strength of reproduction	Remaining quantum residue
Complex state	Yes	Strong formal reconstruction	None specific
Phase	Yes	Strong local and relative-phase model	Mandatory physical coherence
Superposition form	Yes	Formal channel representation	Physical amplitude ontology
Interference form	Yes	Operationally testable	Classical-null rejection
Composite state	Yes	Strong structural representation	Physical subsystem status
Local mixedness	Yes	Explicit through partial trace	Classical purification ambiguity
Conditional collapse	Yes	Explicit internal update	Born rule and no-signalling

Characteristic	Reproduced by Part II?	Strength of reproduction	Remaining quantum residue
Context dependence	Yes	Strong	Irreducible contextuality
Noncommutativity	Yes	Operational instrument order	Fundamental commutator structure
Decoherence-like suppression	Yes	Strong institutional analogue	Physical quantum decoherence
Geometric phase	Yes	Formal and testable	Classical hysteresis distinction
No-signalling	Testable only	Not generally realized	Remains
Bell violation	Testable only	Not established	Remains
Born rule	Formal lift only	Not derived	Remains
No-cloning	No	Weak analogy only	Remains
Monogamy	No general law	Testable resource analogue	Remains
Quantum statistics	No	Not reproduced	Remains

T.18 Strongest defensible claim by evidence level

Evidence achieved	Permitted claim
Complex coordinates defined	“A complex financial representation is available.”
Q improves prediction	“Retained pressure is operationally useful.”
θ predicts transition	“A phase-based state variable improves dynamics.”
Joint model beats local models	“Relational state information matters.”
Off-diagonal model improves held-out prediction	“Candidate operational coherence exists.”
Controlled phase shift succeeds	“Relative phase has intervention-level predictive value.”
Separable states fail	“Representation-relative nonseparability is supported.”
Contextual classical models fail	“Irreducible contextual-style structure is a candidate.”
No-signalling holds	“The joint statistics are no-signalling-compatible.”
Bell assumptions controlled and bound violated	“Tested Bell-classical models fail under the declared protocol.”
Independent physical evidence exists	Only then consider physical quantum interpretation

T.19 Common invalid inference table

Invalid inference	Why invalid	Corrected statement
Complex number → quantum system	Classical systems use complex numbers	Complex representation is quantum-compatible but not quantum-specific
Correlation → entanglement	Separable states can be strongly correlated	Test separability
Contract dependence → quantum nonlocality	Contract provides ordinary causal structure	Describe contractual coupling

Invalid inference	Why invalid	Corrected statement
Conditional update → signalling	Conditional beliefs can change without local marginal change	Test no-signalling separately
Order dependence → quantum contextuality	Memory and disturbance also produce order effects	Reject classical contextual models first
Closed-loop residue → Berry phase	Costs and hysteresis also produce loop effects	Test gauge invariance and area scaling
State-dependent metric → gravity	Many systems have variable metrics	Show source–geometry backreaction
Risk-neutral weights → Born rule	Pricing weights are constructed financially	Born necessity remains unproved
Local mixedness → entanglement	Classical mixtures also produce mixed marginals	Test the separable set
CHSH > 2 → quantum finance	Signalling, memory, and selection may inflate S	Audit assumptions first
Observer ignorance → entanglement	Ignorance explains experience, not global structure	Require objective nonfactorization
Similar equation → same ontology	Functional homology need not imply material identity	Limit the claim to role correspondence

---

T.20 Theory-reduction table

Failed layer	Retain	Remove or demote
A not independently defined	Standard valuation R	Q and θ construction
Q non-predictive	R and A	Q as causal variable
θ unstable	Static R+iQ geometry	θ as internal time
Frame maps fail	Local valuation models	Lorentz-like composition
Metric adds no value	Ordinary state-space model	Curved geometry
Gauge correction fails	Direct frame reconciliation	Gauge connection
Holonomy reduces to cost	Path accounting	Geometric-phase claim
Joint state adds no value	Local or classical joint model	Tensor architecture
Coherence unobservable	Classical mixture	Off-diagonal state
Separable state fits	Contractual coupling	Entanglement claim
No-signalling fails	Causal feedback model	Bell interpretation
Bell bound not violated	Observer-bearing classical world	Bell-nonclassical claim

The correct response to failure is:

Reduce the Theory,
Not Redefine the Evidence. (T.23)

T.21 Recommended reading path through the appendices

Reader goal	Recommended appendices
Understand notation	Appendix A

Reader goal	Recommended appendices
See simplest option example	Appendix B
Understand mixture versus coherence	Appendix C
Understand the three clocks	Appendix D
Understand contract runtime	Appendix E
Compare Lorentz embeddings	Appendix F
Understand metric, gauge, and holonomy	Appendix G
Understand separability and Bell tests	Appendix H
Review assumptions and evidence	Appendix I
Compare Part I and Part II capability	Appendix J
Build a software prototype	Appendix K
Use disciplined publication language	Appendix L
See the full architecture	Appendix M
Follow a numerical worked example	Appendix N
Build a CHSH-style sandbox	Appendix O
Estimate curvature and holonomy	Appendix P
Model observer agreement	Appendix Q
Operate the falsification dashboard	Appendix R
Use the glossary and one-page summary	Appendix S
Compare all layers quickly	Appendix T

T.22 Final comparison table

Question	Final answer
Is CAPM itself quantum?	No
Can CAPM support a complex valuation extension?	Yes
Does R+iQ create entanglement?	No
Do derivatives supply a natural composite-state grammar?	Yes
Can internal observers experience global relations as locally incomplete?	Yes
Can the architecture represent interference form?	Yes
Is financial coherence empirically established?	No
Can clearing and settlement model decoherence-like classicalization?	Yes
Can relative financial frames be modelled hyperbolically?	Yes, under declared invariants
Can global financial geometry be curved?	As an effective model, yes
Is a financial Einstein equation established?	No
Can gauge transport and holonomy be modelled?	Yes, subject to empirical gain
Can finance reproduce no-signalling entanglement?	Not generally
Has Bell nonclassicality been established in finance?	No

Question	Final answer
Does the model derive the Born rule?	No
Does the model reduce quantum mystery?	It narrows which features remain specifically quantum
Is the architecture complete as a formal article?	Yes
What remains?	Editing, diagrams, empirical implementation, and publication formatting

---

T.23 Final appendix proposition

Proposition T.1 — Comparison Discipline

A cross-domain architecture remains coherent when every comparison states:

1. the objects being compared;
2. the mathematical role they share;
3. the role in which they differ;
4. the evidence supporting the correspondence;
5. the interpretation that remains prohibited.

In compact form:

ValidComparison
⇔ SharedRole
∧ DistinctSubstrate
∧ EvidenceBound
∧ ExplicitDifference
∧ InterpretationLimit. (T.24)

With Appendix T added, the article is structurally complete.

Reference

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.